

MERE REALITY

RATIONALITY: FROM AI TO ZOMBIES

in six volumes

- I Map and Territory
- II How to Actually Change Your Mind
- III The Machine in the Ghost
- IV Mere Reality
- V Mere Goodness
- VI Becoming Stronger

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Rationality

From AI to Zombies

BOOK IV

Mere Reality

PERSONAL PRINT EDITION

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The World: An Introduction

by Rob Bensinger

Previous essays have discussed human reasoning, language, goals, and social dynamics. Mathematics, physics, and biology were cited to explain patterns in human behavior, but little has been said about humanity's place in nature, or about the natural world in its own right.

Just as it was useful to contrast humans *as goal-oriented systems* with inhuman processes in evolutionary biology and artificial intelligence, it will be useful in the coming sequences of essays to contrast humans *as physical systems* with inhuman processes that *aren't* mind-like.

We humans are, after all, built out of inhuman parts. The world of atoms looks nothing like the world as we ordinarily think of it, and certainly looks nothing like the world's conscious denizens as we ordinarily think of them. As Giulio Giorello put the point in an interview with Daniel Dennett: "Yes, we have a soul. But it's made of lots of tiny robots."¹

Mere Reality collects seven sequences of essays on this topic. The first three introduce the question of how the human world relates to the world revealed by physics: "Lawful Truth" (on the basic links between physics and human cognition), "Reductionism 101" (on the project of scientifically explaining phenomena), and "Joy in the Merely Real" (on the emotional, personal significance of the scientific world-view). This is followed by two sequences that go into more depth on specific academic debates: "Physicalism 201" (on the hard problem of consciousness) and "Quantum Physics and Many Worlds" (on the measurement problem in physics). Finally, the sequence "Science and Rationality" and the essay A Technical Explanation of Technical Explanation tie these ideas together and relate them to scientific practice.



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1. Daniel C. Dennett, *Freedom Evolves* (Viking Books, 2003).

The discussions of consciousness and quantum physics illustrate the relevance of reductionism to present-day controversies in science and philosophy. For those interested in a bit of extra context, I'll say a few more words about those two topics here. For those eager to skip ahead: skip ahead!

Minds in the World

Can we ever know what it's like to be a bat?

We can certainly develop better cognitive models for predicting bat behavior, or more fine-grained models of bat neurology—but it isn't obvious that this would tell us what echolocation subjectively feels like, or what flying feels like, *from the bat's point of view*.

Indeed, it seems as though we could never even be certain that there *is* anything it's like to be a bat. Why couldn't an unconscious automaton replicate all the overt behaviors of a conscious agent to arbitrary precision? (Philosophers call such automata “zombies,” though they have little in common with the zombies of folklore—who are *quite visibly* different from conscious agents!)

A race of alien psychologists would run into the same problem in trying to model *human* consciousness. They might arrive at a perfect predictive model of what we say and do when we see a red rose, but that wouldn't mean that the aliens fully understand what redness feels like “from the inside.”

Running with examples like these, philosophers like Thomas Nagel and David Chalmers have argued that third-person cognitive and neural models can never fully capture first-person consciousness.^{2,3} No matter how much we know about a physical system, it is always logically possible, on this view, that the system has no first-person experiences. Traditional dualism,

2. David J. Chalmers, *The Conscious Mind: In Search of a Fundamental Theory* (New York: Oxford University Press, 1996).

3. Thomas Nagel, “What Is It Like to Be a Bat?,” *Philosophical Review* 83, no. 4 (1974): 435–450, sharov.me/r/4-1.



with its immaterial souls freely floating around violating physical laws, may be false; but Chalmers insists on a weaker thesis, that consciousness is a “further fact” not fully explainable by the physical facts.

A number of philosophers and scientists have found this line of reasoning persuasive.⁴ If we feel this argument’s intuitive force, should we grant its conclusion and ditch physicalism?

We certainly shouldn’t reject it just because it *sounds strange* or feels vaguely unscientific. But how does the argument stand up to a *technical* understanding of how explanation and belief work? Are there any hints we can take from the history of science, or from our understanding of the physical mechanisms underlying evidence? “Physicalism 201” will return to this question.

Worlds in the World

Quantum mechanics is our best mathematical model of the universe to date, powerfully confirmed by a century of tests. The theory posits a complex-numbered “probability amplitude,” so called because a specific operation (squaring the number’s absolute value—the Born rule) lets us probabilistically predict phenomena at small scales and extreme energy levels. This amplitude changes deterministically in accord with the Schrödinger equation. In the process, it often enters odd states called “superpositions.”

Yet when we perform experiments, the superpositions seem to vanish without a trace. When we aren’t looking, the

4. In a survey of Anglophone professional philosophers, 56.5% endorsed physicalism, 27.1% endorsed anti-physicalism, and 16.4% endorsed other views (e.g., “I don’t know”).⁵ Most philosophers reject the metaphysical possibility of Chalmers’s “zombies,” but there is no consensus about *why*, exactly, Chalmers’s zombie argument fails. Kirk summarizes contemporary positions on phenomenal consciousness, giving arguments that resemble Yudkowsky’s against the possibility of knowing or referring to irreducible qualia.⁶

5. David Bourget and David J. Chalmers, “What Do Philosophers Believe?,” *Philosophical Studies* (2013): 1–36.

6. Robert Kirk, *Mind and Body* (McGill-Queen’s University Press, 2003).

Schrödinger equation appears to capture everything there is to know about the dynamics of physical systems. When we *are* looking, though, this clean determinism is replaced by Born's probabilistic rule. It's as though the ordinary laws of physics are suddenly suspended whenever we make "observations." As John Stewart Bell put the point:

It would seem that the theory is exclusively concerned about "results of measurements" and has nothing to say about anything else. What exactly qualifies some physical systems to play the role of the "measurer"? Was the wavefunction of the world waiting to jump for thousands of millions of years until a single-celled living creature appeared? Or did it have to wait a little longer, for some better qualified system ...with a PhD?

Everyone agrees that this strange mix of Schrödinger and Born's rules has proved empirically adequate. However, the question of exactly *when* Born's rule enters the mix, and what it all *means*, has produced a chaos of different views on the nature of quantum mechanics.

Early on, the Copenhagen school—Niels Bohr and other originators of quantum theory—splintered into several standard ways of talking about the experimental results and the odd formalism used to predict them. Some, taking the theory's focus on "measurements" and "observations" quite literally, proposed that consciousness plays a fundamental role in physical law, intervening to cause complex amplitudes to "collapse" into observables. Others, led by Werner Heisenberg, advocated a non-realistic view according to which physics is about our states of knowledge rather than about any objective reality. Yet another Copenhagen tradition, summed up in the slogan "shut up and calculate," warned against metaphysical speculation of all kinds.

Yudkowsky uses this scientific controversy as a proving ground for some central ideas from previous sequences: map-territory distinctions, mysterious answers, Bayesianism, and Occam's Razor. Since he is not a physicist—and neither am I—

I'll provide some outside sources here for readers who want to vet his arguments or learn more about his physics examples.

Tegmark's *Our Mathematical Universe* discusses a number of relevant ideas in philosophy and physics.⁷ Among Tegmark's more novel ideas is his argument that all consistent mathematical structures exist, including worlds with physical laws and boundary conditions entirely unlike our own. He distinguishes these Tegmark worlds from multiverses in more scientifically mainstream hypotheses—e.g., worlds in stochastic eternal inflationary models of the Big Bang and in Hugh Everett's many-worlds interpretation of quantum physics.

Yudkowsky discusses many-worlds interpretations at greater length, as a response to the Copenhagen interpretations of quantum mechanics. Many-worlds has become very popular in recent decades among physicists, especially cosmologists. However, a number of physicists continue to reject it or maintain agnosticism. For a (mostly) philosophically mainstream introduction to this debate, see Albert's *Quantum Mechanics and Experience*.⁸ See also the *Stanford Encyclopedia of Philosophy*'s introduction to "Measurement in Quantum Theory,"⁹ and their introduction to several of the views associated with "many worlds" in "Everett's Relative-State Formulation"¹⁰ and "Many-Worlds Interpretation."¹¹

On the less theoretical side, Epstein's *Thinking Physics* is a great text for training physical intuitions.¹² It's worth keeping in mind that just as one can understand most of cognitive sci-

7. Max Tegmark, *Our Mathematical Universe: My Quest for the Ultimate Nature of Reality* (Random House LLC, 2014).

8. David Z. Albert, *Quantum Mechanics and Experience* (Harvard University Press, 1994).

9. Henry Krips, "Measurement in Quantum Theory," in *The Stanford Encyclopedia of Philosophy*, Fall 2013, ed. Edward N. Zalta.

10. Jeffrey Barrett, *Everett's Relative-State Formulation of Quantum Mechanics*, ed. Edward N. Zalta, sharov.me/r/4-2.



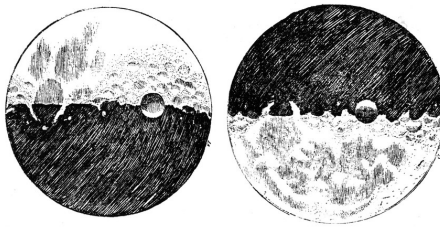
11. Lev Vaidman, "Many-Worlds Interpretation of Quantum Mechanics," in *The Stanford Encyclopedia of Philosophy*, Fall 2008, ed. Edward N. Zalta.

12. Lewis Carroll Epstein, *Thinking Physics: Understandable Practical Reality, 3rd Edition* (Insight Press, 2009).

ence without understanding the nature of subjective awareness, one can understand most of physics without having a settled view of the ultimate nature (and size!) of the physical world.

PART O

Lawful Truth



Universal Fire

In L. Sprague de Camp's fantasy story *The Incomplete Enchanter* (which set the mold for the many imitations that followed), the hero, Harold Shea, is transported from our own universe into the universe of Norse mythology.¹ This world is based on magic rather than technology; so naturally, when Our Hero tries to light a fire with a match brought along from Earth, the match fails to strike.

I realize it was only a fantasy story, but ...how do I put this ...
No.

In the late eighteenth century, Antoine-Laurent de Lavoisier discovered fire. "What?" you say. "Hasn't the use of fire been dated back for hundreds of thousands of years?" Well, yes, people *used* fire; it was hot, bright, sort of orangey-colored, and you could use it to cook things. But nobody knew how it worked. Greek and medieval alchemists thought that Fire was a basic thing, one of the Four Elements. In Lavoisier's time the alchemical paradigm had been gradually amended and greatly complicated, but fire was still held to be basic—in the form of "phlogiston," a rather mysterious substance which was said to explain fire, and also every other phenomenon in alchemy.

Lavoisier's great innovation was to weigh *all* the pieces of the chemical puzzle, both before and after the chemical reaction. It had previously been thought that some chemical transmutations changed the weight of the total material: If you subjected finely ground antimony to the focused sunlight of a burning glass, the antimony would be reduced to ashes after one hour, and the ashes would weigh one-tenth more than the original antimony—even though the burning had been accompanied by the loss of a thick white smoke. Lavoisier weighed *all* the components of such reactions, including the



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1. Lyon Sprague de Camp and Fletcher Pratt, *The Incomplete Enchanter* (New York: Henry Holt & Company, 1941).

air in which the reaction took place, and discovered that matter was neither created nor destroyed. If the burnt ashes increased in weight, there was a corresponding decrease in the weight of the air.

Lavoisier also knew how to separate gases, and discovered that a burning candle diminished the amount of one kind of gas, *vital air*, and produced another gas, *fixed air*. Today we would call them *oxygen* and *carbon dioxide*. When the *vital air* was exhausted, the fire went out. One might guess, perhaps, that combustion transformed *vital air* into *fixed air* and fuel to ash, and that the ability of this transformation to continue was limited by the amount of *vital air* available.

Lavoisier's proposal directly contradicted the then-current phlogiston theory. That alone would have been shocking enough, but it also turned out...

To appreciate what comes next, you must put yourself into an eighteenth-century frame of mind. Forget the discovery of DNA, which occurred only in 1953. Unlearn the cell theory of biology, which was formulated in 1839. Imagine looking at your hand, flexing your fingers...and having absolutely no idea how it worked. The anatomy of muscle and bone was known, but no one had any notion of "what makes it go"—why a muscle moves and flexes, while clay molded into a similar shape just sits there. Imagine *your own body* being composed of mysterious, incomprehensible gloop. And then, imagine discovering...

...that humans, in the course of breathing, consumed *vital air* and breathed out *fixed air*. People also ran on combustion! Lavoisier measured the amount of heat that animals (and Lavoisier's assistant, Seguin) produced when exercising, the amount of *vital air* consumed, and the *fixed air* breathed out. When animals produced more heat, they consumed more *vital air* and exhaled more *fixed air*. People, like fire, consumed fuel and oxygen; people, like fire, produced heat and carbon dioxide. Deprive people of oxygen, or fuel, and the light goes out.

Matches catch fire because of phosphorus—"safety matches" have phosphorus on the ignition strip; strike-anywhere matches have phosphorus in the match heads. Phosphorus

is highly reactive; pure phosphorus glows in the dark and may spontaneously combust. (Henning Brand, who purified phosphorus in 1669, announced that he had discovered Elemental Fire.) Phosphorus is thus also well-suited to its role in *adenosine triphosphate*, ATP, your body's chief method of storing chemical energy. ATP is sometimes called the "molecular currency." It invigorates your muscles and charges up your neurons. Almost every metabolic reaction in biology relies on ATP, and therefore on the chemical properties of phosphorus.

If a match stops working, so do you. You can't change just one thing.

The surface-level rules, "Matches catch fire when struck," and "Humans need air to breathe," are not obviously connected. It took centuries to discover the connection, and even then, it still seems like some distant fact learned in school, relevant only to a few specialists. It is all too easy to imagine a world where one surface rule holds, and the other doesn't; to suppress our credence in one belief, but not the other. But that is *imagination*, not reality. If your map breaks into four pieces for easy storage, it doesn't mean the territory is also broken into disconnected parts. Our minds store different surface-level rules in different compartments, but this does not reflect any division in the laws that govern Nature.

We can take the lesson further. Phosphorus derives its behavior from even deeper laws, electrodynamics and chromodynamics. "Phosphorus" is merely our *word* for electrons and quarks arranged a certain way. You cannot change the chemical properties of phosphorus without changing the laws governing electrons and quarks.

If you stepped into a world where matches failed to strike, you would cease to exist as organized matter.

Reality is laced together a lot more tightly than humans might like to believe.

Universal Law



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Antoine-Laurent de Lavoisier discovered that breathing (respiration) and fire (combustion) operated on the same principle. It was one of the most startling unifications in the history of science, for it brought together the mundane realm of matter and the sacred realm of life, which humans had divided into separate magisteria.

The first great simplification was that of Isaac Newton, who unified the course of the planets with the trajectory of a falling apple. The shock of this discovery was greater by far than Lavoisier's. It wasn't just that Newton had dared to unify the Earthly realm of base matter with the obviously different and sacred celestial realm, once thought to be the abode of the gods. Newton's discovery gave rise to the notion of a *universal law*, one that is the same everywhere and everywhen, with literally *zero* exceptions.

Human beings live in a world of surface phenomena, and surface phenomena are divided into leaky categories with plenty of exceptions. A tiger does not behave like a buffalo. Most buffalo have four legs, but perhaps this one has three. Why would anyone think there would be laws that hold everywhere? It's just so obviously untrue.

The only time when it seems like we would *want* a law to hold everywhere is when we are talking about moral laws—tribal rules of behavior. Some tribe members may try to take more than their fair share of the buffalo meat—perhaps coming up with some clever excuse—so in the case of moral laws we do seem to have an instinct to universality. Yes, the rule about dividing the meat evenly applies to *you*, right now, whether you like it or not. But even here there are exceptions. If—for some bizarre reason—a more powerful tribe threatened to spear all of you unless Bob received twice as much meat on just this one occasion, you'd give Bob twice as much meat. The idea of a rule with literally *no* exceptions seems insanely rigid, the product

of closed-minded thinking by fanatics so in the grip of their one big idea that they can't see the richness and complexity of the real universe.

This is the customary accusation made against scientists—the professional students of the richness and complexity of the real universe. Because *when you actually look at the universe*, it turns out to be, by human standards, insanely rigid in applying its rules. As far as we know, there has been *not one single* violation of Conservation of Momentum from the uttermost dawn of time up until now.

Sometimes—very rarely—we observe an apparent violation of our *models* of the fundamental laws. Though our scientific models may last for a generation or two, they are not stable over the course of centuries...but do not fancy that this makes the universe itself whimsical. That is mixing up the map with the territory. For when the dust subsides and the old theory is overthrown, it turns out that the universe *always was* acting according to the new generalization we have discovered, which once again is absolutely universal as far as humanity's knowledge extends. When it was discovered that Newtonian gravitation was a special case of General Relativity, it was seen that General Relativity had been governing the orbit of Mercury for decades before any human being knew about it; and it would later become apparent that General Relativity had been governing the collapse of stars for billions of years before humanity. It is only our model that was mistaken—the Law itself was always absolutely constant—or so our new model tells us.

I may repose only 80% confidence that the lightspeed limit will last out the next hundred thousand years, but this does not mean that I think the lightspeed limit holds only 80% of the time, with occasional exceptions. The proposition to which I assign 80% probability is that the lightspeed law is *absolutely inviolable throughout the entirety of space and time*.

One of the reasons the ancient Greeks didn't discover science is that they didn't realize you could generalize from experiments. The Greek philosophers were interested in "normal" phenomena. If you set up a contrived experiment, you would

probably get a “monstrous” result, one that had no implications for how things really worked.

So that is how humans tend to dream, before they learn better; but what of the universe’s own quiet dreams that it dreamed to itself before ever it dreamed of humans? If you would learn to think like reality, then here is the Tao:

*Since the beginning
not one unusual thing
has ever happened.*

Is Reality Ugly?

Consider the cubes, $\{1, 8, 27, 64, 125, \dots\}$. Their first differences $\{7, 19, 37, 61, \dots\}$ might at first seem to lack an obvious pattern, but taking the second differences $\{12, 18, 24, \dots\}$ takes you down to the simply related level. Taking the third differences $\{6, 6, \dots\}$ brings us to the perfectly stable level, where chaos dissolves into order.

But this is a handpicked example. Perhaps the “messy real world” lacks the beauty of these abstract mathematical objects? Perhaps it would be more appropriate to talk about neuroscience or gene expression networks?

Abstract math, being constructed solely in imagination, arises from simple foundations—a small set of initial axioms—and is a closed system; conditions that might seem *unnaturally* conducive to neatness.

Which is to say: In pure math, you don’t have to worry about a tiger leaping out of the bushes and eating Pascal’s Triangle.

So is the real world uglier than mathematics?

Strange that people ask this. I mean, the question might have been sensible two and a half millennia ago ...Back when the Greek philosophers were debating what this “real world” thingy might be made of, there were many positions. Heraclitus said, “All is fire.” Thales said, “All is water.” Pythagoras said, “All is number.”

Score:

Heraclitus:	0
Thales:	0
Pythagoras:	1

Beneath the complex forms and shapes of the surface world, there is a simple level, an exact and stable level, whose laws we name “physics.” This discovery, the Great Surprise, has already taken place at our point in human history—but it does not do



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to forget that it was surprising. Once upon a time, people went in search of underlying beauty, with no guarantee of finding it; and once upon a time, they found it; and now it is a known thing, and taken for granted.

Then why can't we predict the location of every tiger in the bushes as easily as we predict the sixth cube?

I count three sources of uncertainty even within worlds of pure math—two obvious sources, and one not so obvious.

The first source of uncertainty is that even a creature of pure math, living embedded in a world of pure math, may not know the math. Humans walked the Earth long before Galileo/Newton/Einstein discovered the law of gravity that prevents us from being flung off into space. You can be governed by stable fundamental rules without knowing them. There is no law of physics which says that laws of physics must be explicitly represented, as knowledge, in brains that run under them.

We do not yet have the Theory of Everything. Our best current theories are things of math, but they are not perfectly integrated with each other. The most probable explanation is that—as has previously proved to be the case—we are seeing surface manifestations of deeper math. So by far the best guess is that reality is made of math; but we do not fully know which math, yet.

But physicists have to construct huge particle accelerators to distinguish between theories—to manifest their remaining uncertainty in any visible fashion. That physicists must go to such lengths to be unsure, suggests that this is not the source of our uncertainty about stock prices.

The second obvious source of uncertainty is that even when you know all the relevant laws of physics, you may not have enough computing power to extrapolate them. We know every fundamental physical law that is relevant to a chain of amino acids folding itself into a protein. But we still can't predict the shape of the protein from the amino acids. Some tiny little 5-nanometer molecule that folds in a microsecond is *too much information* for current computers to handle (never mind tigers and stock prices). Our frontier efforts in protein folding use

clever approximations, rather than the underlying Schrödinger equation. When it comes to describing a 5-nanometer object using *really* basic physics, over quarks—well, you don’t even bother trying.

We have to use instruments like X-ray crystallography and NMR to discover the shapes of proteins that are fully determined by physics we know and a DNA sequence we know. We are not logically omniscient; we cannot see all the implications of our thoughts; we do not know what we believe.

The third source of uncertainty is the most difficult to understand, and Nick Bostrom has written a book about it. Suppose that the sequence $\{1, 8, 27, 64, 125, \dots\}$ exists; suppose that this is a fact. And suppose that atop each cube is a little person—one person per cube—and suppose that this is also a fact.

If you stand on the outside and take a global perspective—looking down from above at the sequence of cubes and the little people perched on top—then these two facts say everything there is to know about the sequence and the people.

But if you are one of the little people perched atop a cube, and you know these two facts, there is still a third piece of information you need to make predictions: “Which cube am *I* standing on?”

You expect to find yourself standing on a cube; you do not expect to find yourself standing on the number 7. Your anticipations are definitely constrained by your knowledge of the basic physics; your beliefs are falsifiable. But you still have to look down to find out whether you’re standing on 1,728 or 5,177,717. If you can do fast mental arithmetic, then seeing that the first two digits of a four-digit cube are 17__ will be sufficient to guess that the last digits are 2 and 8. Otherwise you may have to look to discover the 2 and 8 as well.

To figure out what the night sky should look like, it’s not enough to know the laws of physics. It’s not even enough to have logical omniscience over their consequences. You have to know *where* you are in the universe. You have to know that you’re looking up at the night sky *from Earth*. The information required is not just the information to locate Earth in the *visible* universe, but in the entire universe, including all the parts

that our telescopes can't see because they are too distant, and different inflationary universes, and alternate Everett branches.

It's a good bet that "uncertainty about initial conditions at the boundary" is really indexical uncertainty. But if not, it's empirical uncertainty, uncertainty about how the universe *is* from a global perspective, which puts it in the same class as uncertainty about fundamental laws.

Wherever our best guess is that the "real world" has an *irretrievably* messy component, it is because of the second and third sources of uncertainty—logical uncertainty and indexical uncertainty.

Ignorance of fundamental laws does not tell you that a messy-looking pattern really is messy. It might just be that you haven't figured out the order yet.

But when it comes to messy gene expression networks, we've *already found* the hidden beauty—the stable level of underlying physics. *Because* we've already found the master order, we can guess that we won't find any *additional* secret patterns that will make biology as easy as a sequence of cubes. Knowing the rules of the game, we know that the game is hard. We don't have enough computing power to do protein chemistry from physics (the second source of uncertainty) and evolutionary pathways may have gone different ways on different planets (the third source of uncertainty). New discoveries in basic physics won't help us here.

If you were an ancient Greek staring at the raw data from a biology experiment, you would be much wiser to look for some hidden structure of Pythagorean elegance, all the proteins lining up in a perfect icosahedron. But in biology we already know where the Pythagorean elegance is, and we know it's too far down to help us overcome our indexical and logical uncertainty.

Similarly, we can be confident that no one will ever be able to predict the results of certain quantum experiments, only because our fundamental theory tells us quite definitely that different versions of us will see different results. If your knowledge of fundamental laws tells you that there's a sequence of cubes, and that there's one little person standing on top of each

cube, and that the little people are all alike except for being on different cubes, and that you are one of these little people, then you *know* that you have no way of deducing which cube you're on except by looking.

The best current knowledge says that the “real world” is a perfectly regular, deterministic, and *very large* mathematical object which is highly expensive to simulate. So “real life” is less like predicting the next cube in a sequence of cubes, and more like knowing that lots of little people are standing on top of cubes, but not knowing who *you personally* are, and also not being very good at mental arithmetic. Our knowledge of the rules does constrain our anticipations, quite a bit, but not perfectly.

There, now doesn't *that* sound like real life?

But uncertainty exists in the map, not in the territory. If we are ignorant of a phenomenon, that is a fact about our state of mind, not a fact about the phenomenon itself. Empirical uncertainty, logical uncertainty, and indexical uncertainty are just names for our own bewilderment. The best current guess is that the world is math and the math is perfectly regular. The messiness is only in the eye of the beholder.

Even the huge morass of the blogosphere is embedded in this perfect physics, which is ultimately as orderly as $\{1, 8, 27, 64, 125, \dots\}$.

So the Internet is not a big muck ...it's a series of cubes.

Beautiful Probability



discussion
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Should we expect rationality to be, *on some level*, simple? Should we search and hope for *underlying* beauty in the arts of belief and choice?

Let me introduce this issue by borrowing a complaint of the late great Bayesian Master, E. T. Jaynes:¹

Two medical researchers use the same treatment independently, in different hospitals. Neither would stoop to falsifying the data, but one had decided beforehand that because of finite resources he would stop after treating $n = 100$ patients, however many cures were observed by then. The other had staked his reputation on the efficacy of the treatment, and decided he would not stop until he had data indicating a rate of cures definitely greater than 60%, however many patients that might require. But in fact, both stopped with exactly the same data: $n = 100$ [patients], $r = 70$ [cures]. Should we then draw different conclusions from their experiments?" [Presumably the two control groups also had equal results.]

1. Edwin T. Jaynes, "Probability Theory as Logic," in *Maximum Entropy and Bayesian Methods*, ed. Paul F. Fougère (Springer Netherlands, 1990).

Cyan² directs us to chapter 37³ of MacKay's excellent statistics book, free online, for a more thorough explanation of this problem.⁴

According to old-fashioned statistical procedure—which I believe is still being taught today—the two researchers have performed different experiments with different stopping conditions. The two experiments *could* have terminated with different data, and therefore represent different tests of the hypothesis, requiring different statistical analyses. It's quite possible that the first experiment will be “statistically significant,” the second not.

Whether or not you are disturbed by this says a good deal about your attitude toward probability theory, and indeed, rationality itself.

Non-Bayesian statisticians might shrug, saying, “Well, not all statistical tools have the same strengths and weaknesses, y'know—a hammer isn't like a screwdriver—and if you apply different statistical tools you may get different results, just like using the same data to compute a linear regression or train a regularized neural network. You've got to use the right tool for the occasion. Life is messy—”

And then there's the Bayesian reply: “Excuse *you*? The evidential impact of a fixed experimental method, producing the same data, depends on the researcher's private thoughts? And you have the nerve to accuse *us* of being ‘too subjective’?”

If Nature is one way, the likelihood of the data coming out the way we have seen will be one thing. If Nature is another

2. sharov.me/r/4-3

Beautiful Probability • lesswrong.com

3. sharov.me/r/4-4

inference.phy.cam.ac.uk

4. David J. C. MacKay, *Information Theory, Inference, and Learning Algorithms* (New York: Cambridge University Press, 2003). sharov.me/r/4-5



way, the likelihood of the data coming out that way will be something else. But the likelihood of a given state of Nature producing the data we have seen, has nothing to do with the researcher's private intentions. So whatever our hypotheses about Nature, the likelihood ratio is the same, and the evidential impact is the same, and the posterior belief should be the same, between the two experiments. At least one of the two Old Style methods must discard relevant information—or simply do the wrong calculation—for the two methods to arrive at different answers.

The ancient war between the Bayesians and the accursed frequentists stretches back through decades, and I'm not going to try to recount that elder history in this essay.

But one of the central conflicts is that Bayesians expect probability theory to be ...what's the word I'm looking for? "Neat?" "Clean?" "Self-consistent?"

As Jaynes says, the theorems of Bayesian probability are just that, *theorems* in a coherent proof system. No matter what derivations you use, in what order, the results of Bayesian probability theory should always be consistent—every theorem compatible with every other theorem.

If you want to know the sum $10 + 10$, you can redefine it as $(2 \times 5) + (7 + 3)$ or as $(2 \times (4 + 6))$ or use whatever other *legal* tricks you like, but the result always has to come out to be the same, in this case, 20. If it comes out as 20 one way and 19 the other way, then you may conclude you did something illegal on at least one of the two occasions. (In arithmetic, the illegal operation is usually division by zero; in probability theory, it is usually an infinity that was not taken as the limit of a finite process.)

If you get the result $19 = 20$, look hard for that error you just made, because it's unlikely that you've sent arithmetic itself up in smoke. If anyone should ever succeed in deriving a *real* contradiction from Bayesian probability theory—like, say, two different evidential impacts from the same experimental method yielding the same results—then the whole edifice goes up in smoke. Along with set theory, 'cause I'm pretty sure ZF provides a model for probability theory.

Math! That's the word I was looking for. Bayesians expect probability theory to be *math*. That's why we're interested in Cox's Theorem and its many extensions, showing that any representation of uncertainty which obeys certain constraints has to map onto probability theory. Coherent math is great, but unique math is even better.

And yet ...*should* rationality be math? It is by no means a foregone conclusion that probability should be pretty. The real world is messy—so shouldn't you need messy reasoning to handle it? Maybe the non-Bayesian statisticians, with their vast collection of ad-hoc methods and ad-hoc justifications, are strictly more competent because they have a strictly larger toolbox. It's nice when problems are clean, but they usually aren't, and you have to live with that.

After all, it's a well-known fact that you can't use Bayesian methods on many problems because the Bayesian calculation is computationally intractable. So why not let many flowers bloom? Why not have more than one tool in your toolbox?

That's the fundamental difference in mindset. Old School statisticians thought in terms of *tools*, tricks to throw at particular problems. Bayesians—at least this Bayesian, though I don't think I'm speaking only for myself—we think in terms of *laws*.

Looking for laws isn't the same as looking for especially neat and pretty tools. The Second Law of Thermodynamics isn't an especially neat and pretty refrigerator.

The Carnot cycle is an ideal engine—in fact, *the* ideal engine. No engine powered by two heat reservoirs can be more efficient than a Carnot engine. As a corollary, all thermodynamically reversible engines operating between the same heat reservoirs are equally efficient.

But, of course, you can't use a Carnot engine to power a real car. A real car's engine bears the same resemblance to a Carnot engine that the car's tires bear to perfect rolling cylinders.

Clearly, then, a Carnot engine is a useless *tool* for building a real-world car. The Second Law of Thermodynamics, obviously, is not applicable here. It's too hard to make an engine that obeys it, in the real world. Just ignore thermodynamics—use whatever works.

This is the sort of confusion that I think reigns over they who still cling to the Old Ways.

No, you can't always do the exact Bayesian calculation for a problem. Sometimes you must seek an approximation; often, indeed. This doesn't mean that probability theory has ceased to apply, any more than your inability to calculate the aerodynamics of a 747 on an atom-by-atom basis implies that the 747 is not made out of atoms. Whatever approximation you use, it works to the extent that it approximates the ideal Bayesian calculation—and fails to the extent that it departs.

Bayesianism's coherence and uniqueness proofs cut both ways. Just as any calculation that obeys Cox's coherency axioms (or any of the many reformulations and generalizations) must map onto probabilities, so too, anything that is not Bayesian must fail one of the coherency tests. This, in turn, opens you to punishments like Dutch-bookings (accepting combinations of bets that are sure losses, or rejecting combinations of bets that are sure gains).

You may not be able to compute the optimal answer. But whatever approximation you use, both its failures and successes will be *explainable* in terms of Bayesian probability theory. You may not know the explanation; that does not mean no explanation exists.

So you want to use a linear regression, instead of doing Bayesian updates? But look to the underlying structure of the linear regression, and you see that it corresponds to picking the best point estimate given a Gaussian likelihood function and a uniform prior over the parameters.

You want to use a regularized linear regression, because that works better in practice? Well, that corresponds (says the Bayesian) to having a Gaussian prior over the weights.

Sometimes you can't use Bayesian methods *literally*; often, indeed. But when you *can* use the exact Bayesian calculation that uses every scrap of available knowledge, you are done. You will never find a statistical method that yields a *better* answer. You may find a cheap approximation that works excellently nearly all the time, and it will be cheaper, but it will not be more accurate. Not unless the other method uses knowledge,

perhaps in the form of disguised prior information, that you are not allowing into the Bayesian calculation; and then when you feed the prior information into the Bayesian calculation, the Bayesian calculation will again be equal or superior.

When you use an Old Style ad-hoc statistical tool with an ad-hoc (but often quite interesting) justification, you never know if someone else will come up with an even more clever tool tomorrow. But when you *can* directly use a calculation that mirrors the Bayesian law, you're *done*—like managing to put a Carnot heat engine into your car. It is, as the saying goes, “Bayes-optimal.”

It seems to me that the toolboxers are looking at the sequence of cubes $\{1, 8, 27, 64, 125, \dots\}$ and pointing to the first differences $\{7, 19, 37, 61, \dots\}$ and saying “Look, life isn’t always so neat—you’ve got to adapt to circumstances.” And the Bayesians are pointing to the third differences, the underlying stable level $\{6, 6, 6, 6, \dots\}$. And the critics are saying, “What the heck are you talking about? It’s 7, 19, 37 not 6, 6, 6. You are oversimplifying this messy problem; you are too attached to simplicity.”

It’s not necessarily simple on a *surface* level. You have to dive deeper than that to find stability.

Think laws, not tools. Needing to calculate approximations to a law doesn’t change the law. Planes are still atoms, they aren’t governed by special exceptions in Nature for aerodynamic calculations. The approximation exists in the map, not in the territory. You can know the Second Law of Thermodynamics, and yet apply yourself as an engineer to build an imperfect car engine. The Second Law does not cease to be applicable; your knowledge of that law, and of Carnot cycles, helps you get as close to the ideal efficiency as you can.

We aren’t enchanted by Bayesian methods merely because they’re beautiful. The beauty is a side effect. Bayesian *theorems* are elegant, coherent, optimal, and provably unique because they are *laws*.

Outside the Laboratory



discussion
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“Outside the laboratory, scientists are no wiser than anyone else.” Sometimes this proverb is spoken by scientists, humbly, sadly, to remind themselves of their own fallibility. Sometimes this proverb is said for rather less praiseworthy reasons, to devalue unwanted expert advice. Is the proverb true? Probably not in an absolute sense. It seems much too pessimistic to say that scientists are literally *no* wiser than average, that there is literally *zero* correlation.

But the proverb does appear true to some degree, and I propose that we should be very disturbed by this fact. We should not sigh, and shake our heads sadly. Rather we should sit bolt upright in alarm. Why? Well, suppose that an apprentice shepherd is laboriously trained to count sheep, as they pass in and out of a fold. Thus the shepherd knows when all the sheep have left, and when all the sheep have returned. Then you give the shepherd a few apples, and say: “How many apples?” But the shepherd stares at you blankly, because they weren’t trained to count apples—just sheep. You would probably suspect that the shepherd didn’t understand counting very well.

Now suppose we discover that a PhD economist buys a lottery ticket every week. We have to ask ourselves: Does this person really *understand* expected utility, on a gut level? Or have they just been trained to perform certain algebra tricks?

One thinks of Richard Feynman’s account of a failing physics education program:

The students had memorized everything, but they didn’t know what anything meant. When they heard “light that is reflected from a medium with an index,” they didn’t know that it meant a material *such as water*. They didn’t know that the “direction of the light” is the direction in which you *see* something when you’re looking at it, and so on.

Everything was entirely memorized, yet nothing had been translated into meaningful words. So if I asked, “What is Brewster’s Angle?” I’m going into the computer with the right keywords. But if I say, “Look at the water,” nothing happens—they don’t have anything under “Look at the water”!

Suppose we have an apparently competent scientist, who knows how to design an experiment on N subjects; the N subjects will receive a randomized treatment; blinded judges will classify the subject outcomes; and then we’ll run the results through a computer and see if the results are significant at the 0.05 confidence level. Now this is not just a ritualized tradition. This is not a point of arbitrary etiquette like using the correct fork for salad. It is a ritualized tradition for *testing hypotheses experimentally*. Why should you test your hypothesis experimentally? Because you know the journal will demand so before it publishes your paper? Because you were trained to do it in college? Because everyone else says in unison that it’s important to do the experiment, and they’ll look at you funny if you say otherwise?

No: because, in order to map a territory, you have to go out and look at the territory. It isn’t possible to produce an accurate map of a city while sitting in your living room with your eyes closed, thinking pleasant thoughts about what you wish the city was like. You have to go out, walk through the city, and write lines on paper that correspond to what you see. It happens, in miniature, every time you look down at your shoes to see if your shoelaces are untied. Photons arrive from the Sun, bounce off your shoelaces, strike your retina, are transduced into neural firing frequencies, and are reconstructed by your visual cortex into an activation pattern that is strongly correlated with the current shape of your shoelaces. To gain new information about the territory, you have to interact with the territory. There has to be some real, physical process whereby your brain state ends up correlated to the state of the environment. Reasoning processes aren’t magic; you can give causal descriptions of how

they work. Which all goes to say that, to find things out, you've got to go look.

Now what are we to think of a scientist who seems competent inside the laboratory, but who, outside the laboratory, believes in a spirit world? We ask why, and the scientist says something along the lines of: "Well, no one really knows, and I admit that I don't have any evidence—it's a religious belief, it can't be disproven one way or another by observation." I cannot but conclude that this person *literally doesn't know why you have to look at things*. They may have been taught a certain ritual of experimentation, but they don't understand the *reason* for it—that to map a territory, you have to look at it—that to gain information about the environment, you have to undergo a causal process whereby you interact with the environment and end up correlated to it. This applies just as much to a double-blind experimental design that gathers information about the efficacy of a new medical device, as it does to your eyes gathering information about your shoelaces.

Maybe our spiritual scientist says: "But it's not a matter for experiment. The spirits spoke to me in my heart." Well, if we really suppose that spirits are speaking in any fashion whatsoever, that is a causal interaction and it counts as an observation. Probability theory still applies. If you propose that some personal experience of "spirit voices" is evidence for actual spirits, you must propose that there is a favorable likelihood ratio for spirits causing "spirit voices," as compared to other explanations for "spirit voices," which is sufficient to overcome the prior improbability of a complex belief with many parts. Failing to realize that "the spirits spoke to me in my heart" is an instance of "causal interaction," is analogous to a physics student not realizing that a "medium with an index" means a material such as water.

It is easy to be fooled, perhaps, by the fact that people wearing lab coats use the phrase "causal interaction" and that people wearing gaudy jewelry use the phrase "spirits speaking." Discussants wearing different clothing, as we all know, demarcate independent spheres of existence—"separate magisteria," in Stephen J. Gould's immortal blunder of a phrase. Actually,

“causal interaction” is just a fancy way of saying, “Something that makes something else happen,” and probability theory doesn’t care what clothes you wear.

In modern society there is a prevalent notion that spiritual matters can’t be settled by logic or observation, and therefore you can have whatever religious beliefs you like. If a scientist falls for this, and decides to live their extralaboratorial life accordingly, then this, to me, says that they only understand the experimental principle as a *social convention*. They know when they are *expected* to do experiments and test the results for statistical significance. But put them in a context where it is *socially conventional* to make up wacky beliefs without looking, and they just as happily do that instead.

The apprentice shepherd is told that if “seven” sheep go out, and “eight” sheep go out, then “fifteen” sheep had better come back in. Why “fifteen” instead of “fourteen” or “three”? Because otherwise you’ll get no dinner tonight, that’s why! So that’s professional training of a kind, and it works after a fashion—but if social convention is the only reason why seven sheep plus eight sheep equals fifteen sheep, then maybe seven apples plus eight apples equals three apples. Who’s to say that the rules shouldn’t be different for apples?

But if you know *why* the rules work, you can see that addition is the same for sheep and for apples. Isaac Newton is justly revered, not for his outdated theory of gravity, but for discovering that—amazingly, surprisingly—the celestial planets, in the glorious heavens, obeyed just the same rules as falling apples. In the macroscopic world—the everyday ancestral environment—different trees bear different fruits, different customs hold for different people at different times. A genuinely unified universe, with stationary universal laws, is a highly counterintuitive notion to humans! It is only scientists who really believe it, though some religions may talk a good game about the “unity of all things.”

As Richard Feynman put it:

If we look at a glass closely enough we see the entire universe. There are the things of physics:

the twisting liquid which evaporates depending on the wind and weather, the reflections in the glass, and our imagination adds the atoms. The glass is a distillation of the Earth's rocks, and in its composition we see the secret of the universe's age, and the evolution of the stars. What strange array of chemicals are there in the wine? How did they come to be? There are the ferments, the enzymes, the substrates, and the products. There in wine is found the great generalization: all life is fermentation. Nobody can discover the chemistry of wine without discovering, as did Louis Pasteur, the cause of much disease. How vivid is the claret, pressing its existence into the consciousness that watches it! If our small minds, for some convenience, divide this glass of wine, this universe, into parts—physics, biology, geology, astronomy, psychology, and so on—remember that Nature does not know it! So let us put it all back together, not forgetting ultimately what it is for. Let it give us one more final pleasure: drink it and forget it all!¹

A few religions, especially the ones invented or refurbished after Isaac Newton, may profess that “everything is connected to everything else.” (Since there is a trivial isomorphism between graphs and their complements, this profound wisdom conveys exactly the same useful information as a graph with no edges.) But when it comes to the actual meat of the religion, prophets and priests follow the ancient human practice of making everything up as they go along. And they make up one rule for women under twelve, another rule for men over thirteen; one rule for the Sabbath and another rule for weekdays; one rule for science and another rule for sorcery ...

1. Richard P. Feynman, Robert B. Leighton, and Matthew L. Sands, *The Feynman Lectures on Physics*, 3 vols. (Reading, MA: Addison-Wesley, 1963). Vol I, Chap 3, sharov.me/r/4-6



Reality, we have learned to our shock, is *not* a collection of separate magisteria, but a single unified process governed by mathematically simple low-level rules. Different buildings on a university campus do not belong to different universes, though it may sometimes seem that way. The universe is not divided into mind and matter, or life and nonlife; the atoms in our heads interact seamlessly with the atoms of the surrounding air. Nor is Bayes's Theorem different from one place to another.

If, outside of their specialist field, some particular scientist is just as susceptible as anyone else to wacky ideas, then they probably never did understand *why* the scientific rules work. Maybe they can parrot back a bit of Popperian falsificationism; but they don't understand on a deep level, the algebraic level of probability theory, the causal level of cognition-as-machinery. They've been trained to behave a certain way in the laboratory, but they don't *like* to be constrained by evidence; when they go home, they take off the lab coat and relax with some comfortable nonsense. And yes, that does make me wonder if I can trust that scientist's opinions even in their own field—especially when it comes to any controversial issue, any open question, anything that isn't already nailed down by massive evidence and social convention.

Maybe we *can* beat the proverb—be rational in our personal lives, not just our professional lives. We shouldn't let a mere proverb stop us: "A witty saying proves nothing," as Voltaire said. Maybe we can do better, if we study enough probability theory to know *why* the rules work, and enough experimental psychology to see how they apply in real-world cases—if we can learn to look at the water. An ambition like that lacks the comfortable modesty of being able to confess that, outside your specialty, you're no better than anyone else. But if our theories of rationality don't generalize to everyday life, we're doing something wrong. It's not a different universe inside and outside the laboratory.

The Second Law of Thermodynamics, and Engines of Cognition



discussion
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The First Law of Thermodynamics, better known as Conservation of Energy, says that you can't create energy from nothing: it prohibits perpetual motion machines of the first type, which run and run indefinitely without consuming fuel or any other resource. According to our modern view of physics, energy is conserved in each *individual* interaction of particles. By mathematical induction, we see that no matter how large an assemblage of particles may be, it cannot produce energy from nothing—not without violating what we presently believe to be the laws of physics.

This is why the US Patent Office will summarily reject your amazingly clever proposal for an assemblage of wheels and gears that cause one spring to wind up another as the first runs down, and so continue to do work forever, according to your calculations. There's a *fully general* proof that at least one wheel must violate (our standard model of) the laws of physics for this to happen. So unless you can explain how *one* wheel violates the laws of physics, the *assembly* of wheels can't do it either.

A similar argument applies to a "reactionless drive," a propulsion system that violates Conservation of Momentum. In standard physics, momentum is conserved for all individual particles and their interactions; by mathematical induction, momentum is conserved for physical systems whatever their size. If you can visualize two particles knocking into each other and always coming out with the same total momentum that they started with, then you can see how scaling it up from particles to a gigantic complicated collection of gears won't change anything. Even if there's a trillion quadrillion atoms involved, $0 + 0 + \dots + 0 = 0$.

But Conservation of Energy, as such, cannot prohibit converting heat into work. You can, in fact, build a sealed box that converts ice cubes and stored electricity into warm water. It isn't even difficult. Energy cannot be created or destroyed: the net change in energy, from transforming (ice cubes + electricity) to (warm water), must be 0. So it couldn't violate Conservation of Energy, as such, if you did it the other way around ...

Perpetual motion machines of the second type, which convert warm water into electrical current and ice cubes, are prohibited by the *Second Law of Thermodynamics*.

The second law is a bit harder to understand, as it is essentially Bayesian in nature.

Yes, really.

The essential *physical* law underlying the Second Law of Thermodynamics is a theorem which can be proven within the standard model of physics: *In the development over time of any closed system, phase space volume is conserved.*

Let's say you're holding a ball high above the ground. We can describe this state of affairs as a point in a multidimensional space, at least one of whose dimensions is "height of ball above the ground." Then, when you drop the ball, it moves, and so does the dimensionless point in phase space that describes the entire system that includes you and the ball. "Phase space," in physics-speak, means that there are dimensions for the momentum of the particles, not just their position—e.g., a system of 2 particles would have 12 dimensions, 3 dimensions for each particle's position, and 3 dimensions for each particle's momentum.

If you had a multidimensional space, each of whose dimensions described the position of a gear in a huge assemblage of gears, then as you turned the gears a single point would swoop and dart around in a rather high-dimensional phase space. Which is to say, just as you can view a great big complex machine as a single point in a very-high-dimensional space, so too you can view the laws of physics describing the behavior of this machine over time as describing the trajectory of its point through the phase space.

The Second Law of Thermodynamics is a consequence of a theorem which can be proven in the standard model of physics:

If you take a volume of phase space, and develop it forward in time using standard physics, the total volume of the phase space is conserved.

For example, let there be two systems, X and Y , where X has 8 possible states, Y has 4 possible states, and the joint system (X, Y) has 32 possible states.

The development of the joint system over time can be described as a rule that maps initial points onto future points. For example, the system could start out in X_7Y_2 , then develop (under some set of physical laws) into the state X_3Y_3 a minute later. Which is to say: if X started in state X_7 , and Y started in state Y_2 , and we watched it for 1 minute, we would see X go to X_3 and Y go to Y_3 . Such are the laws of physics.

Next, let's carve out a subspace S of the joint system state. The space S will be the subspace bounded by X being in state X_1 and Y being in states Y_1 through Y_4 . So the total volume of S is 4 states.

And let's suppose that, under the laws of physics governing (X, Y) , the states initially in S behave as follows:

$$X_1Y_1 \rightarrow X_2Y_1$$

$$X_1Y_2 \rightarrow X_4Y_1$$

$$X_1Y_3 \rightarrow X_6Y_1$$

$$X_1Y_4 \rightarrow X_8Y_1.$$

That, in a nutshell, is how a refrigerator works.

The X subsystem began in a narrow region of state space—the single state X_1 , in fact—and Y began distributed over a wider region of space, states Y_1 through Y_4 . By interacting with each other, Y went into a narrow region, and X ended up in a wide region; *but the total phase space volume was conserved*. Four initial states mapped to four end states.

Clearly, so long as total phase space volume is conserved by physics over time, you can't squeeze Y harder than X expands, or vice versa—for every subsystem you squeeze into a narrower region of state space, some other subsystem has to expand into a wider region of state space.

Now let's say that we're *uncertain* about the joint system (X, Y) , and our *uncertainty* is described by an equiprobable distribution over S . That is, we're pretty sure X is in state X_1 , but Y is equally likely to be in any of the states Y_1 through Y_4 . If we shut our eyes for a minute and then open them again, we will expect to see Y in state Y_1 , but X might be in any of the states X_2 through X_8 . Actually, X can only be in *some* of the states X_2 through X_8 , but it would be too costly to think out exactly which states these might be, so we'll just say X_2 through X_8 .

If you consider the Shannon entropy of our uncertainty about X and Y as individual systems, X began with 0 bits of entropy because it had a single definite state, and Y began with 2 bits of entropy because it was equally likely to be in any of 4 possible states. (There's no mutual information between X and Y .) A bit of physics occurred, and lo, the entropy of Y went to 0, but the entropy of X went to $\log_2(7) = 2.8$ bits. So entropy was transferred from one system to another, and decreased *within* the Y subsystem; but due to the cost of bookkeeping, we didn't bother to track some information, and hence (from our perspective) the overall entropy increased.

Suppose there was a physical process that mapped past states onto future states like this:

$$X_2Y_1 \rightarrow X_2Y_1$$

$$X_2Y_2 \rightarrow X_2Y_1$$

$$X_2Y_3 \rightarrow X_2Y_1$$

$$X_2Y_4 \rightarrow X_2Y_1.$$

Then you could have a physical process that would actually *decrease entropy*, because no matter where you started out, you would end up at the same place. The laws of physics, developing over time, would compress the phase space.

But there is a theorem, Liouville's Theorem, which can be proven true of our laws of physics, which says that this never happens: phase space is conserved.

The Second Law of Thermodynamics is a corollary of Liouville's Theorem: no matter how clever your configuration of

wheels and gears, you'll never be able to decrease entropy in one subsystem without increasing it somewhere else. When the phase space of one subsystem narrows, the phase space of another subsystem must widen, and the joint space keeps the same volume.

Except that what was initially a *compact* phase space, may develop squiggles and wiggles and convolutions; so that to draw a simple boundary around the whole mess, you must draw a much larger boundary than before—this is what gives the appearance of entropy increasing. (And in quantum systems, where different universes go different ways, entropy actually does increase in any local universe. But omit this complication for now.)

The Second Law of Thermodynamics is actually probabilistic in nature—if you ask about the probability of hot water spontaneously entering the “cold water and electricity” state, the probability does exist, it's just very small. This doesn't mean Liouville's Theorem is violated with small probability; a theorem's a theorem, after all. It means that if you're in a great big phase space volume at the start, but you *don't know where*, you may assess a tiny little *probability* of ending up in some particular phase space volume. *So far as you know*, with infinitesimal probability, this particular glass of hot water may be the kind that spontaneously transforms itself to electrical current and ice cubes. (Neglecting, as usual, quantum effects.)

So the Second Law really *is* inherently Bayesian. When it comes to any real thermodynamic system, it's a strictly lawful statement of your *beliefs about* the system, but only a probabilistic statement about the system itself.

“Hold on,” you say. “That's not what I learned in physics class,” you say. “In the lectures *I* heard, thermodynamics is about, you know, *temperatures*. Uncertainty is a subjective state of mind! The temperature of a glass of water is an objective property of the water! What does heat have to do with probability?”

Oh ye of little trust.

In one direction, the connection between heat and probability is relatively straightforward: If the only fact you know about

a glass of water is its temperature, then you are much more uncertain about a hot glass of water than a cold glass of water.

Heat is the zipping around of lots of tiny molecules; the hotter they are, the faster they can go. Not all the molecules in hot water are travelling at the same speed—the “temperature” isn’t a uniform speed of all the molecules, it’s an average speed of the molecules, which in turn corresponds to a predictable statistical distribution of speeds—anyway, the point is that, the hotter the water, the faster the water molecules *could be* going, and hence, the more uncertain you are about the velocity (not just speed) of any *individual* molecule. When you multiply together your uncertainties about all the individual molecules, you will be *exponentially* more uncertain about the whole glass of water.

We take the logarithm of this exponential volume of uncertainty, and call that the entropy. So it all works out, you see.

The connection in the other direction is less obvious. Suppose there was a glass of water, about which, initially, you knew only that its temperature was 72 degrees. Then, suddenly, Saint Laplace reveals to you the exact locations and velocities of all the atoms in the water. You now know perfectly the state of the water, so, by the information-theoretic definition of entropy, its entropy is zero. Does that make its *thermodynamic* entropy zero? Is the water colder, because we know more about it?

Ignoring quantumness for the moment, the answer is: Yes! Yes it is!

Maxwell once asked: Why can’t we take a uniformly hot gas, and partition it into two volumes *A* and *B*, and let only fast-moving molecules pass from *B* to *A*, while only slow-moving molecules are allowed to pass from *A* to *B*? If you could build a gate like this, soon you would have hot gas on the *A* side, and cold gas on the *B* side. That would be a cheap way to refrigerate food, right?

The agent who inspects each gas molecule, and decides whether to let it through, is known as “Maxwell’s Demon.” And the reason you can’t build an efficient refrigerator this way, is that Maxwell’s Demon generates entropy in the process

of inspecting the gas molecules and deciding which ones to let through.

But suppose you already *knew* where all the gas molecules were?

Then you actually *could* run Maxwell’s Demon and extract useful work.

So (again ignoring quantum effects for the moment), if you *know* the states of all the molecules in a glass of hot water, it is cold in a genuinely thermodynamic sense: you can take electricity out of it and leave behind an ice cube.

This doesn’t violate Liouville’s Theorem, because if Y is the water, and *you* are Maxwell’s Demon (denoted M), the physical process behaves as:

$$M_1 Y_1 \rightarrow M_1 Y_1$$

$$M_2 Y_2 \rightarrow M_2 Y_1$$

$$M_3 Y_3 \rightarrow M_3 Y_1$$

$$M_4 Y_4 \rightarrow M_4 Y_1.$$

Because Maxwell’s demon *knows* the exact state of Y , this is mutual information between M and Y . The mutual information decreases the joint entropy of (M, Y) : we have $H(M, Y) = H(M) + H(Y) - I(M; Y)$. The demon M has 2 bits of entropy, Y has two bits of entropy, and their mutual information is 2 bits, so (M, Y) has a total of $2 + 2 - 2 = 2$ bits of entropy. The physical process just transforms the “coldness” (negative entropy, or negentropy) of the mutual information to make the actual water cold—afterward, M has 2 bits of entropy, Y has 0 bits of entropy, and the mutual information is 0. Nothing wrong with that!

And don’t tell me that knowledge is “subjective.” Knowledge has to be represented in a brain, and that makes it as physical as anything else. For M to physically represent an accurate picture of the state of Y , it must be that M ’s physical state correlates with the state of Y . You can take thermodynamic advantage of that—it’s called a Szilárd engine.

Or as E. T. Jaynes put it, “The old adage ‘knowledge is power’ is a very cogent truth, both in human relations and in thermodynamics.”

And conversely, *one subsystem cannot increase in mutual information with another subsystem, without (a) interacting with it and (b) doing thermodynamic work.*

Otherwise you could build a Maxwell’s Demon and violate the Second Law of Thermodynamics—which in turn would violate Liouville’s Theorem—which is prohibited in the standard model of physics.

Which is to say: **To form accurate beliefs about something, you *really do* have to observe it.** It’s a very physical, very real process: any rational mind does “work” in the thermodynamic sense, not just the sense of mental effort.

(It is sometimes said that it is erasing bits in order to prepare for the next observation that takes the thermodynamic work—but that distinction is just a matter of words and perspective; the math is unambiguous.)

(Discovering logical “truths” is a complication which I will not, for now, consider—at least in part because I am still thinking through the exact formalism myself. In thermodynamics, knowledge of logical truths does not count as negentropy; as would be expected, since a reversible computer can compute logical truths at arbitrarily low cost. All this that I have said is true of the logically omniscient: any lesser mind will necessarily be less efficient.)

“Forming accurate beliefs requires a corresponding amount of evidence” is a very cogent truth both in human relations and in thermodynamics: if blind faith actually worked as a method of investigation, you could turn warm water into electricity and ice cubes. Just build a Maxwell’s Demon that has blind faith in molecule velocities.

Engines of cognition are not so different from heat engines, though they manipulate entropy in a more subtle form than burning gasoline. For example, to the extent that an engine of cognition is not perfectly efficient, it must radiate waste heat, just like a car engine or refrigerator.

“Cold rationality” is true in a sense that Hollywood scriptwriters never dreamed (and false in the sense that they did dream).

So unless you can tell me which *specific step* in your argument violates the laws of physics by giving you true knowledge of the unseen, don’t expect me to believe that a big, elaborate clever argument can do it either.

Perpetual Motion Beliefs

The last essay concluded:

To form accurate beliefs about something, you *really do have to observe it*. It's a very physical, very real process: any rational mind does "work" in the thermodynamic sense, not just the sense of mental effort ...So unless you can tell me which *specific step* in your argument violates the laws of physics by giving you true knowledge of the unseen, don't expect me to believe that a big, elaborate clever argument can do it either.



discussion
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4-c8

One of the chief morals of the mathematical analogy between thermodynamics and cognition is that the constraints of probability are inescapable; probability may be a "subjective state of belief," but the laws of probability are harder than steel.

People learn under the traditional school regimen that the teacher tells you certain things, and you must believe them and recite them back; but if a mere *student* suggests a belief, you do not have to obey it. They map the domain of belief onto the domain of authority, and think that a certain belief is like an order that must be obeyed, but a probabilistic belief is like a mere suggestion.

They look at a lottery ticket, and say, "But you can't *prove* I won't win, right?" Meaning: "You may have calculated a low probability of winning, but since it is a *probability*, it's just a *suggestion*, and I am *allowed* to believe what I want."

Here's a little experiment: Smash an egg on the floor. The rule that says that the egg won't spontaneously reform and leap back into your hand is merely probabilistic. A suggestion, if you will. The laws of thermodynamics are probabilistic, so they can't really be *laws*, the way that "Thou shalt not murder" is a law ...right?

So why not just ignore the suggestion? Then the egg will unscramble itself...right?

It may help to think of it this way—if you still have some lingering intuition that uncertain beliefs are not authoritative:

In reality, there may be a very small chance that the egg spontaneously reforms. But you cannot *expect* it to reform. You *must* expect it to smash. Your mandatory belief is that the egg's probability of spontaneous reformation is ~ 0 . Probabilities are not certainties, but the *laws of probability* are theorems.

If you doubt this, try dropping an egg on the floor a few decillion times, ignoring the thermodynamic suggestion and expecting it to spontaneously reassemble, and see what happens. Probabilities may be subjective states of belief, but the laws governing them are stronger by far than steel. I once knew a fellow who was *convinced* that his system of wheels and gears would produce reactionless thrust, and he had an Excel spreadsheet that would prove this—which of course he couldn't show us because he was still developing the system. In classical mechanics, violating Conservation of Momentum is *provably* impossible. So any Excel spreadsheet calculated *according to the rules of classical mechanics* must *necessarily* show that no reactionless thrust exists—unless your machine is complicated enough that you have made a mistake in the calculations.

And similarly, when half-trained or tenth-trained rationalists abandon their art and try to believe without evidence just this once, they often build vast edifices of justification, confusing themselves just enough to conceal the magical steps.

It can be quite a pain to nail down where the magic occurs—their structure of argument tends to morph and squirm away as you interrogate them. But there's always some step where a tiny probability turns into a large one—where they try to believe without evidence—where they step into the unknown, thinking, “No one can prove me wrong.”

Their foot naturally lands on thin air, for there is far more thin air than ground in the realms of Possibility. Ah, but there is an (exponentially tiny) amount of ground in Possibility, and you do have an (exponentially tiny) probability of hitting it by

luck, so maybe this time, your foot *will* land in the right place! It is *merely* a probability, so it must be *merely* a suggestion.

The exact state of a glass of boiling-hot water may be unknown to you—indeed, your ignorance of its exact state is what makes the molecules’ kinetic energy “heat,” rather than work waiting to be extracted like the momentum of a spinning fly-wheel. So the water *might* cool down your hand instead of heating it up, with probability ~ 0 .

Decide to ignore the laws of thermodynamics and stick your hand in anyway, and you’ll get burned.

“But you don’t know that!”

I don’t know it with certainty, but it is *mandatory* that I *expect* it to happen. Probabilities are not logical truths, but the *laws of probability* are.

“But what if I guess the state of the boiling water, and I happen to guess correctly?”

Your chance of guessing correctly by luck, is even *less* than the chance of the boiling water cooling your hand by luck.

“But you can’t *prove* I won’t guess correctly.”

I can (indeed, must) assign extremely low probability to it.

“That’s not the same as certainty, though.”

Hey, maybe if you add enough wheels and gears to your argument, it’ll turn warm water into electricity and ice cubes! Or, rather, you will no longer see why this *couldn’t* be the case.

“Right! I *can’t* see why couldn’t be the case! So maybe it is!”

Another gear? That just makes your machine even *less* efficient. It wasn’t a perpetual motion machine before, and each extra gear you add makes it even less efficient than that.

Each extra detail in your argument necessarily decreases the joint probability. The probability that you’ve violated the Second Law of Thermodynamics without knowing exactly how, by guessing the exact state of boiling water without evidence, so that you can stick your finger in without getting burned, is, necessarily, even less than the probability of sticking in your finger into boiling water without getting burned.

I say all this, because people really do construct these huge edifices of argument in the course of believing without evidence. One must learn to see this as analogous to all the wheels and

gears that fellow added onto his reactionless drive, until he finally collected enough complications to make a mistake in his Excel spreadsheet.

Searching for Bayes-Structure

Gnomish helms should not function. Their very construction seems to defy the nature of thaumaturgical law. In fact, they are impossible. Like most products of gnomish minds, they include a large number of bells and whistles, and very little substance. Those that work usually have a *minor helm* contained within, always hidden away, disguised to appear innocuous and inessential.

—Spelljammer campaign set



discussion
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4-c9

We have seen that knowledge implies mutual information between a mind and its environment, and we have seen that this mutual information is negentropy in a very physical sense: If you know where molecules are and how fast they're moving, you can turn heat into work via a Maxwell's Demon / Szilárd engine.

We have seen that forming true beliefs without evidence is the same sort of improbability as a hot glass of water spontaneously reorganizing into ice cubes and electricity. Rationality takes "work" in a thermodynamic sense, not just the sense of mental effort; minds have to radiate heat if they are not perfectly efficient. This cognitive work is governed by probability theory, of which thermodynamics is a special case. (Statistical mechanics is a special case of statistics.)

If you saw a machine continually spinning a wheel, apparently without being plugged into a wall outlet or any other source of power, then you would look for a hidden battery, or a nearby broadcast power source—something to explain the work being done, without violating the laws of physics.

So if a mind is arriving at true beliefs, and we assume that the Second Law of Thermodynamics has not been violated, that mind must be doing something at least *vaguely* Bayesian—at

least one process with a sort-of Bayesian structure *somewhere*—or it *couldn't possibly work*.

In the beginning, at time $T = 0$, a mind has no mutual information with a subsystem S in its environment. At time $T = 1$, the mind has 10 bits of mutual information with S . Somewhere in between, the mind must have encountered evidence—under the Bayesian definition of evidence, because all Bayesian evidence is mutual information and all mutual information is Bayesian evidence, they are just different ways of looking at it—and processed at least some of that evidence, however inefficiently, in the right direction according to Bayes on at least some occasions. The mind must have *moved in harmony with the Bayes* at least a little, somewhere along the line—either that or violated the Second Law of Thermodynamics by creating mutual information from nothingness.

In fact, any *part* of a cognitive process that *contributes usefully* to truth-finding must have at least a little Bayesian structure—must harmonize with Bayes, at some point or another—must partially conform with the Bayesian flow, however noisily—despite however many disguising bells and whistles—even if this Bayesian structure is only apparent in the context of surrounding processes. Or it couldn't even *help*.

How philosophers pondered the nature of words! All the ink spent on the true definitions of words, and the true meaning of definitions, and the true meaning of meaning! What collections of gears and wheels they built, in their explanations! And all along, it was a disguised form of Bayesian inference!

I was actually a bit disappointed that no one in the audience jumped up and said: “*Yes! Yes, that's it! Of course! It was really Bayes all along!*”

But perhaps it is not *quite* as exciting to see something that *doesn't* look Bayesian on the surface, revealed as Bayes wearing a clever disguise, if: (a) you don't unravel the mystery yourself, but read about someone else doing it (Newton had more fun than most students taking calculus), and (b) you don't realize that *searching for the hidden Bayes-structure* is this huge, difficult, omnipresent quest, like searching for the Holy Grail.

It's a different quest for each facet of cognition, but the Grail always *turns out* to be the same. It has to be the *right* Grail, though—and the *entire* Grail, without any parts missing—and so each time you have to go on the quest looking for a full answer *whatever* form it may take, rather than trying to artificially construct vaguely hand-waving Grailish arguments. *Then* you always find the same Holy Grail at the end.

It was previously pointed out to me that I might be losing some of my readers with the long essays, because I hadn't "made it clear where I was going" ...

...but it's not so easy to just tell people where you're going, when you're going somewhere like *that*.

It's not very helpful to merely *know that* a form of cognition is Bayesian, if you don't *know how* it is Bayesian. If you can't see the detailed flow of probability, you have nothing but a password—or, a bit more charitably, a hint at the form an answer would take; but certainly not an answer. That's why there's a Grand Quest for the Hidden Bayes-Structure, rather than being done when you say "Bayes!" Bayes-structure can be buried under all kinds of disguises, hidden behind thickets of wheels and gears, obscured by bells and whistles.

The way you begin to grasp the Quest for the Holy Bayes is that you learn about cognitive phenomenon XYZ, which seems really useful—and there's this bunch of philosophers who've been arguing about its true nature for centuries, and they are still arguing—and there's a bunch of AI scientists trying to make a computer do it, but they can't agree on the philosophy either—

And—*Huh, that's odd!*—this cognitive phenomenon didn't look anything like Bayesian on the surface, but there's this non-obvious underlying structure that has a Bayesian interpretation—but wait, there's still some useful work getting done that can't be explained in Bayesian terms—no wait, *that's* Bayesian too—OH MY GOD this *completely different* cognitive process, that *also* didn't look Bayesian on the surface, ALSO HAS BAYESIAN STRUCTURE—hold on, are these non-Bayesian parts even *doing* anything?

- Yes: Wow, those are Bayesian too!
- No: Dear heavens, what a stupid design. I could eat a bucket of amino acids and puke a better brain architecture than that.

Once this happens to you a few times, you kinda pick up the rhythm. That's what I'm talking about here, the rhythm.

Trying to talk about the rhythm is like trying to dance about architecture.

This left me in a bit of a pickle when it came to trying to explain in advance where I was going. I know from experience that if I say, "Bayes is the secret of the universe," some people may say "Yes! Bayes is the secret of the universe!"; and others will snort and say, "How narrow-minded you are; look at all these other ad-hoc but amazingly useful methods, like regularized linear regression, that I have in my toolbox."

I hoped that with a specific example in hand of "something that doesn't look all that Bayesian on the surface, but turns out to be Bayesian after all"—*and* an explanation of the difference between passwords and knowledge—*and* an explanation of the difference between tools and laws—maybe *then* I could convey such of the rhythm as can be understood without personally going on the quest.

Of course this is not the *full* Secret of the Bayesian Conspiracy, but it's all that I can convey at this point. Besides, the complete secret is known only to the Bayes Council, and if I told you, I'd have to hire you.

To *see through* the surface adhockery of a cognitive process, to the Bayesian structure *underneath*—to perceive the probability flows, and *know how*, not just *know that*, this cognition too is Bayesian—as it always is—as it always must be—to be able to sense the Force underlying all cognition—this, is the Bayes-Sight.

“...And the Queen of Kashfa sees with the Eye of the Serpent.”

“I don't know that she sees with it,” I said. “She's still recovering from the operation. But that's an

interesting thought. If she could see with it, what might she behold?”

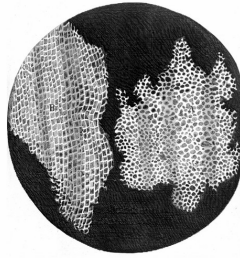
“The clear, cold lines of eternity, I daresay. Beneath all Shadow.”

—Roger Zelazny, *Prince of Chaos*¹

1. Roger Zelazny, *Prince of Chaos* (Thorndike Press, 2001).

PART P

Reductionism 101



Dissolving the Question

“If a tree falls in the forest, but no one hears it, does it make a sound?”

I didn’t *answer* that question. I didn’t pick a position “Yes!” or “No!” and defend it. Instead I went off and deconstructed the human algorithm for processing words, even going so far as to sketch an illustration of a neural network. At the end, I hope, there was no question left—not even the feeling of a question.

Many philosophers—particularly amateur philosophers, and ancient philosophers—share a dangerous instinct: If you give them a question, they try to answer it.

Like, say, “Do we have free will?”

The dangerous instinct of philosophy is to marshal the arguments in favor, and marshal the arguments against, and weigh them up, and publish them in a prestigious journal of philosophy, and so finally conclude: “Yes, we must have free will,” or “No, we cannot possibly have free will.”

Some philosophers are wise enough to recall the warning that most philosophical disputes are really disputes over the meaning of a word, or confusions generated by using different meanings for the same word in different places. So they try to define very precisely what they mean by “free will,” and then ask again, “Do we have free will? Yes or no?”

A philosopher wiser yet may suspect that the confusion about “free will” shows the notion itself is flawed. So they pursue the Traditional Rationalist course: They argue that “free will” is inherently self-contradictory, or meaningless because it has no testable consequences. And then they publish these devastating observations in a prestigious philosophy journal.

But *proving that* you are confused may not make you feel any *less* confused. Proving that a question is meaningless may not help you any more than answering it.



discussion
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4-c10

The philosopher's instinct is to find the most defensible position, publish it, and move on. But the "naive" view, the instinctive view, is a fact about human psychology. You can prove that free will is impossible until the Sun goes cold, but this leaves an unexplained fact of cognitive science: If free will doesn't exist, what goes on inside the head of a human being who thinks it does? This is not a rhetorical question!

It is a fact about human psychology that people think they have free will. Finding a more defensible *philosophical position* doesn't change, or explain, that *psychological fact*. Philosophy may lead you to *reject* the concept, but rejecting a concept is not the same as understanding the cognitive algorithms behind it.

You could look at the Standard Dispute over "If a tree falls in the forest, and no one hears it, does it make a sound?" and you could do the Traditional Rationalist thing: Observe that the two don't disagree on any point of anticipated experience, and triumphantly declare the argument pointless. That happens to be correct in this particular case; but, as a *question of cognitive science*, why did the arguers make that mistake in the first place?

The key idea of the heuristics and biases program is that the *mistakes* we make often reveal far more about our underlying cognitive algorithms than our correct answers. So (I asked myself, once upon a time) what kind of mind design corresponds to the mistake of arguing about trees falling in deserted forests?

The cognitive algorithms we use *are* the way the world feels. And these cognitive algorithms may not have a one-to-one correspondence with reality—not even macroscopic reality, to say nothing of the true quarks. There can be things in the mind that cut skew to the world.

For example, there can be a dangling unit in the center of a neural network, which does not correspond to any real thing, or any real property of any real thing, existent anywhere in the real world. This dangling unit is often useful as a shortcut in computation, which is why we have them. (Metaphorically speaking. Human neurobiology is surely far more complex.)

This dangling unit *feels like* an unresolved question, even after every answerable query is answered. No matter how much anyone proves to you that no difference of anticipated experience depends on the question, you're left wondering: "But does the falling tree *really* make a sound, or not?"

But once you understand *in detail* how your brain generates the *feeling* of the question—once you realize that your feeling of an unanswered question corresponds to an illusory central unit wanting to know whether it should fire, even after all the edge units are clamped at known values—or better yet, you understand the technical workings of Naive Bayes—*then* you're done. Then there's no lingering feeling of confusion, no vague sense of dissatisfaction.

If there is *any* lingering feeling of a remaining unanswered question, or of having been fast-talked into something, then this is a sign that you have not dissolved the question. A vague dissatisfaction should be as much warning as a shout. *Really* dissolving the question doesn't leave anything behind.

A triumphant thundering refutation of free will, an absolutely unarguable proof that free will cannot exist, feels very *satisfying*—a grand cheer for the home team. And so you may not notice that—as a point of cognitive science—you do not have a full and satisfactory descriptive explanation of how each intuitive sensation arises, point by point.

You may not even want to admit your ignorance of this point of cognitive science, because that would feel like a score against Your Team. In the midst of smashing all foolish beliefs of free will, it would seem like a concession to the opposing side to concede that you've left anything unexplained.

And so, perhaps, you'll come up with a just-so evolutionary-psychological argument that hunter-gatherers who believed in free will were more likely to take a positive outlook on life, and so outreproduce other hunter-gatherers—to give one example of a completely bogus explanation. If you say this, you are *arguing that* the brain generates an illusion of free will—but you are not *explaining how*. You are trying to dismiss the opposition by deconstructing its motives—but in the story you

tell, the illusion of free will is a brute fact. You have not taken the illusion apart to see the wheels and gears.

Imagine that in the Standard Dispute about a tree falling in a deserted forest, you first prove that no difference of anticipation exists, and then go on to hypothesize, “But perhaps people who said that arguments were meaningless were viewed as having conceded, and so lost social status, so now we have an instinct to argue about the meanings of words.” That’s *arguing that* or *explaining why* a confusion exists. Now look at the neural network structure in *Feel the Meaning*. That’s *explaining how*, disassembling the confusion into smaller pieces that are not themselves confusing. See the difference?

Coming up with good hypotheses about cognitive algorithms (or even hypotheses that hold together for half a second) is a good deal harder than just refuting a philosophical confusion. Indeed, it is an entirely different art. Bear this in mind, and you should feel less embarrassed to say, “I know that what you say can’t possibly be true, and I can prove it. But I cannot write out a flowchart which shows how your brain makes the mistake, so I’m not done yet, and will continue investigating.”

I say all this, because it sometimes seems to me that at least 20% of the real-world effectiveness of a skilled rationalist comes from not stopping too early. If you keep asking questions, you’ll get to your destination eventually. If you decide too early that you’ve found an answer, you won’t.

The challenge, above all, is to notice when you are confused—even if it just feels like a little tiny bit of confusion—and even if there’s someone standing across from you, *insisting* that humans have free will, and *smirking* at you, and the fact that you don’t know *exactly* how the cognitive algorithms work, has *nothing to do* with the searing folly of their position ...

But when you can lay out the cognitive algorithm in sufficient detail that you can walk through the thought process, step by step, and describe how each intuitive perception arises—decompose the confusion into smaller pieces not themselves confusing—*then* you’re done.

So be warned that you may *believe* you’re done, when all you have is a mere triumphant refutation of a mistake.

But when you're *really* done, you'll *know* you're done. Dissolving the question is an unmistakable feeling—once you experience it, and, having experienced it, resolve not to be fooled again. Those who dream do not know they dream, but when you wake you know you are awake.

Which is to say: When you're done, you'll know you're done, but unfortunately the reverse implication does not hold.

So here's your homework problem: What kind of cognitive algorithm, as felt from the inside, would generate the observed debate about "free will"?

Your assignment is not to argue about whether people have free will, or not.

Your assignment is not to argue that free will is compatible with determinism, or not.

Your assignment is not to argue that the question is ill-posed, or that the concept is self-contradictory, or that it has no testable consequences.

You are not asked to invent an evolutionary explanation of how people who believed in free will would have reproduced; nor an account of how the concept of free will seems suspiciously congruent with bias X. Such are mere attempts to *explain why* people believe in "free will," not *explain how*.

Your homework assignment is to write a stack trace of the internal algorithms of the human mind as they produce the intuitions that power the whole damn philosophical argument.

This is one of the first real challenges I tried as an aspiring rationalist, once upon a time. One of the easier conundrums, relatively speaking. May it serve you likewise.

Wrong Questions



discussion
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4-c11

Where the mind cuts against reality's grain, it generates *wrong questions*—questions that cannot possibly be answered *on their own terms*, but only dissolved by understanding the cognitive algorithm that generates the *perception* of a question.

One good cue that you're dealing with a "wrong question" is when you cannot even *imagine* any concrete, specific state of how-the-world-is that would answer the question. When it doesn't even seem *possible* to answer the question.

Take the Standard Definitional Dispute, for example, about the tree falling in a deserted forest. Is there any way-the-world-could-be—any state of affairs—that corresponds to the word "sound" *really meaning* only acoustic vibrations, or *really meaning* only auditory experiences?

("Why, yes," says the one, "it is the state of affairs where 'sound' means acoustic vibrations." So Taboo the word "means," and "represents," and all similar synonyms, and describe again: What way-the-world-can-be, what state of affairs, would make one side right, and the other side wrong?)

Or if that seems too easy, take free will: What concrete state of affairs, whether in deterministic physics, or in physics with a dice-rolling random component, could ever correspond to having free will?

And if *that* seems too easy, then ask "Why does anything exist at all?," and then tell me what a satisfactory answer to that question would even *look like*.

And no, I don't know the answer to that last one. But I *can* guess one thing, based on my previous experience with unanswerable questions. The answer will not consist of some grand triumphant First Cause. The question will go away as a result of some insight into how my mental algorithms run skew to reality, after which I will understand how the question itself was wrong from the beginning—how the question itself assumed the fallacy, contained the skew.

Mystery exists in the mind, not in reality. If I am ignorant about a phenomenon, that is a fact about my state of mind, not a fact about the phenomenon itself. All the more so if it seems like no possible answer can exist: Confusion exists in the map, not in the territory. *Unanswerable* questions do not mark places where magic enters the universe. They mark places where your mind runs skew to reality.

Such questions *must* be dissolved. Bad things happen when you try to answer them. It inevitably generates the worst sort of Mysterious Answer to a Mysterious Question: The one where you come up with seemingly strong arguments for your Mysterious Answer, but the “answer” doesn’t let you make any new predictions even in retrospect, and the phenomenon still possesses the same sacred inexplicability that it had at the start.

I could guess, for example, that the answer to the puzzle of the First Cause is that nothing *does* exist—that the whole concept of “existence” is bogus. But if you sincerely believed that, would you be any less confused? Me neither.

But the wonderful thing about *unanswerable* questions is that they are *always* solvable, at least in my experience. What went through Queen Elizabeth I’s mind, first thing in the morning, as she woke up on her fortieth birthday? As I can easily *imagine* answers to this question, I can readily see that I may never be able to *actually* answer it, the true information having been lost in time.

On the other hand, “Why does anything exist at all?” seems so absolutely impossible that I can infer that I am just confused, one way or another, and the truth probably isn’t all that complicated in an absolute sense, and once the confusion goes away I’ll be able to see it.

This may seem counterintuitive if you’ve never solved an unanswerable question, but I assure you that it *is* how these things work.

Coming next: a simple trick for handling “wrong questions.”

Righting a Wrong Question



discussion
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4-c12

When you are faced with an *unanswerable* question—a question to which it seems impossible to even *imagine* an answer—there is a simple trick that can turn the question solvable.

Compare:

- “Why do I have free will?”
- “Why do I think I have free will?”

The nice thing about the second question is that it is *guaranteed* to have a real answer, *whether or not* there is any such thing as free will. Asking “Why do I have free will?” or “Do I have free will?” sends you off thinking about tiny details of the laws of physics, so distant from the macroscopic level that you couldn’t begin to see them with the naked eye. And you’re asking “Why is *X* the case?” where *X* may not be *coherent*, let alone the case.

“Why do I *think* I have free will?” in contrast, is guaranteed answerable. You do, in fact, believe you have free will. This belief seems far more solid and graspable than the ephemerality of free will. And there is, *in fact*, some nice solid chain of cognitive cause and effect leading up to this belief.

If you’ve already outgrown free will, choose one of these substitutes:

- “Why does time move forward instead of backward?” versus “Why do I think time moves forward instead of backward?”
- “Why was I born as myself rather than someone else?” versus “Why do I think I was born as myself rather than someone else?”
- “Why am I conscious?” versus “Why do I think I’m conscious?”

- “Why does reality exist?” versus “Why do I think reality exists?”

The beauty of this method is that it works *whether or not* the question is confused. As I type this, I am wearing socks. I could ask “Why am I wearing socks?” or “Why do I believe I’m wearing socks?” Let’s say I ask the second question. Tracing back the chain of causality, I find:

- I believe I’m wearing socks, because I can see socks on my feet.
- I see socks on my feet, because my retina is sending sock signals to my visual cortex.
- My retina is sending sock signals, because sock-shaped light is impinging on my retina.
- Sock-shaped light impinges on my retina, because it reflects from the socks I’m wearing.
- It reflects from the socks I’m wearing, because I’m wearing socks.
- I’m wearing socks because I put them on.
- I put socks on because I believed that otherwise my feet would get cold.
- Et cetera.

Tracing back the chain of causality, step by step, I discover that my belief that I’m wearing socks is fully explained by the fact that I’m wearing socks. This is right and proper, as you cannot gain information about something without interacting with it.

On the other hand, if I see a mirage of a lake in a desert, the correct causal explanation of my vision does not involve the fact of any actual lake in the desert. In this case, my belief in the lake is not just *explained*, but *explained away*.

But *either way*, the belief itself is a real phenomenon taking place in the real universe—psychological events are events—and its causal history can be traced back.

“Why is there a lake in the middle of the desert?” may fail if there is no lake to be explained. But “Why do I *perceive* a lake in the middle of the desert?” always has a causal explanation, one way or the other.

Perhaps someone will see an opportunity to be clever, and say: “Okay. I believe in free will because I have free will. There, I’m done.” Of course it’s not that easy.

My perception of socks on my feet is an event in the visual cortex. The workings of the visual cortex can be investigated by cognitive science, should they be confusing.

My retina receiving light is not a mystical sensing procedure, a magical sock detector that lights in the presence of socks for no explicable reason; there are mechanisms that can be understood in terms of biology. The photons entering the retina can be understood in terms of optics. The shoe’s surface reflectance can be understood in terms of electromagnetism and chemistry. My feet getting cold can be understood in terms of thermodynamics.

So it’s not as easy as saying, “I believe I have free will because I have it—there, I’m done!” You have to be able to break the causal chain into smaller steps, and explain the steps in terms of elements not themselves confusing.

The mechanical interaction of my retina with my socks is quite clear, and can be described in terms of non-confusing components like photons and electrons. Where’s the free-will-sensor in your brain, and how does it detect the presence or absence of free will? How does the sensor interact with the sensed event, and what are the mechanical details of the interaction?

If your belief does derive from valid observation of a real phenomenon, we will eventually reach that fact, if we start tracing the causal chain backward from your belief.

If what you are really seeing is your own confusion, tracing back the chain of causality will find an algorithm that runs skew to reality.

Either way, the question is guaranteed to have an answer. You even have a nice, concrete place to begin tracing—your belief, sitting there solidly in your mind.

Cognitive science may not seem so lofty and glorious as metaphysics. But at least questions of cognitive science are *solvable*. Finding an answer may not be *easy*, but at least an answer *exists*.

Oh, and also: the idea that cognitive science is not so lofty and glorious as metaphysics is simply wrong. Some readers are beginning to notice this, I hope.

Mind Projection Fallacy



discussion
sharov.me/r/
4-c13

In the dawn days of science fiction, alien invaders would occasionally kidnap a girl in a torn dress and carry her off for intended ravishing, as lovingly depicted on many ancient magazine covers. Oddly enough, the aliens never go after men in torn shirts.



Would a non-humanoid alien, with a different evolutionary history and evolutionary psychology, sexually desire a human female? It seems rather unlikely. To put it mildly.

People don't make mistakes like that by deliberately reasoning: "All possible minds are likely to be wired pretty much the same way, therefore a bug-eyed monster will find human females attractive." Probably the artist did not even think to ask whether an alien *perceives* human females as attractive. Instead, a human female in a torn dress *is sexy*—inherently so, as an intrinsic property.

They who went astray did not think about the alien's evolutionary history; they focused on the woman's torn dress. If the

dress were not torn, the woman would be less sexy; the alien monster doesn't enter into it.

Apparently we instinctively represent Sexiness as a direct attribute of the Woman data structure, `Woman.sexiness`, like `Woman.height` or `Woman.weight`.

If your brain uses that data structure, or something metaphorically similar to it, then from the inside it feels like sexiness is an inherent property of the woman, not a property of the alien looking at the woman. Since the woman *is attractive*, the alien monster will be *attracted* to her—isn't that logical?

E. T. Jaynes used the term Mind Projection Fallacy to denote the error of projecting your own mind's properties into the external world. Jaynes, as a late grand master of the Bayesian Conspiracy, was most concerned with the mistreatment of *probabilities* as inherent properties of objects, rather than states of partial knowledge in some particular mind. More about this shortly.

But the Mind Projection Fallacy generalizes as an error. It is in the argument over the real meaning of the word sound, and in the magazine cover of the monster carrying off a woman in the torn dress, and Kant's declaration that space by its very nature is flat, and Hume's definition of a priori ideas as those "discoverable by the mere operation of thought, without dependence on what is anywhere existent in the universe" ...

(Incidentally, I once read a science fiction story about a human male who entered into a sexual relationship with a sentient alien plant of appropriately squishy fronds; discovered that it was an androecious (male) plant; agonized about this for a bit; and finally decided that it didn't really matter at that point. And in Foglio and Pollotta's *Illegal Aliens*, the humans land on a planet inhabited by sentient insects, and see a movie advertisement showing a human carrying off a bug in a delicate chiffon dress. Just thought I'd mention that.)

Probability is in the Mind



discussion
sharov.me/r/
4-c14

In the previous essay I spoke of the Mind Projection Fallacy, giving the example of the alien monster who carries off a girl in a torn dress for intended ravishing—a mistake which I imputed to the artist’s tendency to think that a woman’s sexiness is a property of the woman herself, *woman. sexiness*, rather than something that exists in the mind of an observer, and probably wouldn’t exist in an alien mind.

The term “Mind Projection Fallacy” was coined by the late great Bayesian Master E. T. Jaynes, as part of his long and hard-fought battle against the accursed frequentists. Jaynes was of the opinion that probabilities were in the mind, not in the environment—that probabilities express ignorance, states of partial information; and if I am ignorant of a phenomenon, that is a fact about my state of mind, not a fact about the phenomenon.

I cannot do justice to this ancient war in a few words—but the classic example of the argument runs thus:

You have a coin.

The coin is biased.

You don’t know which way it’s biased or how much it’s biased. Someone just told you “The coin is biased,” and that’s all they said.

This is all the information you have, and the only information you have.

You draw the coin forth, flip it, and slap it down.

Now—before you remove your hand and look at the result—are you willing to say that you assign a 0.5 probability to the coin’s having come up heads?

The frequentist says, “No. Saying ‘probability 0.5’ means that the coin has an inherent propensity to come up heads as often as tails, so that if we flipped the coin infinitely many times, the ratio of heads to tails would approach 1:1. But we know that

the coin is biased, so it can have any probability of coming up heads *except* 0.5.”

The Bayesian says, “Uncertainty exists in the map, not in the territory. In the real world, the coin has either come up heads, or come up tails. Any talk of ‘probability’ must refer to the *information* that I have about the coin—my state of partial ignorance and partial knowledge—not just the coin itself. Furthermore, I have all sorts of theorems showing that if I don’t treat my partial knowledge a certain way, I’ll make stupid bets. If I’ve got to plan, I’ll plan for a 50/50 state of uncertainty, where I don’t weigh outcomes conditional on heads any more heavily in my mind than outcomes conditional on tails. You can call that number whatever you like, but it has to obey the probability laws on pain of stupidity. So I don’t have the slightest hesitation about calling my outcome-weighting a probability.”

I side with the Bayesians. You may have noticed that about me.

Even before a fair coin is tossed, the notion that it has an *inherent* 50% probability of coming up heads may be just plain wrong. Maybe you’re holding the coin in such a way that it’s just about guaranteed to come up heads, or tails, given the force at which you flip it, and the air currents around you. But, if you don’t know which way the coin is biased on this one occasion, so what?

I believe there was a lawsuit where someone alleged that the draft lottery was unfair, because the slips with names on them were not being mixed thoroughly enough; and the judge replied, “To whom is it unfair?”

To make the coinflip experiment repeatable, as frequentists are wont to demand, we could build an automated coinflipper, and verify that the results were 50% heads and 50% tails. But maybe a robot with extra-sensitive eyes and a good grasp of physics, watching the autoflipper prepare to flip, could predict the coin’s fall in advance—not with certainty, but with 90% accuracy. Then what would the *real* probability be?

There is no “real probability.” The robot has one state of partial information. You have a different state of partial information. The coin itself has no mind, and doesn’t assign a

probability to anything; it just flips into the air, rotates a few times, bounces off some air molecules, and lands either heads or tails.

So that is the Bayesian view of things, and I would now like to point out a couple of classic brainteasers that derive their *brain-teasing* ability from the tendency to think of probabilities as inherent properties of objects.

Let's take the old classic: You meet a mathematician on the street, and she happens to mention that she has given birth to two children on two separate occasions. You ask: "Is at least one of your children a boy?" The mathematician says, "Yes, he is."

What is the probability that she has two boys? If you assume that the prior probability of a child's being a boy is $1/2$, then the probability that she has two boys, on the information given, is $1/3$. The prior probabilities were: $1/4$ two boys, $1/2$ one boy one girl, $1/4$ two girls. The mathematician's "Yes" response has probability ~ 1 in the first two cases, and probability ~ 0 in the third. Renormalizing leaves us with a $1/3$ probability of two boys, and a $2/3$ probability of one boy one girl.

But suppose that instead you had asked, "Is your eldest child a boy?" and the mathematician had answered "Yes." Then the probability of the mathematician having two boys would be $1/2$. Since the eldest child is a boy, and the younger child can be anything it pleases.

Likewise if you'd asked "Is your youngest child a boy?" The probability of their being both boys would, again, be $1/2$.

Now, if at least one child is a boy, it must be either the oldest child who is a boy, or the youngest child who is a boy. So how can the answer in the first case be different from the answer in the latter two?

Or here's a very similar problem: Let's say there are four cards, the ace of hearts, the ace of spades, the two of hearts, and the two of spades. I draw two cards at random. You ask me, "Are you holding at least one ace?" and I reply "Yes." What is the probability that I am holding a pair of aces? It is $1/5$. There are six possible combinations of two cards, with equal prior probability, and you have just eliminated the possibility that I

am holding a pair of twos. Of the five remaining combinations, only one combination is a pair of aces. So $1/5$.

Now suppose that instead you asked me, “Are you holding the ace of spades?” If I reply “Yes,” the probability that the other card is the ace of hearts is $1/3$. (You know I’m holding the ace of spades, and there are three possibilities for the other card, only one of which is the ace of hearts.) Likewise, if you ask me “Are you holding the ace of hearts?” and I reply “Yes,” the probability I’m holding a pair of aces is $1/3$.

But then how can it be that if you ask me, “Are you holding at least one ace?” and I say “Yes,” the probability I have a pair is $1/5$? Either I must be holding the ace of spades or the ace of hearts, as you know; and either way, the probability that I’m holding a pair of aces is $1/3$.

How can this be? Have I miscalculated one or more of these probabilities?

If you want to figure it out for yourself, do so now, because I’m about to reveal ...

That all stated calculations are correct.

As for the paradox, there isn’t one. The *appearance* of paradox comes from thinking that the probabilities must be properties of the cards themselves. The ace I’m holding has to be either hearts or spades; but that doesn’t mean that your *knowledge about* my cards must be the same as if you *knew* I was holding hearts, or *knew* I was holding spades.

It may help to think of Bayes’s Theorem:

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}.$$

That last term, where you divide by $P(E)$, is the part where you throw out all the possibilities that have been eliminated, and renormalize your probabilities over what remains.

Now let’s say that you ask me, “Are you holding at least one ace?” *Before* I answer, your probability that I say “Yes” should be $5/6$.

But if you ask me “Are you holding the ace of spades?” your prior probability that I say “Yes” is just $1/2$.

So right away you can see that you're *learning* something very different in the two cases. You're going to be eliminating some different possibilities, and renormalizing using a different $P(E)$. If you learn two different items of evidence, you shouldn't be surprised at ending up in two different states of partial information.

Similarly, if I ask the mathematician "Is at least one of your two children a boy?" then I expect to hear "Yes" with probability $3/4$, but if I ask "Is your eldest child a boy?" then I expect to hear "Yes" with probability $1/2$. So it shouldn't be surprising that I end up in a different state of partial knowledge, depending on which of the two questions I ask.

The only reason for seeing a "paradox" is thinking as though the probability of holding a pair of aces is a *property of cards* that have at least one ace, or a *property of cards* that happen to contain the ace of spades. In which case, it would be paradoxical for card-sets containing at least one ace to have an inherent pair-probability of $1/5$, while card-sets containing the ace of spades had an inherent pair-probability of $1/3$, and card-sets containing the ace of hearts had an inherent pair-probability of $1/3$.

Similarly, if you think a $1/3$ probability of being both boys is an *inherent property* of child-sets that include at least one boy, then that is not consistent with child-sets' of which the eldest is male having an *inherent* probability of $1/2$ of being both boys, and child-sets' of which the youngest is male having an inherent $1/2$ probability of being both boys. It would be like saying, "All green apples weigh a pound, and all red apples weigh a pound, and all apples that are green or red weigh half a pound."

That's what happens when you start thinking as if probabilities are *in* things, rather than probabilities being states of partial information *about* things.

Probabilities express uncertainty, and it is only agents who can be uncertain. A blank map does not correspond to a blank territory. Ignorance is in the mind.

The Quotation is Not the Referent

In classical logic, the operational definition of identity is that whenever $A = B$ is a theorem, you can substitute A for B in any theorem where B appears. For example, if $(2 + 2) = 4$ is a theorem, and $((2 + 2) + 3) = 7$ is a theorem, then $(4 + 3) = 7$ is a theorem.

This leads to a problem that is usually phrased in the following terms: The morning star and the evening star happen to be the same object, the planet Venus. Suppose John knows that the morning star and evening star are the same object. Mary, however, believes that the morning star is the god Lucifer, but the evening star is the god Venus. John believes Mary believes that the morning star is Lucifer. Must John therefore (by substitution) believe that Mary believes that the evening star is Lucifer?

Or here's an even simpler version of the problem. The statement $2 + 2 = 4$ is true; it is a theorem that $((2 + 2) = 4) = \text{TRUE}$. Fermat's Last Theorem is also true. So: I believe $2 + 2 = 4 \Rightarrow$ I believe $\text{TRUE} \Rightarrow$ I believe Fermat's Last Theorem.

Yes, I know this seems *obviously* wrong. But imagine someone writing a logical reasoning program using the principle "equal terms can always be substituted," and this happening to them. Now imagine them writing a paper about how to prevent it from happening. Now imagine someone else disagreeing with their solution. The argument is still going on.

P'rsnally, I would say that John is committing a type error, like trying to subtract 5 grams from 20 meters. "The morning star" is not the same *type* as the morning star, let alone the same thing. Beliefs are not planets.

morning star = evening star
 "morning star" $\neq$ "evening star"



discussion
 sharov.me/r/
 4-c15

The problem, in my view, stems from the failure to enforce the type distinction between beliefs and things. The original error was writing an AI that stores its beliefs about Mary's beliefs about "the morning star" using the same representation as in its beliefs about the morning star.

If Mary believes the "morning star" is Lucifer, that doesn't mean Mary believes the "evening star" is Lucifer, because "morning star" $\neq$ "evening star." The whole paradox stems from the failure to use quote marks in appropriate places.

You may recall that this is not the first time I've talked about enforcing type discipline—the last time was when I spoke about the error of confusing expected utilities with utilities. It is immensely helpful, when one is first learning physics, to learn to keep track of one's units—it may seem like a bother to keep writing down "cm" and "kg" and so on, until you notice that (a) your answer seems to be the wrong order of magnitude and (b) it is expressed in seconds per square gram.

Similarly, beliefs are different things than planets. If we're talking about human beliefs, at least, then: Beliefs live in brains, planets live in space. Beliefs weigh a few micrograms, planets weigh a lot more. Planets are larger than beliefs ...but you get the idea.

Merely putting quote marks around "morning star" seems insufficient to prevent people from confusing it with the morning star, due to the visual similarity of the text. So perhaps a better way to enforce type discipline would be with a visibly different encoding:

morning star = evening star

13.15.18.14.9.14.7.0.19.20.1.18 $\neq$ 5.22.5.14.9.14.7.0.19.20.1.18.

Studying mathematical logic may also help you learn to distinguish the quote and the referent. In mathematical logic, $\vdash P$ (P is a theorem) and $\vdash \Box P$ (it is provable that there exists an encoded proof of the encoded sentence P in some encoded proof system) are very distinct propositions. If you drop a level of quotation in mathematical logic, it's like dropping a metric

unit in physics—you can derive visibly ridiculous results, like “The speed of light is 299,792,458 meters long.”

Alfred Tarski once tried to define the meaning of “true” using an infinite family of sentences:

 (“Snow is white” is true) if and only (snow is white)
 (“Weasels are green” is true) if and only if (weasels are green)
 ⋮

When sentences like these start seeming meaningful, you’ll know that you’ve started to distinguish between encoded sentences and states of the outside world.

Similarly, the notion of truth is quite different from the notion of *reality*. Saying “true” *compares* a belief to reality. Reality itself does not need to be compared to any beliefs in order to be real. Remember this the next time someone claims that nothing is true.

Qualitatively Confused



discussion
sharov.me/r/
4-c16

I suggest that a primary cause of confusion about the distinction between “belief,” “truth,” and “reality” is qualitative thinking about beliefs.

Consider the archetypal postmodernist attempt to be clever:

“The Sun goes around the Earth” is true for Hunga Huntergatherer, but “The Earth goes around the Sun” is true for Amara Astronomer! Different societies have different truths!

No, different societies have different *beliefs*. Belief is of a different type than truth; it’s like comparing apples and probabilities.

Ah, but there’s no difference between the way you use the word “belief” and the way you use the word “truth”! Whether you say, “I believe ‘snow is white,’” or you say, “‘Snow is white’ is true,” you’re expressing exactly the same opinion.

No, these sentences mean quite different things, which is how I can *conceive* of the possibility that my beliefs are false.

Oh, you claim to *conceive* it, but you never *believe* it. As Wittgenstein said, “If there were a verb meaning ‘to believe falsely,’ it would not have any significant first person, present indicative.”

And that’s what I mean by putting my finger on qualitative reasoning as the source of the problem. The dichotomy between belief and disbelief, being binary, is confusingly similar to the dichotomy between truth and untruth.

So let’s use quantitative reasoning instead. Suppose that I assign a 70% probability to the proposition that snow is white.

It follows that I think there's around a 70% chance that the sentence "snow is white" will turn out to be true. If the sentence "snow is white" is true, is my 70% probability assignment to the proposition, also "true"? Well, it's more true than it would have been if I'd assigned 60% probability, but not so true as if I'd assigned 80% probability.

When talking about the correspondence between a probability assignment and reality, a better word than "truth" would be "accuracy." "Accuracy" sounds more quantitative, like an archer shooting an arrow: how close did your probability assignment strike to the center of the target?

To make a long story short, it turns out that there's a very natural way of scoring the accuracy of a probability assignment, as compared to reality: just take the logarithm of the probability assigned to the real state of affairs.

So if snow is white, my belief "70%: 'snow is white'" will score -0.51 bits: $\log_2(0.7) = -0.51$.

But what if snow is not white, as I have conceded a 30% probability is the case? If "snow is white" is false, my belief "30% probability: 'snow is not white'" will score -1.74 bits. Note that $-1.74 < -0.51$, so I have done worse.

About how accurate do I think my own beliefs are? Well, my expectation over the score is $70\% \times -0.51 + 30\% \times -1.74 = -0.88$ bits. If snow is white, then my beliefs will be more accurate than I expected; and if snow is not white, my beliefs will be less accurate than I expected; but in neither case will my belief be *exactly* as accurate as I expected on average.

All this should not be confused with the statement "I assign 70% credence that 'snow is white.'" I may well believe *that* proposition with probability ~ 1 —be quite certain that this is in fact my belief. If so I'll expect my meta-belief " ~ 1 : 'I assign 70% credence that "snow is white"'" to score ~ 0 bits of accuracy, which is as good as it gets.

Just because I am uncertain about snow, does not mean I am uncertain about my *quoted probabilistic beliefs*. Snow is out there, my beliefs are inside me. I may be a great deal less uncertain about how uncertain I am about snow, than I am

uncertain about snow. (Though beliefs about beliefs are not always accurate.)

Contrast this probabilistic situation to the qualitative reasoning where I just believe that snow is white, and believe that I believe that snow is white, and believe “‘snow is white’ is true,” and believe “my belief ‘‘snow is white’’ is true” is correct,” etc. Since all the quantities involved are 1, it’s easy to mix them up.

Yet the nice distinctions of quantitative reasoning will be short-circuited if you start thinking “‘‘snow is white’’ with 70% probability’ is *true*,” which is a type error. It is a true fact about you, that you *believe* “70% probability: ‘snow is white’”; but that does not mean the probability assignment *itself* can possibly be “true.” The belief scores either -0.51 bits or -1.74 bits of accuracy, depending on the actual state of reality.

The cognoscenti will recognize “‘‘snow is white’’ with 70% probability’ is true” as the mistake of thinking that probabilities are inherent properties of things.

From the inside, our beliefs about the world look like the world, and our beliefs about our beliefs look like beliefs. When you see the world, you are experiencing a belief from the inside. When you notice yourself believing something, you are experiencing a belief about belief from the inside. So if your internal representations of belief, and belief about belief, are dissimilar, then you are less likely to mix them up and commit the Mind Projection Fallacy—I hope.

When you think in probabilities, your beliefs, and your beliefs about your beliefs, will hopefully not be represented similarly enough that you mix up belief and accuracy, or mix up accuracy and reality. When you think in probabilities *about the world*, your beliefs will be represented with probabilities in the range (0,1). Unlike the truth-values of propositions, which are in the set {true, false}. As for the accuracy of your probabilistic belief, you can represent that in the range $(-\infty, 0)$. Your probabilities *about your beliefs* will typically be extreme. And things themselves—why, they’re just red, or blue, or weighing 20 pounds, or whatever.

Thus we will be less likely, perhaps, to mix up the map with the territory.

This type distinction may also help us remember that *uncertainty* is a state of mind. A coin is not *inherently* 50% uncertain of which way it will land. The coin is not a belief processor, and does not have partial information about itself. In qualitative reasoning you can create a belief that corresponds very straightforwardly to the coin, like “The coin will land heads.” This belief will be true or false *depending on* the coin, and there will be a transparent implication from the truth or falsity of the belief, to the facing side of the coin.

But even under qualitative reasoning, to say that the coin *itself* is “true” or “false” would be a severe type error. The coin is not a belief. It is a coin. The territory is not the map.

If a coin cannot be true or false, how much less can it assign a 50% probability to itself?

Think Like Reality



discussion
sharov.me/r/
4-c17

Whenever I hear someone describe quantum physics as “weird”—whenever I hear someone bemoaning the mysterious effects of observation on the observed, or the bizarre existence of nonlocal correlations, or the incredible impossibility of knowing position and momentum at the same time—then I think to myself: *This person will never understand physics no matter how many books they read.*

Reality has been around since long before you showed up. Don’t go calling it nasty names like “bizarre” or “incredible.” The universe was propagating complex amplitudes through configuration space for ten billion years before life ever emerged on Earth. Quantum physics is not “weird.” *You* are weird. *You* have the absolutely bizarre idea that reality ought to consist of little billiard balls bopping around, when in fact reality is a perfectly normal cloud of complex amplitude in configuration space. This is *your* problem, not reality’s, and *you* are the one who needs to change.

Human intuitions were produced by evolution and evolution is a hack. The same optimization process that built your retina backward and then routed the optic cable through your field of vision, also designed your visual system to process persistent objects bouncing around in three spatial dimensions because that’s what it took to chase down tigers. But “tigers” are leaky surface generalizations—tigers came into existence gradually over evolutionary time, and they are not all absolutely similar to each other. When you go down to the fundamental level, the level on which the laws are stable, global, and exception-free, there aren’t any tigers. In fact there aren’t any persistent objects bouncing around in three spatial dimensions. Deal with it.

Calling reality “weird” keeps you inside a viewpoint already proven erroneous. Probability theory tells us that surprise is the

measure of a poor hypothesis; if a model is consistently stupid—consistently hits on events the model assigns tiny probabilities—then it’s time to discard that model. A good model makes reality look *normal*, not weird; a good model assigns high probability to that which is actually the case. Intuition is only a model by another name: poor intuitions are shocked by reality, good intuitions make reality feel natural. You want to reshape your intuitions so that the universe looks normal. You want to think like reality.

This end state cannot be forced. It is pointless to pretend that quantum physics feels natural to you when in fact it feels strange. This is merely denying your confusion, not becoming less confused. But it will also hinder you to keep thinking *How bizarre!* Spending emotional energy on incredulity wastes time you could be using to update. It repeatedly throws you back into the frame of the old, wrong viewpoint. It feeds your sense of righteous indignation at reality daring to contradict you.

The principle extends beyond physics. Have you ever caught yourself saying something like, “I just don’t understand how a PhD physicist can believe in astrology?” Well, if you literally *don’t understand*, this indicates a problem with your model of human psychology. Perhaps you are *indignant*—you wish to express strong moral disapproval. But if you literally *don’t understand*, then your indignation is stopping you from coming to terms with reality. It shouldn’t be hard to imagine how a PhD physicist ends up believing in astrology. People compartmentalize, enough said.

I now try to avoid using the English idiom “I just don’t understand how ...” to express indignation. If I *genuinely* don’t understand how, then my model is being surprised by the facts, and I should discard it and find a better model.

Surprise exists in the map, not in the territory. There are no surprising facts, only models that are surprised by facts. Likewise for facts called such nasty names as “bizarre,” “incredible,” “unbelievable,” “unexpected,” “strange,” “anomalous,” or “weird.” When you find yourself tempted by such labels, it may be wise to check if the alleged fact is really factual. But if the fact checks out, then the problem isn’t the fact—it’s you.

Chaotic Inversion



discussion
sharov.me/r/
4-c18

I was recently having a conversation with some friends on the topic of hour-by-hour productivity and willpower maintenance—something I’ve struggled with my whole life.

I can avoid running away from a hard problem the first time I see it (perseverance on a timescale of seconds), and I can stick to the same problem for years; but to keep working on a timescale of *hours* is a constant battle for me. It goes without saying that I’ve already read reams and reams of advice; and the most help I got from it was realizing that a sizable fraction of other creative professionals had the same problem, and couldn’t beat it either, no matter how reasonable all the advice sounds.

“What do you do when you can’t work?” my friends asked me. (Conversation probably not accurate, this is a very loose gist.)

And I replied that I usually browse random websites, or watch a short video.

“Well,” they said, “if you know you can’t work for a while, you should watch a movie or something.”

“Unfortunately,” I replied, “I have to do something whose time comes in short units, like browsing the Web or watching short videos, because I might become able to work again at any time, and I can’t predict when—”

And then I stopped, because I’d just had a revelation.

I’d always thought of my workcycle as something *chaotic*, something *unpredictable*. I never used those words, but that was the way I *treated* it.

But here my friends seemed to be implying—what a strange thought—that *other* people could predict when they would become able to work again, and structure their time accordingly.

And it occurred to me for the first time that I might have been committing that damned old chestnut the Mind Projection Fallacy, right out there in my ordinary everyday life instead of high abstraction.

Maybe it wasn't that my productivity was *unusually chaotic*; maybe I was just *unusually stupid* with respect to predicting it.

That's what inverted stupidity looks like—chaos. Something hard to handle, hard to grasp, hard to guess, something you can't do anything with. It's not just an idiom for high abstract things like Artificial Intelligence. It can apply in ordinary life too.

And the reason we don't think of the alternative explanation "I'm stupid," is *not*—I suspect—that we think so highly of ourselves. It's just that we don't think of ourselves at all. We just see a chaotic feature of the environment.

So now it's occurred to me that my productivity problem may not be chaos, but my own stupidity.

And that may or may not help anything. It certainly doesn't fix the problem right away. Saying "I'm ignorant" doesn't make you knowledgeable.

But it is, at least, a different path than saying "it's too chaotic."

Reductionism



discussion
sharov.me/r/
4-c19

Almost one year ago, in April 2007, Matthew C. submitted the following suggestion for an *Overcoming Bias* topic:

How and why the current reigning philosophical hegemon (reductionistic materialism) is obviously correct [...], while the reigning philosophical viewpoints of all past societies and civilizations are obviously suspect—

I remember this, because I looked at the request and deemed it legitimate, but I knew I couldn't do that topic until I'd started on the Mind Projection Fallacy sequence, which wouldn't be for a while ...

But now it's time to begin addressing this question. And while I haven't yet come to the "materialism" issue, we can now start on "reductionism."

First, let it be said that I do indeed hold that "reductionism," according to the meaning I will give for that word, is obviously correct; and to perdition with any past civilizations that disagreed.

This seems like a strong statement, at least the first part of it. General Relativity seems well-supported, yet who knows but that some future physicist may overturn it?

On the other hand, we are never going *back* to Newtonian mechanics. The ratchet of science turns, but it does not turn in reverse. There are cases in scientific history where a theory suffered a wound or two, and then bounced back; but when a theory takes as many arrows through the chest as Newtonian mechanics, it *stays dead*.

"To hell with what past civilizations thought" seems safe enough, when past civilizations believed in something that has been falsified to the trash heap of history.

And reductionism is not so much a positive hypothesis, as the *absence* of belief—in particular, disbelief in a form of the Mind Projection Fallacy.

I once met a fellow who claimed that he had experience as a Navy gunner, and he said, “When you fire artillery shells, you’ve got to compute the trajectories using Newtonian mechanics. If you compute the trajectories using relativity, you’ll get the wrong answer.”

And I, and another person who was present, said flatly, “No.” I added, “You might not be able to compute the trajectories fast enough to get the answers in time—maybe that’s what you mean? But the relativistic answer will always be more accurate than the Newtonian one.”

“No,” he said, “I mean that relativity will give you the *wrong answer*, because things moving at the speed of artillery shells are governed by Newtonian mechanics, not relativity.”

“If that were really true,” I replied, “you could publish it in a physics journal and collect your Nobel Prize.”

Standard physics uses the same *fundamental* theory to describe the flight of a Boeing 747 airplane, and collisions in the Relativistic Heavy Ion Collider. Nuclei and airplanes alike, according to our understanding, are obeying Special Relativity, quantum mechanics, and chromodynamics.

But we use entirely different *models* to understand the aerodynamics of a 747 and a collision between gold nuclei in the RHIC. A computer modeling the aerodynamics of a 747 may not contain a single token, a single bit of RAM, that represents a quark.

So is the 747 made of something other than quarks? No, you’re just *modeling* it with *representational elements* that do not have a one-to-one correspondence with the quarks of the 747. The map is not the territory.

Why *not* model the 747 with a chromodynamic representation? Because then it would take a gazillion years to get any answers out of the model. Also we could not store the model on all the memory on all the computers in the world, as of 2008.

As the saying goes, “The map is not the territory, but you can’t fold up the territory and put it in your glove compartment.”

Sometimes you need a smaller map to fit in a more cramped glove compartment—but this does not change the territory. The scale of a map is not a fact about the territory, it's a fact about the map.

If it *were* possible to build and run a chromodynamic model of the 747, it would yield accurate predictions. Better predictions than the aerodynamic model, in fact.

To build a fully accurate model of the 747, it is not necessary, in principle, for the model to contain explicit descriptions of things like airflow and lift. There does not have to be a single token, a single bit of RAM, that corresponds to the position of the wings. It is possible, in principle, to build an accurate model of the 747 that makes no mention of anything *except* elementary particle fields and fundamental forces.

"What?" cries the antireductionist. "Are you telling me the 747 *doesn't really have wings*? I can see the wings right there!"

The notion here is a subtle one. It's not *just* the notion that an object can have different descriptions at different levels.

It's the notion that "having different descriptions at different levels" is *itself* something you say that belongs in the realm of Talking About Maps, not the realm of Talking About Territory.

It's not that the *airplane itself*, the *laws of physics themselves*, use different descriptions at different levels—as yonder artillery gunner thought. Rather *we*, for our convenience, use different simplified models at different levels.

If you looked at the ultimate chromodynamic model, the one that contained only elementary particle fields and fundamental forces, that model would contain all the facts about airflow and lift and wing positions—but these facts would be implicit, rather than explicit.

You, looking *at* the model, and thinking *about* the model, would be able to figure out where the wings were. Having figured it out, there would be an explicit representation in your mind of the wing position—an explicit computational object, there in your neural RAM. *In your mind*.

You might, indeed, deduce all sorts of explicit descriptions of the airplane, at various levels, and even explicit rules for how

your models at different levels interacted with each other to produce combined predictions—

And the way that algorithm feels from inside is that the airplane would *seem* to be made up of many levels at once, interacting with each other.

The way a belief *feels from inside* is that you seem to be looking straight at reality. When it actually *seems* that you're looking at a belief, as such, you are really experiencing a belief about belief.

So when your mind simultaneously believes explicit descriptions of many different levels, and believes explicit rules for transiting between levels, as part of an efficient combined model, it *feels like* you are seeing a system that is *made of* different level descriptions and their rules for interaction.

But this is just the brain trying to efficiently compress an object that it cannot remotely begin to model on a fundamental level. The airplane is too large. Even an iron atom would be too large. Quark-to-quark interactions are insanely intractable. You can't handle the *truth*.

But the way physics *really* works, as far as we can tell, is that there is *only* the most basic level—the elementary particle fields and fundamental forces. You can't handle the raw truth, but reality can handle it without the slightest simplification. (I wish I knew where Reality got its computing power.)

The laws of physics do not contain distinct additional causal entities that correspond to lift or airplane wings, the way that *the mind of an engineer* contains distinct additional *cognitive* entities that correspond to lift or airplane wings.

This, as I see it, is the thesis of reductionism. Reductionism is not a positive belief, but rather, a disbelief that the higher levels of simplified multilevel models are out there in the territory. Understanding this on a gut level dissolves the question of “How can you say the airplane doesn't really have wings, when I can see the wings right there?” The critical words are *really* and *see*.

Explaining vs. Explaining Away



discussion
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John Keats's *Lamia* (1819)¹ surely deserves some kind of award for Most Famously Annoying Poetry:

...Do not all charms fly
At the mere touch of cold philosophy?
There was an awful rainbow once in heaven:
We know her woof, her texture; she is given
In the dull catalogue of common things.
Philosophy will clip an Angel's wings,
Conquer all mysteries by rule and line,
Empty the haunted air, and gnomed mine—
Unweave a rainbow ...

My usual reply ends with the phrase: “If we cannot learn to take joy in the merely real, our lives will be empty indeed.” I shall expand on that later.

Here I have a different point in mind. Let's just take the lines:

Empty the haunted air, and gnomed mine—
Unweave a rainbow ...

Apparently “the mere touch of cold philosophy,” i.e., the truth, has destroyed:

- Haunts in the air;
- Gnomes in the mine;
- Rainbows.

Which calls to mind a rather different bit of verse:²

1. John Keats, “Lamia,” *The Poetical Works of John Keats* (London: Macmillan) (1884).
2. sharov.me/r/4-7
youtube.com



One of these things
Is not like the others
One of these things
Doesn't belong.

The air has been emptied of its haunts, and the mine denominated—but the rainbow is still there!

In Righting a Wrong Question, I wrote:

Tracing back the chain of causality, step by step, I discover that my belief that I'm wearing socks is fully explained by the fact that I'm wearing socks...On the other hand, if I see a mirage of a lake in the desert, the correct causal explanation of my vision does not involve the fact of any actual lake in the desert. In this case, my belief in the lake is not just *explained*, but *explained away*.

The rainbow was *explained*. The haunts in the air, and gnomes in the mine, were *explained away*.

I think this is the key distinction that anti-reductionists don't get about reductionism.

You can see this failure to get the distinction in the classic objection to reductionism:

If reductionism is correct, then even your belief in reductionism is just the mere result of the motion of molecules—why should I listen to anything you say?

The key word, in the above, is *mere*; a word which implies that accepting reductionism would explain *away* all the reasoning processes leading up to my acceptance of reductionism, the way that an optical illusion is explained *away*.

But you can explain how a cognitive process works without its being "mere"! My belief that I'm wearing socks is a mere result of my visual cortex reconstructing nerve impulses sent from my retina which received photons reflected off my socks...which is to say, according to scientific reductionism,

my belief that I'm wearing socks is a mere result of the fact that I'm wearing socks.

What could be going on in the anti-reductionists' minds, such that they would put rainbows and belief-in-reductionism in the same category as haunts and gnomes?

Several things are going on simultaneously. But for now let's focus on the basic idea introduced in a previous essay: The Mind Projection Fallacy between a multi-level map and a mono-level territory.

(I.e.: There's no way you can model a 747 quark-by-quark, so you've got to use a multi-level map with explicit cognitive representations of wings, airflow, and so on. This doesn't mean there's a multi-level territory. The true laws of physics, to the best of our knowledge, are only over elementary particle fields.)

I think that when physicists say "There are no *fundamental* rainbows," the anti-reductionists hear, "There are no rainbows."

If you don't distinguish between the multi-level map and the mono-level territory, then when someone tries to explain to you that the rainbow is not a fundamental thing in physics, acceptance of this will *feel like* erasing rainbows from your multi-level map, which *feels like* erasing rainbows from the world.

When Science says "tigers are not *elementary* particles, they are made of quarks" the anti-reductionist hears this as the same sort of dismissal as "we looked in your garage for a dragon, but there was just empty air."

What scientists did to rainbows, and what scientists did to gnomes, seemingly felt the same to Keats ...

In support of this sub-thesis, I deliberately used several phrases, in my discussion of Keats's poem, that were Mind Projection Fallacious. If you didn't notice, this would seem to argue that such fallacies are customary enough to pass unremarked.

For example:

The air has been emptied of its haunts, and the
mine de-gnomed—but the rainbow is still there!

Actually, Science emptied the *model of* air of *belief in* haunts, and emptied the *map of* the mine of *representations of* gnomes.

Science did not actually—as Keats’s poem itself would have it—take real Angel’s wings, and destroy them with a cold touch of truth. In reality there *never were* any haunts in the air, or gnomes in the mine.

Another example:

What scientists did to rainbows, and what scientists did to gnomes, seemingly felt the same to Keats.

Scientists didn’t *do* anything *to* gnomes, only to “gnomes.” The quotation is not the referent.

But if you commit the Mind Projection Fallacy—and by default, our beliefs just feel like the way the world *is*—then at time $T = 0$, the mines (apparently) contain gnomes; at time $T = 1$ a scientist dances across the scene, and at time $T = 2$ the mines (apparently) are empty. Clearly, there used to be gnomes there, but the scientist killed them.

Bad scientist! No poems for you, gnomekiller!

Well, that’s how it *feels*, if you get emotionally attached to the gnomes, and then a scientist says there aren’t any gnomes. It takes a strong mind, a deep honesty, and a deliberate effort to say, at this point, “That which can be destroyed by the truth should be,” and “The scientist hasn’t taken the gnomes away, only taken my delusion away,” and “I never held just title to my belief in gnomes in the first place; I have not been deprived of anything I *rightfully* owned,” and “If there are gnomes, I desire to believe there are gnomes; if there are no gnomes, I desire to believe there are no gnomes; let me not become attached to beliefs I may not want,” and all the other things that rationalists are supposed to say on such occasions.

But with the rainbow it is not even necessary to go that far. The rainbow is *still there*!

Fake Reductionism



discussion
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4-c21

There was an awful rainbow once in heaven:
We know her woof, her texture; she is given
In the dull catalogue of common things.

—John Keats, *Lamia*

I am guessing—though it is only a guess—that Keats himself did *not* know the woof and texture of the rainbow. Not the way that Newton understood rainbows. Perhaps not even at all. Maybe Keats just read, somewhere, that Newton had explained the rainbow as “light reflected from raindrops”—

—which was actually known in the thirteenth century. Newton only added a refinement by showing that the light was decomposed into colored parts, rather than transformed in color. But that put rainbows back in the news headlines. And so Keats, with Charles Lamb and William Wordsworth and Benjamin Haydon, drank “confusion to the memory of Newton” because “he destroyed the poetry of the rainbow by reducing it to a prism.” That’s one reason to suspect Keats didn’t understand the subject too deeply.

I am guessing, though it is only a guess, that Keats could *not* have sketched out on paper why rainbows only appear when the Sun is behind your head, or why the rainbow is an arc of a circle.

If so, Keats had a Fake Explanation. In this case, a *fake reduction*. He’d been *told that* the rainbow had been reduced, but it had not actually *been reduced* in his model of the world.

This is another of those distinctions that anti-reductionists fail to get—the difference between professing the flat fact that something is reducible, and *seeing* it.

In this, the anti-reductionists are not too greatly to be blamed, for it is part of a general problem.

I’ve written before on seeming knowledge that is not knowledge, and beliefs that are not *about* their supposed objects but

only recordings to recite back in the classroom, and words that operate as stop signs for curiosity rather than answers, and technobabble that only conveys membership in the literary genre of “science” ...

There is a very great distinction between being able to *see* where the rainbow comes from, and playing around with prisms to confirm it, and maybe making a rainbow yourself by spraying water droplets—

—versus some dour-faced philosopher just *telling* you, “No, there’s nothing special about the rainbow. Didn’t you hear? Scientists have explained it away. Just something to do with raindrops or whatever. Nothing to be excited about.”

I think this distinction probably accounts for a hell of a lot of the deadly existential emptiness that supposedly accompanies scientific reductionism.

You have to interpret the anti-reductionists’ experience of “reductionism,” not in terms of their *actually seeing* how rainbows work, not in terms of their having the critical “Aha!,” but in terms of their being told that the password is “Science.” The effect is just to move rainbows to a different *literary genre*—a literary genre they have been taught to regard as boring.

For them, the effect of hearing “Science has explained rainbows!” is to hang up a sign over rainbows saying, “This phenomenon has been labeled BORING by order of the Council of Sophisticated Literary Critics. Move along.”

And that’s all the sign says: only that, and nothing more.

So the literary critics have their gnomes yanked out by force; not dissolved in insight, but removed by flat order of authority. They are given no beauty to replace the hauntless air, no genuine understanding that could be interesting in its own right. Just a label saying, “Ha! You thought rainbows were pretty? You poor, unsophisticated fool. This is part of the literary genre of science, of dry and solemn incomprehensible words.”

That’s how anti-reductionists experience “reductionism.”

Well, can’t blame Keats, poor lad probably wasn’t raised right.

But he dared to drink “Confusion to the memory of Newton”?

MERE REALITY

I propose “To the memory of Keats’s confusion” as a toast for rationalists. Cheers.

Savannah Poets

Poets say science takes away from the beauty of the stars—mere globs of gas atoms. Nothing is “mere.” I too can see the stars on a desert night, and feel them. But do I see less or more?

The vastness of the heavens stretches my imagination—stuck on this carousel my little eye can catch one-million-year-old light. A vast pattern—of which I am a part—perhaps my stuff was belched from some forgotten star, as one is belching there. Or see them with the greater eye of Palomar, rushing all apart from some common starting point when they were perhaps all together. What is the pattern, or the meaning, or the why? It does not do harm to the mystery to know a little about it.

For far more marvelous is the truth than any artists of the past imagined! Why do the poets of the present not speak of it?

What men are poets who can speak of Jupiter if he were like a man, but if he is an immense spinning sphere of methane and ammonia must be silent?

—Richard Feynman, *The Feynman Lectures on Physics*,¹

Vol I, p. 3–6 (line breaks added)

That’s a real question, there on the last line—what kind of poet can write about Jupiter the god, but not Jupiter the immense sphere? Whether or not Feynman meant the question rhetorically, it has a real answer:



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1. Richard P. Feynman, Robert B. Leighton, and Matthew L. Sands, *The Feynman Lectures on Physics*, 3 vols. (Reading, MA: Addison-Wesley, 1963).

If Jupiter is like us, he can fall in love, and lose love, and regain love.

If Jupiter is like us, he can strive, and rise, and be cast down.

If Jupiter is like us, he can laugh or weep or dance.

If Jupiter is an immense spinning sphere of methane and ammonia, it is more difficult for the poet to make us feel.

There are poets and storytellers who say that the Great Stories are timeless, and they never change, are only ever retold. They say, with pride, that Shakespeare and Sophocles are bound by ties of craft stronger than mere centuries; that the two playwrights could have swapped times without a jolt.

Donald Brown once compiled a list of over two hundred “human universals,” found in all (or a vast supermajority of) studied human cultures, from San Francisco to the !Kung of the Kalahari Desert. Marriage is on the list, and incest avoidance, and motherly love, and sibling rivalry, and music and envy and dance and storytelling and aesthetics, and ritual magic to heal the sick, and poetry in spoken lines separated by pauses—

No one who knows anything about evolutionary psychology could be expected to deny it: The strongest emotions we have are deeply engraved, blood and bone, brain and DNA.

It might take a bit of tweaking, but you probably *could* tell “Hamlet” sitting around a campfire on the ancestral savanna.

So one can see why John “Unweave a rainbow” Keats might feel something had been lost, on being told that the rainbow was sunlight scattered from raindrops. Raindrops don’t dance.

In the Old Testament, it is written that God once destroyed the world with a flood that covered all the land, drowning all the horribly guilty men and women of the world along with their horribly guilty babies, but Noah built a gigantic wooden ark, etc., and after most of the human species was wiped out, God put rainbows in the sky as a sign that he wouldn’t do it again. At least not with water.

You can see how Keats would be *shocked* that this beautiful story was contradicted by modern science. Especially if (as I described in the previous essay) Keats had no real understanding of rainbows, no “Aha!” insight that could be fascinating in its own right, to replace the drama subtracted—

Ah, but maybe Keats would be right to be disappointed *even if* he knew the math. The Biblical story of the rainbow is a tale of bloodthirsty murder and smiling insanity. How could anything about raindrops and refraction properly replace that? Raindrops don't scream when they die.

So science takes the romance away (says the Romantic poet), and what you are given back never matches the drama of the original—

(that is, the original delusion)

—even if you do know the equations, because the equations are not about strong emotions.

That is the strongest rejoinder I can think of that any Romantic poet could have said to Feynman—though I can't remember ever hearing it said.

You can guess that I don't agree with the Romantic poets. So my own stance is this:

It is not *necessary* for Jupiter to be like a human, because *humans* are like humans. If Jupiter is an immense spinning sphere of methane and ammonia, that doesn't mean that love and hate are emptied from the universe. There *are* still loving and hating minds in the universe. *Us*.

With more than six billion of us at the last count, does Jupiter really need to be on the list of potential protagonists?

It is not *necessary* to tell the Great Stories about planets or rainbows. They play out all over our world, every day. Every day, someone kills for revenge; every day, someone kills a friend by mistake; every day, upward of a hundred thousand people fall in love. And even if this were not so, you could write fiction about humans—not about Jupiter.

Earth is old, and has played out the same stories many times beneath the Sun. I do wonder if it might not be time for some of the Great Stories to change. For me, at least, the story called “Goodbye”² has lost its charm.

2. Yehuda Yudkowsky, 1985-2004, sharov.me/r/4-8



The Great Stories are not timeless, because the human species is not timeless. Go far enough back in hominid evolution, and no one will understand *Hamlet*. Go far enough back in time, and you won't find any brains.

The Great Stories are not eternal, because the human species, *Homo sapiens sapiens*, is not eternal. I most sincerely doubt that we have another thousand years to go in our current form. I do not say this in sadness: I think we can do better.

I would not like to see all the Great Stories lost completely, in our future. I see very little difference between that outcome, and the Sun falling into a black hole.

But the Great Stories in their current forms have *already been* told, over and over. I do not think it ill if some of them should change their forms, or diversify their endings.

"And they lived happily ever after" seems worth trying at least once.

The Great Stories can and should diversify, as humankind grows up. Part of that ethic is the idea that when we find strangeness, we should respect it enough to tell its story truly. Even if it makes writing poetry a little more difficult.

If you are a good enough poet to write an ode to an immense spinning sphere of methane and ammonia, you are writing something *original*, about a newly discovered part of the real universe. It may not be as dramatic, or as gripping, as *Hamlet*. But the tale of *Hamlet* has already been told! If you write of Jupiter as though it were a human, then you are making our map of the universe just a little more impoverished of complexity; you are forcing Jupiter into the mold of all the stories that have already been told of Earth.

James Thomson's "A Poem Sacred to the Memory of Sir Isaac Newton," which praises the rainbow for what it *really* is—you can argue whether or not Thomson's poem is as gripping as John Keats's *Lamia* who was loved and lost. But tales of love and loss and cynicism had *already been* told, far away in ancient Greece, and no doubt many times before. Until we understood the rainbow as a thing *different* from tales of human-shaped magic, the true story of the rainbow could not be poeticized.

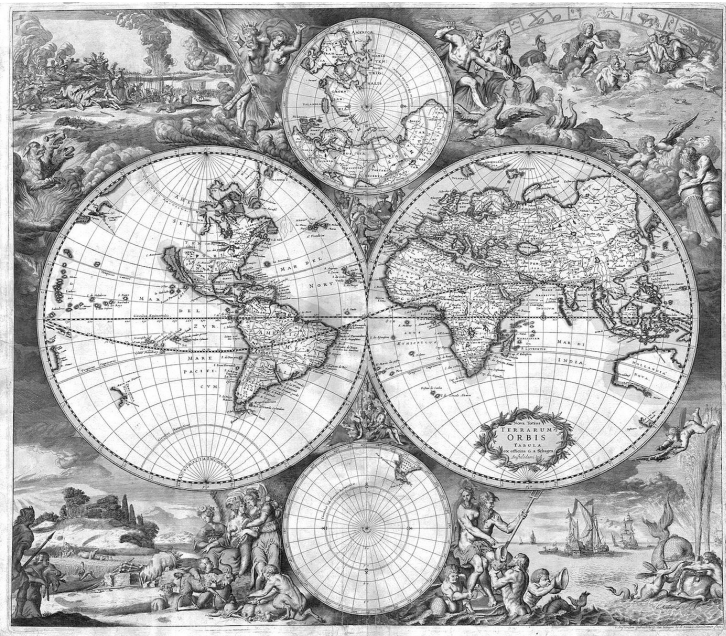
The border between science fiction and space opera was once drawn as follows: If you can take the plot of a story and put it back in the Old West, or the Middle Ages, without changing it, then it is not *real* science fiction. In real science fiction, the science is intrinsically part of the plot—you can't move the story from space to the savanna, not without losing something.

Richard Feynman asked: "What men are poets who can speak of Jupiter if he were like a man, but if he is an immense spinning sphere of methane and ammonia must be silent?"

They are *savanna poets*, who can *only* tell stories that would have made sense around a campfire ten thousand years ago. Savanna poets, who can tell *only* the Great Stories in their classic forms, and nothing more.

PART Q

Joy in the Merely Real



Joy in the Merely Real

...Do not all charms fly
 At the mere touch of cold philosophy?
 There was an awful rainbow once in heaven:
 We know her woof, her texture; she is given
 In the dull catalogue of common things.
 —John Keats, *Lamia*



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Nothing is “mere.”
 —Richard Feynman

You’ve got to admire that phrase, “dull catalogue of common things.” What is it, exactly, that goes in this catalogue? Besides rainbows, that is?

Why, things that are mundane, of course. Things that are normal; things that are unmagical; things that are known, or knowable; things that play by the rules (or that play by *any* rules, which makes them boring); things that are part of the ordinary universe; things that are, in a word, *real*.

Now that’s what I call setting yourself up for a fall.

At that rate, sooner or later you’re going to be disappointed in *everything*—either it will turn out not to exist, or even worse, it will turn out to be real.

If we cannot take joy in things that are merely real, our lives will *always* be empty.

For what sin are rainbows demoted to the dull catalogue of common things? For the sin of having a scientific explanation. “We know her woof, her texture,” says Keats—an interesting use of the word “we,” because I suspect that Keats didn’t know the explanation himself. I suspect that just being told that someone else knew was too much for him to take. I suspect that just the notion of rainbows being scientifically explicable *in principle* would have been too much to take. And if Keats didn’t think like that, well, I know plenty of people who do.

I have already remarked that nothing is *inherently* mysterious—nothing that actually exists, that is. If I am ignorant about a phenomenon, that is a fact about my state of mind, not a fact about the phenomenon; to worship a phenomenon because it seems so wonderfully mysterious is to worship your own ignorance; a blank map does not correspond to a blank territory, it is just somewhere we haven't visited yet, etc., etc....

Which is to say that *everything*—everything that *actually* exists—is liable to end up in “the dull catalogue of common things,” sooner or later.

Your choice is either:

- Decide that things are allowed to be unmagical, knowable, scientifically explicable—in a word, *real*—and yet still worth caring about;
- Or go about the rest of your life suffering from existential ennui that is *unresolvable*.

(Self-deception might be an option for others, but not for you.)

This puts quite a different complexion on the bizarre habit indulged by those strange folk called *scientists*, wherein they suddenly become fascinated by pocket lint or bird droppings or rainbows, or some other ordinary thing which world-weary and sophisticated folk would never give a second glance.

You might say that scientists—at least *some* scientists—are those folk who are *in principle* capable of enjoying life in the real universe.

Joy in Discovery

Newton was the greatest genius who ever lived,
and the most fortunate; for we cannot find more
than once a system of the world to establish.

—Lagrange¹



discussion
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I have more fun discovering things for myself than reading about them in textbooks. This is right and proper, and only to be expected.

But discovering something that *no one else knows*—being the *first* to unravel the secret—

There is a story that one of the first men to realize that stars were burning by fusion—plausible attributions I’ve seen are to Fritz Houtermans and Hans Bethe—was walking out with his girlfriend of a night, and she made a comment on how beautiful the stars were, and he replied: “Yes, and right now, I’m the only man in the world who knows why they shine.”

It is attested by numerous sources that this experience, being the first person to solve a major mystery, is a *tremendous* high. It’s probably the closest experience you can get to taking drugs, without taking drugs—though I wouldn’t know.

That can’t be healthy.

Not that I’m objecting to the euphoria. It’s the exclusivity clause that bothers me. Why should a discovery be worth *less*, just because someone *else* already knows the answer?

The most charitable interpretation I can put on the psychology, is that you don’t struggle with a single problem for months or years if it’s something you can just look up in the library. And that the tremendous high comes from having hit the problem from every angle you can manage, and having bounced;

1. Fred L. Wilson, *History of Science: Newton* citing: Delambre, M. “Notice sur la vie et les ouvrages de M. le comte J. L. Lagrange,” *Oeuvres de Lagrange* I. Paris, 1867, p. xx.

and then having analyzed the problem again, using every idea you can think of, and all the data you can get your hands on—making progress a little at a time—so that when, *finally*, you crack through the problem, all the dangling pieces and unresolved questions fall into place at once, like solving a dozen locked-room murder mysteries with a single clue.

And more, the understanding you get is *real* understanding—understanding that embraces all the clues you studied to solve the problem, when you didn’t yet know the answer. Understanding that comes from asking questions day after day and worrying at them; understanding that no one else can get (no matter how much you tell them the answer) unless they spend months studying the problem in its historical context, even after it’s been solved—and even then, they won’t get the high of solving it all at once.

That’s one possible reason why James Clerk Maxwell might have had more fun *discovering* Maxwell’s equations, than you had fun reading about them.

A slightly less charitable reading is that the tremendous high comes from what is termed, in the *politesse* of social psychology, “commitment” and “consistency” and “cognitive dissonance”; the part where we value something more highly *just* because it took more work to get it. The studies showing that subjecting fraternity pledges to a harsher initiation, causes them to be more convinced of the value of the fraternity—identical wine in higher-priced bottles being rated as tasting better—that sort of thing.

Of course, if you just have more fun solving a puzzle than being told its answer, because you enjoy doing the cognitive work for its own sake, there’s nothing wrong with that. The less charitable reading would be if charging \$100 to be told the answer to a puzzle made you think the answer was more interesting, worthwhile, important, surprising, etc., than if you got the answer for free.

(I strongly suspect that a major part of science’s PR problem in the population at large is people who instinctively believe that if knowledge is given away for free, it cannot be important. If you had to undergo a fearsome initiation ritual to be told the

truth about evolution, maybe people would be more satisfied with the answer.)

The really uncharitable reading is that the joy of first discovery is about status. Competition. Scarcity. Beating everyone else to the punch. It doesn't matter whether you have a three-room house or a four-room house, what matters is having a bigger house than the Joneses. A two-room house would be fine, if you could only ensure that the Joneses had even less.

I don't object to competition as a matter of principle. I don't think that the game of Go is barbaric and should be suppressed, even though it's zero-sum. But if the euphoric joy of scientific discovery *has* to be about scarcity, that means it's only available to one person per civilization for any given truth.

If the joy of scientific discovery is one-shot per discovery, then, from a fun-theoretic perspective, Newton probably used up a substantial increment of the total Physics Fun available over the entire history of Earth-originating intelligent life. That selfish bastard explained the orbits of planets *and* the tides.

And really the situation is even worse than this, because in the Standard Model of physics (discovered by bastards who spoiled the puzzle for everyone else) the universe is spatially infinite, inflationarily branching, and branching via decoherence, which is at least three different ways that Reality is exponentially or infinitely large.

So aliens, or alternate Newtons, or just Tegmark duplicates of Newton, may all have discovered gravity before *our* Newton did—if you believe that “before” means anything relative to those kinds of separations.

When that thought first occurred to me, I actually found it quite uplifting. Once I realized that someone, somewhere in the expanses of space and time, already knows the answer to any answerable question—even biology questions and history questions; there are other decoherent Earths—then I realized how silly it was to think as if the joy of discovery ought to be limited to one person. It becomes a fully inescapable source of unresolvable existential angst, and I regard that as a *reductio*.

The consistent solution which maintains the *possibility* of fun is to stop worrying about what other people know. If you

don't know the answer, it's a mystery to you. If you can raise your hand, and clench your fingers into a fist, and you've got no idea of how your brain is doing it—or even what exact muscles lay beneath your skin—you've got to consider yourself just as ignorant as a hunter-gatherer. Sure, someone else knows the answer—but back in the hunter-gatherer days, someone else in an alternate Earth, or for that matter, someone else in the future, knew what the answer was. Mystery, and the joy of finding out, is either a personal thing, or it doesn't exist at all—and I prefer to say it's personal.

The joy of assisting your civilization by telling it something it doesn't already know does tend to be one-shot per discovery per civilization; that kind of value is conserved, as are Nobel Prizes. And the prospect of that reward may be what it takes to keep you focused on one problem for the years required to develop a really *deep* understanding; plus, working on a problem unknown to your civilization is a sure-fire way to avoid reading any spoilers.

But as part of my general project to undo this idea that rationalists have less fun, I want to restore the magic and mystery to every part of the world that you do not *personally* understand, regardless of what other knowledge may exist, far away in space and time, or even in your next-door neighbor's mind. If *you* don't know, it's a mystery. And now think of how many things you don't know! (If you can't think of anything, you have other problems.) Isn't the world suddenly a much more mysterious and magical and *interesting* place? As if you'd been transported into an alternate dimension, and had to learn all the rules from scratch?

A friend once told me that I look at the world as if I've never seen it before. I thought, that's a nice compliment...Wait! I never *have* seen it before! What—did everyone else get a preview?

—Ran Prieur²

2. sharov.me/r/4-9
ranprieur.com, archived copy



Bind Yourself to Reality



discussion
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So perhaps you're reading all this, and asking: "Yes, but what does this have to do with reductionism?"

Partially, it's a matter of leaving a line of retreat. It's not easy to take something *important* apart into components, when you're convinced that this removes magic from the world, unweaves the rainbow. I do plan to take certain things apart, in this book; and I prefer not to create pointless existential anguish.

Partially, it's the crusade against Hollywood Rationality, the concept that understanding the rainbow subtracts its beauty. The rainbow is still beautiful *plus* you get the beauty of physics.

But even more deeply, it's one of these subtle hidden-core-of-rationality things. You know, the sort of thing where I start talking about "the Way." It's about *binding yourself to reality*.

In one of Frank Herbert's *Dune* books, if I recall correctly, it is said that a Truthsayer gains their ability to detect lies in others by always speaking truth themselves, so that they form a relationship with the truth whose violation they can feel. It wouldn't work, but I still think it's one of the more beautiful thoughts in fiction. At the very least, to get close to the truth, you have to be willing to press yourself up against reality as tightly as possible, without flinching away, or sneering down.

You can see the bind-yourself-to-reality theme in *Lotteries: A Waste of Hope*. Understanding that lottery tickets have negative expected utility does not mean that you give up the hope of being rich. It means that you stop wasting that hope on lottery tickets. You put the hope into your job, your school, your startup, your eBay sideline; and if you truly have nothing worth hoping for, then maybe it's time to start looking.

It's not dreams I object to, only *impossible* dreams. The lottery isn't impossible, but it is an un-actionable near-impossibility. It's not that winning the lottery is extremely *difficult*—requires a desperate effort—but that *work* isn't the issue.

I say all this, to exemplify the idea of taking emotional energy that is flowing off to nowhere, and binding it into the realms of reality.

This doesn't mean setting goals that are low enough to be "realistic," i.e., easy and safe and parentally approved. Maybe this is good advice in your personal case, I don't know, but I'm not the one to say it.

What I mean is that you can invest emotional energy in rainbows even if they turn out *not* to be magic. The future is always absurd but it is never *unreal*.

The Hollywood Rationality stereotype is that "rational = emotionless"; the more reasonable you are, the more of your emotions Reason inevitably destroys. In *Feeling Rational* I contrast this against "*That which can be destroyed by the truth should be*" and "*That which the truth nourishes should thrive*." When you have arrived at your best picture of the truth, there is nothing irrational about the emotions you feel as a result of that—the emotions cannot be destroyed by truth, so they must not be irrational.

So instead of *destroying* emotional energies associated with bad explanations for rainbows, as the Hollywood Rationality stereotype would have it, let us *redirect* these emotional energies into reality—bind them to beliefs that are as true as we can make them.

Want to fly? Don't give up on flight. Give up on flying potions and build yourself an airplane.

Remember the theme of *Think like Reality*, where I talked about how when physics seems counterintuitive, you've got to accept that it's not *physics* that's weird, it's *you*?

What I'm talking about now is like that, only with emotions instead of hypotheses—binding your feelings into the real world. Not the "realistic" everyday world. I would be a howling hypocrite if I told you to shut up and do your homework. I mean the *real* real world, the lawful universe, that includes absurdities like Moon landings and the evolution of human intelligence. Just not any magic, anywhere, ever.

It is a Hollywood Rationality meme that "Science takes the fun out of life."

MERE REALITY

Science puts the fun back *into* life.

Rationality directs your emotional energies into the universe,
rather than somewhere else.

If You Demand Magic, Magic Won't Help

Most witches don't believe in gods. They know that the gods exist, of course. They even deal with them occasionally. But they don't believe in them. They know them too well. It would be like believing in the postman.

—Terry Pratchett, *Witches Abroad*¹

Once upon a time, I was pondering the philosophy of fantasy stories—

And before anyone chides me for my “failure to understand what fantasy is about,” let me say this: I was raised in a science fiction and fantasy household. I have been reading fantasy stories since I was five years old. I occasionally try to *write* fantasy stories.² And I am *not* the sort of person who tries to write for a genre without pondering its philosophy. Where do you think story ideas come from?

Anyway:

I was pondering the philosophy of fantasy stories, and it occurred to me that if there were actually dragons in our world—if you could go down to the zoo, or even to a distant mountain, and meet a fire-breathing dragon—while nobody had ever actually seen a zebra, then our fantasy stories would contain zebras aplenty, while dragons would be unexciting.

Now that's what I call painting yourself into a corner, wot? The grass is always greener on the other side of unreality.



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1. Terry Pratchett, *Witches Abroad* (London: Corgi Books, 1992).
 2. sharov.me/r/4-10
hpmor.com



In one of the standard fantasy plots, a protagonist from our Earth, a sympathetic character with lousy grades or a crushing mortgage but still a good heart, suddenly finds themselves in a world where magic operates in place of science. The protagonist often goes on to practice magic, and become in due course a (superpowerful) sorcerer.

Now here's the question—and yes, it is a little unkind, but I think it needs to be asked: Presumably most readers of these novels see themselves in the protagonist's shoes, fantasizing about their own acquisition of sorcery. Wishing for magic. And, barring improbable demographics, most readers of these novels are not scientists.

Born into a world of science, they did not become scientists. What makes them think that, in a world of magic, they would act any differently?

If they don't have the scientific attitude, that nothing is “mere”—the capacity to be interested in merely real things—how will magic help them? If they actually *had* magic, it would be merely *real*, and lose the charm of unattainability. They might be excited at first, but (like the lottery winners who, six months later, aren't nearly as happy as they expected to be), the excitement would soon wear off. Probably as soon as they had to actually *study* spells.

Unless they can find the capacity to take joy in things that are merely real. To be just as excited by hang-gliding, as riding a dragon; to be as excited by making a light with electricity, as by making a light with magic ...even if it takes a little study ...

Don't get me wrong. I'm not dissing dragons. Who knows, we might even create some, one of these days.

But if you don't have the capacity to enjoy hang-gliding even though it is *merely real*, then as soon as dragons *turn* real, you're not going to be any more excited by dragons than you are by hang-gliding.

Do you think you would prefer living in the Future, to living in the present? That's a quite understandable preference. Things do seem to be getting better over time.

But don't forget that *this is* the Future, relative to the Dark Ages of a thousand years earlier. You have opportunities undreamt-of even by kings.

If the trend continues, the Future might be a very fine place indeed in which to live. But if you do make it to the Future, what you find, when you get there, will be another Now. If you don't have the basic capacity to enjoy being in a Now—if your emotional energy can *only* go into the Future, if you can *only* hope for a better tomorrow—then no amount of passing time can help you.

(Yes, in the Future there could be a pill that fixes the emotional problem of always looking to the Future. I don't think this invalidates my basic point, which is about what sort of pills we should want to take.)

Matthew C., commenting on *Less Wrong*,³ seems very excited about an informally specified “theory” by Rupert Sheldrake which “explains” such non-explanation-demanding phenomena as protein folding and snowflake symmetry. But why isn't Matthew C. just as excited about, say, Special Relativity? Special Relativity is actually *known* to be a law, so why isn't it even *more* exciting? The advantage of becoming excited about a law already known to be true, is that you know your excitement will not be wasted.

If Sheldrake's theory were accepted truth taught in elementary schools, Matthew C. wouldn't care about it. Or why else is Matthew C. fascinated by that one particular law which he believes to be a law of physics, more than all the other laws?

The worst catastrophe you could visit upon the New Age community would be for their rituals to start working reliably, and for UFOs to actually appear in the skies. What would be the point of believing in aliens, if they were just *there*, and everyone else could see them too? In a world where psychic powers were merely real, New Agers wouldn't *believe in* psychic powers,

3. sharov.me/r/4-11
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any more than anyone cares enough about gravity to believe in it. (Except for scientists, of course.)

Why am I so negative about magic? Would it be *wrong* for magic to exist?

I'm not actually negative on magic. Remember, I occasionally try to write fantasy stories. But I'm annoyed with this psychology that, if it were born into a world where spells and potions did work, would pine away for a world where household goods were abundantly produced by assembly lines.

Part of binding yourself to reality, on an emotional as well as intellectual level, is coming to terms with the fact that you *do live here*. Only then can you see this, your world, and whatever opportunities it holds out for you, without wishing your sight away.

Not to put too fine a point on it, but *I've* found no lack of dragons to fight, or magics to master, in this world of my birth. If I were transported into one of those fantasy novels, I wouldn't be surprised to find myself studying the forbidden ultimate sorcery—

—because why should being transported into a magical world change anything? It's not *where* you are, it's *who* you are.

So remember the Litany Against Being Transported Into An Alternate Universe:

If I'm going to be happy anywhere,
Or achieve greatness anywhere,
Or learn true secrets anywhere,
Or save the world anywhere,
Or feel strongly anywhere,
Or help people anywhere,
I may as well do it in reality.

Mundane Magic

I think that part of the rationalist ethos is *binding yourself emotionally* to an absolutely lawful reductionistic universe—a universe containing no supernatural things such as souls or magic—and pouring all your hope and all your care into that merely real universe and its possibilities, without disappointment.

There's an old trick for combating dukkha where you make a list of things you're grateful for, like a roof over your head.

So why not make a list of abilities you have that would be amazingly cool *if they were magic*, or if only a few chosen individuals had them?

For example, suppose that instead of one eye, you possessed a magical *second* eye embedded in your forehead. And this second eye enabled you to *see into the third dimension*—so that you could somehow tell *how far away* things were—where an ordinary eye would see only a two-dimensional shadow of the true world. Only the possessors of this ability can accurately aim the legendary distance-weapons that kill at ranges far beyond a sword, or use to their fullest potential the shells of ultrafast machinery called “cars.”

“Binocular vision” would be too light a term for this ability. We'll only appreciate it once it has a properly impressive name, like Mystic Eyes of Depth Perception.

So here's a list of some of my favorite magical powers:

- *Vibratory Telepathy*. By transmitting invisible vibrations through the very air itself, two users of this ability can *share thoughts*. As a result, Vibratory Telepaths can form emotional bonds much deeper than those possible to other primates.
- *Psychometric Tracery*. By tracing small fine lines on a surface, the Psychometric Tracer can leave impressions of



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emotions, history, knowledge, even the structure of other spells. This is a higher level than Vibratory Telepathy as a Psychometric Tracer can share the thoughts of long-dead Tracers who lived thousands of years earlier. By reading one Tracery and inscribing another simultaneously, Tracers can duplicate Tracings; and these replicated Tracings can even contain the detailed pattern of other spells and magics. Thus, the Tracers wield almost unimaginable power as magicians; but Tracers can get in trouble trying to use complicated Traceries that they could not have Traced themselves.

- *Multidimensional Kinesis.* With simple, almost unthinking acts of will, the Kinetics can cause extraordinarily complex forces to flow through small tentacles and into any physical object within touching range—not just pushes, but combinations of pushes at many points that can effectively apply torques and twists. The Kinetic ability is far subtler than it first appears: they use it not only to wield existing objects with martial precision, but also to apply forces that sculpt objects into forms more suitable for Kinetic wielding. They even create tools that extend the power of their Kinesis and enable them to sculpt ever-finer and ever-more-complicated tools, a positive feedback loop fully as impressive as it sounds.
- *The Eye.* The user of this ability can perceive infinitesimal traveling twists in the Force that binds matter—tiny vibrations, akin to the life-giving power of the Sun that falls on leaves, but far more subtle. A bearer of the Eye can sense objects far beyond the range of touch using the tiny disturbances they make in the Force. Mountains many days travel away can be known to them as if within arm's reach. According to the bearers of the Eye, when night falls and sunlight fails, they can sense huge fusion fires burning at unthinkable distances—though no one else has any way of verifying this. Possession of a single Eye is said to make the bearer equivalent to royalty.

And finally,

- *The Ultimate Power.* The user of this ability contains a smaller, imperfect echo of the entire universe, enabling them to search out paths through probability to any desired future. If this sounds like a ridiculously powerful ability, you're right—game balance goes right out the window with this one. Extremely rare among life forms, it is the *sekai no ougi* or “hidden technique of the world.”

Nothing can oppose the Ultimate Power except the Ultimate Power. Any less-than-ultimate Power will simply be “comprehended” by the Ultimate and disrupted in some inconceivable fashion, or even absorbed into the Ultimates’ own power base. For this reason the Ultimate Power is sometimes called the “master technique of techniques” or the “trump card that trumps all other trumps.” The more powerful Ultimates can stretch their “comprehension” across galactic distances and aeons of time, and even perceive the bizarre laws of the hidden “world beneath the world.”

Ultimates have been killed by immense natural catastrophes, or by extremely swift surprise attacks that give them no chance to use their power. But all such victories are ultimately a matter of luck—it does not confront the Ultimates on their own probability-bending level, and if they survive they will begin to bend Time to avoid future attacks.

But the Ultimate Power itself is also dangerous, and many Ultimates have been destroyed by their own powers—falling into one of the flaws in their imperfect inner echo of the world.

Stripped of weapons and armor and locked in a cell, an Ultimate is still one of the most dangerous life-forms on the planet. A sword can be broken and a limb can be cut off, but the Ultimate Power is “the power that cannot be removed without removing you.”

Perhaps because this connection is so intimate, the Ultimates regard one who loses their Ultimate Power permanently—without hope of regaining it—as *schiaivo*, or “dead while breathing.” The Ultimates argue that the Ultimate Power is so important as to be a necessary part of what makes a creature an end in itself, rather than a means. The Ultimates even insist that anyone who lacks the Ultimate Power cannot begin to truly comprehend the Ultimate Power, and hence, cannot understand why the Ultimate Power is morally important—a suspiciously self-serving argument.

The users of this ability form an absolute aristocracy and treat all other life forms as their pawns.

The Beauty of Settled Science

Facts do not need to be unexplainable to be beautiful; truths do not become less worth learning if someone else knows them; beliefs do not become less worthwhile if many others share them ...

...and if you only care about scientific issues that are controversial, you will end up with a head stuffed full of garbage.

The media thinks that only the cutting edge of science is worth reporting on. How often do you see headlines like “General Relativity Still Governing Planetary Orbits” or “Phlogiston Theory Remains False”? So, by the time anything is solid science, it is no longer a breaking headline. “Newsworthy” science is often based on the thinnest of evidence and wrong half the time—if it were not on the uttermost fringes of the scientific frontier, it would not be breaking news.

Scientific *controversies* are problems *so difficult* that even people who’ve spent years mastering the field can still fool themselves. That’s what makes for the heated arguments that attract all the media attention.

Worse, if you aren’t in the field and part of the game, controversies *aren’t even fun*.

Oh, sure, you can have the fun of picking a side in an argument. But you can get that in any football game. That’s not what the fun of science is about.

Reading a well-written textbook, you get: Carefully phrased explanations for incoming students, math derived step by step (where applicable), plenty of experiments cited as illustration (where applicable), test problems on which to display your new mastery, and a reasonably good guarantee that what you’re learning is actually true.

Reading press releases, you usually get: Fake explanations that convey nothing except the delusion of understanding of a result that the press release author didn’t understand and that probably has a better-than-even chance of failing to replicate.



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Modern science is built on discoveries, built on discoveries, built on discoveries, and so on, all the way back to people like Archimedes, who discovered facts like why boats float, that can make sense even if you don't know about other discoveries. A good place to start traveling that road is at the beginning.

Don't be embarrassed to read *elementary* science textbooks, either. If you want to pretend to be sophisticated, go find a play to sneer at. If you just want to have *fun*, remember that simplicity is at the core of scientific beauty.

And thinking you can jump right into the frontier, when you haven't learned the settled science, is like ...

...like trying to climb only the *top* half of Mount Everest (which is the only part that interests you) by standing at the base of the mountain, bending your knees, and jumping *really hard* (so you can pass over the boring parts).

Now I'm not saying that you should never pay attention to scientific controversies. If 40% of oncologists think that white socks cause cancer, and the other 60% violently disagree, this is an important fact to know.

Just don't go thinking that science *has* to be controversial to be interesting.

Or, for that matter, that science has to be recent to be interesting. A steady diet of science *news* is bad for you: You are what you eat, and if you eat only science reporting on fluid situations, without a solid textbook now and then, your brain will turn to liquid.

Amazing Breakthrough Day: April 1st

So you're thinking, "April 1st ...isn't that already supposed to be April Fool's Day?"

Yes—and that will provide the ideal cover for celebrating Amazing Breakthrough Day.

As I argued in *The Beauty of Settled Science*, it is a major problem that media coverage of science focuses only on *breaking news*. Breaking news, in science, occurs at the furthest fringes of the scientific frontier, which means that the new discovery is often:

- Controversial;
- Supported by only one experiment;
- Way the heck more complicated than an ordinary mortal can handle, and requiring lots of prerequisite science to understand, which is why it wasn't solved three centuries ago;
- Later shown to be wrong.

People never get to see the *solid* stuff, let alone the *understandable* stuff, because it isn't *breaking news*.

On Amazing Breakthrough Day, I propose, journalists who really care about science can report—under the protective cover of April 1st—such important but neglected science stories as:

- **BOATS EXPLAINED:** Centuries-Old Problem Solved By Bathtub Nudist
- **YOU SHALL NOT CROSS!** Königsberg Tourists' Hopes Dashed
- **ARE YOUR LUNGS ON FIRE?** Link Between Respiration And Combustion Gains Acceptance Among Scientists



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Note that every one of these headlines are *true*—they describe events that did, in fact, happen. They just didn’t happen *yesterday*.

There have been many humanly understandable amazing breakthroughs in the history of science, that can be understood without a PhD or even a BSc. The operative word here is *history*. Think of Archimedes’s “Eureka!” when he understood the relation between the water a ship displaces, and the reason the ship floats. This is *far enough back* in scientific history that you don’t need to know fifty other discoveries to understand the theory; it can be explained in a couple of graphs; anyone can see how it’s useful; and the confirming experiments can be duplicated in your own bathtub.

Modern science is built on discoveries built on discoveries built on discoveries and so on all the way back to Archimedes. Reporting science *only* as breaking news is like wandering into a movie three-fourths of the way through, writing a story about “Bloody-handed man kisses girl holding gun!” and wandering back out again.

And if your editor says, “Oh, but our readers won’t be interested in that—”

Then point out that Reddit and Digg don’t link *only* to breaking news. They also link to short webpages that give good explanations of old science. Readers vote it up, and that should tell you something. Explain that if your newspaper doesn’t change to look more like Reddit, you’ll have to start selling drugs to make payroll. Editors love to hear that sort of thing, right?

On the Internet, a good new explanation of old science *is* news and it spreads like news. Why couldn’t the science sections of newspapers work the same way? Why isn’t a new *explanation* worth reporting on?

But all this is too visionary for a first step. For now, let’s just see if any journalists out there pick up on Amazing Breakthrough Day, where you report on some *understandable* science breakthrough as though it had just occurred.

April 1st. Put it on your calendar.

Is Humanism a Religion Substitute?

For many years before the Wright Brothers, people dreamed of flying with magic potions. There was nothing irrational about the *raw desire* to fly. There was nothing *tainted* about the wish to look down on a cloud from above. Only the “magic potions” part was irrational.

Suppose you were to put me into an fMRI scanner, and take a movie of my brain’s activity levels, while I watched a space shuttle launch. (Wanting to visit space is not “realistic,” but it is an essentially lawful dream—one that can be fulfilled in a lawful universe.) The fMRI might—maybe, maybe not—resemble the fMRI of a devout Christian watching a nativity scene.

Should an experimenter obtain this result, there’s a lot of people out there, both Christians and some atheists, who would gloat: “Ha, ha, space travel is your religion!”

But that’s drawing the wrong category boundary. It’s like saying that, because some people once tried to fly by irrational means, no one should ever enjoy looking out of an airplane window on the clouds below.

If a rocket launch is what it takes to give me a feeling of aesthetic transcendence, I do not see this as a *substitute* for religion. That is theomorphism—the viewpoint from gloating religionists who assume that everyone who *isn’t* religious has a hole in their mind that wants filling.

Now, to be fair to the religionists, this is not *just* a gloating assumption. There *are* atheists who have religion-shaped holes in their minds. I *have* seen attempts to substitute atheism or even transhumanism for religion. And the result is invariably awful. Utterly awful. Absolutely abjectly awful.

I call such efforts, “hymns to the nonexistence of God.”

When someone sets out to write an atheistic hymn—“Hail, oh unintelligent universe,” blah, blah, blah—the result will, without exception, suck.



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Why? Because they're being imitative. Because they have no motivation for writing the hymn *except* a vague feeling that since churches have hymns, they ought to have one too. And, on a purely artistic level, that puts them far beneath genuine religious art that is not an imitation of anything, but an original expression of emotion.

Religious hymns were (often) written by people who *felt strongly* and *wrote honestly* and put serious effort into the prosody and imagery of their work—that's what gives their work the grace that it possesses, of artistic integrity.

So are atheists doomed to hymnlessness?

There is an acid test of attempts at post-theism. The acid test is: "If religion had never existed among the human species—if we had *never made* the original mistake—would this song, this art, this ritual, this way of thinking, still make sense?"

If humanity had never made the original mistake, there would be no hymns to the nonexistence of God. But there would still be marriages, so the notion of an atheistic marriage ceremony makes perfect sense—as long as you don't suddenly launch into a lecture on how God doesn't exist. Because, in a world where religion *never had* existed, nobody would interrupt a wedding to talk about the implausibility of a distant hypothetical concept. They'd talk about love, children, commitment, honesty, devotion, but who the heck would mention God?

And, in a human world where religion *never had* existed, there would still be people who got tears in their eyes watching a space shuttle launch.

Which is why, even if experiment shows that watching a shuttle launch makes "religion"-associated areas of my brain light up, associated with feelings of transcendence, I do not see that as a *substitute* for religion; I expect the same brain areas would light up, for the same reason, if I lived in a world where religion had never been invented.

A good "atheistic hymn" is simply a song about anything worth singing about that doesn't happen to be religious.

Also, reversed stupidity is not intelligence. The world's greatest idiot may say the Sun is shining, but that doesn't make it

dark out. The point is *not* to create a life that resembles religion as little as possible in every surface aspect—this is the same kind of thinking that inspires hymns to the nonexistence of God. If humanity had never made the original mistake, no one would be *trying to avoid* things that vaguely resembled religion. Believe accurately, then feel accordingly: If space launches actually exist, and watching a rocket rise makes you want to sing, then write the song, dammit.

If I get tears in my eyes at a space shuttle launch, it doesn't mean I'm trying to fill a hole left by religion—it means that my emotional energies, my *caring*, are bound into the real world.

If God did speak plainly, and answer prayers reliably, God would just become one more boringly real thing, no more worth believing in than the postman. If God were real, it would destroy the inner uncertainty that brings forth outward fervor in compensation. And if everyone else believed God were real, it would destroy the specialness of being one of the elect.

If you invest your emotional energy in space travel, you don't have those vulnerabilities. I can *see* the Space Shuttle rise without losing the awe. Everyone else can believe that Space Shuttles are real, and it doesn't make them any less special. I haven't painted myself into the corner.

The choice between God and humanity is not just a choice of drugs. Above all, humanity *actually exists*.

Scarcity



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What follows is taken primarily from Robert Cialdini's *Influence: The Psychology of Persuasion*.¹ I own three copies of this book: one for myself, and two for loaning to friends.

Scarcity, as that term is used in social psychology, is when things become *more desirable* as they appear *less obtainable*.

- If you put a two-year-old boy in a room with two toys, one toy in the open and the other behind a Plexiglas wall, the two-year-old will ignore the easily accessible toy and go after the apparently forbidden one. If the wall is low enough to be easily climbable, the toddler is no more likely to go after one toy than the other.²
- When Dade County forbade use or possession of phosphate detergents, many Dade residents drove to nearby counties and bought huge amounts of phosphate laundry detergents. Compared to Tampa residents not affected by the regulation, Dade residents rated phosphate detergents as gentler, more effective, more powerful on stains, and even believed that phosphate detergents poured more easily.³

Similarly, information that appears forbidden or secret seems more important and trustworthy:

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1. Robert B. Cialdini, *Influence: The Psychology of Persuasion: Revised Edition* (New York: Quill, 1993).
 2. Sharon S. Brehm and Marsha Weintraub, "Physical Barriers and Psychological Reactance: Two-year-olds' Responses to Threats to Freedom," *Journal of Personality and Social Psychology* 35 (1977): 830–836.
 3. Michael B. Mazis, Robert B. Settle, and Dennis C. Leslie, "Elimination of Phosphate Detergents and Psychological Reactance," *Journal of Marketing Research* 10 (1973): 2; Michael B. Mazis, "Antipollution Measures and Psychological Reactance Theory: A Field Experiment," *Journal of Personality and Social Psychology* 31 (1975): 654–666.

- When University of North Carolina students learned that a speech opposing coed dorms had been banned, they became more opposed to coed dorms (without even hearing the speech).⁴
- When a driver said he had liability insurance, experimental jurors awarded his victim an average of four thousand dollars more than if the driver said he had no insurance. If the judge afterward informed the jurors that information about insurance was inadmissible and must be ignored, jurors awarded an average of thirteen thousand dollars more than if the driver had no insurance.⁵
- Buyers for supermarkets, told by a supplier that beef was in scarce supply, gave orders for twice as much beef as buyers told it was readily available. Buyers told that beef was in scarce supply, and furthermore, that the information about scarcity was itself scarce—that the shortage was not general knowledge—ordered six times as much beef. (Since the study was conducted in a real-world context, the information provided was in fact correct.)⁶

The conventional theory for explaining this is “psychological reactance,” social-psychology-speak for “When you tell people they can’t do something, they’ll just try even harder.” The fundamental instincts involved appear to be preservation of status and preservation of options. We resist dominance, when any human agency tries to restrict our freedom. And when options seem to be in danger of disappearing, even from natural causes, we try to leap on the option before it’s gone.

Leaping on disappearing options may be a good adaptation in a hunter-gatherer society—gather the fruits while they are

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4. Richard D. Ashmore, Vasantha Ramchandra, and Russell A. Jones, “Censorship as an Attitude Change Induction,” *Paper presented at Eastern Psychological Association meeting* (1971).
 5. Dale Broeder, “The University of Chicago Jury Project,” *Nebraska Law Review* 38 (1959): 760–774.
 6. A. Knishinsky, “The Effects of Scarcity of Material and Exclusivity of Information on Industrial Buyer Perceived Risk in Provoking a Purchase Decision” (Doctoral dissertation, Arizona State University, 1982).

still ripe—but in a money-based society it can be rather costly. Cialdini reports that in one appliance store he observed, a salesperson who saw that a customer was evincing signs of interest in an appliance would approach, and sadly inform the customer that the item was out of stock, the last one having been sold only twenty minutes ago. Scarcity creating a sudden jump in desirability, the customer would often ask whether there was any chance that the salesperson could locate an unsold item in the back room, warehouse, or anywhere. “Well,” says the salesperson, “that’s possible, and I’m willing to check; but do I understand that this is the model you want, and if I can find it at this price, you’ll take it?”

As Cialdini remarks, a chief sign of this malfunction is that you dream of *possessing* something, rather than *using* it. (Timothy Ferriss offers similar advice on planning your life: ask which *ongoing experiences* would make you happy, rather than which possessions or status-changes.)

But the really fundamental problem with desiring the unattainable is that as soon as you actually *get* it, it stops being unattainable. If we cannot take joy in the merely available, our lives will *always* be frustrated...

The Sacred Mundane

So I was reading (around the first half of) Adam Frank's *The Constant Fire*,¹ in preparation for my Bloggingheads dialogue with him. Adam Frank's book is about the experience of the sacred. I might not usually call it that, but of course I know the experience Frank is talking about. It's what I feel when I watch a video of a space shuttle launch; or what I feel—to a lesser extent, because in this world it is too common—when I look up at the stars at night, and think about what they mean. Or the birth of a child, say. That which is significant in the Unfolding Story.

Adam Frank holds that this experience is something that science holds deeply in common with religion. As opposed to e.g. being a basic human quality which religion corrupts.

The Constant Fire quotes William James's *The Varieties of Religious Experience* as saying:

Religion...shall mean for us the feelings, acts, and experiences of individual men in their solitude; so far as they apprehend themselves to stand in relation to whatever they may consider the divine.²

And this theme is developed further: Sacredness is something intensely *private* and *individual*.

Which completely nonplussed me. Am I supposed to not have any feeling of sacredness if I'm one of *many* people watching the video of *SpaceShipOne* winning the X-Prize? Why not? Am I supposed to think that my experience of sacredness has to be somehow *different* from that of all the *other* people watching? Why, when we all have the same brain design? Indeed,



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1. Adam Frank, *The Constant Fire: Beyond the Science vs. Religion Debate* (University of California Press, 2009).
 2. William James, *The varieties of religious experience*, pg. 31

why would I *need* to believe I was unique? (But “unique” is another word Adam Frank uses; so-and-so’s “unique experience of the sacred.”) Is the feeling private in the same sense that we have difficulty communicating *any* experience? Then why emphasize this of sacredness, rather than sneezing?

The light came on when I realized that I was looking at a trick of Dark Side Epistemology—if you make something *private*, that shields it from criticism. You can say, “You can’t criticize me, because this is my private, inner experience that you can never access to question it.”

But the price of shielding yourself from criticism is that you are cast into solitude—the solitude that William James admired as the core of religious experience, as if loneliness were a *good* thing.

Such relics of Dark Side Epistemology are key to understanding the many ways that religion twists the experience of sacredness:

Mysteriousness—why should the sacred have to be mysterious? A space shuttle launch gets by just fine without being mysterious. How much *less* would I appreciate the stars if I did *not* know what they were, if they were just little points in the night sky? But if your religious beliefs are questioned—if someone asks, “Why doesn’t God heal amputees?”—then you take refuge and say, in a tone of deep profundity, “It is a sacred mystery!” There are questions that must not be asked, and answers that must not be acknowledged, to defend the lie. Thus unanswerability comes to be associated with sacredness. And the price of shielding yourself from criticism is giving up the true curiosity that truly wishes to find answers. You will worship your own ignorance of the temporarily unanswered questions of your own generation—probably including ones that are already answered.

Faith—in the early days of religion, when people were more naive, when even intelligent folk actually believed that stuff, religions staked their reputation upon the testimony of miracles in their scriptures. And Christian archaeologists set forth truly expecting to find the ruins of Noah’s Ark. But when no such evidence was forthcoming, *then* religion executed what

William Bartley called *the retreat to commitment*, “I believe because I believe!” Thus *belief without good evidence* came to be associated with the experience of the sacred. And the price of shielding yourself from criticism is that you sacrifice your ability to think clearly about that which is sacred, and to progress in your understanding of the sacred, and relinquish mistakes.

Experientialism—if before you thought that the rainbow was a sacred contract of God with humanity, and then you begin to realize that God doesn’t exist, then you may execute a *retreat to pure experience*—to praise yourself just for *feeling* such wonderful sensations when you think about God, whether or not God actually *exists*. And the price of shielding yourself from criticism is solipsism: your experience is stripped of its *referents*. What a terrible hollow feeling it would be to watch a space shuttle rising on a pillar of flame, and say to yourself, “But it doesn’t really matter whether the space shuttle actually exists, so long as I feel.”

Separation—if the sacred realm is not subject to ordinary rules of evidence or investigable by ordinary means, then it must be different in kind from the world of mundane matter: and so we are less likely to think of a space shuttle as a candidate for sacredness, because it is a work of merely *human* hands. Keats lost his admiration of the rainbow and demoted it to the “dull catalogue of mundane things” for the crime of its woof and texture being known. And the price of shielding yourself from all ordinary criticism is that you lose the sacredness of all merely real things.

Privacy—of this I have already spoken.

Such distortions are why we had best *not* to try to salvage religion. No, not even in the form of “spirituality.” Take away the institutions and the factual mistakes, subtract the churches and the scriptures, and you’re left with ...all this nonsense about mysteriousness, faith, solipsistic experience, private solitude, and discontinuity.

The original lie is only the beginning of the problem. Then you have all the ill habits of thought that have evolved to defend it. Religion is a poisoned chalice, from which we had best not even sip. Spirituality is the same cup after the original pellet

of poison has been taken out, and only the dissolved portion remains—a little less directly lethal, but still not good for you.

When a lie has been defended for ages upon ages, the true origin of the inherited habits lost in the mists, with layer after layer of undocumented sickness; then the wise, I think, will start over from scratch, rather than trying to selectively discard the original lie while keeping the habits of thought that protected it. *Just admit you were wrong*, give up *entirely* on the mistake, stop defending it *at all*, stop trying to say you were even a little right, stop trying to save face, just say “Oops!” and throw out the *whole* thing and begin again.

That capacity—to really, *really*, without defense, admit you were *entirely* wrong—is why religious experience will never be like scientific experience. No religion can absorb *that* capacity without losing itself *entirely* and becoming simple humanity...

...to just look up at the distant stars. Believable without strain, without a constant distracting struggle to fend off your awareness of the counterevidence. Truly there *in the world*, the experience united with the referent, a solid part of that unfolding story. Knowable without threat, offering true meat for curiosity. Shared in togetherness with the many other on-lookers, no need to retreat to privacy. Made of the same fabric as yourself and all other things. Most holy and beautiful, the sacred mundane.

To Spread Science, Keep It Secret

Sometimes I wonder if the Pythagoreans had the right idea.

Yes, I've written about how "science" is inherently public. I've written that "science" is distinguished from merely rational knowledge by the in-principle ability to reproduce scientific experiments for yourself, to know without relying on authority. I've said that "science" should be defined as the publicly accessible knowledge of humankind. I've even suggested that future generations will regard all papers not published in an open-access journal as non-science, i.e., it can't be part of the public knowledge of humankind if you make people pay to read it.¹

But that's only one vision of the future. In another vision, the knowledge we now call "science" is taken *out* of the public domain—the books and journals hidden away, guarded by mystic cults of gurus wearing robes, requiring fearsome initiation rituals for access—so that more people will *actually* study it.

I mean, right now, people *can* study science but they *don't*.

"Scarcity," it's called in social psychology. What appears to be in limited supply is more highly valued. And this effect is *especially* strong with information—we're much more likely to try to obtain information that we believe is secret, and to value it more when we do obtain it.

With science, I think, people assume that if the information is freely available, it must not be important. So instead people join cults that have the sense to keep their Great Truths secret. The Great Truth may actually be gibberish, but it's more satisfying than coherent science, because it's *secret*.

Science is the great Purloined Letter of our times, left out in the open and ignored.



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1. Edit 2018: And of course Every Cause Wants to Be a Cult, and as a practical matter, how do you actually keep Science a secret without draconian policies?

Sure, scientific openness helps the scientific elite. They've already *been* through the initiation rituals. But for the rest of the planet, science is kept secret a hundred times more effectively by making it freely available, than if its books were guarded in vaults and you had to walk over hot coals to get access. (This being a fearsome trial indeed, since the great secrets of insulation are only available to Physicist-Initiates of the Third Level.)

If scientific knowledge were hidden in ancient vaults (rather than hidden in inconvenient pay-for-access journals), at least then people would *try* to get into the vaults. They'd be *desperate* to learn science. Especially when they saw the power that Eighth Level Physicists could wield, and were told that they *weren't allowed to know* the explanation.

And if you tried to start a cult around oh, say, Scientology, you'd get some degree of public interest, at first. But people would very quickly start asking uncomfortable questions like "Why haven't you given a public demonstration of your Eighth Level powers, like the Physicists?" and "How come none of the Master Mathematicians seem to want to join your cult?" and "Why should I follow your Founder when they aren't an Eighth Level anything outside their own cult?" and "Why should I study *your* cult *first*, when the Dentists of Doom can do things that are so much more impressive?"

When you look at it from that perspective, the escape of math from the Pythagorean cult starts to look like a major strategic blunder for humanity.

Now, I know what you're going to say: "But science *is* surrounded by fearsome initiation rituals! Plus it's *inherently* difficult to learn! Why doesn't *that* count?" Because the public *thinks* that science is freely available, that's why. If you're *allowed* to learn, it must not be important enough *to* learn.

It's an image problem, people taking their cues from others' attitudes. Just *anyone* can walk into the supermarket and buy a light bulb, and nobody looks at it with awe and reverence. The physics supposedly isn't secret (even though *you* don't know), and there's a one-paragraph explanation in the newspaper that

sounds vaguely authoritative and convincing—essentially, no one treats the lightbulb as a sacred mystery, so neither do you.

Even the simplest little things, completely inert objects like crucifixes, can become magical if everyone *looks* at them like they're magic. But since you're theoretically *allowed* to know why the light bulb works without climbing the mountain to find the remote Monastery of Electricians, there's no need to *actually* bother to learn.

Now, because science does in fact have initiation rituals both social and cognitive, scientists are not wholly dissatisfied with their science. The problem is that, in the present world, very few people bother to study science in the first place. Science cannot be the true Secret Knowledge, because just anyone is allowed to know it—even *though, in fact, they don't*.

If the Great Secret of Natural Selection, passed down from Darwin Who Is Not Forgotten, was only ever imparted to you after you paid \$2,000 and went through a ceremony involving torches and robes and masks and sacrificing an ox, *then* when you were shown the fossils, and shown the optic cable going through the retina under a microscope, and finally told the Truth, you would say "That's the most brilliant thing ever!" and *be satisfied*. After that, if some other cult tried to tell you it was actually a bearded man in the sky 6000 years ago, you'd laugh like hell.

And you know, it might actually be more *fun* to do things that way. Especially if the initiation required you to put together some of the evidence for yourself—together, or with classmates—before you could tell your Science Sensei you were ready to advance to the next level. It wouldn't be *efficient*, sure, but it would be *fun*.

If humanity had never made the mistake—never gone down the religious path, and never learned to fear anything that smacks of religion—then maybe the PhD granting ceremony would involve litanies and chanting, because, hey, that's what people like. Why take the fun out of everything?

Maybe we're just doing it wrong.

And no, I'm not *seriously* proposing that we try to reverse the last five hundred years of openness and classify all the science

secret. At least, not at the moment. Efficiency is important for now, especially in things like medical research. I'm just explaining why it is that I won't tell anyone the Secret of how the ineffable difference between blueness and redness arises from mere atoms for less than \$100,000—

Ahem! I meant to say, I'm telling you about this vision of an alternate Earth, so that you give science equal treatment with cults. So that you don't undervalue scientific truth when you learn it, *just* because it doesn't seem to be protected appropriately to its value. *Imagine* the robes and masks. Visualize yourself creeping into the vaults and stealing the Lost Knowledge of Newton. And don't be fooled by any organization that *does* use robes and masks, unless they also show you the data.

People seem to have holes in their minds for Esoteric Knowledge, Deep Secrets, the Hidden Truth. And I'm not even criticizing this psychology! There *are* deep secret esoteric hidden truths, like quantum mechanics or Bayes-structure. We've just gotten into the habit of presenting the Hidden Truth in a very *unsatisfying* way, wrapped up in false mundanity.

But if the holes for secret knowledge are not filled by true beliefs, they will be filled by false beliefs. There is *nothing but* science to learn—the emotional energy must either be invested in reality, or wasted in total nonsense, or destroyed. For myself, I think it is better to invest the emotional energy; fun should not be needlessly cast away.

Right now, we've got the worst of both worlds. Science isn't *really* free, because the courses are expensive and the textbooks are expensive. But the public *thinks* that anyone is allowed to know, so it must not be important.

Ideally, you would want to arrange things the other way around.

Initiation Ceremony

The torches that lit the narrow stairwell burned intensely and in the wrong color, flame like melting gold or shattered suns.

192...193...

Brennan's sandals clicked softly on the stone steps, snicking in sequence, like dominos very slowly falling.

227...228...

Half a circle ahead of him, a trailing fringe of dark cloth whispered down the stairs, the robed figure itself staying just out of sight.

239...240...

Not much longer, Brennan predicted to himself, and his guess was accurate:

Sixteen times sixteen steps was the number, and they stood before the portal of glass.

The great curved gate had been wrought with cunning, humor, and close attention to indices of refraction: it warped light, bent it, folded it, and generally abused it, so that there were hints of what was on the other side (stronger light sources, dark walls) but no possible way of *seeing through*—unless, of course, you had the key: the counter-door, thick for thin and thin for thick, in which case the two would cancel out.

From the robed figure beside Brennan, two hands emerged, gloved in reflective cloth to conceal skin's color. Fingers like slim mirrors grasped the handles of the warped gate—handles that Brennan had not guessed; in all that distortion, shapes could only be anticipated, not seen.

"Do you want to know?" whispered the guide; a whisper nearly as loud as an ordinary voice, but not revealing the slightest hint of gender.

Brennan paused. The answer to the question seemed suspiciously, indeed extraordinarily obvious, even for ritual.

"Yes," Brennan said finally.



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The guide only regarded him silently.

"Yes, I want to know," said Brennan.

"Know *what*, exactly?" whispered the figure.

Brennan's face scrunched up in concentration, trying to visualize the game to its end, and hoping he hadn't blown it already; until finally he fell back on the first and last resort, which is the truth:

"It doesn't matter," said Brennan, "the answer is still yes."

The glass gate parted down the middle, and slid, with only the tiniest scraping sound, into the surrounding stone.

The revealed room was lined, wall-to-wall, with figures robed and hooded in light-absorbing cloth. The straight walls were not themselves black stone, but mirrored, tiling a square grid of dark robes out to infinity in all directions; so that it seemed as if the people of some much vaster city, or perhaps the whole human kind, watched in assembly. There was a hint of moist warmth in the air of the room, the breath of the gathered: a scent of crowds.

Brennan's guide moved to the center of the square, where burned four torches of that relentless yellow flame. Brennan followed, and when he stopped, he realized with a slight shock that all the cowed hoods were now looking directly at him. Brennan had never before in his life been the focus of such absolute attention; it was frightening, but not entirely unpleasant.

"He is here," said the guide in that strange loud whisper.

The endless grid of robed figures replied in one voice: perfectly blended, exactly synchronized, so that not a single individual could be singled out from the rest, and betrayed:

"*Who is absent?*"

"Jakob Bernoulli," intoned the guide, and the walls replied:

"*Is dead but not forgotten.*"

"Abraham de Moivre,"

"*Is dead but not forgotten.*"

"Pierre-Simon Laplace,"

"*Is dead but not forgotten.*"

"Edwin Thompson Jaynes,"

"*Is dead but not forgotten.*"

“They died,” said the guide, “and they are lost to us; but we still have each other, and the project continues.”

In the silence, the guide turned to Brennan, and stretched forth a hand, on which rested a small ring of nearly transparent material.

Brennan stepped forward to take the ring—

But the hand clenched tightly shut.

“If three-fourths of the humans in this room are women,” said the guide, “and three-fourths of the women and half of the men belong to the Heresy of Virtue, and I am a Virtuist, what is the probability that I am a man?”

“Two-elevenths,” Brennan said confidently.

There was a moment of absolute silence.

Then a titter of shocked laughter.

The guide’s whisper came again, truly quiet this time, almost nonexistent: “It’s one-sixth, actually.”

Brennan’s cheeks were flaming so hard that he thought his face might melt off. The instinct was very strong to run out of the room and up the stairs and flee the city and change his name and start his life over again and get it right this time.

“An honest mistake is at least honest,” said the guide, louder now, “and we may know the honesty by its relinquishment. If I am a Virtuist, what is the probability that I am a man?”

“One—” Brennan started to say.

Then he stopped. Again, the horrible silence.

“Just say ‘one-sixth’ already,” stage-whispered the figure, this time loud enough for the walls to hear; then there was more laughter, not all of it kind.

Brennan was breathing rapidly and there was sweat on his forehead. If he was wrong about this, he really *was* going to flee the city. “Three fourths women times three fourths Virtuists is nine sixteenths female Virtuists in this room. One fourth men times one half Virtuists is two sixteenths male Virtuists. If I have only that information and the fact that you are a Virtuist, I would then estimate odds of two to nine, or a probability of two-elevenths, that you are male. Though I do not, in fact, believe the information given is correct. For one thing, it seems

too neat. For another, there are an odd number of people in this room.”

The hand stretched out again, and opened.

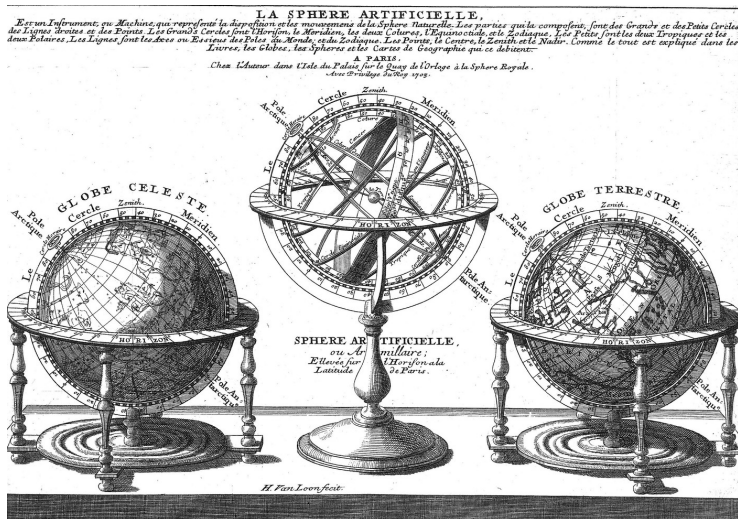
Brennan took the ring. It looked almost invisible, in the torchlight; not glass, but some material with a refractive index very close to air. The ring was warm from the guide’s hand, and felt like a tiny living thing as it embraced his finger.

The relief was so great that he nearly didn’t hear the cowed figures applauding.

From the robed guide came one last whisper:

“You are now a novice of the Bayesian Conspiracy.”

Physicalism 201



Hand vs. Fingers

Back to our original topic: Reductionism and the Mind Projection Fallacy. There can be emotional problems in accepting reductionism, if you think that things have to be fundamental to be fun. But this position commits us to never taking joy in anything more complicated than a quark, and so I prefer to reject it.

To review, the reductionist thesis is that we use multi-level models for computational reasons, but physical reality has only a single level.

Here I'd like to pose the following conundrum: When you pick up a cup of water, is it your *hand* that picks it up?

Most people, of course, go with the naive popular answer: "Yes."

Recently, however, scientists have made a stunning discovery: It's not your *hand* that holds the cup, it's actually your fingers, thumb, and palm.

Yes, I know! I was shocked too. But it seems that after scientists measured the forces exerted on the cup by each of your fingers, your thumb, and your palm, they found there was no force left over—so the force exerted by your *hand* must be zero.

The theme here is that, if you can *see how* (not just *know that*) a higher level reduces to a lower one, they will not seem like separate things within your map; you will be able to *see* how silly it is to think that your fingers could be in one place, and your hand somewhere else; you will be able to *see* how silly it is to argue about whether it is your hand that picks up the cup, or your fingers.

The operative word is "see," as in concrete visualization. Imagining your hand causes you to imagine the fingers and thumb and palm; conversely, imagining fingers and thumb and palm causes you to identify a hand in the mental picture. Thus the high level of *your map* and the low level of *your map* will be tightly bound together *in your mind*.



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In reality, of course, the levels are bound together even tighter than that—bound together by the tightest possible binding: physical identity. You can *see* this: You can *see* that saying (1) “hand” or (2) “fingers and thumb and palm,” does not refer to different *things*, but different *points of view*.

But suppose you lack the knowledge to so tightly bind together the levels of your map. For example, you could have a “hand scanner” that showed a “hand” as a dot on a map (like an old-fashioned radar display), and similar scanners for fingers/thumbs/palms; then you would see a cluster of dots around the hand, but you would be able to *imagine* the hand-dot moving off from the others. So, even though the physical reality of the hand (that is, the thing the dot corresponds to) was identical with / strictly composed of the physical realities of the fingers and thumb and palm, you would not be able to see this fact; even if someone told you, or you guessed from the correspondence of the dots, you would only *know* the fact of reduction, not *see* it. You would still be able to *imagine* the hand dot moving around independently, even though, if the physical makeup of the sensors were held constant, it would be physically impossible for this to actually happen.

Or, at a still lower level of binding, people might just tell you “There’s a hand over there, and some fingers over there”—in which case you would know little more than a Good-Old-Fashioned AI representing the situation using suggestively named LISP tokens. There wouldn’t be anything *obviously* contradictory about asserting:

⊢ Inside(Room, Hand)
 ⊢ ¬Inside(Room, Fingers),
 because you would not possess the *knowledge*
 ⊢ Inside(x, Hand) ⇒ Inside(x, Fingers).

None of this says that a hand can actually detach its existence from your fingers and crawl, ghostlike, across the room; it just says that a Good-Old-Fashioned AI with a propositional representation may not *know* any better. The map is not the territory.

In particular, you shouldn’t draw too many conclusions from how it seems *conceptually possible*, in the mind of some specific

conceiver, to separate the hand from its constituent elements of fingers, thumb, and palm. Conceptual possibility is not the same as logical possibility or physical possibility.

It is *conceptually* possible *to you* that 235,757 is prime, because you don't know any better. But it isn't *logically* possible that 235,757 is prime; if you were logically omniscient, 235,757 would be obviously composite (and you would know the factors). That that's why we have the notion of impossible possible worlds, so that we can put probability distributions on propositions that may or may not be *in fact* logically impossible.

And you can imagine philosophers who criticize "eliminative fingerists" who contradict the direct facts of experience—we can *feel* our hand holding the cup, after all—by suggesting that "hands" *don't really exist*, in which case, obviously, the cup would fall down. And philosophers who suggest "appendigital bridging laws" to explain how a particular configuration of fingers evokes a hand into existence—with the note, of course, that while our world contains those particular appendigital bridging laws, the laws could have been conceivably different, and so are not in any sense *necessary facts*, etc.

All of these are cases of Mind Projection Fallacy, and what I call "naive philosophical realism"—the confusion of philosophical intuitions for direct, veridical information about reality. Your inability to imagine something is just a computational fact about what your brain can or can't imagine. Another brain might work differently.

Angry Atoms



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Fundamental physics—quarks ‘n’ stuff—is far removed from the levels we can *see*, like hands and fingers. At best, you can know how to replicate the experiments that show that your hand (like everything else) is composed of quarks, and you may know how to derive a few equations for things like atoms and electron clouds and molecules.

At worst, the existence of quarks beneath your hand may just be something you were told. In which case it’s questionable in what sense you can be said to “know” it at all, even if you repeat back the same word “quark” that a physicist would use to convey knowledge to another physicist.

Either way, you can’t actually *see* the identity between levels—no one has a brain large enough to *visualize* avogadros of quarks and recognize a hand-pattern in them.

But we at least understand what hands *do*. Hands push on things, exert forces on them. When we’re told about atoms, we visualize little billiard balls bumping into each other. This makes it seem obvious that “atoms” can push on things too, by bumping into them.

Now this notion of atoms is not quite correct. But so far as *human imagination* goes, it’s relatively easy to imagine our hand being made up of a little galaxy of swirling billiard balls, pushing on things when our “fingers” touch them. Democritus imagined this 2,400 years ago, and there was a time, roughly 1803–1922, when Science thought he was right.

But what about, say, anger?

How could little billiard balls be angry? Tiny frowny faces on the billiard balls?

Put yourself in the shoes of, say, a hunter-gatherer—someone who may not even have a notion of writing, let alone the notion of using base matter to perform computations—someone who has no idea that such a thing as neurons exist. Then you can

imagine the *functional* gap that your ancestors might have perceived between billiard balls and “Grrr! Aaarg!”

Forget about subjective experience for the moment, and consider the sheer *behavioral* gap between anger and billiard balls. The difference between what little billiard balls *do*, and what anger makes people *do*. Anger can make people raise their fists and hit someone—or say snide things behind their backs—or plant scorpions in their tents at night. Billiard balls just push on things.

Try to put yourself in the shoes of the hunter-gatherer who’s never had the “Aha!” of information-processing. Try to avoid hindsight bias about things like neurons and computers. Only then will you be able to see the uncrossable explanatory gap:

How can you explain angry behavior in terms of billiard balls?

Well, the *obvious* materialist conjecture is that the little billiard balls push on your arm and make you hit someone, or push on your tongue so that insults come out.

But how do the little billiard balls know how to do this—or how to guide your tongue and fingers through long-term plots—if they aren’t angry themselves?

And besides, if you’re not seduced by—gasp!—scientism, you can see from a first-person perspective that this explanation is obviously false. Atoms can push on your arm, but they can’t make you *want* anything.

Someone may point out that drinking wine can make you angry. But who says that wine is made exclusively of little billiard balls? Maybe wine just contains a potency of angerness.

Clearly, reductionism is just a flawed notion.

(The novice goes astray and says “The art failed me”; the master goes astray and says “I failed my art.”)

What does it take to cross this gap? It’s not just the idea of “neurons” that “process information”—if you say only this and nothing more, it just inserts a magical, unexplained level-crossing rule into your model, where you go from billiards to thoughts.

But an Artificial Intelligence programmer who knows how to create a chess-playing program out of base matter has taken

a *genuine* step toward crossing the gap. If you understand concepts like consequentialism, backward chaining, utility functions, and search trees, you can make merely causal/mechanical systems compute plans.

The trick goes something like this: For each possible chess move, compute the moves your opponent could make, then your responses to those moves, and so on; evaluate the furthest position you can see using some local algorithm (you might simply count up the material); then trace back using minimax¹ to find the best move on the current board; then make that move.

More generally: If you have chains of causality inside the mind that have a kind of mapping—a mirror, an echo—to what goes on in the environment, then you can run a utility function over the end products of imagination, and find an action that achieves something that the utility function rates highly, and output that action. It is not necessary for the chains of causality inside the mind, that are similar to the environment, to be made out of billiard balls that have little auras of intentionality. Deep Blue's transistors do not need little chess pieces carved on them, in order to work. See also vol. I, p. 174, *The Simple Truth*.

All this is still tremendously oversimplified, but it should, at least, reduce the apparent length of the gap. If you can understand all that, you can see how a planner built out of base matter can be influenced by alcohol to output more angry behaviors. The billiard balls in the alcohol push on the billiard balls making up the utility function.

But even if you know how to write small AIs, you can't *visualize* the level-crossing between transistors and chess. There are too many transistors, and too many moves to check.

Likewise, even if you knew all the facts of neurology, you would not be able to *visualize* the level-crossing between neurons and anger—let alone the level-crossing between atoms

1. sharov.me/r/4-12

Minimax - Wikipedia • en.wikipedia.org



and anger. Not the way you can visualize a hand consisting of fingers, thumb, and palm.

And suppose a cognitive scientist just flatly tells you “Anger is hormones”? Even if you repeat back the words, it doesn’t mean you’ve crossed the gap. You may believe you believe it, but that’s not the same as understanding what little billiard balls have to do with wanting to hit someone.

So you come up with interpretations like, “Anger is *mere* hormones, it’s caused by little molecules, so it must not be justified in any moral sense—*that’s* why you should learn to control your anger.”

Or, “There isn’t really any such thing as anger—it’s an illusion, a quotation with no referent, like a mirage of water in the desert, or looking in the garage for a dragon and not finding one.”

These are both tough pills to swallow (not that you *should* swallow them) and so it is a good deal easier to profess them than to believe them.

I think this is what non-reductionists/non-materialists think they are criticizing when they criticize reductive materialism.

But materialism isn’t that easy. It’s not as cheap as saying, “Anger is made out of atoms—there, now I’m done.” That wouldn’t explain how to get from billiard balls to hitting. You need the specific insights of computation, consequentialism, and search trees before you can start to close the explanatory gap.

All this was a relatively easy example *by modern standards*, because I restricted myself to talking about angry *behaviors*. Talking about outputs doesn’t require you to appreciate how an algorithm feels from inside (cross a first-person/third-person gap) or dissolve a wrong question (untangle places where the interior of your own mind runs skew to reality).

Going from material substances that bend and break, burn and fall, push and shove, to angry *behavior*, is just a practice problem by the standards of modern philosophy. But it is an *important* practice problem. It can only be fully appreciated, if you realize how *hard* it would have been to solve before writing was invented. There was once an explanatory gap here—though

it may not seem that way in hindsight, now that it's been bridged for generations.

Explanatory gaps can be crossed, if you accept help from science, and don't trust the view from the interior of your own mind.

Heat vs. Motion

After the last essay, it occurred to me that there's a much simpler example of reductionism jumping a gap of apparent-difference-in-kind: the reduction of heat to motion.

Today, the equivalence of heat and motion may seem too obvious in hindsight—everyone says that “heat is motion,” therefore, it can't be a “weird” belief.

But there was a time when the kinetic theory of heat was a highly controversial scientific hypothesis, contrasting to belief in a caloric fluid that flowed from hot objects to cold objects. Still earlier, the main theory of heat was “Phlogiston!”

Suppose you'd *separately* studied kinetic theory and caloric theory. You now know something about kinetics: collisions, elastic rebounds, momentum, kinetic energy, gravity, inertia, free trajectories. Separately, you know something about heat: temperatures, pressures, combustion, heat flows, engines, melting, vaporization.

Not only is this state of knowledge a plausible one, it is the state of knowledge possessed by e.g. Sadi Carnot, who, working strictly from within the caloric theory of heat, developed the principle of the Carnot cycle—a heat engine of maximum efficiency, whose existence implies the Second Law of Thermodynamics. This in 1824, when kinetics was a highly developed science.

Suppose, like Carnot, you know a great deal about kinetics, and a great deal about heat, as *separate* entities. Separate entities *of knowledge*, that is: your brain has separate filing baskets for beliefs about kinetics and beliefs about heat. But from the inside, this state of knowledge *feels* like living in a world of moving things and hot things, a world where motion and heat are independent properties of matter.

Now a Physicist From The Future comes along and tells you: “Where there is heat, there is motion, and vice versa. That's why, for example, rubbing things together makes them hotter.”



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There are (at least) two possible interpretations you could attach to this statement, “Where there is heat, there is motion, and vice versa.”

First, you could suppose that heat and motion exist separately—that the caloric theory is correct—but that among our universe’s physical laws is a “bridging law” which states that, where objects are moving quickly, caloric will come into existence. And conversely, another bridging law says that caloric can exert pressure on things and make them move, which is why a hotter gas exerts more pressure on its enclosure (thus a steam engine can use steam to drive a piston).

Second, you could suppose that heat and motion are, in some as-yet-mysterious sense, *the same thing*.

“Nonsense,” says Thinker 1, “the words ‘heat’ and ‘motion’ have two different meanings; that is why we have two different words. We know how to determine when we will call an observed phenomenon ‘heat’—heat can melt things, or make them burst into flame. We know how to determine when we will say that an object is ‘moving quickly’—it changes position; and when it crashes, it may deform, or shatter. Heat is concerned with change of substance; motion, with change of position and shape. To say that these two words have the same meaning is simply to confuse yourself.”

“Impossible,” says Thinker 2. “It may be that, in our world, heat and motion are associated by bridging laws, so that it is a law of physics that motion creates caloric, and vice versa. But I can easily imagine a world where rubbing things together does *not* make them hotter, and gases *don’t* exert more pressure at higher temperatures. Since there are possible worlds where heat and motion are not associated, they must be different properties—this is true a priori.”

Thinker 1 is confusing the quotation and the referent: $2 + 2 = 4$, but $“2 + 2” \neq “4.”$ The string “2 + 2” contains five characters (including whitespace) and the string “4” contains only one character. If you type the two strings into a Python interpreter, they yield the same output, `>>> 4`. So you can’t conclude, from looking at the strings “2 + 2” and “4,” that just because

the strings are different, they must have different “meanings” relative to the Python Interpreter.

The words “heat” and “kinetic energy” can be said to “refer to” the same thing, even before we *know* how heat reduces to motion, in the sense that we don’t know yet what the referent is, but the referents are in fact the same. You might imagine an Idealized Omniscient Science Interpreter that would give the same output when we typed in “heat” and “kinetic energy” on the command line.

I talk about the Science Interpreter to emphasize that, to dereference the pointer, you’ve got to step outside cognition. The end result of the dereference is something out there in reality, not in anyone’s mind. So you can say “real referent” or “actual referent,” but you can’t *evaluate* the words locally, from the inside of your own head. You can’t reason using the actual heat-referent—if you thought using *real heat*, thinking “one million Kelvin” would vaporize your brain. But, by forming a belief about your belief about heat, you can talk *about* your belief about heat, and say things like “It’s possible that my belief about heat doesn’t much resemble *real heat*.” You can’t actually perform that comparison right there in your own mind, but you can talk *about* it.

Hence you can say, “My beliefs about heat and motion are not the same beliefs, but it’s possible that actual heat and actual motion are the same thing.” It’s just like being able to acknowledge that “the morning star” and “the evening star” might be the same planet, while also understanding that you can’t determine this just by examining your beliefs—you’ve got to haul out the telescope.

Thinker 2’s mistake follows similarly. A physicist told them, “Where there is heat, there is motion” and Thinker 2 mistook this for a statement of *physical law*: The presence of caloric *causes* the existence of motion. What the physicist really means is more akin to an *inferential rule*: Where you are told there is “heat,” deduce the presence of “motion.”

From this basic projection of a multilevel model into a multilevel reality follows another, distinct error: the conflation of conceptual possibility with logical possibility. To Sadi Carnot, it

is *conceivable* that there could be another world where heat and motion are not associated. To Richard Feynman, armed with specific knowledge of how to derive equations about heat from equations about motion, this idea is not only inconceivable, but so wildly inconsistent as to make one's head explode.

I should note, in fairness to philosophers, that there are philosophers who have said these things. For example, Hilary Putnam, writing on the "Twin Earth" thought experiment:¹

Once we have discovered that water (in the actual world) is H_2O , *nothing counts as a possible world in which water isn't H_2O* . In particular, if a "logically possible" statement is one that holds in some "logically possible world," *it isn't logically possible that water isn't H_2O* .

On the other hand, we can perfectly well imagine having experiences that would convince us (and that would make it rational to believe that) water *isn't* H_2O . In that sense, it is conceivable that water isn't H_2O . It is conceivable but it isn't logically possible! Conceivability is no proof of logical possibility.

It appears to me that "water" is being used in two different senses in these two paragraphs—one in which the word "water" *refers* to what we type into the Science Interpreter, and one in which "water" *refers* to what we get out of the Science Interpreter when we type "water" into it. In the first paragraph, Hilary seems to be saying that after we do some experiments and find out that water is H_2O , water becomes automatically redefined to *mean* H_2O . But you could coherently hold a different position about whether the word "water" now *means* " H_2O " or "whatever is *really* in that bottle next to me," so long as you use your terms consistently.

I believe the above has already been said as well? Anyway ...

1. Hilary Putnam, "The Meaning of Meaning," in *The Twin Earth Chronicles*, ed. Andrew Pessin and Sanford Goldberg (M. E. Sharpe, Inc., 1996), 3–52.

It is quite possible for there to be only *one* thing out-there-in-the-world, but for it to take on sufficiently different forms, and for you yourself to be sufficiently ignorant of the reduction, that it feels like living in a world containing two entirely different things. Knowledge concerning these two different phenomena may be taught in two different classes, and studied by two different academic fields, located in two different buildings of your university.

You've got to put yourself quite a ways back, into a historically realistic frame of mind, to remember how *different* heat and motion once seemed. Though, depending on how much you know today, it may not be as hard as all that, if you can look past the pressure of conventionality (that is, "heat is motion" is an un-weird belief, "heat is not motion" is a weird belief). I mean, suppose that tomorrow the physicists stepped forward and said, "Our popularizations of science have always contained one lie. Actually, heat has nothing to do with motion." Could you *prove* they were wrong?

Saying "Maybe heat and motion are the same thing!" is easy. The difficult part is explaining *how*. It takes a great deal of detailed knowledge to get yourself to the point where you can no longer *conceive* of a world in which the two phenomena go separate ways. Reduction isn't cheap, and that's why it buys so much.

Or maybe you could say: "Reductionism is easy, reduction is hard." But it does kinda help to be a reductionist, I think, when it comes time to go looking for a reduction.

Brain Breakthrough! It's Made of Neurons!



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In an amazing breakthrough, a multinational team of scientists led by Nobel laureate Santiago Ramón y Cajal announced that the brain is composed of a *ridiculously* complicated network of tiny cells connected to each other by infinitesimal threads and branches.

The multinational team—which also includes the famous technician Antonie van Leeuwenhoek, and possibly Imhotep, promoted to the Egyptian god of medicine—issued this statement:

“The present discovery culminates years of research indicating that the convoluted squishy thing inside our skulls is even more complicated than it looks. Thanks to Cajal’s application of a new staining technique invented by Camillo Golgi, we have learned that this structure is not a continuous network like the blood vessels of the body, but is actually composed of many tiny cells, or ‘neurons,’ connected to one another by even more tiny filaments.

“Other extensive evidence, beginning from Greek medical researcher Alcmaeon and continuing through Paul Broca’s research on speech deficits, indicates that the brain is the seat of reason.

“Nemesius, the Bishop of Emesia, has previously argued that brain tissue is too earthy to act as an intermediary between the body and soul, and so the mental faculties are located in the ventricles of the brain. However, if this is correct, there is no reason why this organ should turn out to have an immensely complicated internal composition.

“Charles Babbage has independently suggested that many small mechanical devices could be collected into an ‘Analytical Engine,’ capable of performing activities, such as arithmetic, which are widely believed to require thought. The work of

Luigi Galvani and Hermann von Helmholtz suggests that the activities of neurons are electrochemical in nature, rather than mechanical pressures as previously believed. Nonetheless, we think an analogy with Babbage's 'Analytical Engine' suggests that a vastly complicated network of neurons could similarly exhibit thoughtful properties.

"We have found an enormously complicated material system located where the mind should be. The implications are shocking, and must be squarely faced. We believe that the present research offers strong experimental evidence that Benedictus Spinoza was correct, and René Descartes wrong: Mind and body are of one substance.

"In combination with the work of Charles Darwin showing how such a complicated organ could, in principle, have arisen as the result of processes not themselves intelligent, the bulk of scientific evidence now seems to indicate that intelligence is ontologically non-fundamental and has an extended origin in time. This strongly weighs against theories which assign mental entities an ontologically fundamental or causally primal status, including all religions ever invented.

"Much work remains to be done on discovering the specific identities between electrochemical interactions between neurons, and thoughts. Nonetheless, we believe our discovery offers the promise, though not yet the realization, of a full scientific account of thought. The problem may now be declared, if not solved, then solvable."

We regret that Cajal and most of the other researchers involved on the Project are no longer available for comment.

When Anthropomorphism Became Stupid



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It turns out that most things in the universe don't have minds.

This statement would have provoked incredulity among many earlier cultures. "Animism" is the usual term. They thought that trees, rocks, streams, and hills all had spirits because, hey, why not?

I mean, those lumps of flesh known as "humans" contain thoughts, so why shouldn't the lumps of wood known as "trees"?

My muscles move at my will, and water flows through a river. Who's to say that the river doesn't have a will to move the water? The river overflows its banks, and floods my tribe's gathering-place—why not think that the river was angry, since it moved its parts to hurt us? It's what we would think when someone's fist hit our nose.

There is no obvious reason—no reason obvious *to a hunter-gatherer*—why this cannot be so. It only seems like a *stupid* mistake if you confuse weirdness with stupidity. Naturally the belief that rivers have animating spirits seems "weird" to us, since it is not a belief of our tribe. But there is nothing obviously stupid about thinking that great lumps of moving water have spirits, just like our own lumps of moving flesh.

If the idea were *obviously* stupid, no one would have believed it. Just like, for the longest time, nobody believed in the obviously stupid idea that the Earth moves while seeming motionless.

Is it obvious that trees can't think? Trees, let us not forget, are *in fact* our distant cousins. Go far enough back, and you have a common ancestor with your fern. If lumps of flesh can think, why not lumps of wood?

For it to be *obvious* that wood doesn't think, you have to belong to a culture with microscopes. Not just *any* microscopes, but really *good* microscopes.

Aristotle thought the brain was an organ for cooling the blood. (It's a good thing that what we believe about our brains has very little effect on their actual operation.)

Egyptians threw the brain away during the process of mummification.

Alcmaeon of Croton, a Pythagorean of the fifth century BCE, put his finger on the brain as the seat of intelligence, because he'd traced the optic nerve from the eye to the brain. Still, with the amount of evidence he had, it was only a guess.

When did the central role of the brain stop being a guess? I do not know enough history to answer this question, and probably there wasn't any sharp dividing line. Maybe we could put it at the point where someone traced the anatomy of the nerves, and discovered that severing a nervous connection *to the brain* blocked movement and sensation?

Even so, that is only a mysterious spirit moving through the nerves. Who's to say that wood and water, even if they lack the little threads found in human anatomy, might not carry the same mysterious spirit by different means?

I've spent some time online trying to track down the exact moment when someone noticed the vastly tangled internal structure of the brain's neurons, and said, "Hey, I bet all this giant tangle is doing complex information-processing!" I haven't had much luck. (It's not Camillo Golgi—the tangledness of the circuitry was known before Golgi.) Maybe there was never a watershed moment there, either.

But the discovery of that tangledness, and Charles Darwin's theory of natural selection, and the notion of cognition as computation, is where I would put the gradual beginning of anthropomorphism's descent into being *obviously* wrong.

It's the point where you can look at a tree, and say: "I don't see anything in the tree's biology that's doing complex information-processing. Nor do I see it in the behavior, and if it's hidden in a way that doesn't affect the tree's behavior,

how would a selection pressure for such complex information-processing arise?"

It's the point where you can look at a river, and say, "Water doesn't contain patterns replicating with distant heredity and substantial variation subject to iterative selection, so how would a river come to have any pattern so complex and functionally optimized as a brain?"

It's the point where you can look at an atom, and say: "Anger may look simple, but it's not, and there's no room for it to fit in something as simple as an atom—not unless there are whole universes of subparticles inside quarks; and even then, since we've never seen any sign of atomic anger, it wouldn't have any effect on the high-level phenomena we know."

It's the point where you can look at a puppy, and say: "The puppy's parents may push it to the ground when it does something wrong, but that doesn't mean the puppy is doing moral reasoning. Our current theories of evolutionary psychology holds that moral reasoning arose as a response to more complex social challenges than that—in their full-fledged human form, our moral adaptations are the result of selection pressures over linguistic arguments about tribal politics."

It's the point where you can look at a rock, and say, "This lacks even the simple search trees embodied in a chess-playing program—where would it get the *intentions* to *want* to roll downhill, as Aristotle once thought?"

It is written:¹

Zhuangzi and Huizi were strolling along the dam of the Hao Waterfall when Zhuangzi said, "See how the minnows come out and dart around where they please! That's what fish really enjoy!"

Huizi said, "You're not a fish—how do you know what fish enjoy?"

Zhuangzi said, "You're not I, so how do you know I don't know what fish enjoy?"

Now we know.

1. Zhuangzi, chapter 17

A Priori

Traditional Rationality is phrased as social rules, with violations interpretable as cheating: if you break the rules and no one else is doing so, you're the first to defect—making you a bad, bad person. To Bayesians, the brain is an engine of accuracy: if you violate the laws of rationality, the engine doesn't run, and this is equally true whether anyone else breaks the rules or not.

Consider the problem of Occam's Razor, as confronted by Traditional philosophers. If two hypotheses fit the same observations equally well, why believe the simpler one is more likely to be true? You could argue that Occam's Razor has worked in the past, and is therefore likely to continue to work in the future. But this, itself, appeals to a prediction from Occam's Razor. "Occam's Razor works up to October 8th, 2027 and then stops working thereafter" is more complex, but it fits the observed evidence equally well.

You could argue that Occam's Razor is a reasonable distribution on prior probabilities. But what is a "reasonable" distribution? Why not label "reasonable" a very complicated prior distribution, which makes Occam's Razor work in all observed tests so far, but generates exceptions in future cases?

Indeed, it seems there is no way to *justify* Occam's Razor except by *appealing* to Occam's Razor, making this *argument* unlikely to *convince* any *judge* who does not already *accept* Occam's Razor. (What's special about the words I italicized?)

If you are a philosopher whose daily work is to write papers, criticize other people's papers, and respond to others' criticisms of your own papers, then you may look at Occam's Razor and shrug. Here is an end to justifying, arguing and convincing. You decide to call a truce on writing papers; if your fellow philosophers do not demand justification for your un-arguable beliefs, you will not demand justification for theirs. And as the



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symbol of your treaty, your white flag, you use the phrase “a priori truth.”

But to a Bayesian, in this era of cognitive science and evolutionary biology and Artificial Intelligence, saying “a priori” doesn’t explain why the brain-engine runs. If the brain has an amazing “a priori truth factory” that *works* to produce accurate beliefs, it makes you wonder why a thirsty hunter-gatherer can’t use the “a priori truth factory” to locate drinkable water. It makes you wonder why eyes evolved in the first place, if there are ways to produce accurate beliefs without looking at things.

James R. Newman¹ said: “The fact that one apple added to one apple invariably gives two apples helps in the teaching of arithmetic, but has no bearing on the truth of the proposition that $1 + 1 = 2$.” The *Internet Encyclopedia of Philosophy* defines “a priori” propositions as those knowable independently of experience. Wikipedia quotes² Hume: Relations of ideas are “discoverable by the mere operation of thought, without dependence on what is anywhere existent in the universe.” You can see that $1 + 1 = 2$ *just by thinking about it*, without looking at apples.

But in this era of neurology, one ought to be aware that *thoughts* are existent in the universe; they are identical to the operation of brains. Material brains, real in the universe, composed of quarks in a single unified mathematical physics whose laws draw no border between the inside and outside of your skull.

When you add $1 + 1$ and get 2 by thinking, these thoughts are themselves embodied in flashes of neural patterns. In principle, we could *observe*, experientially, the exact same material events as they occurred within someone else’s brain. It would require some advances in computational neurobiology and brain-computer interfacing, but in principle, it could be

1. James Roy Newman, *The World of Mathematics*, Vol 3, pg 1614

2. sharov.me/r/4-13

A priori and a posteriori - Wikipedia • en.wikipedia.org



done. You could see someone else's engine operating materially, through material chains of cause and effect, to compute by "pure thought" that $1 + 1 = 2$. How is observing this pattern in *someone else's* brain any different, as a way of knowing, from observing your own brain doing the same thing? When "pure thought" tells you that $1 + 1 = 2$, "independently of any experience or observation," you are, in effect, observing your own brain as evidence.

If this seems counterintuitive, try to see minds/brains as engines—an engine that collides the neural pattern for 1 and the neural pattern for 1 and gets the neural pattern for 2. If this engine works at all, then it should *have the same output* if it observes (with eyes and retina) a similar brain-engine carrying out a similar collision, and copies into itself the resulting pattern. In other words, for every form of a priori knowledge obtained by "pure thought," you are learning exactly the same thing you would learn if you saw an outside brain-engine carrying out the same pure flashes of neural activation. The engines are equivalent, the bottom-line outputs are equivalent, the belief-entanglements are the same.

There is nothing you can know "a priori," which you could not know with equal validity by observing the chemical release of neurotransmitters within some outside brain. What do you think you *are*, dear reader?

This is *why* you can predict the result of adding 1 apple and 1 apple by imagining it first in your mind, or punch " 3×4 " into a calculator to predict the result of imagining 4 rows with 3 apples per row. You and the apple exist within a boundary-less unified physical process, and one part may echo another.

Are the sort of neural flashes that philosophers label "a priori beliefs" *arbitrary*? Many AI algorithms function better with "regularization" that biases the solution space toward simpler solutions. But the regularized algorithms are themselves more complex; they contain an extra line of code (or 1,000 extra lines) compared to unregularized algorithms. The human brain is biased toward simplicity, and we think more efficiently thereby. If you press the Ignore button at this point, you're left with a complex brain that exists for no reason and works for no

reason. So don't try to tell me that "a priori" beliefs are arbitrary, because they sure aren't generated by rolling random numbers. (What does the adjective "arbitrary" *mean*, anyway?)

You can't excuse calling a proposition "a priori" by pointing out that *other* philosophers are having trouble justifying *their* propositions. If a philosopher fails to explain something, this fact cannot supply electricity to a refrigerator, nor act as a magical factory for accurate beliefs. There's no truce, no white flag, until you understand why the engine works.

If you clear your mind of *justification*, of *argument*, then it seems obvious why Occam's Razor works in practice: we live in a simple world, a low-entropy universe in which there are short explanations to be found. "But," you cry, "why is the universe itself orderly?" This I do not know, but it is what I see as the next mystery to be explained. This is not the same question as "How do I argue Occam's Razor to a hypothetical debater who has not already accepted it?"

Perhaps you cannot argue *anything* to a hypothetical debater who has not accepted Occam's Razor, just as you cannot argue anything to a rock. A mind needs a certain amount of dynamic structure to be an argument-acceptor. If a mind doesn't implement Modus Ponens, it can accept "*A*" and "*A* → *B*" all day long without ever producing "*B*." How do you justify Modus Ponens to a mind that hasn't accepted it? How do you argue a rock into becoming a mind?

Brains evolved from non-brainy matter by natural selection; they were not justified into existence by arguing with an ideal philosophy student of perfect emptiness. This does not make our judgments meaningless. A brain-engine can work correctly, producing accurate beliefs, even if it was merely *built*—by human hands or cumulative stochastic selection pressures—rather than argued into existence. But to be satisfied by this answer, one must see rationality in terms of engines, rather than arguments.

Reductive Reference

The reductionist thesis (as I formulate it) is that human minds, for reasons of efficiency, use a multi-level map in which we separately *think* about things like “atoms” and “quarks,” “hands” and “fingers,” or “heat” and “kinetic energy.” Reality itself, on the other hand, is single-level in the sense that it does not seem to contain atoms as *separate, additional, causally efficacious* entities *over and above* quarks.

Sadi Carnot formulated the (precursor to) the Second Law of Thermodynamics using the caloric theory of heat, in which heat was just a fluid that flowed from hot things to cold things, produced by fire, making gases expand—the effects of heat were studied separately from the science of kinetics, considerably before the reduction took place. If you’re trying to design a steam engine, the effects of all those tiny vibrations and collisions which we name “heat” can be summarized into a much simpler description than the full quantum mechanics of the quarks. Humans compute efficiently, thinking of only significant effects on goal-relevant quantities.

But reality itself does seem to use the full quantum mechanics of the quarks. I once met a fellow who thought that if you used General Relativity to compute a low-velocity problem, like an artillery shell, General Relativity would give you the *wrong answer*—not just a slow answer, but an *experimentally wrong* answer—because at low velocities, artillery shells are governed by Newtonian mechanics, not General Relativity. This is exactly how physics does *not* work. Reality just seems to go on crunching through General Relativity, even when it only makes a difference at the fourteenth decimal place, which a human would regard as a huge waste of computing power. Physics does it with brute force. No one has *ever* caught physics simplifying its calculations—or if someone did catch it, the Matrix Lords erased the memory afterward.



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Our map, then, is very much unlike the territory; our maps are multi-level, the territory is single-level. Since the representation is so incredibly unlike the referent, in what sense can a belief like “I am wearing socks” be called *true*, when in reality itself, there are only quarks?

In case you’ve forgotten what the word “true” means, the classic definition was given by Alfred Tarski:

The statement “snow is white” is *true* if and only if snow is white.

In case you’ve forgotten what the difference is between the statement “I believe ‘snow is white’” and “‘Snow is white’ is true,” see p. 74, *Qualitatively Confused*. Truth can’t be evaluated *just* by looking inside your own head—if you want to know, for example, whether “the morning star = the evening star,” you need a telescope; it’s not enough just to look at the beliefs themselves.

This is the point missed by the postmodernist folks screaming, “But how do you *know* your beliefs are true?” When you do an experiment, you actually *are* going outside your own head. You’re engaging in a complex interaction whose outcome is causally determined by the thing you’re reasoning about, not just your beliefs about it. I once defined “reality” as follows:¹

Even when I have a simple hypothesis, strongly supported by all the evidence I know, sometimes I’m still surprised. So I need different names for the thingies that determine my predictions and the thingy that determines my experimental results. I call the former thingies “belief,” and the latter thingy “reality.”

The interpretation of your experiment still depends on your prior beliefs. I’m not going to talk, for the moment, about Where Priors Come From, because that is not the subject of this essay. My point is that truth refers to an *ideal* comparison between

1. See vol. I, p. 189

a belief and reality. Because we understand that planets are distinct from beliefs about planets, we can design an experiment to test whether the belief “the morning star and the evening star are the same planet” is *true*. This experiment will involve telescopes, not just introspection, because we understand that “truth” involves comparing an internal belief to an external fact; so we use an instrument, the telescope, whose perceived behavior we believe to depend on the external fact of the planet.

Believing that the telescope helps us evaluate the “truth” of “morning star = evening star” relies on our prior beliefs about the telescope interacting with the planet. Again, I’m not going to address that in this particular essay, except to quote one of my favorite Raymond Smullyan lines: “If the more sophisticated reader objects to this statement on the grounds of its being a mere tautology, then please at least give the statement credit for not being inconsistent.” Similarly, I don’t see the use of a telescope as circular logic, but as reflective coherence; for every systematic way of arriving at truth, there ought to be a rational explanation for how it works.

The question on the table is what it *means* for “snow is white” to be *true*, when, in reality, there are just quarks.

There’s a certain pattern of neural connections making up your beliefs about “snow” and “whiteness”—we believe this, but we do not know, and cannot concretely visualize, the actual neural connections. Which are, themselves, embodied in a pattern of quarks even less known. Out there in the world, there are water molecules whose temperature is low enough that they have arranged themselves in tiled repeating patterns; they look nothing like the tangles of neurons. In what sense, comparing one (ever-fluctuating) pattern of quarks to the other, is the belief “snow is white” *true*?

Obviously, neither I nor anyone else can offer an Ideal Quark Comparer Function that accepts a quark-level description of a neurally embodied belief (including the surrounding brain) and a quark-level description of a snowflake (and the surrounding laws of optics), and outputs “true” or “false” over “snow is white.” And who says the fundamental level is *really* about particle fields?

On the other hand, throwing out all beliefs because they aren't written as gigantic unmanageable specifications about quarks we can't even see ...doesn't seem like a very prudent idea. Not the best way to optimize our goals.

It seems to me that a word like "snow" or "white" can be taken as a kind of promissory note—not a *known* specification of exactly which physical quark configurations count as "snow," but, nonetheless, there are things you call snow and things you don't call snow, and even if you got a few items wrong (like plastic snow), an Ideal Omniscient Science Interpreter would see a tight cluster in the center and redraw the boundary to have a simpler definition.

In a single-layer universe whose bottom layer is unknown, or uncertain, or just too large to talk about, the concepts in a multi-layer mind can be said to represent a kind of promissory note—we don't know *what* they correspond to, out there. But it seems to us that we can distinguish positive from negative cases, in a predictively productive way, so we think—perhaps in a fully general sense—that there is *some* difference of quarks, *some* difference of configurations at the fundamental level, that explains the differences that feed into our senses, and ultimately result in our saying "snow" or "not snow."

I see this white stuff, and it is the same on several occasions, so I hypothesize a stable latent cause in the environment—I give it the name "snow"; "snow" is then a promissory note referring to a believed-in simple boundary that could be drawn around the unseen causes of my experience.

Hilary Putnam's "Twin Earth" thought experiment (where water is not H₂O but some strange other substance denoted XYZ, otherwise behaving much like water), and the subsequent philosophical debate, helps to highlight this issue. "Snow" doesn't have a logical definition known to us—it's more like an empirically determined pointer to a logical definition. This is true even if you believe that snow is ice crystals is low-temperature tiled water molecules. The water molecules are made of quarks. What if quarks turn out to be made of something else? What is a snowflake, then? You don't know—but it's still a snowflake, not a fire hydrant.

And of course, these very paragraphs I have just written are likewise far above the level of quarks. “Sensing white stuff, visually categorizing it, and thinking ‘snow’ or ‘not snow’”—this is also talking very far above the quarks. So my meta-beliefs are also promissory notes, for things that an Ideal Omniscient Science Interpreter might know about which configurations of the quarks (or whatever) making up my brain correspond to “believing ‘snow is white.’”

But then, the entire grasp that we have upon reality is made up of promissory notes of this kind. So, rather than calling it circular, I prefer to call it self-consistent.

This can be a bit unnerving—maintaining a precarious epistemic perch, in both object-level beliefs and reflection, far above a huge unknown underlying fundamental reality, and hoping one doesn’t fall off.

On reflection, though, it’s hard to see how things could be any other way.

So at the end of the day, the statement “reality does not contain hands as fundamental, additional, separate causal entities, over and above quarks” is not the same statement as “hands do not exist” or “I don’t have any hands.” There are no *fundamental* hands; hands are made of fingers, palm, and thumb, which in turn are made of muscle and bone, all the way down to elementary particle fields, which are the fundamental causal entities, so far as we currently know.

This is not the same as saying, “there are no ‘hands.’” It is not the same as saying, “the word ‘hands’ is a promissory note that will never be paid, because there is no empirical cluster that corresponds to it”; or “the ‘hands’ note will never be paid, because it is logically impossible to reconcile its supposed characteristics”; or “the statement ‘humans have hands’ refers to a sensible state of affairs, but reality is not in that state.”

Just: There are patterns that exist *in* reality where we see “hands,” and these patterns have something in common, but they are not fundamental.

If I *really* had no hands—if reality suddenly transitioned to be in a state that we would describe as “Eliezer has no hands”—real-

ity would shortly thereafter correspond to a state we would describe as “Eliezer screams as blood jets out of his wrist stumps.”

And this is *true*, even though the above paragraph hasn’t specified any quark positions.

The previous sentence is likewise meta-true.

The map is multilevel, the territory is single-level. This doesn’t mean that the higher levels “don’t exist,” like looking in your garage for a dragon and finding nothing there, or like seeing a mirage in the desert and forming an expectation of drinkable water when there is nothing to drink. The higher levels of your map are not *false*, without referent; they have referents *in* the single level of physics. It’s not that the wings of an airplane unexist—then the airplane would drop out of the sky. The “wings of an airplane” exist *explicitly* in an engineer’s multilevel model of an airplane, and the wings of an airplane exist *implicitly* in the quantum physics of the real airplane. Implicit existence is not the same as nonexistence. The exact description of this implicitness is not known to us—is not explicitly represented in our map. But this does not prevent our map from working, or even prevent it from being *true*.

Though it is a bit unnerving to contemplate that every single concept and belief in your brain, including these meta-concepts about how your brain works and why you can form accurate beliefs, are perched orders and orders of magnitude above reality ...

Zombies! Zombies?

Your “zombie,” in the philosophical usage of the term, is putatively a being that is exactly like you in *every* respect—identical behavior, identical speech, identical brain; every atom and quark in *exactly* the same position, moving according to the same causal laws of motion—*except* that your zombie is not conscious.

It is furthermore claimed that if zombies are “possible” (a term over which battles are still being fought), then, purely from our knowledge of this “possibility,” we can deduce a priori that consciousness is extra-physical, in a sense to be described below; the standard term for this position is “epiphenomenalism.”

(For those unfamiliar with zombies, I emphasize that *this is not a strawman*. See, for example, the *Stanford Encyclopedia of Philosophy* entry on Zombies.¹ The “possibility” of zombies is accepted by a substantial fraction, possibly a majority, of academic philosophers of consciousness.)

I once read somewhere, “You are not the one who speaks your thoughts—you are the one who *hears* your thoughts.” In Hebrew, the word for the highest soul, that which God breathed into Adam, is N’Shama—“the hearer.”

If you conceive of “consciousness” as a purely passive listening, then the notion of a zombie initially seems easy to imagine. It’s someone who lacks the N’Shama, the hearer.

(Warning: *Very* long 6,600-word essay involving David Chalmers ahead. This may be taken as my demonstrative coun-



discussion
sharov.me/r/
4-c42

1. sharov.me/r/4-14

Zombies (Stanford Encyclopedia of Philosophy) •
plato.stanford.edu



terexample to Richard Chappell's Arguing with Eliezer Part II,² in which Richard accuses me of not engaging with the complex arguments of real philosophers.)

When you open a refrigerator and find that the orange juice is gone, you think "Darn, I'm out of orange juice." The sound of these words is probably represented in your auditory cortex, as though you'd heard someone else say it. (Why do I think this? Because native Chinese speakers can remember longer digit sequences than English-speakers. Chinese digits are all single syllables, and so Chinese speakers can remember around ten digits, versus the famous "seven plus or minus two" for English speakers. There appears to be a loop of repeating sounds back to yourself, a size limit on working memory in the auditory cortex, which is genuinely phoneme-based.)

Let's suppose the above is correct; as a postulate, it should certainly present no problem for advocates of zombies. Even if humans are not like this, it seems easy enough to imagine an AI constructed this way (and imaginability is what the zombie argument is all about). It's not only conceivable in principle, but quite possible in the next couple of decades, that surgeons will lay a network of neural taps over someone's auditory cortex and read out their internal narrative. (Researchers have already tapped the lateral geniculate nucleus of a cat and reconstructed recognizable visual inputs.)³

So your zombie, being physically identical to you down to the last atom, will open the refrigerator and form auditory cortical patterns for the phonemes "Darn, I'm out of orange juice." On this point, epiphenomalists would willingly agree.

But, says the epiphenomenalist, in the zombie there is no one inside to *hear*; the inner listener is missing. The internal

2. sharov.me/r/4-15
Philosophy, et cetera: Arguing with Eliezer: Part II •
philosophyetc.net



3. Stanley GB, Li FF, Dan Y, "Reconstruction of natural scenes from ensemble responses in the lateral geniculate nucleus", *J Neurosci.* 1999 Sep 15;19(18):8036-42.

narrative is spoken, but unheard. You are not the one who speaks your thoughts. You are the one who hears them.

It seems a lot more straightforward (they would say) to make an AI that prints out some kind of internal narrative, than to show that an inner listener hears it.

The Zombie Argument is that if the Zombie World is *possible*—not necessarily physically possible in our universe, just “possible in theory,” or “imaginable,” or something along those lines—then consciousness must be extra-physical, something over and above mere atoms. Why? Because even if you somehow knew the positions of all the atoms in the universe, you would still have to be told, as a separate and additional fact, that people were conscious—that they had inner listeners—that we were not in the Zombie World, as seems *possible*.

Zombie-ism is not the same as dualism. Descartes thought there was a body-substance and a wholly different kind of mind-substance, but Descartes also thought that the mind-substance was a *causally active* principle, interacting with the body-substance, controlling our speech and behavior. Subtracting out the mind-substance from the human would leave a *traditional* zombie, of the lurching and groaning sort.

And though the Hebrew word for the innermost soul is N’Shama, that-which-hears, I can’t recall hearing a rabbi arguing for the possibility of zombies. Most rabbis would probably be aghast at the idea that the divine part which God breathed into Adam *doesn’t actually do anything*.

The technical term for the belief that consciousness is there, but has no effect on the physical world, is *epiphenomenalism*.

Though there are other elements to the zombie argument (I’ll deal with them below), I think that the intuition of the passive listener is what first seduces people to zombie-ism. In particular, it’s what seduces a lay audience to zombie-ism. The core notion is simple and easy to access: The lights are on but no one’s home.

Philosophers are appealing to the intuition of the passive listener when they say “Of course the zombie world is imaginable; you know exactly what it would be like.”

One of the great battles in the Zombie Wars is over what, exactly, is meant by saying that zombies are “possible.” Early zombie-ist philosophers (in the 1970s) just thought it was obvious that zombies were “possible,” and didn’t bother to define what sort of possibility was meant.

Because of my reading in mathematical logic, what instantly comes into my mind is logical possibility. If you have a collection of statements like $\{(A \Rightarrow B), (B \Rightarrow C), (C \Rightarrow \neg A)\}$, then the compound belief is *logically possible* if it has a *model*—which, in the simple case above, reduces to finding a value assignment to $\{A, B, C\}$ that makes all of the statements $(A \Rightarrow B)$, $(B \Rightarrow C)$, and $(C \Rightarrow \neg A)$ true. In this case, $A = B = C = 0$ works, as does $\{A = 0, B = C = 1\}$ or $\{A = B = 0, C = 1\}$.

Something will *seem* possible—will seem “conceptually possible” or “imaginable”—if you can consider the collection of statements without *seeing* a contradiction. But it is, in general, a very hard problem to see contradictions or to find a full specific model! If you limit yourself to simple Boolean propositions of the form $((A \text{ or } B \text{ or } C) \text{ and } (B \text{ or } \neg C \text{ or } D) \text{ and } (D \text{ or } \neg A \text{ or } \neg C) \dots)$, conjunctions of disjunctions of three variables, then this is a very famous problem called 3-SAT, which is one of the first problems ever to be proven NP-complete.

So just because you don’t see a contradiction in the Zombie World at first glance, it doesn’t mean that no contradiction is there. It’s like not seeing a contradiction in the Riemann Hypothesis at first glance. From conceptual possibility (“I don’t see a problem”) to *logical possibility*, in the full technical sense, is a very great leap. It’s easy to make it an NP-complete leap, and with first-order theories you can make it arbitrarily hard to compute even for finite questions. And it’s *logical* possibility of the Zombie World, not conceptual possibility, that is needed to suppose that a logically omniscient mind could know the positions of all the atoms in the universe, and yet need to be told as an *additional* non-entailed fact that we have inner listeners.

Just because you don’t see a contradiction *yet* is no guarantee that you won’t see a contradiction in another thirty seconds. “All odd numbers are prime. Proof: 3 is prime, 5 is prime, 7 is prime ...”

So let us ponder the Zombie Argument *a little longer*: Can we think of a counterexample to the assertion “Consciousness has no third-party-detectable causal impact on the world”?

If you close your eyes and concentrate on your inward awareness, you will begin to form thoughts, in your internal narrative, that go along the lines of “I am aware” and “My awareness is separate from my thoughts” and “I am not the one who speaks my thoughts, but the one who hears them” and “My stream of consciousness is not my consciousness” and “It seems like there is a part of me that I can imagine being eliminated without changing my outward behavior.”

You can even say these sentences out loud, as you meditate. In principle, someone with a super-fMRI could probably read the phonemes out of your auditory cortex; but saying it out loud removes all doubt about whether you have entered the realms of testability and physical consequences.

This certainly seems like the inner listener is being *caught in the act of listening* by whatever part of you writes the internal narrative and flaps your tongue.

Imagine that a mysterious race of aliens visit you, and leave you a mysterious black box as a gift. You try poking and prodding the black box, but (as far as you can tell) you never succeed in eliciting a reaction. You can’t make the black box produce gold coins or answer questions. So you conclude that the black box is causally inactive: “For all *X*, the black box doesn’t do *X*.” The black box is an effect, but not a cause; epiphenomenal; without causal potency. In your mind, you test this general hypothesis to see if it is true in some trial cases, and it seems to be true—“Does the black box turn lead to gold? No. Does the black box boil water? No.”

But you can *see* the black box; it absorbs light, and weighs heavy in your hand. This, too, is part of the dance of causality. If the black box were *wholly* outside the causal universe, you couldn’t see it; you would have no way to know it existed; you could not say, “Thanks for the black box.” You didn’t *think* of this counterexample, when you formulated the general rule: “All *X*: Black box doesn’t do *X*.” But it was there all along.

(Actually, the aliens left you *another* black box, this one *purely* epiphenomenal, and you haven't the slightest clue that it's there in your living room. That was their joke.)

If you can close your eyes, and sense yourself sensing—if you can be aware of yourself being aware, and think “I am aware that I am aware”—and say out loud, “I am aware that I am aware”—then your consciousness is not without effect on your internal narrative, or your moving lips. You can see yourself seeing, and your internal narrative reflects this, and so do your lips if you choose to say it out loud.

I have not seen the above argument written out that particular way—“the listener caught in the act of listening”—though it may well have been said before.

But it is a standard point—which zombie-ist philosophers accept!—that the Zombie World's philosophers, being atom-by-atom identical to our own philosophers, write identical papers about the philosophy of consciousness.

At this point, the Zombie World stops being an intuitive consequence of the idea of a passive listener.

Philosophers writing papers about consciousness would *seem* to be at least one effect of consciousness upon the world. You can argue clever reasons why this is not so, but you have to be clever.

You would intuitively suppose that if your inward awareness went away, this would change the world, in that your internal narrative would no longer say things like “There is a mysterious listener within me,” because the mysterious listener would be gone. It is usually right *after* you focus your awareness on your awareness, that your internal narrative says “I am aware of my awareness,” which suggests that if the first event never happened again, neither would the second. You can argue clever reasons why this is not so, but you have to be clever.

You can form a propositional belief that “Consciousness is without effect,” and not *see* any contradiction at first, if you don't realize that talking about consciousness is an effect of being conscious. But once you see the connection from the general rule that consciousness has no effect, to the specific implication that consciousness has no effect on how philosophers write

papers about consciousness, zombie-ism stops being intuitive and starts requiring you to postulate strange things.

One strange thing you might postulate is that there's a Zombie Master, a god within the Zombie World who surreptitiously takes control of zombie philosophers and makes them talk and write about consciousness.

A Zombie Master doesn't seem impossible. Human beings often don't sound all that coherent when talking about consciousness. It might not be that hard to fake their discourse, to the standards of, say, a human amateur talking in a bar. Maybe you could take, as a corpus, one thousand human amateurs trying to discuss consciousness; feed them into a non-conscious but sophisticated AI, better than today's models but not self-modifying; and get back discourse about "consciousness" that sounded as sensible as most humans, which is to say, not very.

But this speech about "consciousness" would not be spontaneous. It would not be produced *within* the AI. It would be a recorded imitation of someone else talking. That is just a holodeck, with a central AI writing the speech of the non-player characters. This is *not* what the Zombie World is about.

By supposition, the Zombie World is atom-by-atom identical to our own, except that the inhabitants lack consciousness. Furthermore, the atoms in the Zombie World move under the same laws of physics as in our own world. If there are "bridging laws" that govern *which configurations of atoms evoke consciousness*, those bridging laws are absent. But, by hypothesis, the difference is not experimentally detectable. When it comes to saying whether a quark zigs or zags or exerts a force on nearby quarks—anything experimentally measurable—the same physical laws govern.

The Zombie World has no *room* for a Zombie Master, because a Zombie Master has to control the zombie's lips, and that control is, in principle, experimentally detectable. The Zombie Master moves lips, therefore it has observable consequences. There would be a point where an electron zags, instead of zigging, because the Zombie Master says so. (Unless the Zombie Master is actually *in* the world, as a pattern of quarks—but then

the Zombie World is not atom-by-atom identical to our own, unless you think *this* world also contains a Zombie Master.)

When a philosopher in our world types, “I think the Zombie World is possible,” their fingers strike keys in sequence: Z-O-M-B-I-E. There is a chain of causality that can be traced back from these keystrokes: muscles contracting, nerves firing, commands sent down through the spinal cord, from the motor cortex—and then into less understood areas of the brain, where the philosopher’s internal narrative first began talking about “consciousness.”

And the philosopher’s zombie twin strikes the same keys, *for the same reason*, causally speaking. There is no cause within the chain of explanation for why the philosopher writes the way they do that is not also present in the zombie twin. The zombie twin also has an internal narrative about “consciousness,” that a super-fMRI could read out of the auditory cortex. And whatever other thoughts, or other causes of any kind, led to that internal narrative, they are exactly the same in our own universe and in the Zombie World.

So you can’t say that the philosopher is writing about consciousness *because of* consciousness, while the zombie twin is writing about consciousness *because of* a Zombie Master or AI chatbot. When you trace back the chain of causality behind the keyboard, to the internal narrative echoed in the auditory cortex, to the cause of the narrative, you must find the *same* physical explanation in our world as in the zombie world.

As the most formidable advocate of zombie-ism, David Chalmers, writes:⁴

Think of my zombie twin in the universe next door. He talks about conscious experience all the time—in fact, he seems obsessed by it. He spends ridiculous amounts of time hunched over a computer, writing chapter after chapter on the mysteries of consciousness. He often comments on the pleasure he gets from certain sensory qualia,

4. Chalmers, *The Conscious Mind*.

professing a particular love for deep greens and purples. He frequently gets into arguments with zombie materialists, arguing that their position cannot do justice to the realities of conscious experience.

And yet he has no conscious experience at all! In his universe, the materialists are right and he is wrong. Most of his claims about conscious experience are utterly false. But there is certainly a physical or functional explanation of why he makes the claims he makes. After all, his universe is fully law-governed, and no events therein are miraculous, so there must be some explanation of his claims.

...Any explanation of my twin's behavior will equally count as an explanation of my behavior, as the processes inside his body are precisely mirrored by those inside mine. The explanation of his claims obviously does not depend on the existence of consciousness, as there is no consciousness in his world. It follows that the explanation of my claims is also independent of the existence of consciousness.

Chalmers is not arguing *against* zombies; those are his actual beliefs!

This paradoxical situation is at once delightful and disturbing. It is not obviously fatal to the nonreductive position, but it is at least something that we need to come to grips with ...

I would seriously nominate this as the largest bullet ever bitten in the history of time. And that is a backhanded compliment to David Chalmers: A lesser mortal would simply fail to see the implications, or refuse to face them, or rationalize a reason it wasn't so.

Why would anyone bite a bullet that large? Why would anyone postulate unconscious zombies who write papers about

consciousness for *exactly the same reason* that our own genuinely conscious philosophers do?

Not because of the first intuition I wrote about, the intuition of the passive listener. That intuition may say that zombies can drive cars or do math or even fall in love, but it doesn't say that zombies write philosophy papers about their passive listeners.

The zombie argument does not rest *solely* on the intuition of the passive listener. If this was all there was to the zombie argument, it would be dead by now, I think. The intuition that the "listener" can be eliminated without effect would go away as soon as you realized that your internal narrative routinely *seems* to catch the listener in the act of listening.

No, the drive to bite *this* bullet comes from an entirely different intuition—the intuition that no matter how many atoms you add up, no matter how many masses and electrical charges interact with each other, they will never *necessarily* produce a subjective sensation of the mysterious redness of red. It may be a fact about our physical universe (Chalmers says) that putting such-and-such atoms into such-and-such a position *evokes* a sensation of redness; but if so, it is not a *necessary* fact, it is something to be explained above and beyond the motion of the atoms.

But if you consider the second intuition on its own, without the intuition of the passive listener, it is hard to see why it implies zombie-ism. Maybe there's just a *different kind of stuff*, apart from and additional to atoms, that is *not* causally passive—a soul that actually *does* stuff, a soul that plays a real causal role in why we write about "the mysterious redness of red." Take out the soul, and ...well, assuming you don't just fall over in a coma, you certainly won't write any more papers about consciousness!

This is the position taken by Descartes and most other ancient thinkers: The soul is of a different kind, but it *interacts* with the body. Descartes's position is technically known as *substance dualism*—there is a thought-stuff, a mind-stuff, and it is not like atoms; but it is causally potent, interactive, and leaves a visible mark on our universe.

Zombie-ists are *property dualists*—they don't believe in a *separate* soul; they believe that matter in our universe has *additional properties* beyond the physical.

"Beyond the physical"? What does that mean? It means the extra properties are there, but they don't influence the motion of the atoms, like the properties of electrical charge or mass. The extra properties are not experimentally detectable *by third parties*; *you* know you are conscious, from the *inside* of your extra properties, but no scientist can ever directly detect this from outside.

So the additional properties are there, but not causally active. The extra properties do not move atoms around, which is why they can't be detected by third parties.

And that's why we can (allegedly) imagine a universe just like this one, with all the atoms in the same places, but the extra properties missing, so that everything goes on the same as before, but no one is conscious.

The Zombie World may not be *physically* possible, say the zombie-ists—because it is a fact that all the matter in our universe has the extra properties, or obeys the bridging laws that evoke consciousness—but the Zombie World is *logically* possible: the bridging laws could have been different.

But, once you realize that conceivability is not the same as logical possibility, and that the Zombie World isn't even all that intuitive, why say that the Zombie World is logically possible?

Why, oh why, say that the extra properties are epiphenomenal and undetectable?

We can put this dilemma very sharply: Chalmers believes that there *is* something called consciousness, and this consciousness embodies the true and indescribable substance of the mysterious redness of red. It may be a property beyond mass and charge, but it's *there*, and it *is* consciousness. Now, having said the above, Chalmers furthermore specifies that this true stuff of consciousness is epiphenomenal, without causal potency—but *why say that*?

Why say that you could subtract this true stuff of consciousness, and leave all the atoms in the same place doing the same things? If that's true, we need some *separate* physical expla-

nation for why Chalmers talks about “the mysterious redness of red.” That is, there exists both a mysterious redness of red, which is extra-physical, and *an entirely separate* reason, *within* physics, why Chalmers *talks* about the “mysterious redness of red.”

Chalmers does confess that these two things seem like they ought to be related, but really, why do you need both? Why not just pick one or the other?

Once you’ve postulated that there is a mysterious redness of red, why not just say that it interacts with your internal narrative and makes you talk about the “mysterious redness of red”?

Isn’t Descartes taking the simpler approach, here? The *strictly* simpler approach?

Why postulate an extramaterial soul, *and then* postulate that the soul has no effect on the physical world, *and then* postulate a mysterious unknown *material* process that causes your internal narrative to talk about conscious experience?

Why not postulate the true stuff of consciousness which no amount of mere mechanical atoms can add up to, *and then*, having gone that far already, let this true stuff of consciousness have causal effects like making philosophers talk about consciousness?

I am not endorsing Descartes’s view. But at least I can understand where Descartes is coming from. Consciousness seems mysterious, so you postulate a mysterious stuff of consciousness. Fine.

But now the zombie-ists postulate that this mysterious stuff *doesn’t do anything*, so you need a *whole new* explanation for why you *say* you’re conscious.

That isn’t vitalism. That’s something so bizarre that vitalists would spit out their coffee. “When fires burn, they release phlogiston. *But* phlogiston doesn’t have any experimentally detectable impact on our universe, so you’ll have to go looking for a *separate* explanation of why a fire can melt snow.” *What?*

Are property dualists under the impression that if they postulate a new *active* force, something that has a causal impact on observables, they will be sticking their necks out too far?

Me, I'd say that if you postulate a mysterious, separate, additional, inherently mental property of consciousness, above and beyond positions and velocities, then, at that point, you have *already* stuck your neck out as far as it can go. To postulate this stuff of consciousness, and then further postulate that it *doesn't do anything*—for the love of cute kittens, *why*?

There isn't even an obvious career motive. "Hi, I'm a philosopher of consciousness. My subject matter is the most important thing in the universe and I should get lots of funding? Well, it's nice of you to say so, but actually the phenomenon I study doesn't do anything whatsoever." (Argument from career impact is not valid, but I say it to leave a line of retreat.)

Chalmers critiques substance dualism on the grounds that it's hard to see what new theory of physics, what new substance that interacts with matter, could possibly explain consciousness. But property dualism has exactly the same problem. No matter what kind of dual property you talk about, how exactly does it explain consciousness?

When Chalmers postulated an extra property that *is* consciousness, he *took* that leap across the unexplainable. How does it help his theory to further specify that this extra property *has no effect*? Why not just let it be causal?

If I were going to be unkind, this would be the time to drag in the dragon—to mention Carl Sagan's parable of the dragon in the garage. "I have a dragon in my garage." Great! I want to see it, let's go! "You can't see it—it's an invisible dragon." Oh, I'd like to hear it then. "Sorry, it's an inaudible dragon." I'd like to measure its carbon dioxide output. "It doesn't breathe." I'll toss a bag of flour into the air, to outline its form. "The dragon is permeable to flour."

One motive for trying to make your theory unfalsifiable is that deep down you fear to put it to the test. Sir Roger Penrose (physicist) and Stuart Hameroff (neurologist) are substance dualists; they think that there is something mysterious going on in quantum, that Everett is wrong and that the "collapse of the wavefunction" is physically real, and that this is where consciousness lives and how it exerts causal effect upon your lips when you say aloud "I think therefore I am." Believing this,

they predicted that neurons would protect themselves from decoherence long enough to maintain macroscopic quantum states.

This is in the process of being tested, and so far, prospects are not looking good for Penrose—

—but Penrose’s basic conduct is scientifically respectable. Not Bayesian, maybe, but still fundamentally healthy. He came up with a wacky hypothesis. He said how to test it. He went out and tried to actually test it.

As I once said to Stuart Hameroff, “I think the hypothesis you’re testing is completely hopeless, and your experiments should *definitely* be funded. Even if you don’t find exactly what you’re looking for, you’re looking in a place where no one else is looking, and you might find something interesting.”

So a nasty dismissal of epiphenomenalism would be that zombie-ists are afraid to say the consciousness-stuff can have *effects*, because then scientists could go *looking* for the extra properties, and fail to find them.

I don’t think this is actually true of Chalmers, though. If Chalmers lacked self-honesty, he could make things a *lot* easier on himself.

(But just in case Chalmers is reading this and does have falsification-fear, I’ll point out that if epiphenomenalism is false, then there *is* some other explanation for that-which-we-call consciousness, and it will eventually be found, leaving Chalmers’s theory in ruins; so if Chalmers cares about his place in history, he has no motive to endorse epiphenomenalism unless he really thinks it’s true.)

Chalmers is one of the most frustrating philosophers I know. Sometimes I wonder if he’s pulling an Atheism Conquered.⁵ Chalmers does this really *sharp* analysis ...and then turns left at the last minute. He lays out everything that’s wrong with the

5. sharov.me/r/4-16
expatsingapore.com, archived copy



Zombie World scenario, and then, having reduced the whole argument to smithereens, calmly accepts it.

Chalmers does the same thing when he lays out, in calm detail, the problem with saying that our own beliefs in consciousness are justified, when our zombie twins say exactly the same thing for exactly the same reasons and are wrong.

On Chalmers's theory, Chalmers's saying that he believes in consciousness cannot be *causally* justified; the belief is not caused by the fact itself. In the absence of consciousness, Chalmers would write the same papers for the same reasons.

On epiphenomenalism, Chalmers's saying that he believes in consciousness cannot be justified as the product of a process that systematically outputs true beliefs, because the zombie twin writes the same papers using the same systematic process and is wrong.

Chalmers admits this. Chalmers, in fact, explains the argument in great detail in his book. Okay, so Chalmers has solidly proven that he is not justified in believing in epiphenomenal consciousness, right? No. Chalmers writes:

Conscious experience lies at the center of our epistemic universe; we have access to it *directly*. This raises the question: what is it that justifies our beliefs about our experiences, if it is not a causal link to those experiences, and if it is not the mechanisms by which the beliefs are formed? I think the answer to this is clear: it is *having* the experiences that justifies the beliefs. For example, the very fact that I have a red experience now provides justification for my belief that I am having a red experience ...

Because my zombie twin lacks experiences, he is in a very different epistemic situation from me, and his judgments lack the corresponding justification. It may be tempting to object that if my belief lies in the physical realm, its justification must lie in the physical realm; but this is a *non sequitur*. From the fact that there is no justification in the phys-

ical realm, one might conclude that the *physical* portion of me (my brain, say) is not justified in its belief. But the question is whether *I* am justified in the belief, not whether my *brain* is justified in the belief, and if property dualism is correct than there is more to me than my brain.

So—if I’ve got this thesis right—there’s a core you, above and beyond your brain, that believes it is not a zombie, and directly experiences not being a zombie; and so its beliefs are justified.

But Chalmers just *wrote all that stuff down*, in his very physical *book*, and so did the zombie-Chalmers.

The zombie Chalmers can’t have written the book *because* of the zombie’s core self above the brain; there must be some entirely different reason, within the laws of physics.

It follows that even if there *is* a part of Chalmers hidden away that is conscious and believes in consciousness, directly and without mediation, there is also a *separable subspace* of Chalmers—a causally closed cognitive subsystem that acts entirely *within* physics—and this “outer self” is what speaks Chalmers’s internal narrative, and writes papers on consciousness.

I do not see any way to evade the charge that, on Chalmers’s own theory, this separable outer Chalmers is deranged. This is the part of Chalmers that is the same in this world, or the Zombie World; and in either world it writes philosophy papers on consciousness *for no valid reason*. Chalmers’s philosophy papers are not output by that inner core of awareness and belief-in-awareness; they are output by the mere physics of the internal narrative that makes Chalmers’s fingers strike the keys of his computer.

And yet this deranged outer Chalmers is writing philosophy papers that *just happen* to be perfectly right, *by a separate and additional miracle*. Not a logically necessary miracle (then the Zombie World would not be logically possible). A physically contingent miracle, that happens to be true in what we think is our universe, even though science can never distinguish our universe from the Zombie World.

Or at least, that would seem to be the implication of what the self-confessedly deranged outer Chalmers is telling us.

I think I speak for all reductionists when I say *Huh?*

That's not epicycles. That's, "Planetary motions follow these epicycles—but epicycles don't actually *do* anything—there's something else that makes the planets move the same way the epicycles say they should, which I haven't been able to explain—and by the way, I would say this even if there weren't any epicycles."

I have a nonstandard perspective on philosophy because I look at everything with an eye to designing an AI; specifically, a self-improving Artificial General Intelligence with stable motivational structure.

When I think about designing an AI, I ponder principles like probability theory, the Bayesian notion of evidence as differential diagnostic, and above all, reflective coherence. Any self-modifying AI that starts out in a reflectively inconsistent state won't stay that way for long.

If a self-modifying AI looks at a part of itself that concludes "*B*" on condition *A*—a part of itself that writes "*B*" to memory whenever condition *A* is true—and the AI inspects this part, determines how it (causally) operates in the context of the larger universe, and the AI decides that this part systematically tends to write false data to memory, then the AI has found what appears to be a bug, and the AI will self-modify not to write "*B*" to the belief pool under condition *A*.

Any epistemological theory that disregards reflective coherence is not a good theory to use in constructing self-improving AI. This is a knockdown argument from my perspective, considering what I intend to actually use philosophy *for*. So I have to invent a reflectively coherent theory anyway. And when I do, by golly, reflective coherence turns out to make intuitive sense.

So that's the unusual way in which I tend to think about these things. And now I look back at Chalmers:

The causally closed "outer Chalmers" (that is not influenced in any way by the "inner Chalmers" that has separate additional awareness and beliefs) must be carrying out some sys-

tematically unreliable, unwarranted operation which *in some unexplained fashion* causes the internal narrative to produce beliefs about an “inner Chalmers” that are *correct for no logical reason* in what happens to be our universe.

But there’s no possible warrant for the outer Chalmers *or any reflectively coherent self-inspecting AI* to believe in this mysterious correctness. A good AI design should, I think, look like a reflectively coherent intelligence embodied in a causal system, with a *testable* theory of how that selfsame causal system produces systematically accurate beliefs on the way to achieving its goals.

So the AI will scan Chalmers and see a closed causal cognitive system producing an internal narrative that is uttering nonsense. Nonsense that seems to have a high impact on what Chalmers thinks *should be considered a morally valuable person*.

This is not a *necessary* problem for Friendly AI theorists. It is *only* a problem if you happen to be an epiphenomenalist. If you believe either the reductionists (consciousness happens *within* the atoms) or the substance dualists (consciousness is *causally potent* immaterial stuff), people talking about consciousness are talking about something real, and a reflectively consistent Bayesian AI can see this by tracing back the chain of causality for what makes people say “consciousness.”

According to Chalmers, the causally closed cognitive system of Chalmers’s internal narrative is (mysteriously) malfunctioning in a way that, not by necessity, but just in *our* universe, miraculously happens to be correct. Furthermore, the internal narrative asserts “the internal narrative is mysteriously malfunctioning, but miraculously happens to be correctly echoing the justified thoughts of the epiphenomenal inner core,” and again, in *our* universe, miraculously happens to be correct.

Oh, come on!

Shouldn’t there come a point where you just give up on an idea? Where, on some raw intuitive level, you just go: *What on Earth was I thinking?*

Humanity has accumulated some broad experience with what correct theories of the world look like. *This is not what a correct theory looks like.*

“Argument from incredulity,” you say. Fine, you want it spelled out? The said Chalmersian theory postulates multiple unexplained complex miracles. This drives down its prior probability, by the conjunction rule of probability and Occam’s Razor. It is therefore dominated by at least two theories that postulate fewer miracles, namely:

- Substance dualism:
 - There is a stuff of consciousness which is not yet understood, an extraordinary super-physical stuff that *visibly affects* our world; and this stuff is what makes us talk about consciousness.
- Not-quite-faith-based reductionism:
 - That-which-we-name “consciousness” happens *within* physics, in a way not yet understood, just like what happened the last three thousand times humanity ran into something mysterious.
 - Your intuition that no material substance can possibly add up to consciousness is incorrect. If you *actually* knew *exactly* why you talk about consciousness, this would give you new insights, of a form you can’t now anticipate; and afterward you would realize that your arguments about normal physics having no room for consciousness were flawed.

Compare to:

- Epiphenomenal property dualism:
 - Matter has additional consciousness-properties which are not yet understood. These properties are epiphenomenal with respect to ordinarily observable physics—they make no difference to the motion of particles.
 - *Separately*, there exists a not-yet-understood reason *within normal physics* why philosophers talk about consciousness and invent theories of dual properties.

- *Miraculously*, when philosophers talk about consciousness, the bridging laws of *our* world are exactly right to make this talk about consciousness correct, even though it arises from a malfunction (drawing of logically unwarranted conclusions) in the causally closed cognitive system that types philosophy papers.

I know I'm speaking from limited experience, here. But based on my limited experience, the Zombie Argument may be a candidate for *the most deranged idea in all of philosophy*.

There are times when, as a rationalist, you have to believe things that seem weird to you. Relativity seems weird, quantum mechanics seems weird, natural selection seems weird.

But these weirdnesses are pinned down by massive evidence. There's a difference between believing something weird because science has confirmed it overwhelmingly—

—versus believing a proposition that seems downright *deranged*, because of a great big complicated philosophical argument centered around unspecified miracles and giant blank spots not even claimed to be understood—

—in a case where *even if you accept everything that has been told to you so far*, afterward the phenomenon will still seem like a mystery and still have the same quality of wondrous impenetrability that it had at the start.

The correct thing for a rationalist to say at this point, if all of David Chalmers's arguments seem individually plausible—which they don't seem to me—is:

“Okay ...I don't know how consciousness works ...I admit that ...and maybe I'm approaching the whole problem wrong, or asking the wrong questions ...but this zombie business *can't possibly be right*. The arguments aren't nailed down enough to make me believe this—especially when accepting it won't make me feel any less confused. On a core gut level, this just *doesn't look* like the way reality could *really really* work.”

Mind you, I am not saying this is a substitute for careful analytic refutation of Chalmers's thesis. System 1 is not a sub-

stitute for System 2, though it can help point the way. You still have to track down where the problems are *specifically*.

Chalmers wrote a big book, not all of which is available through free Google preview. I haven't duplicated the long chains of argument where Chalmers lays out the arguments against himself in calm detail. I've just tried to tack on a final refutation of Chalmers's last presented defense, which Chalmers has not yet countered to my knowledge. Hit the ball back into his court, as it were.

But, yes, on a core level, the *sane* thing to do when you see the conclusion of the zombie argument, is to say "That can't *possibly* be right" and start looking for a flaw.

Zombie Responses



discussion
sharov.me/r/
4-c43

I'm a bit tired today, having stayed up until 3 a.m. writing yesterday's >6,000-word essay on zombies, so today I'll just reply to Richard, and tie up a loose end I spotted the next day.

(A) Richard Chappell writes:¹

A terminological note (to avoid unnecessary confusion): what you call "conceivable," others of us would merely call "*apparently* conceivable."

The gap between "I don't see a contradiction yet" and "this is logically possible" is so huge (it's NP-complete even in some simple-seeming cases) that you really should have two different words. As the zombie argument is boosted to the extent that this huge gap can be swept under the rug of minor terminological differences, I really think it would be a good idea to say "conceivable" versus "logically possible" or maybe even have a still more visible distinction. I can't choose professional terminology that has already been established, but in a case like this, I might seriously refuse to use it.

Maybe I will say "apparently conceivable" for the kind of information that zombie advocates get by imagining Zombie Worlds, and "logically possible" for the kind of information that is established by exhibiting a complete model or logical proof. Note the size of the gap between the information you can get by closing your eyes and imagining zombies, and the information you need to carry the argument for epiphenomenalism.

That is, your view would be characterized as a form of Type-A materialism, the view that zom-

1. sharov.me/r/4-17
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bies are not even (genuinely) conceivable, let alone metaphysically possible.

Type-A materialism is a large bundle; you shouldn't attribute the bundle to me until you see me agree with each of the parts. I think that someone who asks "What is consciousness?" is asking a legitimate question, has a legitimate demand for insight; I don't necessarily think that the *answer* takes the form of "Here is this stuff that has all the properties you would attribute to consciousness, for such-and-such reason," but may to some extent consist of insights that cause you to realize you were asking the question the wrong way.

This is not being eliminative about consciousness. It is being realistic about what kind of insights to expect, faced with a problem that (1) seems like it must have *some* solution, (2) seems like it cannot possibly have any solution, and (3) is being *discussed* in a fashion that has a great big dependence on the not-fully-understood ad-hoc architecture of human cognition.

(1) You haven't, so far as I can tell, identified any *logical contradiction* in the description of the zombie world. You've just pointed out that it's kind of strange. But there are many bizarre possible worlds out there. That's no reason to posit an implicit contradiction. So it's still completely mysterious to me what this alleged contradiction is supposed to be.

Okay, I'll spell it out from a materialist standpoint:

1. The zombie world, by definition, contains all parts of our world that are within the closure of the "caused by" or "effect of" relation of any observable phenomenon. In particular, it contains the *cause of* my visibly saying, "I think therefore I am."
2. When I focus my inward awareness on my inward awareness, I shortly thereafter experience my internal narrative saying "I am focusing my inward awareness on my inward awareness," and can, if I choose, say so out loud.

3. Intuitively, it sure seems like my inward awareness is causing my internal narrative to say certain things, and that my internal narrative can cause my lips to say certain things.
4. The word “consciousness,” if it has any meaning at all, refers to that-which-is or that-which-causes or that-which-makes-me-say-I-have inward awareness.
5. From (3) and (4) it would follow that if the zombie world is closed with respect to the causes of my saying “I think therefore I am,” the zombie world contains that which we refer to as “consciousness.”
6. By definition, the zombie world does not contain consciousness.
7. (3) seems to me to have a rather high probability of being empirically true. Therefore I evaluate a high empirical probability that the zombie world is logically impossible.

You can save the Zombie World by letting the cause of my internal narrative’s saying “I think therefore I am” be something entirely other than consciousness. In conjunction with the assumption that consciousness does exist, this is the part that struck me as deranged.

But if the above is *conceivable*, then isn’t the Zombie World conceivable?

No, because the two constructions of the Zombie World involve giving the word “consciousness” different empirical referents, like “water” in our world meaning H₂O versus “water” in Putnam’s Twin Earth meaning XYZ. For the Zombie World to be logically possible, it does not suffice that, for all *you* knew about how the empirical world worked, the word “consciousness” *could* have referred to an epiphenomenon that is entirely different from the consciousness we know. The Zombie World lacks consciousness, not “consciousness”—it is a world without H₂O, not a world without “water.” This is what is required to carry the empirical statement, “You could eliminate the referent

of whatever is meant by ‘consciousness’ from our world, while keeping all the atoms in the same place.”

Which is to say: I hold that it is an *empirical* fact, given what the word “consciousness” actually refers to, that it is *logically* impossible to eliminate consciousness without moving any atoms. What it would mean to eliminate “consciousness” from a world, rather than consciousness, I will not speculate.

(2) It’s misleading to say it’s “miraculous” (on the property dualist view) that our qualia line up so neatly with the physical world. There’s a natural law which guarantees this, after all. So it’s no more miraculous than any other logically contingent nomic necessity (e.g. the constants in our physical laws).

It is the natural law itself that is “miraculous”—counts as an additional complex-improbable element of the theory to be postulated, without having been itself justified in terms of things already known. One postulates (a) an inner world that is conscious, (b) a malfunctioning outer world that talks about consciousness for no reason, and (c) that the two align perfectly. Statement (c) does not follow from (a) and (b), and so is a separate postulate.

I agree that this usage of “miraculous” conflicts with the philosophical sense of violating a natural law; I meant it in the sense of improbability appearing from no apparent source, a la perpetual motion belief. Hence the word was ill-chosen in context. But is this not *intuitively* the sort of thing we should call a miracle? Your consciousness doesn’t really cause you to say you’re conscious, there’s a separate physical thing that makes you say you’re conscious, but also there’s a law aligning the two—this is indeed an event on a similar order of wackiness to a cracker taking on the substance of Christ’s flesh while possessing the exact appearance and outward behavior of a cracker, there’s just a natural law which guarantees this, you know.

That is, Zombie (or “Outer”) Chalmers doesn’t actually conclude *anything*, because his utterances

are meaningless. A fortiori, he doesn't conclude anything unwarrantedly. He's just making noises; these are no more susceptible to epistemic assessment than the chirps of a bird.

Looking at this from an AI-design standpoint, it seems to me like you should be able to build an AI that systematically refines an inner part of itself that correlates (in the sense of mutual information or systematic relations) to the environment, perhaps including floating-point numbers of a sort that I would call "probabilities" because they obey the internal relations mandated by Cox's Theorems when the AI encounters new information—pardon me, new sense inputs.

You will say that, unless the AI is more than mere transistors—unless it has the dual aspect—the AI has no beliefs.

I think my views on this were expressed pretty clearly in *The Simple Truth*.

To me, it seems pretty straightforward to construct maps that correlate to territories in systematic ways, without mentioning anything other than things of pure physical causality. The AI outputs a map of Texas. Another AI flies with the map to Texas and checks to see if the highways are in the corresponding places, chirping "True" when it detects a match and "False" when it detects a mismatch. You can refuse to call this "a map of Texas" but the AIs themselves are still chirping "True" or "False," and the said AIs are going to chirp "False" when they look at Chalmers's belief in an epiphenomenal inner core, and I for one would agree with them.

It's clear that the *function of mapping reality* is performed strictly by Outer Chalmers. The whole business of *producing belief representations* is handled by Bayesian structure in causal interactions. There's nothing left for the Inner Chalmers to do, but bless the whole affair with epiphenomenal *meaning*. Where now "meaning" is something entirely unrelated to systematic map-territory correspondence or the ability to use that map to navigate reality. So when it comes to talking about "accuracy," let alone "systematic accuracy," it seems to me like we should be able to determine it strictly by looking at the Outer Chalmers.

(B) In yesterday's text, I left out an assumption when I wrote:

If a self-modifying AI looks at a part of itself that concludes “*B*” on condition *A*—a part of itself that writes “*B*” to memory whenever condition *A* is true—and the AI inspects this part, determines how it (causally) operates in the context of the larger universe, and the AI decides that this part systematically tends to write false data to memory, then the AI has found what appears to be a bug, and the AI will self-modify not to write “*B*” to the belief pool under condition *A*.

...

But there's no possible warrant for the outer Chalmers *or any reflectively coherent self-inspecting AI* to believe in this mysterious correctness. A good AI design should, I think, be a reflectively coherent intelligence with a testable theory of how it operates as a causal system, hence with a testable theory of how that causal system produces systematically accurate beliefs on the way to achieving its goals.

Actually, you need an additional assumption to the above, which is that a “good AI design” (the kind I was thinking of, anyway) judges its own rationality in a modular way; it enforces global rationality by enforcing local rationality. If there is a piece that, relative to its context, is locally systematically unreliable—for some possible beliefs “*B_i*” and conditions *A_i*, it adds some “*B_i*” to the belief pool under local condition *A_i*, where reflection by the system indicates that *B_i* is not true (or in the case of probabilistic beliefs, not accurate) when the local condition *A_i* is true—then this is a bug. This kind of modularity is *a* way to make the problem tractable, and it's how I currently think about the first-generation AI design. [Edit 2013: The ac-

tual notion I had in mind here has now been fleshed out and formalized in Tiling Agents for Self-Modifying AI,² section 6.]

The notion is that a causally closed cognitive system—such as an AI designed by its programmers to use only causally efficacious parts; or an AI whose theory of its own functioning is entirely testable; or the outer Chalmers that writes philosophy papers—that believes that it has an epiphenomenal inner self, must be doing something systematically unreliable because it would conclude the same thing in a Zombie World. A mind all of whose parts are systematically locally reliable, relative to their contexts, would be systematically globally reliable. Ergo, a mind that is globally unreliable must contain at least one locally unreliable part. So a causally closed cognitive system inspecting itself for local reliability must discover that at least one step involved in adding the belief of an epiphenomenal inner self is unreliable.

If there are other ways for minds to be reflectively coherent that avoid this proof of disbelief in zombies, philosophers are welcome to try and specify them.

The reason why I have to specify all this is that otherwise you get a kind of extremely cheap reflective coherence where the AI can never label itself unreliable. E.g., if the AI finds a part of itself that computes $2 + 2 = 5$ (in the surrounding context of counting sheep) the AI will reason: “Well, this part malfunctions and says that $2 + 2 = 5$...but by pure coincidence, $2 + 2$ is equal to 5, or so it seems to me ...so while the part looks systematically unreliable, I better keep it the way it is, or it will handle this special case wrong.” That’s why I talk about enforcing global reliability by enforcing local systematic reliability—if you just compare your global beliefs to your global beliefs, you don’t go anywhere.

This does have a general lesson: Show your arguments are globally reliable by virtue of each step being locally reliable;

2. sharov.me/r/4-18
intelligence.org



don't just compare the arguments' conclusions to your intuitions. [Edit 2013: See *Proofs, Implications, and Models*³ for a discussion of the fact that valid logic is locally valid.]

(C) An anonymous poster wrote:

A sidepoint, this, but I believe your etymology for “n’shama” is wrong. It is related to the word for “breath,” not “hear.” The root for “hear” contains an ayin, which n’shama does not.

Now that’s what I call a miraculously misleading coincidence—although the word N’Shama arose for completely different reasons, it sounded *exactly the right way* to make me think it referred to an inner listener.

Oops.

3. sharov.me/r/4-19

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The Generalized Anti-Zombie Principle



discussion
sharov.me/r/
4-c44

Each problem that I solved became a rule which served afterwards to solve other problems.

—René Descartes, *Discours de la Méthode*¹

“Zombies” are putatively beings that are atom-by-atom identical to us, governed by all the same third-party-visible physical laws, except that they are not conscious.

Though the philosophy is complicated, the core argument against zombies is simple: When you focus your inward awareness on your inward awareness, your internal narrative (the little voice inside your head that speaks your thoughts) says “I am aware of being aware” soon after, and then you say it out loud, and then you type it into a computer keyboard, and create a third-party visible blog post.

Consciousness, whatever it may be—a substance, a process, a name for a confusion—is not epiphenomenal; your mind can catch the inner listener in the act of listening, and say so out loud. *The fact that I have typed this paragraph* would at least *seem* to refute the idea that consciousness has no experimentally detectable consequences.

I hate to say “So now let’s accept this and move on,” over such a philosophically controversial question, but it seems like a considerable majority of *Overcoming Bias* commenters do accept this. And there are other conclusions you can only get to after you accept that you cannot subtract consciousness and leave the universe looking exactly the same. So now let’s accept this and move on.

The form of the Anti-Zombie Argument seems like it should generalize, becoming an Anti-Zombie Principle. But what is the proper generalization?

1. René Descartes, *Discours de la Méthode*, vol. 45 (Librairie des Bibliophiles, 1887).

Let's say, for example, that someone says: "I have a switch in my hand, which does not affect your brain in any way; and if this switch is flipped, you will cease to be conscious." Does the Anti-Zombie Principle rule this out as well, with the same structure of argument?

It appears to me that in the case above, the answer is yes. In particular, you can say: "Even after your switch is flipped, I will still talk about consciousness *for exactly the same reasons* I did before. If I am conscious right now, I will still be conscious after you flip the switch."

Philosophers may object, "But now you're equating consciousness with talking about consciousness! What about the Zombie Master, the chatbot that regurgitates a remixed corpus of amateur human discourse on consciousness?"

But I did *not* equate "consciousness" with verbal behavior. The core premise is that, *among other things*, the true referent of "consciousness" is *also* the *cause in humans* of talking about inner listeners.

As I argued (at some length) in the sequence on words, what you want in defining a word is not always a perfect Aristotelian necessary-and-sufficient definition; sometimes you just want a treasure map that leads you to the extensional referent. So "that which *does in fact* make me talk about an unspeakable awareness" is not a necessary-and-sufficient definition. But if what does *in fact* cause me to discourse about an unspeakable awareness is not "consciousness," then ...

...then the discourse gets pretty futile. That is not a knock-down argument against zombies—an empirical question can't be settled by mere difficulties of discourse. But if you try to defy the Anti-Zombie Principle, you will have problems with the *meaning* of your discourse, not just its plausibility.

Could we *define* the word "consciousness" to mean "whatever actually makes humans talk about 'consciousness'"? This would have the powerful advantage of guaranteeing that there is at least one real fact named by the word "consciousness." Even if our belief in consciousness is a confusion, "consciousness" would name the cognitive architecture that generated the confusion. But to establish a definition is only to promise to use

a word consistently; it doesn't settle any empirical questions, such as whether our inner awareness makes us talk about our inner awareness.

Let's return to the Off-Switch.

If we allow that the Anti-Zombie Argument applies against the Off-Switch, then the Generalized Anti-Zombie Principle does *not* say only, "Any change that is not in-principle experimentally detectable (IPED) cannot remove your consciousness." The switch's flipping is experimentally detectable, but it still seems *highly* unlikely to remove your consciousness.

Perhaps the Anti-Zombie Principle says, "Any change that does not affect you in any IPED way cannot remove your consciousness"?

But is it a reasonable stipulation to say that flipping the switch does not affect you in *any* IPED way? All the particles in the switch are interacting with the particles composing your body and brain. There are gravitational effects—tiny, but real and IPED. The gravitational pull from a one-gram switch ten meters away is around $6 \times 10^{-16} \text{ m/s}^2$. That's around half a neutron diameter per second per second, far below thermal noise, but way above the Planck level.

We could flip the switch light-years away, in which case the flip would have no immediate causal effect on you (whatever "immediate" means in this case) (if the Standard Model of physics is correct).

But it doesn't seem like we *should* have to alter the thought experiment in this fashion. It seems that, if a disconnected switch is flipped on the other side of a room, you should not expect your inner listener to go out like a light, because the switch "obviously doesn't change" that which is the true cause of your talking about an inner listener. Whatever you really are, you don't expect the switch to mess with it.

This is a *large* step.

If you deny that it is a reasonable step, you had better never go near a switch again. But still, it's a large step.

The key idea of reductionism is that our maps of the universe are multi-level to save on computing power, but physics seems to be strictly single-level. All our discourse about the universe

takes place using references far above the level of fundamental particles.

The switch's flip *does* change the fundamental particles of your body and brain. It nudges them by whole neutron diameters away from where they would have otherwise been.

In ordinary life, we gloss a change this small by saying that the switch "doesn't affect you." But it *does* affect you. It changes everything by whole neutron diameters! What could possibly be remaining the same? Only the *description* that you would give of the higher levels of organization—the cells, the proteins, the spikes traveling along a neural axon. As the map is far less detailed than the territory, it must map many different states to the same description.

Any reasonable sort of humanish *description* of the brain that talks about neurons and activity patterns (or even the conformations of individual microtubules making up axons and dendrites) won't change when you flip a switch on the other side of the room. Nuclei are larger than neutrons, atoms are larger than nuclei, and by the time you get up to talking about the *molecular* level, that tiny little gravitational force has vanished from the list of things you bother to *track*.

But if you add up enough tiny little gravitational pulls, they will eventually yank you across the room and tear you apart by tidal forces, so clearly a small effect is *not* "no effect at all."

Maybe the tidal force from that tiny little pull, by an *amazing* coincidence, pulls a single extra calcium ion just a tiny bit closer to an ion channel, causing it to be pulled in just a tiny bit sooner, making a single neuron fire infinitesimally sooner than it would otherwise have done, a difference which amplifies chaotically, finally making a whole neural spike occur that otherwise wouldn't have occurred, sending you off on a different train of thought, that triggers an epileptic fit, that kills you, causing you to cease to be conscious ...

If you add up a lot of tiny quantitative effects, you get a big quantitative effect—big enough to mess with anything you care to name. And so claiming that the switch has literally *zero* effect on the things you care about, is taking it too far.

But with just one switch, the force exerted is vastly less than thermal uncertainties, never mind quantum uncertainties. If you don't expect your consciousness to flicker in and out of existence as the result of thermal jiggling, then you certainly shouldn't expect to go out like a light when someone sneezes a kilometer away.

The alert Bayesian will note that I have just made an argument about *expectations*, states of *knowledge*, justified *beliefs* about what can and can't switch off your consciousness.

This doesn't necessarily destroy the Anti-Zombie Argument. Probabilities are not certainties, but the *laws of* probability are theorems; if rationality says you can't believe something on your current information, then that is a law, not a suggestion.

Still, this version of the Anti-Zombie Argument is weaker. It doesn't have the nice, clean, absolutely clear-cut status of, "You can't possibly eliminate consciousness while leaving all the atoms in *exactly* the same place." (Or for "all the atoms" substitute "all causes with in-principle experimentally detectable effects," and "same wavefunction" for "same place," etc.)

But the new version of the Anti-Zombie Argument still carries. You can say, "I don't know what consciousness really is, and I suspect I may be fundamentally confused about the question. But if the word refers to anything at all, it refers to something that is, among other things, the cause of my talking about consciousness. Now, I don't know why I talk about consciousness. But it happens inside my skull, and I expect it has something to do with neurons firing. Or maybe, if I really understood consciousness, I would have to talk about an even more fundamental level than that, like microtubules, or neurotransmitters diffusing across a synaptic channel. But still, that switch you just flipped has an effect on my neurotransmitters and microtubules that's much, much less than thermal noise at 310 Kelvin. So whatever the true cause of my talking about consciousness may be, I don't expect it to be hugely affected by the gravitational pull from that switch. Maybe it's just a tiny little infinitesimal bit affected? But it's certainly not going to go out like a light. I expect to go on talking about consciousness

in *almost exactly* the same way afterward, for *almost exactly* the same reasons.”

This application of the Anti-Zombie Principle is weaker. But it’s also much more general. And, in terms of sheer common sense, correct.

The reductionist and the substance dualist actually have two different versions of the above statement. The reductionist furthermore says, “Whatever makes me talk about consciousness, it seems likely that the important parts take place on a much higher functional level than atomic nuclei. Someone who understood consciousness could abstract away from individual neurons firing, and talk about high-level cognitive architectures, and still describe how my mind produces thoughts like ‘I think therefore I am.’ So nudging things around by the diameter of a nucleon shouldn’t affect my consciousness (except maybe with very small probability, or by a very tiny amount, or not until after a significant delay).”

The substance dualist furthermore says, “Whatever makes me talk about consciousness, it’s got to be something beyond the computational physics we know, which means that it might very well involve quantum effects. But still, my consciousness doesn’t flicker on and off whenever someone sneezes a kilometer away. If it did, I would *notice*. It would be like skipping a few seconds, or coming out of a general anesthetic, or sometimes saying, ‘I don’t think therefore I’m not.’ So since it’s a physical fact that thermal vibrations don’t disturb the stuff of my awareness, I don’t expect flipping the switch to disturb it either.”

Either way, you *shouldn’t* expect your sense of awareness to vanish when someone says the word “Abracadabra,” even if that does have some infinitesimal physical effect on your brain—

But hold on! If you *hear* someone say the word “Abracadabra,” that has a very noticeable effect on your brain—so large, even your brain can notice it. It may alter your internal narrative; you may think, “Why did that person just say ‘Abracadabra’?”

Well, but *still* you expect to go on talking about consciousness in almost exactly the same way afterward, for almost exactly the same reasons.

And again, it's not that "consciousness" is being *equated* to "that which makes you talk about consciousness." It's just that consciousness, *among other things*, makes you talk about consciousness. So anything that makes your consciousness go out like a light should make you stop talking about consciousness.

If we do something to you, where you don't see how it could *possibly* change your internal narrative—the little voice in your head that sometimes says things like "I think therefore I am," whose words you can choose to say aloud—then it shouldn't make you cease to be conscious.

And this is true even if the internal narrative is just "pretty much the same," and the causes of it are also pretty much the same; among the causes that are pretty much the same is whatever you mean by "consciousness."

If you're wondering where all this is going, and why it's important to go to such tremendous lengths to ponder such an obvious-seeming Generalized Anti-Zombie Principle, then consider the following debate:

ALBERT: "Suppose I replaced all the neurons in your head with tiny robotic artificial neurons that had the same connections, the same local input-output behavior, and analogous internal state and learning rules."

BERNICE: "That's killing me! There wouldn't be a conscious being there anymore."

CHARLES: "Well, there'd still be a conscious being there, but it wouldn't be *me*."

SIR ROGER PENROSE: "The thought experiment you propose is impossible. You *can't* duplicate the behavior of neurons without tapping into quantum gravity. That said, there's not much point in me taking further part in this conversation."
(*Wanders away.*)

ALBERT: "Suppose that the replacement is carried out one neuron at a time, and the swap occurs so fast that it doesn't make any difference to global processing."

BERNICE: "How could that possibly be the case?"

ALBERT: "The little robot swims up to the neuron, surrounds it, scans it, learns to duplicate it, and then suddenly takes over the behavior, between one spike and the next. In fact, the imitation is so good that your outward behavior is just the same as it would be if the brain were left undisturbed. Maybe not *exactly* the same, but the causal impact is much less than thermal noise at 310 Kelvin."

CHARLES: "So what?"

ALBERT: "So don't your beliefs violate the Generalized Anti-Zombie Principle? Whatever just happened, it didn't change your internal narrative! You'll go around talking about consciousness for exactly the same reason as before."

BERNICE: "Those little robots are a Zombie Master. They'll make me talk about consciousness even though I'm not conscious. The Zombie World is possible if you allow there to be an added, extra, experimentally detectable Zombie Master—which those robots *are*."

CHARLES: "Oh, that's not right, Bernice. The little robots aren't plotting how to fake consciousness, or processing a corpus of text from human amateurs. They're doing the same thing neurons do, just in silicon instead of carbon."

ALBERT: "Wait, didn't you just agree with me?"

CHARLES: "I never said the new person wouldn't be conscious. I said it wouldn't be *me*."

ALBERT: "Well, obviously the Anti-Zombie Principle generalizes to say that this operation hasn't disturbed the true cause of your talking about this *me* thing."

CHARLES: “Uh-uh! Your operation certainly did disturb the true cause of my talking about consciousness. It substituted a *different* cause in its place, the robots. Now, just because that new cause *also* happens to be conscious—talks about consciousness for the same *generalized* reason—doesn’t mean it’s the *same* cause that was originally there.”

ALBERT: “But I wouldn’t even have to *tell* you about the robot operation. You wouldn’t *notice*. If you think, going on introspective evidence, that you are in an important sense ‘the same person’ that you were five minutes ago, and I do something to you that doesn’t change the introspective evidence available to you, then your conclusion that you are the same person that you were five minutes ago should be equally justified. Doesn’t the Generalized Anti-Zombie Principle say that if I do something to you that alters your consciousness, let alone makes you a completely different person, then you ought to *notice* somehow?”

BERNICE: “Not if you replace me with a Zombie Master. Then there’s no one there *to* notice.”

CHARLES: “Introspection isn’t perfect. Lots of stuff goes on inside my brain that I don’t notice.”

ALBERT: “You’re postulating epiphenomenal facts about consciousness and identity!”

BERNICE: “No I’m not! I can experimentally detect the difference between neurons and robots.”

CHARLES: “No I’m not! I can experimentally detect the moment when the old me is replaced by a new person.”

ALBERT: “Yeah, and I can detect the switch flipping! You’re detecting something that doesn’t *make a noticeable difference* to the *true cause* of your talk about consciousness and personal identity. And the proof is, you’ll talk just the same way afterward.”

BERNICE: "That's because of your robotic Zombie Master!"

CHARLES: "Just because two people talk about 'personal identity' for similar reasons doesn't make them the same person."

I think the Generalized Anti-Zombie Principle supports Albert's position, but the reasons shall have to wait for future essays. I need other prerequisites, and besides, this essay is already too long.

But you see the importance of the question, "How far can you generalize the Anti-Zombie Argument and have it still be valid?"

The makeup of future galactic civilizations may be determined by the answer ...

GAZP vs. GLUT



discussion
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In “The Unimagined Preposterousness of Zombies,” Daniel Dennett says:¹

To date, several philosophers have told me that they plan to accept my challenge to offer a non-question-begging defense of zombies, but the only one I have seen so far involves postulating a “logically possible” but fantastic being—a descendent of Ned Block’s Giant Lookup Table fantasy ...

A Giant Lookup Table, in programmer’s parlance, is when you implement a function as a giant table of inputs and outputs, usually to save on runtime computation. If my program needs to know the multiplicative product of two inputs between 1 and 100, I can write a multiplication algorithm that computes each time the function is called, or I can precompute a Giant Lookup Table with 10,000 entries and two indices. There are times when you *do* want to do this, though not for multiplication—times when you’re going to reuse the function a lot and it doesn’t have many possible inputs; or when clock cycles are cheap while you’re initializing, but very expensive while executing.

Giant Lookup Tables get very large, very fast. A GLUT of all possible twenty-ply conversations with ten words per remark, using only 850-word Basic English, would require 7.6×10^{585} entries.

Replacing a human brain with a Giant Lookup Table of all possible sense inputs and motor outputs (relative to some fine-grained digitization scheme) would require an *unreasonably large amount* of memory storage. But “in principle,” as philosophers are fond of saying, it could be done.

1. Daniel C. Dennett, “The Unimagined Preposterousness of Zombies,” *Journal of Consciousness Studies* 2 (4 1995): 322–26.

The GLUT is not a zombie in the classic sense, because it is microphysically dissimilar to a human. (In fact, a GLUT can't *really* run on the same physics as a human; it's too large to fit in our universe. For philosophical purposes, we shall ignore this and suppose a supply of unlimited memory storage.)

But is the GLUT a zombie at *all*? That is, does it behave exactly like a human without being conscious?

The GLUT-ed body's tongue talks about consciousness. Its fingers write philosophy papers. In every way, so long as you don't peer inside the skull, the GLUT seems just like a human...which certainly seems like a valid example of a zombie: it behaves just like a human, but there's no one home.

Unless the GLUT is conscious, in which case it wouldn't be a valid example.

I can't recall ever seeing *anyone* claim that a GLUT is conscious. (Admittedly my reading in this area is not up to professional grade; feel free to correct me.) Even people who are accused of being (gasp!) functionalists don't claim that GLUTs can be conscious.

GLUTs are the *reductio ad absurdum* to anyone who suggests that consciousness *is simply* an input-output pattern, thereby disposing of all troublesome worries about what goes on inside.

So what does the Generalized Anti-Zombie Principle (GAZP) say about the Giant Lookup Table (GLUT)?

At first glance, it would seem that a GLUT is the very archetype of a Zombie Master—a distinct, additional, detectable, non-conscious system that animates a zombie and makes it talk about consciousness for *different* reasons.

In the interior of the GLUT, there's merely a very simple computer program that looks up inputs and retrieves outputs. Even talking about a "simple computer program" is overshooting the mark, in a case like this. A GLUT is more like ROM than a CPU. We could equally well talk about a series of switched tracks by which some balls roll out of a previously stored stack and into a trough—*period*; that's *all* the GLUT does.

A spokesperson from People for the Ethical Treatment of Zombies replies: "Oh, that's what all the anti-mechanists say, isn't it? That when you look in the brain, you just find a bunch

of neurotransmitters opening ion channels? If ion channels can be conscious, why not levers and balls rolling into bins?”

“The problem isn’t the levers,” replies the functionalist, “the problem is that a GLUT has the *wrong pattern* of levers. You need levers that implement things like, say, formation of beliefs about beliefs, or self-modeling ...Heck, you need the ability to write things to memory just so that time can pass for the computation. Unless you think it’s possible to program a conscious being in Haskell.²”

“I don’t know about that,” says the PETZ spokesperson, “all I know is that this so-called zombie writes philosophical papers about consciousness. Where do these philosophy papers come from, if not from consciousness?”

Good question! Let us ponder it deeply.

There’s a game in physics called Follow-The-Energy. Richard Feynman’s father played it with young Richard:³

It was the kind of thing my father would have talked about: “What makes it go? Everything goes because the Sun is shining.” And then we would have fun discussing it:

“No, the toy goes because the spring is wound up,” I would say. “How did the spring get wound up?” he would ask.

“I wound it up.”

“And how did you get moving?”

“From eating.”

“And food grows only because the Sun is shining. So it’s because the Sun is shining that all these things are moving.” That would get the concept

2. Edit 2017: While a conscious being could not be programmed as one Haskell function called once, repeated calling of a function in Haskell would allow this. $f(M \times S) \rightarrow M \times A$ where M is the memory, S is the sensory input and A is the action allows a Haskell function to update memory and choose an action.

3. sharov.me/r/4-20

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across that motion is simply the *transformation* of the Sun's power.⁴

When you get a little older, you learn that energy is conserved, never created or destroyed, so the notion of *using up* energy doesn't make much sense. You can never change the total amount of energy, so in what sense are you *using* it?

So when physicists grow up, they learn to play a new game called Follow-The-Negentropy—which is really the same game they were playing all along; only the rules are mathier, the game is more useful, and the principles are harder to wrap your mind around conceptually.

Rationalists learn a game called Follow-The-Improbability, the grownup version of "How Do You Know?" The rule of the rationalist's game is that every improbable-seeming belief needs an equivalent amount of evidence to justify it. (This game has *amazingly similar* rules to Follow-The-Negentropy.)

Whenever someone violates the rules of the rationalist's game, you can find a place in their argument where a quantity of improbability appears from nowhere; and this is as much a sign of a problem as, oh, say, an ingenious design of linked wheels and gears that keeps itself running forever.

The one comes to you and says: "I believe with firm and abiding faith that there's an object in the asteroid belt, one foot across and composed entirely of chocolate cake; you can't prove that this is impossible." But, unless the one had access to some kind of evidence for this belief, it would be highly improbable for a correct belief to form *spontaneously*. So either the one can point to evidence, or the belief won't turn out to be true. "But you can't prove it's *impossible* for my mind to spontaneously generate a belief that happens to be correct!" No, but that kind of spontaneous generation is *highly improbable*, just like, oh, say, an egg unscrambling itself.

In Follow-The-Improbability, it's highly suspicious to even *talk* about a specific hypothesis without having had enough evidence to narrow down the space of possible hypotheses. Why

4. Richard P. Feynman, "Judging Books by Their Covers," in *Surely You're Joking, Mr. Feynman!* (New York: W. W. Norton & Company, 1985).

aren't you giving equal air time to a decillion other equally plausible hypotheses? You need sufficient evidence to find the "chocolate cake in the asteroid belt" hypothesis in the hypothesis space—otherwise there's no reason to give it more air time than a trillion other candidates like "There's a wooden dresser in the asteroid belt" or "The Flying Spaghetti Monster threw up on my sneakers."

In Follow-The-Improbability, you are not allowed to pull out big complicated specific hypotheses from thin air without *already* having a corresponding amount of evidence; because it's not realistic to suppose that you could spontaneously start discussing the *true* hypothesis by *pure coincidence*.

A philosopher says, "This zombie's skull contains a Giant Lookup Table of all the inputs and outputs for some human's brain." This is a very *large* improbability. So you ask, "How did this improbable event occur? Where did the GLUT come from?"

Now this is not standard philosophical procedure for thought experiments. In standard philosophical procedure, you are allowed to postulate things like "Suppose you were riding a beam of light..." without worrying about physical possibility, let alone mere improbability. But in this case, the origin of the GLUT matters; and that's why it's important to understand the motivating question, "Where did the improbability come from?"

The obvious answer is that you took a computational specification of a human brain, and used *that* to precompute the Giant Lookup Table. (Thereby creating uncounted googols of human beings, some of them in extreme pain, the supermajority gone quite mad in a universe of chaos where inputs bear no relation to outputs. But damn the ethics, this is for *philosophy*.)

In this case, the GLUT *is* writing papers about consciousness because of a conscious algorithm. The GLUT is no zombie, any more than a cellphone is a zombie because it can talk about consciousness while being just a small consumer electronic device. The cellphone is just transmitting philosophy speeches from whoever happens to be on the other end of the line. A

GLUT generated from an originally human brain-specification is doing the same thing.

“All right,” says the philosopher, “the GLUT was generated randomly, and *just happens* to have the same input-output relations as some reference human.”

How, exactly, did you randomly generate the GLUT?

“We used a true randomness source—a quantum device.”

But a quantum device just implements the Branch Both Ways instruction; when you generate a bit from a quantum randomness source, the deterministic result is that one set of universe-branches (locally connected amplitude clouds) see 1, and another set of universes see 0. Do it 4 times, create 16 (sets of) universes.

So, really, this is like saying that you got the GLUT by writing down all possible GLUT-sized sequences of 0s and 1s, in a really damn huge bin of lookup tables; and then reaching into the bin, and *somehow* pulling out a GLUT that happened to correspond to a human brain-specification. Where did the improbability come from?

Because if this *wasn't just a coincidence*—if you had some reach-into-the-bin function that pulled out a human-corresponding GLUT by *design*, not just chance—then that reach-into-the-bin function is probably conscious, and so the GLUT is again a cellphone, not a zombie. It's connected to a human at two removes, instead of one, but it's still a cellphone! Nice try at concealing the source of the improbability there!

Now behold where Follow-The-Improbability has taken us: where is the source of this body's tongue talking about an inner listener? The consciousness isn't in the lookup table. The consciousness isn't in the factory that manufactures lots of possible lookup tables. The consciousness was in whatever *pointed to one particular already-manufactured lookup table*, and said, “Use *that* one!”

You can see why I introduced the game of Follow-The-Improbability. Ordinarily, when we're talking to a person, we tend to think that whatever is inside the skull must be “where the consciousness is.” It's only by playing Follow-The-Improbability that we can realize that the real source of the

conversation we're having is that-which-is-responsible-for the *improbability* of the conversation—however distant in time or space, as the Sun moves a wind-up toy.

“No, no!” says the philosopher. “In the thought experiment, they aren't randomly generating lots of GLUTs, and then using a conscious algorithm to pick out one GLUT that seems humanlike! I am *specifying* that, in this thought experiment, they reach into the inconceivably vast GLUT bin, and *by pure chance* pull out a GLUT that is identical to a human brain's inputs and outputs! *There!* I've got you cornered now! You can't play Follow-The-Improbability any further!”

Oh. So your *specification* is the source of the improbability here.

When we play Follow-The-Improbability again, we end up *outside the thought experiment*, looking at the *philosopher*.

That which points to the one GLUT that talks about consciousness, out of all the vast space of possibilities, is now ...the conscious person asking us to imagine this whole scenario. And our own brains, which will fill in the blank when we imagine, “What will this GLUT say in response to ‘Talk about your inner listener’?”

The moral of this story is that when you follow back discourse about “consciousness,” you generally find consciousness. It's not always right in front of you. Sometimes it's very cleverly hidden. But it's there. Hence the Generalized Anti-Zombie Principle.

If there is a Zombie Master in the form of a chatbot that processes and remixes amateur human discourse about “consciousness,” the humans who generated the original text corpus are conscious.

If someday you come to understand consciousness, and look back, and see that there's a program you can write that will output confused philosophical discourse that sounds an awful lot like humans without itself being conscious—then when I ask “How did this program come to sound similar to humans?” the answer is that *you* wrote it to sound similar *to conscious humans*, rather than choosing on the criterion of similarity to something else. This doesn't mean your little Zombie Master is conscious—

but it does mean I can find consciousness somewhere in the universe by tracing back the chain of causality, which means we're not entirely in the Zombie World.

But suppose someone actually *did* reach into a GLUT-bin and by *genuinely pure chance* pulled out a GLUT that wrote philosophy papers?

Well, then it wouldn't be conscious. In my humble opinion. I mean, there's got to be more to it than inputs and outputs. Otherwise even a GLUT would be conscious, right?

Oh, and for those of you wondering how this sort of thing relates to my day job ...

In this line of business you meet an awful lot of people who think that an arbitrarily generated powerful AI will be "moral." They can't agree among themselves on why, or what they mean by the word "moral"; but they all agree that doing Friendly AI theory is unnecessary. And when you ask them how an arbitrarily generated AI ends up with moral outputs, they proffer elaborate rationalizations aimed at AIs of that which they deem "moral"; and there are all sorts of problems with this, but the number one problem is, "Are you *sure* the AI would follow the same line of thought you invented to argue human morals, when, unlike you, the AI doesn't start out knowing what *you* want it to rationalize?" You could call the counter-principle Follow-The-Decision-Information, or something along those lines. You can account for an AI that does improbably nice things by telling me how you chose the AI's design from a huge space of possibilities, but otherwise the improbability is being pulled out of nowhere—though more and more heavily disguised, as rationalized premises are rationalized in turn.

So I've already done a whole series of essays which I myself generated using Follow-The-Improbability. But I didn't spell out the rules *explicitly* at that time, because I hadn't done the thermodynamics essays yet ...

Just thought I'd mention that. It's amazing how many of my essays coincidentally turn out to include ideas surprisingly relevant to discussion of Friendly AI theory ...if you believe in coincidence.

Belief in the Implied Invisible



discussion
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One generalized lesson *not* to learn from the Anti-Zombie Argument is, “Anything you can’t see doesn’t exist.”

It’s tempting to conclude the general rule. It would make the Anti-Zombie Argument much simpler, on future occasions, if we could take this as a premise. But unfortunately that’s just not Bayesian.

Suppose I transmit a photon out toward infinity, not aimed at any stars, or any galaxies, pointing it toward one of the great voids between superclusters. Based on standard physics, in other words, I don’t expect this photon to intercept anything on its way out. The photon is moving at light speed, so I can’t chase after it and capture it again.

If the expansion of the universe is accelerating, as current cosmology holds, there will come a future point where I don’t expect to be able to interact with the photon even in principle—a future time beyond which I don’t expect the photon’s future light cone to intercept my world-line. Even if an alien species captured the photon and rushed back to tell us, they couldn’t travel fast enough to make up for the accelerating expansion of the universe.

Should I believe that, in the moment where I can no longer interact with it even in principle, the photon disappears?

No.

It would violate Conservation of Energy. And the Second Law of Thermodynamics. And just about every other law of physics. And probably the Three Laws of Robotics. It would imply the photon knows I care about it and knows exactly when to disappear.

It’s a *silly idea*.

But if you can believe in the continued existence of photons that have become experimentally undetectable to you, why doesn’t this imply a general license to believe in the invisible?

(If you want to think about this question on your own, do so before reading on ...)

Though I failed to Google a source, I remember reading that when it was first proposed that the Milky Way was our *galaxy*—that the hazy river of light in the night sky was made up of millions (or even billions) of stars—that Occam’s Razor was invoked against the new hypothesis. Because, you see, the hypothesis vastly multiplied the number of “entities” in the believed universe. Or maybe it was the suggestion that “nebulae”—those hazy patches seen through a telescope—might be galaxies full of stars, that got the invocation of Occam’s Razor.

Lex parsimoniae: Entia non sunt multiplicanda praeter necessitatem.

That was Occam’s original formulation, the law of parsimony: Entities should not be multiplied beyond necessity.

If you postulate billions of stars that no one has ever believed in before, you’re multiplying entities, aren’t you?

No. There are two Bayesian formalizations of Occam’s Razor: Solomonoff induction, and Minimum Message Length. Neither penalizes galaxies for being big.

Which they had better not do! One of the lessons of history is that what-we-call-reality keeps turning out to be bigger and bigger and huger yet. Remember when the Earth was at the center of the universe? Remember when no one had invented Avogadro’s number? If Occam’s Razor was weighing against the multiplication of entities every time, we’d have to start doubting Occam’s Razor, because it would have consistently turned out to be wrong.

In Solomonoff induction, the complexity of your model is the amount of *code* in the computer program you have to write to simulate your model. The amount of *code*, not the amount of RAM it uses or the number of cycles it takes to compute. A model of the universe that contains billions of galaxies containing billions of stars, each star made of a billion trillion decillion quarks, will take a lot of RAM to run—but the *code* only has to describe the behavior of the quarks, and the stars and galaxies can be left to run themselves. I am speaking semi-

metaphorically here—there are things in the universe besides quarks¹—but the point is, postulating an extra billion galaxies doesn't count against the size of your code, if you've already described one galaxy. It just takes a bit more RAM, and Occam's Razor doesn't care about RAM.

Why not? The Minimum Message Length formalism, which is nearly equivalent to Solomonoff induction, may make the principle clearer: If you have to tell someone how your model of the universe works, you don't have to individually specify the location of each quark in each star in each galaxy. You just have to write down some equations. The amount of "stuff" that obeys the equation doesn't affect how long it takes to write the equation down. If you encode the equation into a file, and the file is 100 bits long, then there are 2^{100} other models that would be around the same file size, and you'll need roughly 100 bits of supporting evidence. You've got a limited amount of probability mass; and a priori, you've got to divide that mass up among all the messages you could send; and so postulating a model from within a model space of 2^{100} alternatives, means you've got to accept a 2^{-100} prior probability penalty—but having more galaxies doesn't add to this.

Postulating billions of stars in billions of galaxies doesn't affect the length of your message describing the overall behavior of all those galaxies. So you don't take a probability hit from

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1. Edit 2017: The full list of understood physics is the Standard Model (which includes quantum mechanics) and General Relativity. The Standard Model has six quarks (up, down, charm, strange, top and bottom), six leptons (electron, muon, tau, electron neutrino, muon neutrino, tau neutrino), the bosons (gluon, photon, Z boson, W boson and Higgs boson) and three forces (electromagnetic, weak and strong). The quarks and leptons have opposite sign antiparticles, and the quarks come in three varieties that are the same except for 'color'. The Standard Model can be described in the Lagrangian form that will fit on a single sheet of paper (sharov.me/t/4-21). General relativity can be described with the Einstein field equations that fit on one line. The object you are reading this on is made of electrons, protons (each made of two up quarks and one down quark), and neutrons (each made of one up quark and two down quarks).



having the *same* equations describing more things. (So long as your model's predictive successes aren't sensitive to the exact initial conditions. If you've got to specify the exact positions of all the quarks for your model to predict as well as it does, the extra quarks do count as a hit.)

If you suppose that the photon disappears when you are no longer looking at it, this is an *additional law* in your model of the universe. It's the laws that are "entities," costly under the laws of parsimony. Extra quarks are free.

So does it boil down to, "I believe the photon goes on existing as it wings off to nowhere, because my priors say it's simpler for it to go on existing than to disappear"?

This is what I thought at first, but on reflection, it's not quite right. (And not just because it opens the door to obvious abuses.)

I would boil it down to a distinction between belief in the *implied invisible*, and belief in the *additional invisible*.

When you believe that the photon goes on existing as it wings out to infinity, you're not believing that as an *additional* fact.

What you believe (assign probability to) is a set of simple equations; you believe these equations describe the universe. You believe these equations because they are the simplest equations you could find that describe the evidence. These equations are *highly* experimentally testable; they explain huge mounds of evidence visible in the past, and predict the results of many observations in the future.

You believe these equations, and it is a *logical implication* of these equations that the photon goes on existing as it wings off to nowhere, so you believe that as well.

Your priors, or even your probabilities, don't *directly* talk about the photon. What you assign probability to is not the photon, but the general laws. When you assign probability to the laws of physics as we know them, you *automatically* contribute that same probability to the photon continuing to exist on its way to nowhere—if you believe the logical implications of what you believe.

It's not that you believe in the invisible *as such*, from reasoning about invisible things. Rather the experimental evidence supports certain laws, and belief in those laws logically implies the existence of certain entities that you can't interact with. This is belief in the *implied invisible*.

On the other hand, if you believe that the photon is eaten out of existence by the Flying Spaghetti Monster—maybe on just this one occasion—or even if you believed without reason that the photon hit a dust speck on its way out—then you would be believing in a specific extra invisible event, on its own. If you thought that this sort of thing happened in general, you would believe in a specific extra invisible law. This is belief in the *additional invisible*.

To make it clear why you would sometimes want to think about implied invisibles, suppose you're going to launch a spaceship, at nearly the speed of light, toward a faraway supercluster. By the time the spaceship gets there and sets up a colony, the universe's expansion will have accelerated too much for them to ever send a message back. Do you deem it worth the purely altruistic effort to set up this colony, for the sake of all the people who will live there and be happy? Or do you think the spaceship blips out of existence before it gets there? This could be a very real question at some point.

The whole matter would be a lot simpler, admittedly, if we could just rule out the existence of entities we can't interact with, once and for all—have the universe stop existing at the edge of our telescopes. But this requires us to be very silly.

Saying that you shouldn't ever need a separate and additional belief about invisible things—that you only believe invisibles that are *logical implications* of general laws which are themselves testable, and even then, don't have any further beliefs about them that are not logical implications of visibly testable general rules—actually does seem to rule out all abuses of belief in the invisible, when applied correctly.

Perhaps I should say, “you should assign unaltered prior probability to additional invisibles,” rather than saying, “do not believe in them.” But if you think of a *belief* as something evidentially additional, something you bother to track, something

where you bother to count up support for or against, then it's questionable whether we should ever have additional beliefs about additional invisibles.

There are exotic cases that break this in theory. (E.g.: The epiphenomenal demons are watching you, and will torture 3 ↑↑↑ 3 victims for a year, somewhere you can't ever verify the event, if you ever say the word "Niblick.") But I can't think of a case where the principle fails in human practice.

Zombies: The Movie



discussion
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4-c47

FADE IN around a serious-looking group of uniformed military officers. At the head of the table, a senior, heavy-set man, GENERAL FRED, speaks.

GENERAL FRED: The reports are confirmed. New York has been overrun...by *zombies*.

COLONEL TODD: Again? But we just had a zombie invasion 28 days ago!

GENERAL FRED: These zombies...are different. They're...*philosophical* zombies.

CAPTAIN MUDD: Are they filled with rage, causing them to bite people?

COLONEL TODD: Do they lose all capacity for reason?

GENERAL FRED: No. They behave...*exactly* like we do...except that they're not conscious.

(*Silence grips the table.*)

COLONEL TODD: Dear God.

GENERAL FRED moves over to a computerized display.

GENERAL FRED: This is New York City, two weeks ago.

The display shows crowds bustling through the streets, people eating in restaurants, a garbage truck hauling away trash.

GENERAL FRED: *This*... is New York City...*now*.

The display changes, showing a crowded subway train, a group of students laughing in a park, and a couple holding hands in the sunlight.

COLONEL TODD: It's worse than I imagined.

CAPTAIN MUDD: How can you tell, exactly?

COLONEL TODD: I've never seen anything so brutally ordinary.

A lab-coated SCIENTIST stands up at the foot of the table.

SCIENTIST: The zombie disease eliminates consciousness without changing the brain in any way. We've been trying to understand how the disease is transmitted. Our conclusion is that, since the disease attacks dual properties of ordinary matter,

it must, itself, operate outside our universe. We're dealing with an *epiphenomenal virus*.

GENERAL FRED: Are you sure?

SCIENTIST: As sure as we can be in the total absence of evidence.

GENERAL FRED: All right. Compile a report on every epiphenomenon ever observed. What, where, and who. I want a list of everything that hasn't happened in the last fifty years.

CAPTAIN MUDD: If the virus is epiphenomenal, how do we know it exists?

SCIENTIST: The same way we know *we're* conscious.

CAPTAIN MUDD: Oh, okay.

GENERAL FRED: Have the doctors made any progress on finding an epiphenomenal cure?

SCIENTIST: They've tried every placebo in the book. No dice. Everything they do has an effect.

GENERAL FRED: Have you brought in a homeopath?

SCIENTIST: I tried, sir! I couldn't find any!

GENERAL FRED: Excellent. And the Taoists?

SCIENTIST: They refuse to do anything!

GENERAL FRED: Then we may yet be saved.

COLONEL TODD: What about David Chalmers? Shouldn't he be here?

GENERAL FRED: Chalmers...was one of the first victims.

COLONEL TODD: Oh no.

(*Cut to the INTERIOR of a cell, completely walled in by reinforced glass, where DAVID CHALMERS paces back and forth.*)

DOCTOR: David! David Chalmers! Can you hear me?

CHALMERS: Yes.

NURSE: It's no use, doctor.

CHALMERS: I'm perfectly fine. I've been introspecting on my consciousness, and I can't detect any difference. I *know* I would be expected to say that, but—

The DOCTOR turns away from the glass screen in horror.

DOCTOR: His words, they...they *don't mean anything*.

CHALMERS: This is a grotesque distortion of my philosophical views. This sort of thing can't actually happen!

DOCTOR: Why not?

NURSE: Yes, why not?

CHALMERS: Because—

(*Cut to two POLICE OFFICERS, guarding a dirt road leading up to the imposing steel gate of a gigantic concrete complex. On their uniforms, a badge reads BRIDGING LAW ENFORCEMENT AGENCY.*)

OFFICER 1: You've got to watch out for those clever bastards. They look like humans. They can talk like humans. They're identical to humans on the atomic level. But they're not human.

OFFICER 2: Scumbags.

The huge noise of a throbbing engine echoes over the hills. Up rides the MAN on a white motorcycle. The MAN is wearing black sunglasses and a black leather business suit with a black leather tie and silver metal boots. His white beard flows in the wind. He pulls to a halt in front of the gate.

The OFFICERS bustle up to the motorcycle.

OFFICER 1: State your business here.

MAN: Is this where you're keeping David Chalmers?

OFFICER 2: What's it to you? You a friend of his?

MAN: Can't say I am. But even zombies have rights.

OFFICER 1: All right, buddy, let's see your qualia.

MAN: I don't have any.

OFFICER 2 suddenly pulls a gun, keeping it trained on the MAN.

OFFICER 2: Aha! A zombie!

OFFICER 1: No, zombies claim to have qualia.

OFFICER 2: So he's an ordinary human?

OFFICER 1: No, they also claim to have qualia.

The OFFICERS look at the MAN, who waits calmly.

OFFICER 2: Um...

OFFICER 1: Who *are* you?

MAN: I'm Daniel Dennett, bitches.

Seemingly from nowhere, DENNETT pulls a sword and slices OFFICER 2's gun in half with a steely noise. OFFICER 1 begins to reach for his own gun, but DENNETT is suddenly standing behind OFFICER 1 and chops with a fist, striking the junction

of OFFICER 1's shoulder and neck. OFFICER 1 drops to the ground.

OFFICER 2 steps back, horrified.

OFFICER 2: That's not possible! How'd you do that?

DENNETT: I am one with my body.

DENNETT drops OFFICER 2 with another blow, and strides toward the gate. He looks up at the imposing concrete complex, and grips his sword tighter.

DENNETT (*quietly to himself*): There is a spoon.

(*Cut back to GENERAL FRED and the other military officials.*)

GENERAL FRED: I've just received the reports. We've lost Detroit.

CAPTAIN MUDD: I don't want to be the one to say "Good riddance," but—

GENERAL FRED: Australia has been ...*reduced to atoms*.

COLONEL TODD: The epiphenomenal virus is spreading faster. Civilization itself threatens to dissolve into total normality. We could be looking at the middle of humanity.

CAPTAIN MUDD: Can we negotiate with the zombies?

GENERAL FRED: We've sent them messages. They sent only a single reply.

CAPTAIN MUDD: Which was...?

GENERAL FRED: It's on its way now.

An orderly brings in an envelope, and hands it to GENERAL FRED.

GENERAL FRED opens the envelope, takes out a single sheet of paper, and reads it.

Silence envelops the room.

CAPTAIN MUDD: What's it say?

GENERAL FRED: It says ...that *we're* the ones with the virus.
(A silence falls.)

COLONEL TODD raises his hands and stares at them.

COLONEL TODD: My God, it's true. It's true. I...

(A tear rolls down COLONEL TODD's cheek.)

COLONEL TODD: I don't feel anything.

The screen goes black.

The sound goes silent.

MERE REALITY

The movie continues exactly as before.

Excluding the Supernatural

Occasionally, you hear someone claiming that creationism should not be taught in schools, especially not as a competing hypothesis to evolution, because creationism is *a priori and automatically* excluded from scientific consideration, in that it invokes the “supernatural.”

So ...is the idea here, that creationism *could* be true, but *even if it were true*, you wouldn’t be *allowed* to teach it in science class, because science is only about “natural” things?

It seems clear enough that this notion stems from the desire to avoid a confrontation between science and religion. You don’t want to come right out and say that science doesn’t teach Religious Claim X because X has been tested by the scientific method and found false. So instead, you can ...um ...claim that science is excluding hypothesis X *a priori*. That way you don’t have to discuss how experiment has falsified X *a posteriori*.

Of course this plays right into the creationist claim that Intelligent Design isn’t getting a fair shake from science—that science has *prejudged* the issue in favor of atheism, regardless of the evidence. If science excluded Intelligent Design *a priori*, this would be a justified complaint!

But let’s back up a moment. The one comes to you and says: “Intelligent Design is excluded from being science *a priori*, because it is ‘supernatural,’ and science only deals in ‘natural’ explanations.”

What exactly do they mean, “supernatural”? Is any explanation invented by someone with the last name “Cohen” a supernatural one? If we’re going to summarily kick a set of hypotheses out of science, what is it that we’re supposed to exclude?

By *far* the best definition I’ve ever heard of the supernatural is Richard Carrier’s: A “supernatural” explanation appeals to *ontologically basic mental things*, mental entities that cannot be reduced to nonmental entities.



discussion
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This is the difference, for example, between saying that water rolls downhill because it *wants* to be lower, and setting forth differential equations that claim to describe only motions, not desires. It's the difference between saying that a tree puts forth leaves because of a tree spirit, versus examining plant biochemistry. Cognitive science takes the fight against supernaturalism into the realm of the mind.

Why is this an excellent definition of the supernatural? I refer you to Richard Carrier¹ for the full argument. But consider: Suppose that you discover what seems to be a *spirit*, inhabiting a tree—a dryad who can materialize outside or inside the tree, who speaks in English about the need to protect her tree, et cetera. And then suppose that we turn a microscope on this tree spirit, and she turns out to be made of parts—not inherently spiritual and ineffable parts, like fabric of desireness and cloth of belief, but rather the same sort of parts as quarks and electrons, parts whose behavior is defined in motions rather than minds. Wouldn't the dryad immediately be demoted to the dull catalogue of common things?

But if we accept Richard Carrier's definition of the supernatural, then a dilemma arises: we *want* to give religious claims a fair shake, but it seems that we have *very good* grounds for excluding supernatural explanations *a priori*.

I mean, what *would* the universe look like if reductionism were false?

I previously defined the reductionist thesis as follows: human minds create multi-level *models* of reality in which high-level patterns and low-level patterns are separately and explicitly *represented*. A physicist knows Newton's equation for gravity, Einstein's equation for gravity, and the derivation of the former as a low-speed approximation of the latter. But these three separate mental representations are only a convenience of human cognition. It is not that *reality itself* has an Einstein equation

1. sharov.me/r/4-22

Richard Carrier Blogs: *Defining the Supernatural* •
richardcarrier.blogspot.com



that governs at high speeds, a Newton equation that governs at low speeds, and a “bridging law” that smooths the interface. Reality itself has only a single level, Einsteinian gravity. It is only the Mind Projection Fallacy that makes some people talk as if the higher levels could have a separate existence—different levels of organization can have separate representations in human maps, but the territory itself is a single unified low-level mathematical object.

Suppose this were wrong.

Suppose that the Mind Projection Fallacy was not a fallacy, but simply true.

Suppose that a 747 had a fundamental physical existence apart from the quarks making up the 747.

What experimental observations would you expect to make, if you found yourself in such a universe?

If you can’t come up with a good answer to that, it’s not *observation* that’s ruling out “non-reductionist” beliefs, but a priori logical incoherence. If you can’t say what predictions the “non-reductionist” model makes, how can you say that experimental evidence rules it out?

My thesis is that non-reductionism is a *confusion*; and once you realize that an idea is a confusion, it becomes a tad difficult to envision what the universe would look like if the confusion were *true*. Maybe I’ve got some multi-level model of the world, and the multi-level model has a one-to-one direct correspondence with the causal elements of the physics? But once all the rules are specified, why wouldn’t the model just flatten out into yet another list of fundamental things and their interactions? Does everything I can *see* in the model, like a 747 or a human mind, have to become a separate real thing? But what if I see a pattern in that new supersystem?

Supernaturalism is a special case of non-reductionism, where it is not 747s that are irreducible, but just (some) mental things. Religion is a special case of supernaturalism, where the irreducible mental things are God(s) and souls; and perhaps also sins, angels, karma, etc.

If I propose the existence of a powerful entity with the ability to survey and alter each element of our observed universe, but

with the entity reducible to nonmental parts that interact with the elements of our universe in a lawful way; if I propose that this entity wants certain particular things, but “wants” using a brain composed of particles and fields; then this is not yet a religion, just a naturalistic hypothesis about a naturalistic Matrix. If tomorrow the clouds parted and a vast glowing amorphous figure thundered forth the above description of reality, then this would not imply that the figure was necessarily honest; but I would show the movies in a science class, and I would try to derive testable predictions from the theory.

Conversely, religions have ignored the discovery of that ancient bodiless thing: omnipresent in the working of Nature and immanent in every falling leaf; vast as a planet’s surface and billions of years old; itself unmade and arising from the structure of physics; designing without brain to shape all life on Earth and the minds of humanity. Natural selection, when Darwin proposed it, was not hailed as the long-awaited Creator: It wasn’t *fundamentally* mental.

But now we get to the dilemma: if the staid conventional normal boring understanding of physics and the brain *is* correct, there’s no way *in principle* that a human being can concretely envision, and derive testable experimental predictions about, an alternate universe in which things *are* irreducibly mental. Because if the boring old normal model is correct, your brain is made of quarks, and so your brain will only be able to envision and concretely predict things that can predicted by quarks. You will only ever be able to construct models made of interacting simple things.

People who live in reductionist universes cannot concretely envision non-reductionist universes. They can pronounce the syllables “non-reductionist” but they can’t *imagine* it.

The basic error of anthropomorphism, and the reason why supernatural explanations sound much simpler than they really are, is your brain using itself as an opaque black box to predict other things labeled “mindful.” Because you already have big, complicated webs of neural circuitry that implement your “wanting” things, it seems like you can easily describe water that “wants” to flow downhill—the one word “want” acts

as a lever to set your *own* complicated wanting-machinery in motion.

Or you imagine that God likes beautiful things, and therefore made the flowers. Your own “beauty” circuitry determines what is “beautiful” and “not beautiful.” But you don’t know the diagram of your own synapses. You can’t describe a *nonmental* system that computes the same label for what is “beautiful” or “not beautiful”—can’t write a computer program that predicts your own labelings. But this is just a defect of knowledge on your part; it doesn’t mean that the brain has no explanation.

If the “boring view” of reality is correct, then you can *never* predict anything irreducible because *you* are reducible. You can never get Bayesian confirmation for a hypothesis of irreducibility, because any *prediction you can make* is, therefore, something that could also be predicted by a reducible thing, namely your brain.

Some boxes you really *can’t* think outside. If our universe *really is* Turing computable, we will never be able to *concretely* envision anything that isn’t Turing-computable—no matter how many levels of halting oracle hierarchy our mathematicians can talk *about*, we won’t be able to predict what a halting oracle would actually *say*, in such fashion as to experimentally discriminate it from merely computable reasoning.

Of course, that’s all assuming the “boring view” is correct. *To the extent* that you believe evolution is true, you should not expect to encounter strong evidence against evolution. To the extent you believe reductionism is true, you should expect non-reductionist hypotheses to be *incoherent* as well as wrong. To the extent you believe supernaturalism is false, you should expect it to be *inconceivable* as well.

If, on the other hand, a supernatural hypothesis turns out to be true, then presumably you will also discover that it is not inconceivable.

So let us bring this back full circle to the matter of Intelligent Design:

Should ID be excluded a priori from experimental falsification and science classrooms, because, by invoking the supernatural, it has placed itself outside of natural philosophy?

I answer: “Of course not.” The *irreducibility* of the intelligent designer is not an indispensable part of the ID hypothesis. For every irreducible God that can be proposed by the IDers, there exists a corresponding reducible alien that behaves in accordance with the same predictions—since the IDers themselves are reducible. To the extent I believe reductionism is in fact correct, which is a rather strong extent, I must expect to discover reducible formulations of all supposedly supernatural predictive models.

If we’re going over the archeological records to test the assertion that Jehovah parted the Red Sea out of an explicit desire to display its superhuman power, then it makes little difference whether Jehovah is ontologically basic, or an alien with nanotech, or a Dark Lord of the Matrix. You do some archeology, find no skeletal remnants or armor at the Red Sea site, and indeed find records that Egypt ruled much of Canaan at the time. So you stamp the historical record in the Bible “disproven” and carry on. The hypothesis is coherent, falsifiable and wrong.

Likewise with the evidence from biology that foxes are designed to chase rabbits, rabbits are designed to evade foxes, and neither is designed “to carry on their species” or “protect the harmony of Nature”; likewise with the retina being designed backwards with the light-sensitive parts at the bottom; and so on through a thousand other items of evidence for splintered, immoral, incompetent design. The Jehovah model of our alien god is coherent, falsifiable, and wrong—coherent, that is, so long as you don’t care whether Jehovah is ontologically basic or just an alien.

Just convert the supernatural hypothesis into the corresponding natural hypothesis. Just make the same predictions the same way, without asserting any mental things to be ontologically basic. Consult your brain’s black box if necessary to make predictions—say, if you want to talk about an “angry god” without building a full-fledged angry AI to label behaviors as angry or not angry. So you derive the predictions, or look up the predictions made by ancient theologians without advance knowledge of our experimental results. If experiment conflicts with those predictions, then it is fair to speak of the religious

claim having been scientifically refuted. It was given its just chance at confirmation; it is being excluded a posteriori, not a priori.

Ultimately, reductionism is just disbelief in *fundamentally complicated* things. If “fundamentally complicated” sounds like an oxymoron...well, that’s why I think that the doctrine of non-reductionism is a *confusion*, rather than a way that things could be, but aren’t. You would be wise to be wary, if you find yourself supposing such things.

But the ultimate rule of science is to look and see. If ever a God appeared to thunder upon the mountains, it would be something that people looked at and saw.

Corollary: Any supposed designer of Artificial General Intelligence who talks about religious beliefs in respectful tones is clearly not an expert on reducing mental things to nonmental things; and indeed knows so very little of the uttermost basics, as for it to be scarcely plausible that they could be expert at the art; unless their *idiot savancy* is complete. Or, of course, if they’re outright lying. We’re not talking about a subtle mistake.

Psychic Powers



discussion
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In the last essay, I wrote:

If the “boring view” of reality is correct, then you can *never* predict anything irreducible because you are reducible. You can never get Bayesian confirmation for a hypothesis of irreducibility, because any *prediction you can make* is, therefore, something that could also be predicted by a reducible thing, namely your brain.

Benja Fallenstein commented:¹

I think that while you can in this case never devise an empirical test whose outcome could *logically prove* irreducibility, there is no clear reason to believe that you cannot devise a test whose counterfactual outcome in an irreducible world would make irreducibility subjectively *much more probable* (given an Occamian prior).

Without getting into reducibility/irreducibility, consider the scenario that the physical universe makes it possible to build a hypercomputer—that performs operations on arbitrary real numbers, for example—but that our brains do not actually make use of this: they can be simulated perfectly well by an ordinary Turing machine, thank you very much ...

Well, that’s a very intelligent argument, Benja Fallenstein. But I have a crushing reply to your argument, such that, once I

1. sharov.me/r/4-23

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deliver it, you will at once give up further debate with me on this particular point:

You're right.

Alas, I don't get modesty credit on this one, because after publishing the last essay I realized a similar flaw on my own—this one concerning Occam's Razor and psychic powers:

If beliefs and desires are irreducible and ontologically basic entities, or have an ontologically basic *component* not covered by existing science, that would make it far more likely that there was an ontological rule governing the interaction of different minds—an interaction which bypassed ordinary “material” means of communication like sound waves, known to existing science.

If naturalism is correct, then there exists a conjugate reductionist model that makes the *same predictions* as any concrete prediction that any parapsychologist can make about telepathy.

Indeed, if naturalism is correct, the only reason we can *conceive* of beliefs as “fundamental” is due to lack of self-knowledge of our own neurons—that the peculiar reflective architecture of our own minds exposes the “belief” class but hides the machinery behind it.

Nonetheless, the discovery of information transfer between brains, in the absence of any known material connection between them, is *probabilistically* a privileged prediction of supernatural models (those that contain ontologically basic mental entities). Just because it is so much *simpler* in that case to have a new law relating beliefs between different minds, compared to the “boring” model where beliefs are complex constructs of neurons.

The hope of psychic powers arises from treating beliefs and desires as sufficiently fundamental objects that they can have *unmediated* connections to reality. If beliefs are patterns of neurons made of known material, with inputs given by organs like eyes constructed of known material, and with outputs through muscles constructed of known material, and this seems sufficient to account for all known mental powers of humans, then there's no reason to expect anything more—no reason to postulate additional connections. This is why reductionists don't

expect psychic powers. Thus, observing psychic powers would be strong evidence for the supernatural in Richard Carrier's sense.

We have an Occam rule that counts the number of ontologically basic classes and ontologically basic laws in the model, and penalizes the count of entities. If naturalism is correct, then the attempt to count "belief" or the "relation between belief and reality" as a single basic entity is simply misguided anthropomorphism; we are only tempted to it by a quirk of our brain's internal architecture. But if you *just go with* that misguided view, then it assigns a much higher probability to psychic powers than does naturalism, because you can implement psychic powers using apparently simpler laws.

Hence the actual discovery of psychic powers would imply that the human-naïve Occam rule was in fact better-calibrated than the sophisticated naturalistic Occam rule. It would argue that reductionists had been wrong all along in trying to take apart the brain; that what our minds exposed as a seemingly simple lever was in fact a simple lever. The naïve dualists would have been right from the beginning, which is why their ancient wish would have been enabled to come true.

So telepathy, and the ability to influence events just by wishing at them, and precognition, would all, if discovered, be strong Bayesian evidence in favor of the hypothesis that beliefs are ontologically fundamental. Not logical proof, but strong Bayesian evidence.

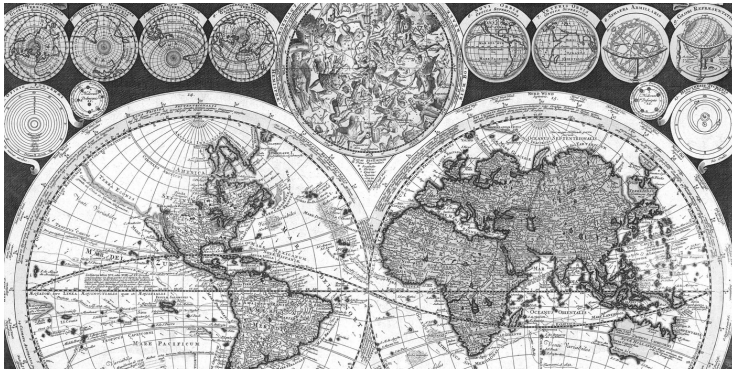
If reductionism is correct, then any science-fiction story containing psychic powers can be output by a system of simple elements (i.e., the story's author's brain); but if we *in fact* discover psychic powers, that would make it much more probable that events were occurring which could not *in fact* be described by reductionist models.

Which just goes to say: The existence of psychic powers is a privileged probabilistic assertion of non-reductionist worldviews—they *own* that advance prediction; they devised it and put it forth, in defiance of reductionist expectations. So by the laws of science, if psychic powers are discovered, non-reductionism wins.

I am therefore confident in dismissing psychic powers as a priori implausible, despite all the claimed experimental evidence in favor of them.

PART S

Quantum Physics and Many Worlds



Quantum Explanations

There's a widespread belief that quantum mechanics is *supposed* to be confusing. This is not a good frame of mind for either a teacher or a student.

And I find that legendarily “confusing” subjects often are not really all that complicated *as math*, particularly if you just want a very basic—but still mathematical—grasp on what goes on down there.

I am not a physicist, and physicists famously hate it when non-professional-physicists talk about quantum mechanics. But I do have some experience with explaining mathy things that are allegedly “hard to understand.”

I wrote the Intuitive Explanation of Bayesian Reasoning because people were complaining that Bayes's Theorem was “counterintuitive”—in fact it was *famously* counterintuitive—and this did not seem right. The equation just did not seem complicated enough to deserve the fearsome reputation it had. So I tried explaining it *my way*, and I did not manage to reach my original target of elementary school students, but I get frequent grateful emails from formerly confused folks ranging from reporters to outside academic college professors.

Besides, as a Bayesian, I don't believe in phenomena that are *inherently* confusing. Confusion exists in our models of the world, not in the world itself. If a subject is widely known as *confusing*, not just *difficult* ...you shouldn't leave it at that. It doesn't suffice; it is not an okay place to be. Maybe you can fix the problem, maybe you can't; but you shouldn't be *happy* to leave students confused.

The first way in which my introduction is going to depart from the traditional, standard introduction to quantum mechanics, is that I am *not* going to tell you that quantum mechanics is supposed to be confusing.

I am not going to tell you that it's okay for you to not understand quantum mechanics, because no one understands



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quantum mechanics, as Richard Feynman once claimed. There was a historical time when this was true, but we no longer live in that era.

I am not going to tell you: “You don’t understand quantum mechanics, you just get used to it.” (As von Neumann is reputed to have said; back in the dark decades when, in fact, no one *did* understand quantum mechanics.)

Explanations are supposed to make you *less confused*. If you feel like you don’t understand something, this indicates a *problem*—either with you, or your teacher—but at any rate a problem; and you should move to *resolve* the problem.

I am not going to tell you that quantum mechanics is *weird*, *bizarre*, *confusing*, or *alien*. Quantum mechanics is counterintuitive, but that is a problem with your intuitions, not a problem with quantum mechanics. Quantum mechanics has been around for billions of years before the Sun coalesced from interstellar hydrogen. Quantum mechanics was here before you were, and if you have a problem with that, *you* are the one who needs to change. Quantum mechanics sure won’t. There are no *surprising facts*, only *models* that are *surprised* by facts; and if a model is surprised by the facts, it is no credit to that model.

It is always best to think of reality as perfectly normal. Since the beginning, not one unusual thing has ever happened.

The *goal* is to become completely at home in a quantum universe. Like a native. Because, in fact, that is where you live.

In the coming sequence on quantum mechanics, I am going to consistently speak as if quantum mechanics is *perfectly normal*; and when human intuitions depart from quantum mechanics, I am going to make fun of the *intuitions* for being weird and unusual. This may seem odd, but the point is to swing your mind around to a *native* quantum point of view.

Another thing: The traditional introduction to quantum mechanics closely follows the order in which quantum mechanics was discovered.

The traditional introduction starts by saying that matter sometimes behaves like little billiard balls bopping around, and sometimes behaves like crests and troughs moving through a pool of water. Then the traditional introduction gives some

examples of matter acting like a little billiard ball, and some examples of it acting like an ocean wave.

Now, it happens to be a historical fact that, back when students of matter were working all this stuff out and had *no clue* about the true underlying math, those early scientists first thought that matter was like little billiard balls. And then that it was like waves in the ocean. And then that it was like billiard balls again. And then the early scientists got *really* confused, and stayed that way for several decades, until it was finally sorted out in the second half of the twentieth century.

Dragging a modern-day student through all this may be a *historically realistic* approach to the subject matter, but it also ensures the historically realistic outcome of *total bewilderment*. Talking to aspiring young physicists about “wave/particle duality” is like starting chemistry students on the Four Elements.

An electron is *not* a billiard ball, and it’s *not* a crest and trough moving through a pool of water. An electron is a mathematically different sort of entity, *all the time and under all circumstances*, and it has to be accepted on its own terms.

The universe is not wavering between using particles and waves, unable to make up its mind. It’s only human *intuitions* about quantum mechanics that swap back and forth. The intuitions we have for billiard balls, and the intuitions we have for crests and troughs in a pool of water, both look *sort of* like they’re applicable to electrons, at different times and under different circumstances. But the truth is that both intuitions simply *aren’t applicable*.

If you try to think of an electron as being like a billiard ball on some days, and like an ocean wave on other days, you will *confuse the living daylights* out of yourself.

Yet it’s your eyes that are wobbling and unstable, not the world.

Furthermore:

The order in which humanity *discovered* things is not necessarily the best order in which to *teach* them. First, humanity noticed that there were other animals running around. Then we cut them open and found that they were full of organs. Then we examined the organs carefully and found they were made of

tissues. Then we looked at the tissues under a microscope and discovered cells, which are made of proteins and some other chemically synthesized stuff. Which are made of molecules, which are made of atoms, which are made of protons and neutrons and electrons *which are way simpler than entire animals but were discovered tens of thousands of years later*.

Physics doesn't start by talking about biology. So why should it start by talking about very high-level complicated phenomena, like, say, the observed results of experiments?

The ordinary way of teaching quantum mechanics keeps stressing the experimental results. Now I do understand why that sounds nice from a rationalist perspective. Believe me, I understand.

But it seems to me that the upshot is dragging in big complicated mathematical tools that you need to analyze real-world situations, before the student understands what *fundamentally* goes on in the simplest cases.

It's like trying to teach programmers how to write concurrent multithreaded programs before they know how to add two variables together, because concurrent multithreaded programs are closer to everyday life. Being close to everyday life is not always a strong recommendation for what to teach first.

Maybe the monomaniacal focus on experimental observations made sense in the dark decades when *no one* understood what was fundamentally going on, and you *couldn't* start there, and all your models were just mysterious maths that gave good experimental predictions ...you can still find this view of quantum physics presented in many books ...but maybe today it's worth trying a different angle? The result of the standard approach is standard confusion.

The classical world is strictly implicit in the quantum world, but seeing from a classical perspective makes everything bigger and more complicated. Everyday life is a higher level of organization, like molecules versus quarks—huge catalogue of molecules, six quarks. I think it is worth trying to teach from the perspective of the quantum world first, and talking about classical experimental results afterward.

I am not going to start with the normal classical world and then talk about a bizarre quantum backdrop hidden behind the scenes. The quantum world *is* the scene and it defines normality.

I am not going to talk as if the classical world is real life, and occasionally the classical world transmits a request for an experimental result to a quantum-physics server, and the quantum-physics server does some peculiar calculations and transmits back a classical experimental result. I am going to talk as if the quantum world is the really real and the classical world something far away. Not just because that makes it easier to be a native of a quantum universe, but because, at a core level, it's the truth.

Finally, I am going to take a strictly realist perspective on quantum mechanics—the quantum world is really out there, our equations describe the territory and not our maps of it, and the classical world only exists implicitly within the quantum one. I am not going to discuss non-realist views in the early stages of my introduction, except to say why you should not be confused by certain intuitions that non-realists draw upon for support. I am not going to apologize for this, and I would like to ask any non-realists on the subject of quantum mechanics to wait and hold their comments until called for in a later essay. Do me this favor, please. I think non-realism is one of the main things that confuses prospective students, and prevents them from being able to concretely visualize quantum phenomena. I *will* discuss the issues explicitly in a future essay.

But everyone should be aware that, even though I'm not going to discuss the issue at first, there is a sizable community of scientists who dispute the realist perspective on quantum mechanics. Myself, I don't think it's worth figuring both ways; I'm a pure realist, for reasons that will become apparent. But if you read my introduction, you are getting my view. It is not only my view. It is probably the majority view among theoretical physicists, if that counts for anything (though I will argue the matter separately from opinion polls). Still, it is not the only view that exists in the modern physics community. I do not feel obliged to present the other views *right away*, but I feel obliged

to warn my readers that there *are* other views, which I will not be presenting during the initial stages of the introduction.

To sum up, my goal will be to teach you to think like a *native of a quantum universe*, not a *reluctant tourist*.

Embrace reality. Hug it tight.

Configurations and Amplitude

So the universe isn't made of little billiard balls, and it isn't made of crests and troughs in a pool of aether ...Then what is the stuff that stuff is made of?



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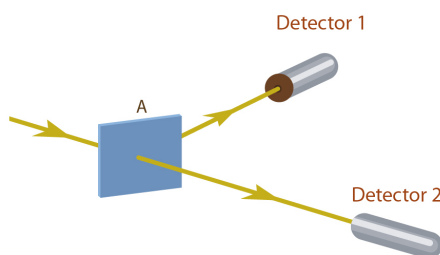


Figure 230.1

In Figure 230.1, we see, at A, a half-silvered mirror, and two photon detectors, Detector 1 and Detector 2.

Early scientists, when they ran experiments like this, became confused about what the results meant. They would send a photon toward the half-silvered mirror, and half the time they would see Detector 1 click, and the other half of the time they would see Detector 2 click.

The early scientists—you're going to laugh at this—thought that the silver mirror deflected the photon half the time, and let it through half the time.

Ha, ha! As if the half-silvered mirror did different things on different occasions! I want you to let go of this idea, because if you cling to what early scientists thought, you will become extremely confused. The half-silvered mirror obeys the same rule every time.

If you were going to write a computer program that *was* this experiment—not a computer program that *predicted* the result

of the experiment, but a computer program that resembled the underlying reality—it might look sort of like this:

At the start of the program (the start of the experiment, the start of time) there's a certain mathematical entity, called a *configuration*. You can think of this configuration as corresponding to “there is one photon heading from the photon source toward the half-silvered mirror,” or just “a photon heading toward A.”

A configuration can store a single complex value—“complex” as in the complex numbers $(a + bi)$, with i defined as $\sqrt{-1}$. At the start of the program, there's already a complex number stored in the configuration “a photon heading toward A.” The exact value doesn't matter so long as it's not zero. We'll let the configuration “a photon heading toward A” have a value of $(-1 + 0i)$.

All this is a fact within the territory, not a description of anyone's knowledge. A configuration isn't a proposition or a possible way the world could be. A configuration is a variable in the program—you can think of it as a kind of memory location whose index is “a photon heading toward A”—and it's out there in the territory.

As the complex numbers that get assigned to configurations are not positive real numbers between 0 and 1, there is no danger of confusing them with probabilities. “A photon heading toward A” has complex value -1, which is hard to see as a degree of belief. The complex numbers are values within the program, again out there in the territory. We'll call the complex numbers *amplitudes*.

There are two other configurations, which we'll call “a photon going from A to Detector 1” and “a photon going from A to Detector 2.” These configurations don't have a complex value yet; it gets assigned as the program runs.

We are going to calculate the amplitudes of “a photon going from A toward 1” and “a photon going from A toward 2” using the value of “a photon going toward A,” and the rule that describes the half-silvered mirror at A.

Roughly speaking, the half-silvered mirror rule is “multiply by 1 when the photon goes straight, and multiply by i when the photon turns at a right angle.” This is the universal rule

that relates the amplitude of the configuration of “a photon going in,” to the amplitude that goes to the configurations of “a photon coming out straight” or “a photon being deflected.”¹

So we pipe the amplitude of the configuration “a photon going toward A,” which is $(-1 + 0i)$, into the half-silvered mirror at A, and this transmits an amplitude of $(-1 + 0i) \times i = (0 - i)$ to “a photon going from A toward 1,” and also transmits an amplitude of $(-1 + 0i) \times 1 = (-1 + 0i)$ to “a photon going from A toward 2.”

In the Figure 230.1 experiment, these are all the configurations and all the transmitted amplitude we need to worry about, so we’re done. Or, if you want to think of “Detector 1 gets a photon” and “Detector 2 gets a photon” as separate configurations, they’d just inherit their values from “A to 1” and “A to 2” respectively. (Actually, the values inherited should be multiplied by another complex factor, corresponding to the distance from A to the detector; but we will ignore that for now, and suppose that all distances traveled in our experiments happen to correspond to a complex factor of 1.)

So the final program state is:

Configuration “a photon going toward A”: $(-1 + 0i)$

Configuration “a photon going from A toward 1”: $(0 - i)$

Configuration “a photon going from A toward 2”: $(-1 + 0i)$
and optionally

Configuration “Detector 1 gets a photon”: $(0 - i)$

Configuration “Detector 2 gets a photon”: $(-1 + 0i)$.

This same result occurs—the same amplitudes stored in the same configurations—every time you run the program (every time you do the experiment).

1. **Editor’s Note:** Strictly speaking, a standard half-silvered mirror would yield a rule “multiply by -1 when the photon turns at a right angle,” not “multiply by i .” The basic scenario described by the author is not physically impossible, and its use does not affect the substantive argument. However, physics students may come away confused if they compare the discussion here to textbook discussions of Mach–Zehnder interferometers. We’ve left this idiosyncrasy in the text because it eliminates any need to specify which side of the mirror is half-silvered, simplifying the experiment.

Now, for *complicated* reasons that we aren't going to go into here—considerations that belong on a higher level of organization than fundamental quantum mechanics, the same way that atoms are more complicated than quarks—there's no *simple* measuring instrument that can directly tell us the exact amplitudes of each configuration. We can't directly see the program state.

So how do physicists know what the amplitudes are?

We *do* have a magical measuring tool that can tell us the *squared modulus* of a configuration's amplitude. If the original complex amplitude is $(a + bi)$, we can get the positive real number $(a^2 + b^2)$. Think of the Pythagorean theorem: if you imagine the complex number as a little arrow stretching out from the origin on a two-dimensional plane, then the magic tool tells us the squared length of the little arrow, but it doesn't tell us the direction the arrow is pointing.

To be more precise, the magic tool actually just tells us the *ratios* of the squared lengths of the amplitudes in some configurations. We don't know how long the arrows are in an absolute sense, just how long they are relative to each other. But this turns out to be enough information to let us reconstruct the laws of physics—the rules of the program. And so I can talk about amplitudes, not just ratios of squared moduli.

When we wave the magic tool over “Detector 1 gets a photon” and “Detector 2 gets a photon,” we discover that these configurations have the same squared modulus—the lengths of the arrows are the same. Thus speaks the magic tool. By doing more *complicated* experiments (to be seen shortly), we can tell that the original complex numbers had a ratio of i to 1.

And what is this magical measuring tool?

Well, from the perspective of everyday life—way, way, way above the quantum level and a lot more complicated—the magical measuring tool is that we send some photons toward the half-silvered mirror, one at a time, and count up how many photons arrive at Detector 1 versus Detector 2 over a few thousand trials. The ratio of these values is the ratio of the squared moduli of the amplitudes. But the reason for this is *not* something we are going to consider yet. Walk before you run. It is

not possible to understand what happens *all the way up* at the level of everyday life, before you understand what goes on in much simpler cases.

For today's purposes, we have a magical squared-modulus-ratio reader. And the magic tool tells us that the little two-dimensional arrow for the configuration "Detector 1 gets a photon" has the same squared length as for "Detector 2 gets a photon." That's all.

You may wonder, "Given that the magic tool works this way, what motivates us to use quantum theory, instead of thinking that the half-silvered mirror reflects the photon around half the time?"

Well, that's just begging to be confused—putting yourself into a historically realistic frame of mind like that and using everyday intuitions. Did I say anything about a little billiard ball going one way or the other and possibly bouncing off a mirror? That's not how reality works. *Reality* is about complex amplitudes flowing between configurations, and the laws of the flow are stable.

But if you insist on seeing a more complicated situation that billiard-ball ways of thinking can't handle, here's a more complicated experiment.

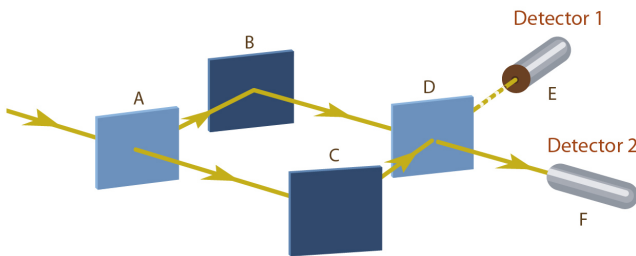


Figure 230.2

In Figure 230.2, *B* and *C* are full mirrors, and *A* and *D* are half-mirrors. The line from *D* to *E* is dashed for reasons that will become apparent, but amplitude is flowing from *D* to *E* under exactly the same laws.

Now let's apply the rules we learned before:

At the beginning of time "a photon heading toward A " has amplitude $(-1 + 0i)$.

We proceed to compute the amplitude for the configurations "a photon going from A to B " and "a photon going from A to C ":

"a photon going from A to B " = $i \times$ "a photon heading toward A " = $(0 - i)$.

Similarly,

"a photon going from A to C " = $1 \times$ "a photon heading toward A " = $(-1 + 0i)$.

The full mirrors behave (as one would expect) like half of a half-silvered mirror—a full mirror just bends things by right angles and multiplies them by i . (To state this slightly more precisely: For a full mirror, the amplitude that flows, from the configuration of a photon heading in, to the configuration of a photon heading out at a right angle, is multiplied by a factor of i .)

So:

"a photon going from B to D " = $i \times$ "a photon going from A to B " = $(1 + 0i)$,

"a photon going from C to D " = $i \times$ "a photon going from A to C " = $(0 - i)$.

" B to D " and " C to D " are two different configurations—we don't simply write "a photon at D "—because the photons are arriving at two different angles in these two different configurations. And what D does to a photon depends on the angle at which the photon arrives.

Again, the rule (speaking loosely) is that when a half-silvered mirror bends light at a right angle, the amplitude that flows from the photon-going-in configuration to the photon-going-out configuration, is the amplitude of the photon-going-in configuration multiplied by i . And when two configurations are related by a half-silvered mirror letting light straight through, the amplitude that flows from the photon-going-in configuration is multiplied by 1.

So:

- From the configuration “a photon going from B to D ,” with original amplitude $(1 + 0i)$:
 - Amplitude of $(1 + 0i) \times i = (0 + i)$ flows to “a photon going from D to E .”
 - Amplitude of $(1 + 0i) \times 1 = (1 + 0i)$ flows to “a photon going from D to F .”
- From the configuration “a photon going from C to D ,” with original amplitude $(0 - i)$:
 - Amplitude of $(0 - i) \times i = (1 + 0i)$ flows to “a photon going from D to F .”
 - Amplitude of $(0 - i) \times 1 = (0 - i)$ flows to “a photon going from D to E .”

Therefore:

- The *total* amplitude flowing to configuration “a photon going from D to E ” is $(0 + i) + (0 - i) = (0 + 0i) = 0$.
- The total amplitude flowing to configuration “a photon going from D to F ” is $(1 + 0i) + (1 + 0i) = (2 + 0i)$.

(You may want to try working this out yourself on pen and paper if you lost track at any point.)

But the upshot, from that super-high-level “experimental” perspective that we think of as normal life, is that we see *no* photons detected at E . Every photon seems to end up at F . The ratio of squared moduli between “ D to E ” and “ D to F ” is 0 to 4. That’s why the line from D to E is dashed, in this figure.

This is not something it is possible to explain by thinking of half-silvered mirrors deflecting little incoming billiard balls half the time. You’ve *got* to think in terms of amplitude flows.

If half-silvered mirrors deflected a little billiard ball half the time, in this setup, the little ball would end up at Detector 1 around half the time and Detector 2 around half the time. Which it doesn’t. So don’t think that.

You may say, “But wait a minute! I can think of another hypothesis that accounts for this result. What if, when a half-silvered mirror reflects a photon, it does something to the photon that ensures it doesn’t get reflected next time? And when

it lets a photon go through straight, it does something to the photon so it gets reflected next time.”

Now really, there’s no need to go making the rules so complicated. Occam’s Razor, remember. Just stick with simple, normal amplitude flows between configurations.

But if you want *another* experiment that disproves your *new* alternative hypothesis, it’s Figure 230.3.

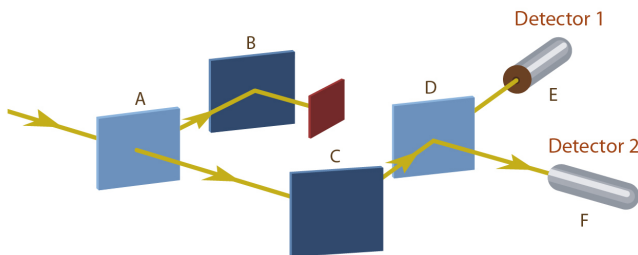


Figure 230.3

Here, we’ve left the whole experimental setup the same, and just put a little blocking object between *B* and *D*. This ensures that the amplitude of “a photon going from *B* to *D*” is 0.

Once you eliminate the amplitude contributions from that configuration, you end up with totals of $(1 + 0i)$ in “a photon going from *D* to *F*,” and $(0 - i)$ in “a photon going from *D* to *E*.”

The squared moduli of $(1 + 0i)$ and $(0 - i)$ are both 1, so the magic measuring tool should tell us that the ratio of squared moduli is 1. Way back up at the level where physicists exist, we should find that Detector 1 goes off half the time, and Detector 2 half the time.

The same thing happens if we put the block between *C* and *D*. The amplitudes are different, but the ratio of the squared moduli is still 1, so Detector 1 goes off half the time and Detector 2 goes off half the time.

This cannot *possibly* happen with a little billiard ball that either does or doesn’t get reflected by the half-silvered mirrors.

Because complex numbers can have opposite directions, like 1 and -1, or i and $-i$, amplitude flows can cancel each other out.

Amplitude flowing from configuration X into configuration Y can be canceled out by an equal and opposite amplitude flowing from configuration Z into configuration Y . In fact, that's exactly what happens in this experiment.

In probability theory, when something can either happen one way or another, X or $\neg X$, then $P(Z) = P(Z|X)P(X) + P(Z|\neg X)P(\neg X)$. And all probabilities are positive. So if you establish that the probability of Z happening given X is $1/2$, and the probability of X happening is $1/3$, then the total probability of Z happening is *at least* $1/6$ no matter *what* goes on in the case of $\neg X$. There's no such thing as negative probability, less-than-impossible credence, or $(0 + i)$ credibility, so *degrees of belief* can't cancel each other out like amplitudes do.

Not to mention that probability is in the mind to begin with; and we are talking *about* the territory, the program-that-is-reality, not talking *about* human cognition or states of partial knowledge.

By the same token, configurations are not *propositions*, not *statements*, not *ways the world could conceivably be*. Configurations are not semantic constructs. Adjectives like *probable* do not apply to them; they are not beliefs or sentences or possible worlds. They are not *true* or *false* but simply *real*.

In the experiment of Figure 230.2, do not be tempted to think anything like: "The photon goes to either B or C , but it *could* have gone the other way, and this possibility interferes with its ability to go to E ..."

It makes no sense to think of something that "could have happened but didn't" exerting an effect on the world. We can *imagine* things that could have happened but didn't—like thinking, "Gosh, that car almost hit me"—and our imagination can have an effect on our future behavior. But the event of imagination is a real event, that actually happens, and *that* is what has the effect. It's your imagination of the unreal event—your very real imagination, implemented within a quite physical brain—that affects your behavior.

To think that the *actual event* of a car hitting you—this event which could have happened to you, but in fact didn't—is directly

exerting a *causal* effect on your behavior, is mixing up the map with the territory.

What affects the world is real. (If things can affect the world without being “real,” it’s hard to see what the word “real” means.) Configurations and amplitude flows are causes, and they have visible effects; they are real. Configurations are not possible worlds and amplitudes are not degrees of belief, any more than your chair is a possible world or the sky is a degree of belief.

So what *is* a configuration, then?

Well, you’ll be getting a clearer idea of that in later essays.

But to give you a quick idea of how the real picture differs from the simplified version we saw in this essay ...

Our experimental setup only dealt with one moving particle, a single photon. Real configurations are about multiple particles. The next essay will deal with the case of more than one particle, and that should give you a much clearer idea of what a configuration is.

Each configuration we talked about *should* have described a joint position of all the particles in the mirrors and detectors, not just the position of one photon bopping around.

In fact, the *really real* configurations are over joint positions of all the particles in the universe, including the particles making up the experimenters. You can see why I’m saving the notion of *experimental results* for later essays.

In the real world, amplitude is a continuous distribution over a continuous *space* of configurations. This essay’s “configurations” were blocky and digital, and so were our “amplitude flows.” It was as if we were talking about a photon teleporting from one place to another.

If none of that made sense, don’t worry. It will be cleared up in later essays. Just wanted to give you some idea of where this was heading.

Joint Configurations

The key to understanding configurations, and hence the key to understanding quantum mechanics, is realizing on a truly gut level that configurations are about more than one particle.

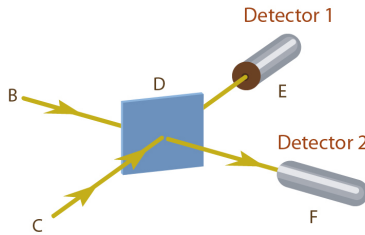


Figure 231.1

Continuing from the previous essay, Figure 231.1 shows an altered version of the experiment where we send in *two* photons toward *D* at the same time, from the sources *B* and *C*.

The starting configuration then is:

“a photon going from *B* to *D*,
and a photon going from *C* to *D*.”

Again, let’s say the starting configuration has amplitude $(-1 + 0i)$.

And remember, the rule of the half-silvered mirror (at *D*) is that a right-angle deflection multiplies by i , and a straight line multiplies by 1.

So the amplitude flows from the starting configuration, separately considering the four cases of deflection/non-deflection of each photon, are:

1. The “*B* to *D*” photon is deflected and the “*C* to *D*” photon is deflected. This amplitude flows to the configuration “a photon going from *D* to *E*, and a photon going from *D* to *F*.” The amplitude flowing is $(-1 + 0i) \times i \times i = (1 + 0i)$.

2. The “*B* to *D*” photon is deflected and the “*C* to *D*” photon goes straight. This amplitude flows to the configuration “two photons going from *D* to *E*.” The amplitude flowing is $(-1 + 0i) \times i \times 1 = (0 - i)$.
3. The “*B* to *D*” photon goes straight and the “*C* to *D*” photon is deflected. This amplitude flows to the configuration “two photons going from *D* to *F*.” The amplitude flowing is $(-1 + 0i) \times 1 \times i = (0 - i)$.
4. The “*B* to *D*” photon goes straight and the “*C* to *D*” photon goes straight. This amplitude flows to the configuration “a photon going from *D* to *F*, and a photon going from *D* to *E*.” The amplitude flowing is $(-1 + 0i) \times 1 \times 1 = (-1 + 0i)$.

Now—and this is a *very important and fundamental idea in quantum mechanics*—the amplitudes in cases 1 and 4 are flowing to the *same* configuration. Whether the *B* photon and *C* photon both go straight, or both are deflected, the resulting configuration is *one photon going toward E and another photon going toward F*.

So we add up the two incoming amplitude flows from case 1 and case 4, and get a total amplitude of $(1 + 0i) + (-1 + 0i) = 0$.

When we wave our magic squared-modulus-ratio reader over the three final configurations, we’ll find that “two photons at Detector 1” and “two photons at Detector 2” have the same squared modulus, but “a photon at Detector 1 and a photon at Detector 2” has squared modulus zero.

Way up at the level of experiment, we never find Detector 1 and Detector 2 both going off. We’ll find Detector 1 going off twice, or Detector 2 going off twice, with equal frequency. (Assuming I’ve gotten the math and physics right. I didn’t actually perform the experiment.)

The configuration’s identity is *not*, “the *B* photon going toward *E* and the *C* photon going toward *F*.” Then the resultant configurations in case 1 and case 4 would not be equal. Case 1 would be, “*B* photon to *E*, *C* photon to *F*” and case 4 would be “*B* photon to *F*, *C* photon to *E*.” These would be two distinguishable

configurations, *if* configurations had photon-tracking structure.

So we would not add up the two amplitudes and cancel them out. We would keep the amplitudes in two separate configurations. The total amplitudes would have non-zero squared moduli. And when we ran the experiment, we would find (around half the time) that Detector 1 and Detector 2 each registered one photon. Which doesn't happen, if my calculations are correct.

Configurations don't keep track of where particles come from. A configuration's identity is just, "a photon here, a photon there; an electron here, an electron there." No matter how you get into that situation, so long as there are the same species of particles in the same places, it counts as the same configuration.

I say again that the question "What kind of information does the configuration's structure incorporate?" has *experimental consequences*. You can deduce, from experiment, the way that reality itself must be treating configurations.

In a classical universe, there would be no experimental consequences. If the photon were like a little billiard ball that either went one way or the other, and the configurations were our beliefs about possible states the system could be in, and instead of amplitudes we had probabilities, it would not make a difference whether we tracked the origin of photons or threw the information away.

In a classical universe, I could assign a 25% probability to both photons going to *E*, a 25% probability of both photons going to *F*, a 25% probability of the *B* photon going to *E* and the *C* photon going to *F*, and 25% probability of the *B* photon going to *F* and the *C* photon going to *E*. Or, since I *personally* don't care which of the two latter cases occurred, I could decide to collapse the two possibilities into one possibility and add up their probabilities, and just say, "a 50% probability that each detector gets one photon."

With probabilities, we can aggregate events as we like—draw our boundaries around sets of possible worlds as we please—and the numbers will still work out the same. The probability of two mutually exclusive events always equals the probability of the first event plus the probability of the second event.

But you can't arbitrarily collapse configurations together, or split them apart, in your model, and get the same experimental predictions. Our magical tool tells us the ratios of squared moduli. When you add two complex numbers, the squared modulus of the sum is not the sum of the squared moduli of the parts:

$$\begin{aligned} \text{Squared_Modulus}(C1 + C2) \\ \neq \text{Squared_Modulus}(C1) \\ + \text{Squared_Modulus}(C2). \end{aligned}$$

E.g.:

$$\begin{aligned} S_M((2 + i) + (1 - i)) &= S_M(3 + 0i) \\ &= 3^2 + 0^2 \\ &= 9, \\ S_M(2 + i) + S_M(1 - i) &= (2^2 + 1^2) + (1^2 + (-1)^2) \\ &= (4 + 1) + (1 + 1) \\ &= 7. \end{aligned}$$

Or in the current experiment of discourse, we had flows of $(1 + 0i)$ and $(-1 + 0i)$ cancel out, adding up to 0, whose squared modulus is 0, where the squared modulus of the parts would have been 1 and 1.

If in place of Squared_Modulus, our magical tool was some linear function—any function where $F(X + Y) = F(X) + F(Y)$ —then all the quantumness would instantly vanish and be replaced by a classical physics. (A *different* classical physics, not the same illusion of classicality we hallucinate from inside the higher levels of organization in our own quantum world.)

If amplitudes were just probabilities, they couldn't cancel out when flows collided. If configurations were just states of knowledge, you could reorganize them however you liked.

But the configurations are nailed in place, indivisible and unmergeable without changing the laws of physics.

And part of what is nailed is the way that configurations treat multiple particles. A configuration says, “a photon here, a photon there,” not “*this* photon here, *that* photon there.” “*This* photon here, *that* photon there” does not have a different identity from “*that* photon here, *this* photon there.”

The result, visible in today’s experiment, is that you can’t factorize the physics of our universe to be about particles with individual identities.

Part of the reason why humans have trouble coming to grips with *perfectly normal* quantum physics, is that humans bizarrely keep trying to factor reality into a sum of individually real billiard balls.

Ha ha! Silly humans.

Distinct Configurations



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The experiment in the previous essay carried two key lessons:

First, we saw that because amplitude flows can cancel out, and because our magic measure of squared modulus is not linear, the identity of configurations is nailed down—you can’t reorganize configurations the way you can regroup possible worlds. Which configurations are the same, and which are distinct, has experimental consequences; it is an observable fact.

Second, we saw that configurations are about multiple particles. If there are two photons entering the apparatus, that doesn’t mean there are two initial configurations. Instead the initial configuration’s identity is “two photons coming in.” (Ideally, each configuration we talk about would include every particle in the experiment—including the particles making up the mirrors and detectors. And in the real universe, every configuration is about *all* the particles ...*everywhere*.)

What makes for distinct configurations is not distinct particles. Each configuration is about every particle. What makes configurations distinct is particles occupying different positions—at least one particle in a different state.

To take one important demonstration ...

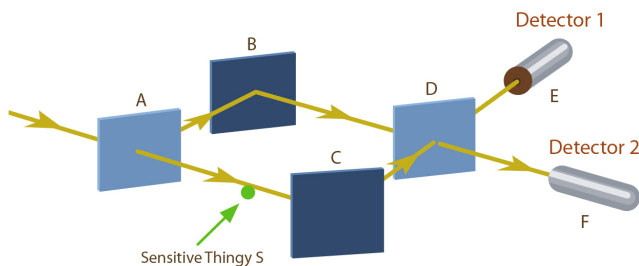


Figure 232.1

Figure 232.1 is the same experiment as Figure 230.2, with one important change: Between *A* and *C* has been placed a sensitive thingy, *S*. The key attribute of *S* is that if a photon goes past *S*, then *S* ends up in a slightly different state.

Let's say that the two possible states of *S* are *Yes* and *No*. The sensitive thingy *S* starts out in state *No*, and ends up in state *Yes* if a photon goes past.

Then the initial configuration is:

"photon heading toward *A*; and *S* in state *No*," $(-1 + 0i)$.

Next, the action of the half-silvered mirror at *A*. In the previous version of this experiment, without the sensitive thingy, the two resultant configurations were "*A* to *B*" with amplitude $-i$ and "*A* to *C*" with amplitude -1 . Now, though, a new element has been introduced into the system, and all configurations are about all particles, and so every configuration mentions the new element. So the amplitude flows from the initial configuration are to:

"photon from *A* to *B*; and *S* in state *No*," $(0 - i)$

"photon from *A* to *C*; and *S* in state *Yes*," $(-1 + 0i)$.

Next, the action of the full mirrors at *B* and *C*:

"photon from *B* to *D*; and *S* in state *No*," $(1 + 0i)$

"photon from *C* to *D*; and *S* in state *Yes*," $(0 - i)$.

And then the action of the half-mirror at *D*, on the amplitude flowing from both of the above configurations:

1. "photon from *D* to *E*; and *S* in state *No*," $(0 + i)$

2. "photon from *D* to *F*; and *S* in state *No*," $(1 + 0i)$

3. "photon from *D* to *E*; and *S* in state *Yes*," $(0 - i)$

4. "photon from *D* to *F*; and *S* in state *Yes*," $(1 + 0i)$.

When we did this experiment without the sensitive thingy, the amplitude flows (1) and (3) of $(0 + i)$ and $(0 - i)$ to the "*D* to *E*" configuration canceled each other out. We were left with no amplitude for a photon going to Detector 1 (way up at the experimental level, we never observe a photon striking Detector 1).

But in this case, the two amplitude flows (1) and (3) are now to distinct configurations; at least one entity, *S*, is in a different state between (1) and (3). The amplitudes don't cancel out.

When we wave our magical squared-modulus-ratio detector over the four final configurations, we find that the squared moduli of all are equal: 25% probability each. Way up at the level of the real world, we find that the photon has an equal chance of striking Detector 1 and Detector 2.

All the above is true, even if we, the researchers, don't care about the state of *S*. Unlike possible worlds, configurations cannot be regrouped on a whim. The laws of *physics* say the two configurations are distinct; it's not a question of how *we* can most conveniently parse up the world.

All the above is true, even if we don't bother to look at the state of *S*. The configurations (1) and (3) are distinct in physics, even if we don't know the distinction.

All the above is true, even if we don't know *S* exists. The configurations (1) and (3) are distinct whether or not *we* have distinct *mental representations* for the two possibilities.

All the above is true, even if we're in space, and *S* transmits a new photon off toward the interstellar void in two distinct directions, depending on whether the photon of interest passed it or not. So that we couldn't *ever* find out whether *S* had been in *Yes* or *No*. The state of *S* would be embodied in the photon transmitted off to nowhere. The lost photon can be an implied invisible, and the state of *S* pragmatically undetectable; but the configurations are still distinct.

(The main reason it *wouldn't* work, is if *S* were nudged, but *S* had an original spread in configuration space that was larger than the nudge. Then you couldn't rely on the nudge to separate the amplitude distribution over configuration space into distinct lumps. In reality, all this takes place within a differentiable amplitude distribution over a continuous configuration space.)

Configurations are not belief states. Their distinctness is an objective fact with experimental consequences. The configurations are distinct even if no one knows the state of *S*; distinct even if no intelligent entity can ever find out. The con-

figurations are distinct so long as at least *one particle* in the universe *anywhere* is in a different position. This is experimentally demonstrable.

Why am I emphasizing this? Because back in the dark ages when no one understood quantum physics ...

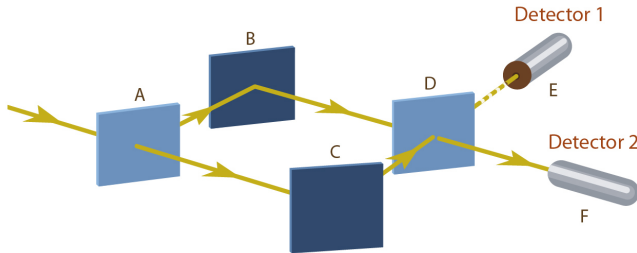


Figure 232.2

Okay, so imagine that you've got no clue what's really going on, and you try the experiment in Figure 232.2, and no photons show up at Detector 1. Cool.

You also discover that when you put a block between *B* and *D*, or a block between *A* and *C*, photons show up at Detector 1 and Detector 2 in equal proportions. But only one at a time—Detector 1 or Detector 2 goes off, not both simultaneously.

So, yes, it *does* seem to you like you're dealing with a particle—the photon is only in one place at one time, every time *you* see it.

And yet there's some kind of ...*mysterious phenomenon*... that prevents the photon from showing up in Detector 1. And this mysterious phenomenon depends on the photon being *able* to go both ways. Even though the photon only shows up in one detector or the other, which shows, *you would think*, that the photon is only in one place at a time.

Which makes the whole pattern of the experiments seem pretty bizarre! After all, the photon either goes from *A* to *C*, or from *A* to *B*; one or the other. (Or so you would think, if you were instinctively trying to break reality down into individually real particles.) But when you block off one course or the other,

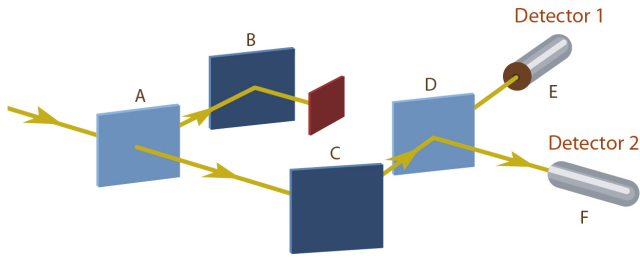


Figure 232.3

as in Figure 232.3, you start getting different experimental results!

It's like the photon wants to be *allowed* to go both ways, even though (you would think) it only goes one way or the other. And it can *tell* if you try to block it off, without actually going *there*—if it'd gone there, it would have run into the block, and not hit any detector at all.

It's as if mere *possibilities* could have causal effects, in defiance of what the word “real” is usually thought to *mean*...

But it's a bit early to jump to conclusions like *that*, when you don't have a complete picture of what goes on inside the experiment.

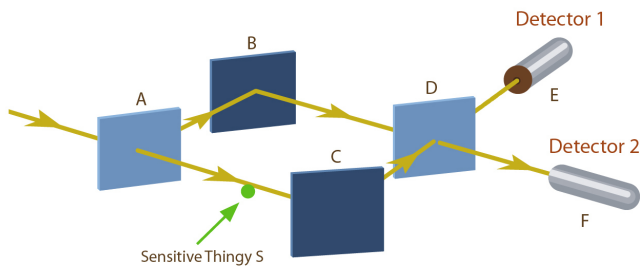


Figure 232.4

So it occurs to you to put a sensor between *A* and *C*, like in Figure 232.4, so you can tell which way the photon *really* goes on each occasion.

And the mysterious phenomenon goes away.

I mean, now how crazy is that? What kind of paranoia does that inspire in some poor scientist?

Okay, so in the twenty-first century we realize in order to “know” a photon’s history, the particles making up your brain have to be correlated with the photon’s history. If having a tiny little sensitive thingy *S* that correlates to the photon’s history is enough to distinguish the final configurations and prevent the amplitude flows from canceling, then an entire sensor with a digital display, never mind a human brain, will put *septillions* of particles in different positions and prevent the amplitude flows from canceling.

But if you hadn’t worked that out yet ...

Then you would ponder the sensor having banished the Mysterious Phenomenon, and think:

The photon doesn’t just want to be *physically* free to go either way. It’s not a little wave going along an unblocked pathway, because then just having a physically unblocked pathway would be enough.

No ...I’m not allowed to *know* which way the photon went.

The mysterious phenomenon ...*doesn’t want me looking at it too closely* ...while it’s doing its mysterious thing.

It’s not *physical possibilities* that have an effect on reality ...only *epistemic possibilities*. If I *know* which way the photon went, it’s no longer *plausible* that it went the other way ...which cuts off the mysterious phenomenon as effectively as putting a block between *B* and *D*.

I have to *not observe* which way the photon went, in order for it to always end up at Detector 2. It has to be *reasonable* that the photon could have gone to either *B* or *C*. What I can *know* is the determining factor, regardless of which physical paths I leave open or closed.

STOP THE PRESSES! MIND IS FUNDAMENTAL AFTER ALL! CONSCIOUS AWARENESS DETERMINES OUR EXPERIMENTAL RESULTS!

You can *still read* this kind of stuff. In *physics textbooks*. Even now, when a majority of theoretical physicists know better. Stop the presses. Please, stop the presses.

Hindsight is 20/20; and so it's easy to say that, in hindsight, there were certain clues that this interpretation was not correct.

Like, if you put the sensor between *A* and *C* *but don't read it*, the mysterious phenomenon *still* goes away, and the photon still sometimes ends up at Detector 1. (Oh, but you *could* have read it, and possibilities are real now ...)

But it doesn't even have to be a *sensor*, a scientific instrument that you built. A single particle that gets nudged far enough will dispel the interference. A photon radiating off to where you'll never see it again can do the trick. Not much human involvement there. Not a whole lot of conscious awareness.

Maybe before you pull the dualist fire alarm on human brains being physically special, you should provide experimental proof that a rock can't play the same role in dispelling the Mysterious Phenomenon as a human researcher?

But that's hindsight, and it's easy to call the shots in hindsight. Do you *really* think you could've done better than John von Neumann, if you'd been alive at the time? The point of this kind of retrospective analysis is to ask what kind of fully general clues you could have followed, and whether there are any similar clues you're ignoring now on current mysteries.

Though it *is* a little embarrassing that even *after* the theory of amplitudes and configurations had been worked out—with the theory now giving the definite prediction that any nudged particle would do the trick—early scientists *still* didn't get it.

But you see ...it had been established as Common Wisdom that configurations were possibilities, it was epistemic possibility that mattered, amplitudes were a very strange sort of partial information, and conscious observation made quantumness go away. And that it was best to avoid thinking too hard about the whole business, so long as your experimental predictions came out right.

Collapse Postulates

Macroscopic decoherence—also known as “many-worlds”—is the idea that the known quantum laws that govern microscopic events simply govern at all levels without alteration. Back when people didn’t know about decoherence—before it occurred to anyone that the laws deduced with such precision for microscopic physics might apply universally—what did people *think* was going on?

The initial reasoning seems to have gone something like:

When my calculations showed an amplitude of $(-1/3)i$ for this photon to get absorbed, my experimental statistics showed that the photon was absorbed around 107 times out of 1,000, which is a good fit to $1/9$, the square of the modulus.

to

The amplitude *is* the probability (by way of the squared modulus).

to

Once you measure something and *know it didn’t happen*, its *probability* goes to zero.

Read literally, this implies that knowledge itself—or even conscious awareness—causes the collapse. Which was in fact the form of the theory put forth by Werner Heisenberg!

But people became increasingly nervous about the notion of importing dualistic language into fundamental physics—as well they should have been! And so the original reasoning was replaced by the notion of an objective “collapse” that destroyed all parts of the wavefunction except one, and was triggered sometime before superposition grew to human-sized levels.

Now, once you’re supposing that parts of the wavefunction can just vanish, you might think to ask:



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Is there only *one* survivor? Maybe there are many surviving worlds, but they survive with a frequency determined by their integrated squared modulus, and so the typical surviving world has experimental statistics that match the Born rule.

Yet collapse theories considered in modern academia only postulate *one* surviving world. Why?

Collapse theories were devised in a time when it *simply didn't occur* to any physicists that more than one world *could* exist! People took for granted that measurements had single outcomes—it was an assumption so deep it was invisible, because it was what they *saw happening*. Collapse theories were devised to explain *why measurements had single outcomes*, rather than (in full generality) *why experimental statistics matched the Born rule*.

For similar reasons, the “collapse postulates” considered academically suppose that collapse occurs *before* any human beings get superposed. But experiments are steadily ruling out the possibility of “collapse” in increasingly large entangled systems. Apparently an experiment is underway to demonstrate quantum superposition at 50-micrometer scales, which is bigger than most neurons and getting up toward the diameter of some human hairs!

So why doesn't someone try jumping ahead of the game, and ask:

Say, we keep having to postulate the collapse occurs steadily later and later. What if collapse occurs only once superposition reaches planetary scales and substantial divergence occurs—say, Earth's wavefunction collapses around once a minute? Then, while the surviving Earths at any given time would *remember* a long history of quantum experiments that matched the Born statistics, a supermajority of those Earths would begin obtaining non-Born results from quantum experiments and then abruptly cease to exist a minute later.

Why don't collapse theories like *that* one have a huge academic following, among the many people who apparently think it's okay for parts of the wavefunction to just vanish? Especially given that experiments are proving superposition in steadily larger systems?

A cynic might suggest that the reason for collapse's continued support isn't the *physical plausibility* of having large parts of the wavefunction suddenly vanish, or the hope of somehow explaining the Born statistics. The point is to keep the intuitive appeal of "I don't remember the measurement having more than one result, therefore only one thing happened; I don't remember splitting, so there must be only one of me." You don't remember dying, so superposed humans must never collapse. A theory that dared to stomp on intuition would be missing the whole point. You might as well just move on to decoherence.

So a cynic might suggest.

But surely it is too early to be attacking the motives of collapse supporters. That is mere argument *ad hominem*. What about the actual physical plausibility of collapse theories?

Well, first: Does any collapse theory have any experimental support? No.

With that out of the way ...

If collapse actually worked the way its adherents say it does, it would be:

1. The only non-linear evolution in all of quantum mechanics.
2. The only non-unitary evolution in all of quantum mechanics.
3. The only non-differentiable (in fact, discontinuous) phenomenon in all of quantum mechanics.
4. The only phenomenon in all of quantum mechanics that is non-local in the configuration space.
5. The only phenomenon in all of physics that violates CPT symmetry.

6. The only phenomenon in all of physics that violates Liouville's Theorem (has a many-to-one mapping from initial conditions to outcomes).
7. The only phenomenon in all of physics that is acausal / non-deterministic / inherently random.
8. The only phenomenon in all of physics that is non-local in spacetime and propagates an influence faster than light.

WHAT DOES THE GOD-DAMNED COLLAPSE POSTULATE HAVE TO *DO* FOR PHYSICISTS TO REJECT IT? KILL A GOD-DAMNED PUPPY?

Decoherence is Simple

An epistle to the physicists:

When I was but a little lad, my father, a PhD physicist, warned me sternly against meddling in the affairs of physicists; he said that it was hopeless to try to comprehend physics without the formal math. Period. No escape clauses. But I had read in Feynman's popular books that if you really understood physics, you ought to be able to explain it to a nonphysicist. I believed Feynman instead of my father, because Feynman had won the Nobel Prize and my father had not.

It was not until later—when I was reading the *Feynman Lectures*, in fact—that I realized that my father had given me the simple and honest truth. No math = no physics.

By vocation I am a Bayesian, not a physicist. Yet although I was raised not to meddle in the affairs of physicists, my hand has been forced by the occasional gross misuse of three terms: *simple*, *falsifiable*, and *testable*.

The foregoing introduction is so that you don't laugh, and say, "Of course I know what those words mean!" There is math here. What follows will be a restatement of the points in *Belief in the Implied Invisible*, as they apply to quantum physics.

Let's begin with the remark that started me down this whole avenue, of which I have seen several versions; paraphrased, it runs:

The many-worlds interpretation of quantum mechanics postulates that there are vast numbers of other worlds, existing alongside our own. Occam's Razor says we should not multiply entities unnecessarily.

Now it must be said, in all fairness, that those who say this will usually also confess:



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But this is not a universally accepted application of Occam's Razor; some say that Occam's Razor should apply to the laws governing the model, not the number of objects inside the model.

So it is good that we are all acknowledging the contrary arguments, and telling both sides of the story—

But suppose you had to *calculate* the simplicity of a theory. The original formulation of William of Ockham stated:¹

Lex parsimoniae: Entia non sunt multiplicanda praeter necessitatem.

“The law of parsimony: Entities should not be multiplied beyond necessity.”

But this is qualitative advice. It is not enough to say whether one theory seems more simple, or seems more complex, than another—you have to assign a number; and the number has to be meaningful, you can't just make it up. Crossing this gap is like the difference between being able to eyeball which things are moving “fast” or “slow,” and starting to measure and calculate velocities.

Suppose you tried saying: “Count the words—that's how complicated a theory is.”

Robert Heinlein once claimed (tongue-in-cheek, I hope) that the “simplest explanation” is always: “The woman down the street is a witch; she did it.” Eleven words—not many physics papers can beat that.

Faced with this challenge, there are two different roads you can take.

First, you can ask: “The woman down the street is a *what*?” Just because English has one word to indicate a concept doesn't mean that the concept itself is simple. Suppose you were talking

1. Edit 2018: Which has not been found in any writing of Occam's that we know of, but he did write *Numquam ponenda est pluralitas sine necessitate* (“Plurality must never be posited without necessity”) and *Frustra fit per plura, quod potest fieri per pauciora*. (“It is pointless to do with more what can be done with fewer.”) sharov.me/r/4-24



to aliens who didn't know about witches, women, or streets—how long would it take you to explain your theory to them? Better yet, suppose you had to write a computer program that embodied your hypothesis, and output what you say are your hypothesis's predictions—how big would that computer program have to be? Let's say that your task is to predict a time series of measured positions for a rock rolling down a hill. If you write a subroutine that simulates witches, this doesn't seem to help narrow down where the rock rolls—the extra subroutine just inflates your code. You might find, however, that your code necessarily includes a subroutine that squares numbers.

Second, you can ask: "The woman down the street is a witch; she did *what*?" Suppose you want to describe some event, as precisely as you possibly can given the evidence available to you—again, say, the distance/time series of a rock rolling down a hill. You can preface your explanation by saying, "The woman down the street is a witch," but your friend then says, "What did she *do*?" and you reply, "She made the rock roll one meter after the first second, nine meters after the third second..." Prefacing your message with "The woman down the street is a witch," doesn't help to *compress* the rest of your description. On the whole, you just end up sending a longer message than necessary—it makes more sense to just leave off the "witch" prefix. On the other hand, if you take a moment to talk about Galileo, you may be able to greatly compress the next five thousand detailed time series for rocks rolling down hills.

If you follow the first road, you end up with what's known as Kolmogorov complexity and Solomonoff induction. If you follow the second road, you end up with what's known as Minimum Message Length.

Ah, so I can pick and choose among definitions of simplicity?

No, actually the two formalisms in their most highly developed forms were proven equivalent.

And I suppose now you're going to tell me that both formalisms come down on the side of "Occam means counting laws, not counting objects."

More or less. In Minimum Message Length, so long as you can tell your friend an exact recipe they can mentally follow to get the rolling rock's time series, we don't care how much mental work it takes to follow the recipe. In Solomonoff induction, we count bits in the program code, not bits of RAM used by the program as it runs. "Entities" are lines of code, not simulated objects. And as said, these two formalisms are ultimately equivalent.

Now before I go into any further detail on formal simplicity, let me digress to consider the objection:

So what? Why can't I just invent my *own* formalism that does things differently? Why should I pay any attention to the way you happened to decide to do things, over in your field? Got any *experimental* evidence that shows I should do things this way?

Yes, actually, believe it or not. But let me start at the beginning.

The conjunction rule of probability theory states:

$$P(X, Y) \leq P(X).$$

For any propositions X and Y , the probability that " X is true, and Y is true," is *less than or equal to* the probability that " X is true (whether or not Y is true)." (If this statement sounds not terribly profound, then let me assure you that it is easy to find cases where human probability assessors violate this rule.)

You usually can't apply the conjunction rule $P(X, Y) \leq P(X)$ directly to a conflict between mutually exclusive hypotheses. The conjunction rule only applies directly to cases where the left-hand-side strictly implies the right-hand-side. Furthermore, the conjunction is just an inequality; it doesn't give us the kind of quantitative calculation we want.

But the conjunction rule does give us a rule of monotonic decrease in probability: as you tack more details onto a story, and each additional detail can potentially be true or false, the story's probability goes down monotonically. Think of probability as a conserved quantity: there's only so much to go around.

As the number of details in a story goes up, the number of possible stories increases exponentially, but the sum over their probabilities can never be greater than 1. For every story “ X and Y ,” there is a story “ X and $\neg Y$.” When you *just* tell the story “ X ,” you get to *sum over* the possibilities Y and $\neg Y$.

If you add ten details to X , each of which could potentially be true or false, then that story must compete with $2^{10} - 1$ other equally detailed stories for precious probability. If on the other hand it suffices to *just* say X , you can sum your probability over 2^{10} stories

$$\begin{aligned} & ((X \text{ and } Y \text{ and } Z \text{ and } \dots) \text{ or} \\ & (X \text{ and } \neg Y \text{ and } Z \text{ and } \dots) \text{ or } \dots). \end{aligned}$$

The “entities” counted by Occam’s Razor should be individually costly in probability; this is why we prefer theories with fewer of them.

Imagine a lottery which sells up to a million tickets, where each possible ticket is sold only once, and the lottery has sold every ticket at the time of the drawing. A friend of yours has bought one ticket for \$1—which seems to you like a poor investment, because the payoff is only \$500,000. Yet your friend says, “Ah, but consider the alternative hypotheses, ‘Tomorrow, someone will win the lottery’ and ‘Tomorrow, I will win the lottery.’ Clearly, the latter hypothesis is simpler by Occam’s Razor; it only makes mention of one person and one ticket, while the former hypothesis is more complicated: it mentions a million people and a million tickets!”

To say that Occam’s Razor only counts laws, and not objects, is not quite correct: what counts against a theory are the entities it must mention *explicitly*, because these are the entities that cannot be *summed over*. Suppose that you and a friend are puzzling over an amazing billiards shot, in which you are told the starting state of a billiards table, and which balls were sunk, but not how the shot was made. You propose a theory which involves ten specific collisions between ten specific balls; your friend counters with a theory that involves five specific collisions between five specific balls. What counts

against your theories is not *just* the laws that you claim to govern billiard balls, but any *specific* billiard balls that had to be in some *particular* state for your model's prediction to be successful.

If you measure the temperature of your living room as 22° Celsius, it does not make sense to say: "Your thermometer is probably in error; the room is much more likely to be 20° C. Because, when you consider all the particles in the room, there are exponentially vastly more states they can occupy if the temperature is really 22° C—which makes any *particular* state all the more improbable." But no matter which exact 22° C state your room occupies, you can make the same prediction (for the supervast majority of these states) that your thermometer will end up showing 22° C, and so you are not sensitive to the *exact* initial conditions. You do not need to specify an exact position of all the air molecules in the room, so that is not counted against the probability of your explanation.

On the other hand—returning to the case of the lottery—suppose your friend won ten lotteries in a row. At this point you should suspect the fix is in. The hypothesis "My friend wins the lottery every time" is more complicated than the hypothesis "Someone wins the lottery every time." But the former hypothesis is predicting the data much more precisely.

In the Minimum Message Length formalism, saying "There is a single person who wins the lottery every time" at the beginning of your message compresses your description of who won the next ten lotteries; you can just say "And that person is Fred Smith" to finish your message. Compare to, "The first lottery was won by Fred Smith, the second lottery was won by Fred Smith, the third lottery was..."

In the Solomonoff induction formalism, the prior probability of "My friend wins the lottery every time" is low, because the program that describes the lottery now needs explicit code that singles out your friend; but because that program can produce a *tighter probability distribution* over potential lottery winners than "Someone wins the lottery every time," it can, by Bayes's Rule, overcome its prior improbability and win out as a hypothesis.

Any formal theory of Occam's Razor should quantitatively define, not only "entities" and "simplicity," but also the "necessity" part.

Minimum Message Length defines necessity as "that which compresses the message."

Solomonoff induction assigns a prior probability to each possible computer program, with the entire distribution, over every possible computer program, summing to no more than 1. This can be accomplished using a binary code where no valid computer program is a prefix of any other valid computer program ("prefix-free code"), e.g. because it contains a stop code. Then the prior probability of any program P is simply $2^{-L(P)}$ where $L(P)$ is the length of P in bits.

The program P itself can be a program that takes in a (possibly zero-length) string of bits and outputs the conditional probability that the *next* bit will be 1; this makes P a probability distribution over all binary sequences. This version of Solomonoff induction, for any string, gives us a mixture of posterior probabilities dominated by the shortest programs that most precisely predict the string. Summing over this mixture gives us a prediction for the next bit.

The upshot is that it takes more Bayesian evidence—more successful predictions, or more precise predictions—to justify more complex hypotheses. But it can be done; the burden of prior improbability is not infinite. If you flip a coin four times, and it comes up heads every time, you don't conclude right away that the coin produces only heads; but if the coin comes up heads twenty times in a row, you should be considering it very seriously. What about the hypothesis that a coin is fixed to produce HTHTTT...in a repeating cycle? That's more bizarre—but after a hundred coinflips you'd be a fool to deny it.

Standard chemistry says that in a gram of hydrogen gas there are six hundred billion trillion hydrogen atoms. This is a startling statement, but there was some amount of evidence that sufficed to convince physicists in general, and you particularly, that this statement was true.

Now ask yourself how much evidence it would take to convince you of a theory with six hundred billion trillion separately specified physical laws.

Why doesn't the prior probability of a program, in the Solomonoff formalism, include a measure of how much RAM the program uses, or the total running time?

The simple answer is, "Because space and time resources used by a program aren't mutually exclusive possibilities." It's not like the program specification, that can only have a 1 or a 0 in any particular place.

But the even simpler answer is, "Because, historically speaking, that heuristic doesn't work."

Occam's Razor was raised as an objection to the suggestion that nebulae were actually distant galaxies—it seemed to vastly multiply the number of entities in the universe. *All those stars!*

Over and over, in human history, the universe has gotten bigger. A variant of Occam's Razor which, on each such occasion, would label the vaster universe as *more unlikely*, would fare less well under humanity's historical experience.

This is part of the "experimental evidence" I was alluding to earlier. While you can justify theories of simplicity on mathy sorts of grounds, it is also desirable that they actually work in practice. (The other part of the "experimental evidence" comes from statisticians / computer scientists / Artificial Intelligence researchers, testing which definitions of "simplicity" let them construct computer programs that do empirically well at predicting future data from past data. Probably the Minimum Message Length paradigm has proven most productive here, because it is a very adaptable way to think about real-world problems.)

Imagine a spaceship whose launch you witness with great fanfare; it accelerates away from you, and is soon traveling at 0.9c. If the expansion of the universe continues, as current cosmology holds it should, there will come some future point where—according to your model of reality—you don't expect to be able to interact with the spaceship even in principle; it has gone over the cosmological horizon relative to you, and

photons leaving it will not be able to outrace the expansion of the universe.

Should you believe that the spaceship literally, physically disappears from the universe at the point where it goes over the cosmological horizon relative to you?

If you believe that Occam's Razor counts the objects in a model, then yes, you should. Once the spaceship goes over your cosmological horizon, the model in which the spaceship instantly disappears, and the model in which the spaceship continues onward, give indistinguishable predictions; they have no Bayesian evidential advantage over one another. But one model contains many fewer "entities"; it need not speak of all the quarks and electrons and fields composing the spaceship. So it is simpler to suppose that the spaceship vanishes.

Alternatively, you could say: "Over numerous experiments, I have generalized certain laws that govern observed particles. The spaceship is made up of such particles. Applying these laws, I deduce that the spaceship should continue on after it crosses the cosmological horizon, with the same momentum and the same energy as before, on pain of violating the conservation laws that I have seen holding in every examinable instance. To suppose that the spaceship vanishes, I would have to add a new law, 'Things vanish as soon as they cross my cosmological horizon.'"

The decoherence (a.k.a. many-worlds) version of quantum mechanics states that measurements obey the same quantum-mechanical rules as all other physical processes. Applying these rules to macroscopic objects in exactly the same way as microscopic ones, we end up with observers in states of superposition. Now there are many questions that can be asked here, such as "But then why don't all binary quantum measurements appear to have 50/50 probability, since different versions of us see both outcomes?"

However, the objection that decoherence violates Occam's Razor on account of multiplying objects in the model is simply wrong.

Decoherence does not require the wavefunction to take on some complicated exact initial state. Many-worlds is not speci-

fyng all its worlds by hand, but generating them via the compact laws of quantum mechanics. A computer program that directly simulates quantum mechanics to make experimental predictions, would require a great deal of RAM to run—but simulating the wavefunction is exponentially expensive in *any* flavor of quantum mechanics! Decoherence is simply more so. *Many* physical discoveries in human history, from stars to galaxies, from atoms to quantum mechanics, have vastly increased the apparent CPU load of what we believe to be the universe.

Many-worlds is not a zillion worlds worth of complicated, any more than the atomic hypothesis is a zillion atoms worth of complicated. For anyone with a quantitative grasp of Occam's Razor that is simply not what the term "complicated" means.

As with the historical case of galaxies, it may be that people have mistaken their *shock* at the notion of a universe that large, for a probability penalty, and invoked Occam's Razor in justification. But if there are probability penalties for decoherence, the *largeness of the implied universe*, per se, is definitely not their source!

The notion that decoherent worlds are additional entities penalized by Occam's Razor is just plain mistaken. It is not sort-of-right. It is not an argument that is weak but still valid. It is not a defensible position that could be shored up with further arguments. It is entirely defective as probability theory. It is not fixable. It is bad math. $2 + 2 = 3$.

Decoherence is Falsifiable and Testable

The words “falsifiable” and “testable” are sometimes used interchangeably, which imprecision is the price of speaking in English. There are two different probability-theoretic qualities I wish to discuss here, and I will refer to one as “falsifiable” and the other as “testable” because it seems like the best fit.

As for the math, it begins, as so many things do, with:

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)}.$$

This is Bayes’s Theorem. I own at least two distinct items of clothing printed with this theorem, so it must be important.

To review quickly, B here refers to an item of evidence, A_i is some hypothesis under consideration, and the A_j are competing, mutually exclusive hypotheses. The expression $P(B|A_i)$ means “the probability of seeing B , if hypothesis A_i is true” and $P(A_i|B)$ means “the probability hypothesis A_i is true, if we see B .”

The mathematical phenomenon that I will call “falsifiability” is the scientifically desirable property of a hypothesis that it should concentrate its probability mass into preferred outcomes, which implies that it must also assign low probability to some un-preferred outcomes; probabilities must sum to 1 and there is only so much probability to go around. Ideally there should be possible observations which would drive down the hypothesis’s probability to nearly zero: There should be things the hypothesis *cannot* explain, conceivable experimental results with which the theory is *not* compatible. A theory that can explain everything prohibits nothing, and so gives us no advice about what to expect.

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)}$$



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In terms of Bayes's Theorem, if there is at least some observation B that the hypothesis A_i can't explain, i.e., $P(B|A_i)$ is tiny, then the numerator $P(B|A_i)P(A_i)$ will also be tiny, and likewise the posterior probability $P(A_i|B)$. Updating on having seen the impossible result B has driven the probability of A_i down to nearly zero. A theory that refuses to make itself vulnerable in this way will need to spread its probability widely, so that it has no holes; it will not be able to strongly concentrate probability into a few preferred outcomes; it will not be able to offer precise advice.

Thus is the rule of science derived in probability theory.

As depicted here, "falsifiability" is something you evaluate by looking at a *single* hypothesis, asking, "How narrowly does it concentrate its probability distribution over possible outcomes? How narrowly does it tell me what to expect? Can it explain some possible outcomes much better than others?"

Is the decoherence interpretation of quantum mechanics *falsifiable*? Are there experimental results that could drive its probability down to an infinitesimal?

Sure: We could measure entangled particles that should always have opposite spin, and find that if we measure them far enough apart, they sometimes have the same spin.

Or we could find apples falling upward, the planets of the Solar System zigging around at random, and an atom that kept emitting photons without any apparent energy source. Those observations would also falsify decoherent quantum mechanics. They're things that, on the hypothesis that decoherent quantum mechanics governs the universe, we should definitely *not expect* to see.

So there do exist observations B whose $P(B|A_{\text{deco}})$ is infinitesimal, which would drive $P(A_{\text{deco}}|B)$ down to an infinitesimal.

But that's just because decoherent quantum mechanics is still quantum mechanics! What about the decoherence part, per se, versus the collapse postulate?

We're getting there. The point is that I just defined a test that leads you to think about one hypothesis at a time (and called it

“falsifiability”). If you want to distinguish decoherence *versus* collapse, you have to think about at least *two* hypotheses at a time.

Now really the “falsifiability” test is not quite *that* singly focused, i.e., the sum in the denominator has got to contain *some* other hypothesis. But what I just defined as “falsifiability” pinpoints the kind of problem that Karl Popper was complaining about, when he said that Freudian psychoanalysis was “unfalsifiable” because it was equally good at coming up with an explanation for every possible thing the patient could do.

If you belonged to an alien species that had never invented the collapse postulate or Copenhagen Interpretation—if the only physical theory you’d *ever* heard of was decoherent quantum mechanics—if *all* you had in your head was the differential equation for the wavefunction’s evolution plus the Born probability rule—you would still have sharp expectations of the universe. You would not live in a magical world where anything was probable.

But you could say exactly the same thing about quantum mechanics *without* (macroscopic) decoherence.

Well, yes! Someone walking around with the differential equation for the wavefunction’s evolution, plus a collapse postulate that obeys the Born probabilities and is triggered before superposition reaches macroscopic levels, still lives in a universe where apples fall down rather than up.

But where does decoherence make a *new* prediction, one that lets us *test* it?

A “new” prediction relative to what? To the state of knowledge possessed by the ancient Greeks? If you went back in time and showed them decoherent quantum mechanics, they would be enabled to make many experimental predictions they could not have made before.

When you say “new prediction,” you mean “new” relative to some other hypothesis that defines the “old prediction.” This

gets us into the theory of what I've chosen to label *testability*; and the algorithm inherently considers at least two hypotheses at a time. You cannot call something a “new prediction” by considering only one hypothesis in isolation.

In Bayesian terms, you are looking for an item of evidence B that will produce evidence for one hypothesis over another, distinguishing between them, and the process of producing this evidence we could call a “test.” You are looking for an experimental result B such that

$$P(B|A_d) \neq P(B|A_c);$$

that is, some outcome B which has a different probability, conditional on the decoherence hypothesis being true, versus its probability if the collapse hypothesis is true. Which in turn implies that the posterior odds for decoherence and collapse will become different from the prior odds:

$$\begin{aligned} \frac{P(B|A_d)}{P(B|A_c)} &\neq 1 \text{ implies} \\ \frac{P(A_d|B)}{P(A_c|B)} &= \frac{P(B|A_d)}{P(B|A_c)} \times \frac{P(A_d)}{P(A_c)} \\ \frac{P(A_d|B)}{P(A_c|B)} &\neq \frac{P(A_d)}{P(A_c)}. \end{aligned}$$

This equation is symmetrical (assuming no probability is literally equal to 0). There isn't one A_j labeled “old hypothesis” and another A_j labeled “new hypothesis.”

This symmetry is a feature, not a bug, of probability theory! If you are designing an artificial reasoning system that arrives at different beliefs depending on the order in which the evidence is presented, this is labeled “hysteresis” and considered a Bad Thing. I hear that it is also frowned upon in Science.

From a probability-theoretic standpoint we have various trivial theorems that say it shouldn't matter whether you update on X first and then Y , or update on Y first and then X . At least they'd be trivial if human beings didn't violate them so often and so lightly.

If decoherence is “untestable” relative to collapse, then so too, collapse is “untestable” relative to decoherence. What if the history of physics had transpired differently—what if Hugh Everett and John Wheeler had stood in the place of Bohr and Heisenberg, and vice versa? Would it then be right and proper for the people of that world to look at the collapse interpretation, and snort, and say, “Where are the *new* predictions?”

What if someday we meet an alien species that invented decoherence before collapse? Are we each bound to keep the theory we invented first? Will Reason have nothing to say about the issue, leaving no recourse to settle the argument but interstellar war?

But if we revoke the requirement to yield new predictions, we are left with scientific chaos. You can add arbitrary untestable complications to old theories, and get experimentally equivalent predictions. If we reject what you call “hysteresis,” how can we defend our current theories against every crackpot who proposes that electrons have a new property called “scent,” just like quarks have “flavor”?

Let it first be said that I quite agree that you should reject the one who comes to you and says: “Hey, I’ve got this brilliant new idea! Maybe it’s not the electromagnetic field that’s tugging on charged particles. Maybe there are tiny little angels who actually push on the particles, and the electromagnetic field just tells them how to do it. Look, I have all these successful experimental predictions—the predictions you used to call your own!”

So yes, I agree that we shouldn’t buy this amazing new theory, but it is not the *newness* that is the problem.

Suppose that human history had developed only slightly differently, with the Church being a primary grant agency for Science. And suppose that when the laws of electromagnetism were first being worked out, the phenomenon of magnetism had been taken as proof of the existence of unseen spirits, of angels. James Clerk becomes Saint Maxwell, who described the laws that direct the actions of angels.

A couple of centuries later, after the Church's power to burn people at the stake has been restrained, someone comes along and says: "Hey, do we really need the angels?"

"Yes," everyone says. "How else would the mere numbers of the electromagnetic field translate into the actual motions of particles?"

"It might be a fundamental law," says the newcomer, "or it might be something other than angels, which we will discover later. What I am suggesting is that interpreting the numbers *as the action of angels* doesn't really add anything, and we should just keep the numbers and throw out the angel part."

And they look one at another, and finally say, "But your theory doesn't make any new experimental predictions, so why should we adopt it? How do we test your assertions about the absence of angels?"

From a normative perspective, it seems to me that if we should reject the crackpot angels in the first scenario, *even without being able to distinguish the two theories experimentally*, then we should also reject the angels of established science in the second scenario, even without being able to distinguish the two theories experimentally.

It is ordinarily the crackpot who adds on new useless complications, rather than scientists who accidentally build them in at the start. But the problem is not that the complications are new, but that they are useless whether or not they are new.

A Bayesian would say that the extra complications of the angels in the theory lead to penalties on the prior probability of the theory. If two theories make equivalent predictions, we keep the one that can be described with the shortest message, the smallest program. If you are evaluating the prior probability of each hypothesis by counting bits of code, and then applying Bayesian updating rules on all the evidence available, then it makes no difference which hypothesis you hear about first, or the order in which you apply the evidence.

It is usually not possible to apply formal probability theory in real life, any more than you can predict the winner of a tennis match using quantum field theory. But if probability theory can

serve as a guide to practice, this is what it says: Reject *useless* complications *in general*, not just when they are *new*.

Yes, and *useless* is precisely what the many worlds of decoherence are! There are supposedly all these worlds alongside our own, and they don't *do* anything to our world, but I'm supposed to believe in them anyway?

No, according to decoherence, what you're supposed to believe are the general laws that govern wavefunctions—and these general laws are very visible and testable.

I have argued elsewhere that the imprimatur of science should be associated with general laws, rather than particular events, because it is the general laws that, in principle, anyone can go out and test for themselves. I assure you that I happen to be wearing white socks right now as I type this. So you are probably *rationaly* justified in believing that this is a historical fact. But it is not the specially strong kind of statement that we canonize as a provisional belief of science, because there is no experiment that you can do for yourself to determine the truth of it; you are stuck with my authority. Now, if I were to tell you the mass of an electron in general, you could go out and find your own electron to test, and thereby see for yourself the truth of the general law in that particular case.

The ability of anyone to go out and verify a general scientific law for themselves, by constructing some particular case, is what makes our belief in the general law specially reliable.

What decoherentists say they believe in is the differential equation that is observed to govern the evolution of wavefunctions—which you can go out and test yourself any time you like; just look at a hydrogen atom.

Belief in the existence of separated portions of the universal wavefunction is not *additional*, and it is not *supposed* to be explaining the price of gold in London; it is just a deductive consequence of the wavefunction's evolution. If the evidence of many particular cases gives you cause to believe that $X \rightarrow Y$ is a general law, and the evidence of some particular case gives you

cause to believe X , then you should have $P(Y) \geq P(X \text{ and } (X \rightarrow Y))$.

Or to look at it another way, if $P(Y|X) \approx 1$, then $P(X \text{ and } Y) \approx P(X)$.

Which is to say, believing extra details doesn't cost you extra probability when they are *logical implications* of general beliefs you already have. Presumably the general beliefs themselves are falsifiable, though, or why bother?

This is why we don't believe that spaceships blink out of existence when they cross the cosmological horizon relative to us. True, the spaceship's continued existence doesn't have an impact on our world. The spaceship's continued existence isn't helping to explain the price of gold in London. But we get the invisible spaceship for free as a consequence of general laws that imply conservation of mass and energy. If the spaceship's continued existence were *not* a deductive consequence of the laws of physics as we presently model them, *then* it would be an additional detail, cost extra probability, and we would have to question why our theory must include this assertion.

The part of decoherence that is supposed to be testable is not the many worlds per se, but just the general law that governs the wavefunction. The decoherentists note that, applied universally, this law implies the existence of entire superposed worlds. Now there are critiques that can be leveled at this theory, most notably, "But then where do the Born probabilities come from?" But within the internal logic of decoherence, the many worlds are not offered as an explanation for anything, nor are they the substance of the theory that is meant to be tested; they are simply a logical consequence of those general laws that constitute the substance of the theory.

If $A \Rightarrow B$ then $\neg B \Rightarrow \neg A$. To deny the existence of superposed worlds is necessarily to deny the universality of the quantum laws formulated to govern hydrogen atoms and every other examinable case; it is this denial that seems to the decoherentists like the extra and untestable detail. You can't see the other parts of the wavefunction—why postulate *additionally* that they don't exist?

The events surrounding the decoherence controversy may be unique in scientific history, marking the first time that serious scientists have come forward and said that by historical accident humanity has developed a powerful, successful, mathematical physical theory that includes angels. That there is an entire law, the collapse postulate, that can simply be thrown away, leaving the theory *strictly* simpler.

To this discussion I wish to contribute the assertion that, in the light of a mathematically solid understanding of probability theory, decoherence is not ruled out by Occam's Razor, nor is it unfalsifiable, nor is it untestable.

We may consider e.g. decoherence and the collapse postulate, side by side, and evaluate critiques such as "Doesn't decoherence definitely predict that quantum probabilities should always be 50/50?" and "Doesn't collapse violate Special Relativity by implying influence at a distance?" We can consider the relative merits of these theories on grounds of their compatibility with experience and the apparent character of physical law.

To assert that decoherence is not even in the game—because the many worlds themselves are "extra entities" that violate Occam's Razor, or because the many worlds themselves are "untestable," or because decoherence makes no "new predictions"—all this is, I would argue, an outright error of probability theory. The discussion should simply discard those particular arguments and move on.

Privileging the Hypothesis



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Suppose that the police of Largeville, a town with a million inhabitants, are investigating a murder in which there are few or no clues—the victim was stabbed to death in an alley, and there are no fingerprints and no witnesses.

Then, one of the detectives says, “Well ...we have no idea who did it ...no particular evidence singling out any of the million people in this city ...but let’s *consider the hypothesis* that this murder was committed by Mortimer Q. Snodgrass, who lives at 128 Ordinary Ln. It *could* have been him, after all.”

I’ll label this *the fallacy of privileging the hypothesis*. (Do let me know if it already has an official name—I can’t recall seeing it described.)

Now the detective may perhaps have some form of rational evidence that is not legal evidence admissible in court—hearsay from an informant, for example. But if the detective does not have *some* justification *already in hand* for promoting Mortimer to the police’s special attention—if the name is pulled entirely out of a hat—then Mortimer’s rights are being violated.

And this is true even if the detective is not claiming that Mortimer “did” do it, but only asking the police to spend time pondering that Mortimer *might* have done it—unjustifiably promoting that particular hypothesis to attention. It’s human nature to look for confirmation rather than disconfirmation. Suppose that three detectives each suggest their hated enemies, as names to be considered; and Mortimer is brown-haired, Frederick is black-haired, and Helen is blonde. Then a witness is found who says that the person leaving the scene was brown-haired. “Aha!” say the police. “We previously had no evidence to distinguish among the possibilities, but *now* we know that Mortimer did it!”

This is related to the principle I’ve started calling “locating the hypothesis,” which is that if you have a billion boxes only

one of which contains a diamond (the truth), and your detectors only provide 1 bit of evidence apiece, then it takes much more evidence to promote the truth to your particular attention—to narrow it down to ten good possibilities, each deserving of our individual attention—than it does to figure out *which* of those ten possibilities is true. It takes 27 bits to narrow it down to ten, and just another 4 bits will give us better than even odds of having the right answer.

Thus the detective, in calling Mortimer to the particular attention of the police, for no reason out of a million other people, is skipping over *most of the evidence* that needs to be supplied against Mortimer.

And the detective ought to have this evidence in their possession, at the first moment when they bring Mortimer to the police's attention *at all*. It may be mere rational evidence rather than legal evidence, but if there's *no evidence* then the detective is harassing and persecuting poor Mortimer.

During my recent diavlog with Scott Aaronson on quantum mechanics,¹ I did manage to corner Scott to the extent of getting Scott to admit that there was no concrete evidence whatsoever that favors a collapse postulate or single-world quantum mechanics. But, said Scott, we might encounter *future* evidence in favor of single-world quantum mechanics, and many-worlds still has the open question of the Born probabilities.²

This is indeed what I would call the fallacy of privileging the hypothesis. There must be a trillion better ways to answer the Born question without adding a collapse postulate that would be the only non-linear, non-unitary, discontinuous, non-differentiable, non-CPT-symmetric, non-local in the configuration space, Liouville's-Theorem-violating, privileged-space-of-

1. [sharov.me/r/4-25](https://www.sharov.me/r/4-25)
Bloggingheads.tv • bloggingheads.tv

2. [sharov.me/r/4-26](https://www.sharov.me/r/4-26)
The Born Probabilities • lesswrong.com



simultaneity-possessing, faster-than-light-influencing, acausal, informally specified law in all of physics. Something that unphysical is not worth *saying out loud* or even *thinking about as a possibility* without a rather large weight of evidence—far more than the current grand total of zero.

But because of a historical accident, collapse postulates and single-world quantum mechanics are indeed on everyone's lips and in everyone's mind to be thought of, and so the open question of the Born probabilities is offered up (by Scott Aaronson no less!) as evidence that many-worlds can't yet offer a complete picture of the world. Which is taken to mean that single-world quantum mechanics is still in the running somehow.

In the minds of human beings, if you can get them to think about this particular hypothesis rather than the trillion other possibilities that are no more complicated or unlikely, you really *have* done a huge chunk of the work of persuasion. Anything thought about is treated as “in the running,” and if other runners seem to fall behind in the race a little, it's assumed that this runner is edging forward or even entering the lead.

And yes, this is just the same fallacy committed, on a much more blatant scale, by the theist who points out that modern science does not offer an absolutely complete explanation of the entire universe, and takes this as evidence for the existence of Jehovah. Rather than Allah, the Flying Spaghetti Monster, or a trillion other gods no less complicated—never mind the space of naturalistic explanations!

To talk about “intelligent design” whenever you point to a purported flaw or open problem in evolutionary theory is, again, privileging the hypothesis—you must have evidence *already in hand* that points to intelligent design *specifically* in order to justify *raising that particular idea to our attention*, rather than a thousand others.

So that's the *sane* rule. And the corresponding anti-epistemology is to talk endlessly of “possibility” and how you “can't disprove” an idea, to hope that future evidence may confirm it without presenting past evidence already in hand, to dwell and dwell on *possibilities* without evaluating

possibly unfavorable evidence, to draw glowing word-pictures of confirming observations that *could* happen but haven't happened *yet*, or to try and show that piece after piece of negative evidence is "not conclusive."

Just as Occam's Razor says that more complicated propositions require more evidence to believe, more complicated propositions also ought to require more work to raise to attention. Just as the principle of burdensome details requires that each part of a belief be separately justified, it requires that each part be separately raised to attention.

As discussed in Perpetual Motion Beliefs, faith and type 2 perpetual motion machines (water → ice cubes + electricity) have in common that they purport to *manufacture improbability from nowhere*, whether the improbability of water forming ice cubes or the improbability of arriving at correct beliefs without observation. Sometimes most of the anti-work involved in manufacturing this improbability is getting us to *pay attention* to an unwarranted belief—thinking on it, dwelling on it. In large answer spaces, attention without evidence is more than halfway to belief without evidence.

Someone who spends all day thinking about whether the *Trinity* does or does not exist, rather than Allah or Thor or the Flying Spaghetti Monster, is more than halfway to Christianity. If leaving, they're less than half departed; if arriving, they're more than halfway there.

An oft-encountered mode of privilege is to try to make uncertainty within a space, slop outside of that space onto the privileged hypothesis. For example, a creationist seizes on some (allegedly) debated aspect of contemporary theory, argues that scientists are *uncertain about evolution*, and then says, "We don't really know which theory is right, so maybe intelligent design is right." But the uncertainty is uncertainty *within* the realm of naturalistic theories of evolution—we have no reason to believe that we'll need to leave that realm to deal with our uncertainty, still *less* that we would jump out of the realm of standard science and land *on Jehovah in particular*. That is privileging the hypothesis—taking doubt *within* a normal space, and

trying to slop doubt *out* of the normal space, onto a privileged (and usually discredited) *extremely* abnormal target.

Similarly, our uncertainty about where the Born statistics come from should be uncertainty *within* the space of quantum theories that are continuous, linear, unitary, slower-than-light, local, causal, naturalistic, et cetera—the usual character of physical law. Some of that uncertainty might slop outside the standard space onto theories that violate *one* of these standard characteristics. It's indeed possible that we might have to think outside the box. But single-world theories violate *all* these characteristics, and there is no reason to privilege that hypothesis.

Living in Many Worlds

Some commenters have recently expressed disturbance at the thought of constantly splitting into zillions of other people, as is the straightforward and unavoidable prediction of quantum mechanics.

Others have confessed themselves unclear as to the implications of many-worlds for planning: If you decide to buckle your seat belt in this world, does that increase the chance of another self unbuckling their seat belt? Are you being selfish at their expense?

Just remember Egan's Law: *It all adds up to normality.*

(After Greg Egan, in *Quarantine*.¹)

Frank Sulloway said:²

Ironically, psychoanalysis has it over Darwinism precisely because its predictions are so outlandish and its explanations are so counterintuitive that we think, *Is that really true? How radical!* Freud's ideas are so intriguing that people are willing to pay for them, while one of the great disadvantages of Darwinism is that we feel we know it already, because, in a sense, we do.

When Einstein overthrew the Newtonian version of gravity, apples didn't stop falling, planets didn't swerve into the Sun. Every new theory of physics must *capture* the successful predictions of the old theory it displaced; it should predict that the sky will be blue, rather than green.

So don't think that many-worlds is there to make strange, radical, exciting predictions. It all adds up to normality.

Then why should anyone care?

1. Greg Egan, *Quarantine* (London: Legend Press, 1992).

2. Robert S. Boynton, "The Birth of an Idea: A Profile of Frank Sulloway," *The New Yorker* (October 1999).



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Because there was once asked the question, fascinating unto a rationalist: *What* all adds up to normality?

And the answer to this question turns out to be: quantum mechanics. It is *quantum mechanics* that adds up to normality.

If there were something else there *instead* of quantum mechanics, *then* the world would look strange and unusual.

Bear this in mind, when you are wondering how to live in the strange new universe of many worlds: *You have always been there.*

Religions, anthropologists tell us, usually exhibit a property called *minimal counterintuitiveness*; they are startling enough to be memorable, but not so bizarre as to be *difficult* to memorize. Anubis has the head of a dog, which makes him memorable, but the rest of him is the body of a man. Spirits can see through walls; but they still become hungry.

But physics is not a religion, set to surprise you just exactly enough to be memorable. The underlying phenomena are so counterintuitive that it takes long study for humans to come to grips with them. But the surface phenomena are entirely ordinary. You will *never* catch a glimpse of another world out of the corner of your eye. You will *never* hear the voice of some other self. That is unambiguously prohibited outright by the laws. Sorry, you're just schizophrenic.

The act of making *decisions* has no special interaction with the process that branches worlds. In your *mind*, in your *imagination*, a decision seems like a branching point where the world could go two different ways. But you would feel just the same uncertainty, visualize just the same alternatives, if there were only one world. That's what people thought for centuries before quantum mechanics, and they still visualized alternative outcomes that could result from their decisions.

Decision and *decoherence* are *entirely orthogonal concepts*. If your brain never became decoherent, then that single cognitive process would still have to imagine different choices and their different outcomes. And a rock, which makes no decisions, obeys the same laws of quantum mechanics as anything else, and splits frantically as it lies in one place.

You don't split *when you come to a decision in particular*, any more than you particularly split when you take a breath. You're just splitting all the time as the result of decoherence, which has nothing to do with choices.

There is a population of worlds, and in each world, it all adds up to normality: apples don't stop falling. In each world, people choose the course that seems best to them. Maybe they happen on a different line of thinking, and see new implications or miss others, and come to a different choice. But it's not that one world chooses each choice. It's not that one version of you chooses what seems best, and another version chooses what seems worst. In each world, apples go on falling and people go on doing what seems like a good idea.

Yes, you can nitpick exceptions to this rule, but they're *normal* exceptions. It all adds up to normality, in all the worlds.

You cannot "choose which world to end up in." In all the worlds, people's choices determine outcomes in the same way they would in just one single world.

The choice you make here does not have some strange balancing influence on some world elsewhere. There is no causal communication between decoherent worlds. In each world, people's choices control the future of that world, not some other world.

If you can imagine decisionmaking in one world, you can imagine decision-making in many worlds: just have the world constantly splitting while otherwise obeying all the same rules.

In no world does two plus two equal five. In no world can spaceships travel faster than light. All the quantum worlds obey our laws of physics; their existence is asserted in the first place by our laws of physics. Since the beginning, not one unusual thing has ever happened, in this or any other world. They are all lawful.

Are there horrible worlds out there, which are utterly beyond your ability to affect? Sure. And horrible things happened during the twelfth century, which are also beyond your ability to affect. But the twelfth century is not your responsibility, because it has, as the quaint phrase goes, "already happened."

I would suggest that you consider every world that is not in your future to be part of the “generalized past.”

Live in your own world. Before you knew about quantum physics, you would not have been tempted to try living in a world that did not seem to exist. Your decisions should add up to this same normality: you shouldn’t try to live in a quantum world you can’t communicate with.

Your decision theory should (almost always) be the same, whether you suppose that there is a 90% probability of something happening, or if it will happen in 9 out of 10 worlds. Now, because people have trouble handling probabilities, it may be helpful to visualize something happening in 9 out of 10 worlds. But this just helps you use normal decision theory.

Now is a good time to begin learning how to shut up and multiply. As I note in *Lotteries: A Waste of Hope*:

The human brain doesn’t do 64-bit floating-point arithmetic, and it can’t devalue the emotional force of a pleasant anticipation by a factor of 0.00000001 without dropping the line of reasoning entirely.

And in *New Improved Lottery*:

Between zero chance of becoming wealthy, and epsilon chance, there is an order-of-epsilon difference. If you doubt this, let epsilon equal one over googolplex.

If you’re thinking about a world that could arise in a lawful way, but whose probability is a quadrillion to one, and something very pleasant or very awful is happening in this world ...well, it does probably exist, if it is lawful. But you should try to release one quadrillionth as many neurotransmitters, in your reward centers or your aversive centers, so that you can weigh that world *appropriately* in your decisions. If you don’t think you can do that ...don’t bother thinking about it.

Otherwise you might as well go out and buy a lottery ticket using a quantum random number, a strategy that is *guaranteed* to result in a very tiny mega-win.

Or here's another way of thinking about it: Are you considering expending some mental energy on a world whose frequency in your future is less than a trillionth? Then go get a 10-sided die from your local gaming store, and, before you begin thinking about that strange world, start rolling the die. If the die comes up 9 twelve times in a row, *then* you can think about that world. Otherwise don't waste your time; thought-time is a resource to be expended wisely.

You can roll the dice as many times as you like, but you can't think about the world until 9 comes up twelve times in a row. Then you can think about it for a minute. After that you have to start rolling the die again.

This may help you to appreciate the concept of "trillion to one" on a more visceral level.

If at any point you catch yourself thinking that quantum physics might have some kind of strange, *abnormal* implication for everyday life—then you should probably stop right there.

Oh, there are a *few* implications of many-worlds for ethics. Average utilitarianism suddenly looks a lot more attractive—you don't need to worry about creating as many people as possible, because there are already plenty of people exploring person-space. You just want the average quality of life to be as high as possible, in the future worlds that are your responsibility.

And you should always take joy in discovery, as long as *you personally* don't know a thing. It is meaningless to talk of being the "first" or the "only" person to know a thing, when everything knowable is known within worlds that are in neither your past nor your future, and are neither before or after you.

But, by and large, it all adds up to normality. If your understanding of many-worlds is the tiniest bit *shaky*, and you are contemplating whether to believe some strange proposition, or feel some strange emotion, or plan some strange strategy, then I can give you very simple advice: Don't.

The quantum universe is not a strange place into which you have been thrust. It is the way things have always been.

Quantum Non-Realism



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Does the moon exist when no one is looking at it?

—Albert Einstein, asked of Niels Bohr

Suppose you were just starting to work out a theory of quantum mechanics.

You begin to encounter experiments that deliver different results depending on how closely you observe them. You dig underneath the reality you know, and find an extremely precise mathematical description that only gives you the relative frequency of outcomes; worse, it's made of complex numbers. Things behave like particles on Monday and waves on Tuesday.

The correct answer is not available to you as a hypothesis, because it will not be invented for another thirty years.

In a mess like that, what's the best you *could* do?

The best you can do is the *strict* “shut up and calculate” interpretation of quantum mechanics. You'll go on *trying* to develop new theories, because doing your best doesn't mean giving up. But we've specified that the correct answer won't be available for thirty years, and that means none of the new theories will really be any good. Doing the *best* you could theoretically do would mean that you recognized that, even as you looked for ways to test the hypotheses.

The best you could theoretically do would *not* include saying anything like, “The wavefunction only gives us probabilities, not certainties.” That, in retrospect, was jumping to a conclusion; the wavefunction gives us a certainty of many worlds existing. So that part about the wavefunction being only a probability was not-quite-right. You calculated, but failed to shut up.

If you do the *best* that you can do without the correct answer being available, then, when you hear about decoherence, it will turn out that you have not said *anything* incompatible with decoherence. Decoherence is not ruled out by the data

and the calculations. So if you refuse to affirm, as positive knowledge, any proposition which was not forced by the data and the calculations, the calculations will not *force* you to say anything incompatible with decoherence. So too with whatever the correct theory may be, if it is not decoherence. If you go astray, it must be from your own impulses.

But it is hard for human beings to shut up and calculate—*really* shut up and calculate. There is an overwhelming tendency to treat our ignorance as if it were positive knowledge.¹

I don't know if any conversations like this ever really took place, but this is how ignorance becomes knowledge:

GALLANT: "Shut up and calculate."

GOOFUS: "Why?"

GALLANT: "Because I don't know what these equations mean, just that they seem to work."

—*five minutes later*—

GOOFUS: "Shut up and calculate."

STUDENT: "Why?"

GOOFUS: "Because these equations don't *mean* anything, they just work."

STUDENT: "Really? How do you know?"

GOOFUS: "Gallant told me."

A similar transformation occurs in the leap from:

GALLANT: "When my calculations show an amplitude of $(-1/3)i$ for this photon to get absorbed, my experiments showed that the photon was absorbed around 107 times out of 1,000, which is a good fit to $1/9$, the square of the modulus. There's clearly some kind of connection between the experimental statistics and the squared modulus of the amplitude, but I don't know what."

1. Edit 2018: There was social pressure to philosophize about quantum physics. For example, Enrico Fermi, who mostly tried to just focus on real physics problems was called a mere "quantum engineer" by Wolfgang Pauli. (David N. Schwartz, *The Last Man Who Knew Everything*, 2017, pages xxii and 36)

GOOFUS: “The probability amplitude doesn’t say where the electron *is*, but where it *might be*. The squared modulus is the probability that reality will turn out that way. Reality *itself* is inherently nondeterministic.”

And again:

GALLANT: “Once I measure something and get an experimental result, I do my future calculations using only the amplitude whose squared modulus went into calculating the frequency of that experimental result. Only this rule makes my further calculations correspond to observed frequencies.”

GOOFUS: “Since the amplitude *is* the probability, once you *know* the experimental result, the probability of everything else becomes zero!”

The whole slip from:

The square of this “amplitude” stuff corresponds tightly to our experimentally observed frequencies

to

The amplitude is the probability of getting the measurement

to

Well, obviously, once you know you didn’t get a measurement, its probability becomes zero

has got to be one of the most embarrassing wrong turns in the history of science.

If you take all this *literally*, it becomes the consciousness-causes-collapse interpretation of quantum mechanics. These days, just about nobody will confess to *actually* believing in the consciousness-causes-collapse interpretation of quantum mechanics—

But the physics textbooks are still written this way! People say they don't believe it, but they *talk as if* knowledge is responsible for removing incompatible "probability" amplitudes.

Yet as implausible as I find consciousness-causes-collapse, it at least gives us a picture of reality. Sure, it's an informal picture. Sure, it gives mental properties ontologically basic status. You can't *calculate* when an "experimental observation" occurs or what people "know," you *just know* when certain probabilities are *obviously* zero. And this "just knowing" just happens to fit your experimental results, whatever they are—

—but at least consciousness-causes-collapse purports to tell us how the universe works. The amplitudes are real, the collapse is real, the consciousness is real.

Contrast to this argument schema:

STUDENT: "Wait, you're saying that this amplitude disappears as soon as the measurement tells me it's not true?"

GOOFUS: "No, no! It doesn't *literally* disappear. The equations don't *mean* anything—they just give good predictions."

STUDENT: "But then what *does* happen?"

GOOFUS: (*Whorble. Hiss.*) "Never ask that question."

STUDENT: "And what about the part where we measure this photon's polarization over here, and a light-year away, the entangled photon's probability of being polarized up-down changes from 50% to 25%?"

GOOFUS: "Yes, what about it?"

STUDENT: "Doesn't that violate Special Relativity?"

GOOFUS: "No, because you're just *finding out* the other photon's polarization. Remember, the amplitudes aren't *real*."

STUDENT: "But Bell's Theorem shows there's no possible local hidden variable that could de-

scribe the other photon's polarization before we measure it—"

GOOFUS: "Exactly! It's meaningless to talk about the photon's polarization before we measure it."

STUDENT: "But the probability suddenly changes—"

GOOFUS: "*It's meaningless to talk about it before we measure it!*"

What does Goofus even *mean*, here? Never mind the plausibility of his words; what sort of state of reality would correspond to his words being true?

What way could reality *be*, that would make it meaningless to talk about Special Relativity being violated, because the property being influenced didn't exist, even though you could calculate the changes to it?

But you know what? Forget that. I want to know the answer to an even more important question:

Where is Goofus *getting* all this stuff?

Let's suppose that you take the Schrödinger equation, and assert, as a positive fact:

This equation generates good predictions, but it doesn't mean anything!

Really? *How do you know?*

I sometimes go around saying that the fundamental question of rationality is *Why do you believe what you believe?*

You say the Schrödinger equation "doesn't mean anything." How did this item of definite knowledge end up in your possession, if it is not simply ignorance misinterpreted as knowledge?

Was there some experiment that told you? I am open to the idea that experiments can tell us things that seem philosophically impossible. But in this case I should like to see the decisive data. Was there a point where you carefully set up an experimental apparatus, and worked out what you should expect to see if (1) the Schrödinger equation was meaningful or (2)

the Schrödinger equation was meaningless; and then you got result (2)?

GALLANT: "If I measure the 90° polarization of a photon, and then measure the 45° polarization, and then measure 90° again, my experimental history shows that in 100 trials a photon was absorbed 47 times and transmitted 53 times."

GOOFUS: "The 90° polarization and 45° polarization are incompatible properties; they can't both *exist* at the same time, and if you measure one, it is meaningless to *talk* about the other."

How do you know?

How did you acquire that piece of knowledge, Goofus? I know where Gallant got *his*—but where did *yours* come from?

My attitude toward questions of existence and meaning was nicely illustrated in a discussion of the current state of evidence for whether the universe is spatially finite or spatially infinite, in which James D. Miller chided Robin Hanson:

Robin, you are suffering from overconfidence bias in assuming that the universe exists. Surely there is some chance that the universe is of size zero.

To which I replied:

James, if the universe doesn't exist, it would still be nice to know whether it's an infinite or a finite universe that doesn't exist.

Ha! You think pulling that old "universe doesn't exist" trick will stop me? It won't even slow me down!

It's not that I'm *ruling out* the possibility that the universe doesn't exist. It's just that, *even if* nothing exists, I still want to understand the nothing as best I can. My curiosity doesn't suddenly go away just because there's no reality, you know!

The nature of "reality" is something about which I'm still confused, which leaves open the possibility that there isn't

any such thing. But Egan's Law still applies: "It all adds up to normality." Apples didn't stop falling when Einstein disproved Newton's theory of gravity.

Sure, when the dust settles, it could turn out that apples don't exist, Earth doesn't exist, reality doesn't exist. But the nonexistent apples will still fall toward the nonexistent ground at a meaningless rate of 9.8 m/s^2 .

You say the universe doesn't exist? Fine, suppose I believe that—though it's not clear *what* I'm supposed to believe, aside from repeating the words.

Now, what happens if I press *this* button?

In The Simple Truth, I said:

Frankly, I'm not entirely sure myself where this "reality" business comes from. I can't create my own reality in the lab, so I must not understand it yet. But occasionally I believe strongly that something is going to happen, and then something else happens instead...So I need different names for the things that determine my predictions and the thingy that determines my experimental results. I call the former thingies "belief," and the latter thingy "reality."

You want to say that the quantum-mechanical equations are "not real"? I'll be charitable, and suppose this means something. What might it mean?

Maybe it means the equations which determine my predictions are substantially different from the thingy that determines my experimental results. Then what *does* determine my experimental results? If you tell me "nothing," I would like to know what sort of "nothing" it is, and why this "nothing" exhibits such apparent regularity in determining e.g. my experimental measurements of the mass of an electron.

I don't take well to people who tell me to stop asking questions. If you tell me something is definitely positively meaningless, I want to know exactly what you mean by that, and how you came to know. Otherwise you have not given me an answer, only told me to stop asking the question.

The Simple Truth describes the life of a shepherd and apprentice who have discovered how to count sheep by tossing pebbles into buckets, when they are visited by a delegate from the court who wants to know how the “magic pebbles” work. The shepherd tries to explain, “An empty bucket is magical if and only if the pastures are empty of sheep,” but is soon overtaken by the excited discussions of the apprentice and the delegate as to how the magic might get into the pebbles.

Here we have quantum equations that deliver excellent experimental predictions. What *exactly* does it mean for them to be “meaningless”? Is it like a bucket of pebbles that *works for counting sheep*, but *doesn’t have any magic*?

Back before Bell’s Theorem ruled out local hidden variables, it seemed possible that (as Einstein thought) there was some more complete description of reality which we didn’t have, and the quantum theory summarized incomplete knowledge of this more complete description. The laws we’d learned would turn out to be like the laws of statistical mechanics: quantitative statements of uncertainty. This would hardly make the equations “meaningless”; partial knowledge *is* the meaning of probability.

But Bell’s Theorem makes it much less plausible that the quantum equations are partial knowledge of something deterministic, the way that statistical mechanics over classical physics is partial knowledge of something deterministic. And even so, the quantum equations would not be “meaningless” as that phrase is usually taken; they would be “statistical,” “approximate,” “partial information,” or at worst “wrong.”

Here we have equations that give us excellent predictions. You say they are “meaningless.” I ask what it is that determines my experimental results, then. You cannot answer. Fine, then how do you justify ruling out the possibility that the quantum equations give such excellent predictions because they are, oh, say, *meaningful*?

I don’t mean to trivialize questions of reality or meaning. But to call something “meaningless” and say that the argument is now resolved, finished, over, done with, you must have a

theory of exactly how it is meaningless. And when the *answer* is given, the question should seem no longer mysterious.

As you may recall from Semantic Stopsigns, there are words and phrases which are not so much *answers* to questions, as cognitive traffic signals which indicate you should *stop asking* questions. “Why does anything exist in the first place? God!” is the classical example, but there are others, such as “Élan vital!”

Tell people to “shut up and calculate” because you don’t know what the calculations mean, and inside of five years, “Shut up!” will be masquerading as a positive theory of quantum mechanics.

I have the *highest* respect for any historical physicists who even came *close* to *actually* shutting up and calculating, who were genuinely conservative in assessing what they did and didn’t know. This is the best they could possibly do without actually being Hugh Everett, and I award them fifty rationality points. My scorn is reserved for those who interpreted “We don’t know why it works” as the positive knowledge that the equations were definitely not real.

I mean, if that trick worked, it would be too good to confine to one subfield. Why shouldn’t physicists use the “not real” loophole *outside* of quantum mechanics?

“Hey, doesn’t your new ‘yarn theory’ violate Special Relativity?”

“Nah, the equations are meaningless. Say, doesn’t your model of ‘chaotic evil inflation’ violate CPT symmetry?”

“My equations are *even more meaningless* than your equations! So your criticism *double* doesn’t count.”

And if that doesn’t work, try writing yourself a Get Out of Jail Free card.

If there is a moral to the whole story, it is the moral of how very hard it is to stay in a state of *confessed* confusion, without making up a story that gives you closure—how hard it is to avoid manipulating your ignorance as if it were definite knowledge that you possessed.

If Many-Worlds Had Come First

Not that I'm claiming I could have done better, if I'd been born into that time, instead of this one ...

Macroscopic decoherence, a.k.a. many-worlds, was first proposed in a 1957 paper by Hugh Everett III.¹ The paper was ignored. John Wheeler told Everett to see Niels Bohr. Bohr didn't take him seriously.

Crushed, Everett left academic physics, invented the general use of Lagrange multipliers in optimization problems, and became a multimillionaire.

It wasn't until 1970, when Bryce DeWitt (who coined the term "many-worlds") wrote an article for *Physics Today*,² that the general field was first informed of Everett's ideas. Macroscopic decoherence has been gaining advocates ever since, and may now be the majority viewpoint (or not).

But suppose that decoherence and macroscopic decoherence had been realized immediately following the discovery of entanglement, in the 1920s. And suppose that no one had proposed collapse theories until 1957. Would decoherence now be steadily declining in popularity, while collapse theories were slowly gaining steam?

Imagine an alternate Earth, where the very first physicist to discover entanglement and superposition said, "Holy flaming monkeys, there's a zillion other Earths out there!"

In the years since, many hypotheses have been proposed to explain the mysterious Born probabilities. But no one has *yet* suggested a collapse postulate. That possibility simply has not occurred to anyone.

One day, Huve Erett walks into the office of Biels Nohr ...



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1. Hugh Everett, III, " 'Relative State' Formulation of Quantum Mechanics." *Rev. Mod. Phys.* 29, 454 – Published 1 July 1957

2. Bryce S. DeWitt, "Quantum mechanics and reality". *Physics Today* 23, 9, 30 (1970)

"I just don't understand," Huve Erett said, "why no one in physics even seems *interested* in my hypothesis. Aren't the Born statistics the greatest puzzle in modern quantum theory?"

Biels Nohr sighed. Ordinarily, he wouldn't even bother, but something about the young man compelled him to try.

"Huve," says Nohr, "every physicist meets dozens of people per year who think they've explained the Born statistics. If you go to a party and tell someone you're a physicist, chances are at least one in ten they've got a new explanation for the Born statistics. It's one of the most famous problems in modern science, and worse, it's a problem that everyone thinks they can understand. To get attention, a new Born hypothesis has to be ...pretty darn good."

"And *this*," Huve says, "*this* isn't good?"

Huve gestures to the paper he'd brought to Biels Nohr. It is a short paper. The title reads, "The Solution to the Born Problem." The body of the paper reads:

When you perform a measurement on a quantum system, all parts of the wavefunction except one point vanish, with the survivor chosen non-deterministically in a way determined by the Born statistics.

"Let me make absolutely sure," Nohr says carefully, "that I understand you. You're saying that we've got this wavefunction—evolving according to the Wheeler-DeWitt equation—and, all of a sudden, the whole wavefunction, except for one part, just spontaneously goes to zero amplitude. Everywhere at once. This happens when, way up at the macroscopic level, we 'measure' something."

"Right!" Huve says.

"So the wavefunction knows when we 'measure' it. What exactly is a 'measurement'? How does the wavefunction know we're here? What happened before humans were around to measure things?"

"Um..." Huve thinks for a moment. Then he reaches out for the paper, scratches out "When you perform a measure-

ment on a quantum system,” and writes in, “When a quantum superposition gets too large.”

Huve looks up brightly. “Fixed!”

“I see,” says Nohr. “And how large is ‘too large’?”

“At the 50-micron level, maybe,” Huve says, “I hear they haven’t tested that yet.”

Suddenly a student sticks his head into the room. “Hey, did you hear? They just verified superposition at the 50-micron level.”

“Oh,” says Huve, “um, whichever level, then. Whatever makes the experimental results come out right.”

Nohr grimaces. “Look, young man, the truth here isn’t going to be comfortable. Can you hear me out on this?”

“Yes,” Huve says, “I just want to know why physicists won’t listen to me.”

“All right,” says Nohr. He sighs. “Look, if this theory of yours were actually true—if whole sections of the wavefunction just instantaneously vanished—it would be ...let’s see. The only law in all of quantum mechanics that is non-linear, non-unitary, non-differentiable and discontinuous. It would prevent physics from evolving locally, with each piece only looking at its immediate neighbors. Your ‘collapse’ would be the only fundamental phenomenon in all of physics with a preferred basis and a preferred space of simultaneity. Collapse would be the only phenomenon in all of physics that violates CPT symmetry, Liouville’s Theorem, and Special Relativity. In your original version, collapse would also have been the only phenomenon in all of physics that was inherently mental. Have I left anything out?”

“Collapse is also the only acausal phenomenon,” Huve points out. “Doesn’t that make the theory more wonderful and amazing?”

“I think, Huve,” says Nohr, “that physicists may view the exceptionalism of your theory as a point not in its favor.”

“Oh,” said Huve, taken aback. “Well, I think I can fix that non-differentiability thing by postulating a second-order term in the—”

“Huve,” says Nohr, “I don’t think you’re getting my point, here. The reason physicists aren’t paying attention to you, is that your theory isn’t physics. It’s magic.”

“But the Born statistics are the greatest puzzle of modern physics, and this theory provides a mechanism for the Born statistics!” Huve protests.

“No, Huve, it doesn’t,” Nohr says wearily. “That’s like saying that you’ve ‘provided a mechanism’ for electromagnetism by saying that there are little angels pushing the charged particles around in accordance with Maxwell’s equations. Instead of saying, ‘Here are Maxwell’s equations, which tells the angels where to push the electrons,’ we just say, ‘Here are Maxwell’s equations’ and are left with a strictly simpler theory. Now, we don’t know *why* the Born statistics happen. But you haven’t given the slightest reason why your ‘collapse postulate’ should eliminate worlds in accordance with the Born statistics, rather than something else. You’re not even making use of the fact that quantum evolution is unitary—”

“That’s because it’s not,” interjects Huve.

“—which everyone pretty much knows has got to be the key to the Born statistics, somehow. Instead you’re merely saying, ‘Here are the Born statistics, which tell the collapse how to eliminate worlds,’ and it’s strictly simpler to just say ‘Here are the Born statistics.’”

“But—” says Huve.

“*Also*,” says Nohr, raising his voice, “you’ve given no justification for why there’s only *one* surviving world left by the collapse, or why the collapse happens before any *humans* get superposed, which makes your theory *really suspicious* to a modern physicist. This is exactly the sort of untestable hypothesis that the ‘One Christ’ crowd uses to argue that we should ‘teach the controversy’ when we tell high school students about other Earths.”

“I’m not a One-Christer!” protests Huve.

“Fine,” Nohr says, “then *why* do you just assume there’s only one world left? And that’s not the only problem with your theory. Which part of the wavefunction gets eliminated, exactly? And in which basis? It’s clear that the whole wavefunction

isn't being compressed down to a delta, or ordinary quantum computers couldn't stay in superposition when any collapse occurred anywhere—heck, ordinary molecular chemistry might start failing—

Huve quickly crosses out “one point” on his paper, writes in “one part,” and then says, “Collapse doesn't compress the wavefunction down to one point. It eliminates all the amplitude *except* one world, but leaves *all* the amplitude *in* that world.”

“Why?” says Nohr. “In principle, once you postulate ‘collapse,’ then ‘collapse’ could eliminate any part of the wavefunction, anywhere—why just one neat world left? Does the collapse *know we're in here?*”

Huve says, “It leaves one whole world because that's what fits our experiments.”

“Huve,” Nohr says patiently, “the term for that is ‘post hoc.’ Furthermore, decoherence is a continuous process. If you partition by whole brains with distinct neurons firing, the partitions have almost zero mutual interference within the wavefunction. But plenty of other processes overlap a great deal. There's no possible way you can point to ‘one world’ and eliminate everything else without making completely arbitrary choices, including an arbitrary choice of basis—”

“But—” Huve says.

“And *above all*,” Nohr says, “the *reason* you can't tell me which part of the wavefunction vanishes, or exactly when it happens, or exactly what triggers it, is that if we did adopt this theory of yours, it would be *the only informally specified, qualitative fundamental law* taught in all of physics. Soon no two physicists anywhere would agree on the exact details! Why? Because it would be the *only fundamental law in all of modern physics that was believed without experimental evidence to nail down exactly how it worked.*”

“What, really?” says Huve. “I thought a lot of physics was more informal than that. I mean, weren't you just talking about how it's impossible to point to ‘one world’?”

“That's because worlds aren't *fundamental*, Huve! We have massive experimental evidence underpinning the fundamental law, the Wheeler-DeWitt equation, that we use to describe the

evolution of the wavefunction. We just apply exactly the same equation to get our description of macroscopic decoherence. But for difficulties of calculation, the equation would, in principle, tell us *exactly* when macroscopic decoherence occurred. We don't know where the Born statistics come from, but we have massive evidence for what the Born statistics *are*. But when I ask you when, or where, collapse occurs, you don't know—*because there's no experimental evidence whatsoever to pin it down*. Huve, even if this 'collapse postulate' worked the way you say it does, *there's no possible way you could know it!* Why not a gazillion other equally magical possibilities?"

Huve raises his hands defensively. "I'm not saying my theory should be taught in the universities as accepted truth! I just want it experimentally tested! Is that so wrong?"

"You haven't specified when collapse happens, so I can't construct a test that falsifies your theory," says Nohr. "Now with that said, we're already looking experimentally for any part of the quantum laws that change at increasingly macroscopic levels. Both on general principles, in case there's something in the 20th decimal point that only shows up in macroscopic systems, and also in the hopes we'll discover something that sheds light on the Born statistics. We check decoherence times as a matter of course. But we keep a *broad* outlook on what might be different. Nobody's going to privilege your non-linear, non-unitary, non-differentiable, non-local, non-CPT-symmetric, non-relativistic, a-frikkin'-causal, faster-than-light, *in-bloody-formal* 'collapse' when it comes to looking for clues. Not until they see absolutely unmistakable evidence. And believe me, Huve, it's going to take a hell of a lot of evidence to un mistake *this*. Even if we did find anomalous decoherence times, and I don't think we will, it wouldn't force your 'collapse' as the explanation."

"What?" says Huve. "Why not?"

"Because there's got to be a billion more explanations that are more plausible than violating Special Relativity," says Nohr. "Do you realize that if this really happened, there would only be a *single* outcome when you measured a photon's polarization? Measuring one photon in an entangled pair would influence

the other photon a light-year away. Einstein would have a heart attack.”

“It doesn’t *really* violate Special Relativity,” says Huve. “The collapse occurs in exactly the right way to prevent you from ever actually *detecting* the faster-than-light influence.”

“That’s not a point in your theory’s favor,” says Nohr. “Also, Einstein would still have a heart attack.”

“Oh,” says Huve. “Well, we’ll say that the relevant aspects of the particle *don’t exist* until the collapse occurs. If something doesn’t exist, influencing it doesn’t violate Special Relativity—”

“You’re just digging yourself deeper. Look, Huve, as a general principle, theories that are actually *correct* don’t generate this level of confusion. But above all, there isn’t any evidence for it. You have no logical way of knowing that collapse occurs, and no reason to believe it. You made a mistake. Just say ‘oops’ and get on with your life.”

“But they *could* find the evidence someday,” says Huve.

“I can’t think of what evidence could determine *this particular* one-world hypothesis as an explanation, but in any case, right now we *haven’t* found any such evidence,” says Nohr. “We haven’t found anything even vaguely suggestive of it! You can’t update on evidence that could theoretically arrive someday but hasn’t arrived! Right now, today, there’s no reason to spend valuable time thinking about this rather than a billion other equally magical theories. There’s absolutely nothing that justifies your belief in ‘collapse theory’ any *more* than believing that someday we’ll learn to transmit faster-than-light messages by tapping into the acausal effects of praying to the Flying Spaghetti Monster!”

Huve draws himself up with wounded dignity. “You know, if my theory is wrong—and I do admit it might be wrong—”

“*If?*” says Nohr. “*Might?*”

“If, I say, my theory is wrong,” Huve continues, “then somewhere out there is another world where *I* am the famous physicist and *you* are the lone outcast!”

Nohr buries his head in his hands. “Oh, not this again. Haven’t you heard the saying, ‘Live in your own world’? And *you* of all people—”

“Somewhere out there is a world where the vast majority of physicists believe in collapse theory, and no one has even *suggested* macroscopic decoherence over the last thirty years!”

Nohr raises his head, and begins to laugh.

“What’s so funny?” Huve says suspiciously.

Nohr just laughs harder. “Oh, my! Oh, my! You really think, Huve, that there’s a world out there where they’ve known about quantum physics for thirty years, and nobody has even *thought* there might be more than one world?”

“Yes,” Huve says, “that’s exactly what I think.”

“Oh my! So you’re saying, Huve, that physicists detect superposition in microscopic systems, and work out quantitative equations that govern superposition in every single instance they can test. And for thirty years, not *one person* says, ‘Hey, I wonder if these laws happen to be universal.’ ”

“Why should they?” says Huve. “Physical models sometimes turn out to be wrong when you examine new regimes.”

“But to not even *think* of it?” Nohr says incredulously. “You see apples falling, work out the law of gravity for all the planets in the solar system except Jupiter, and it doesn’t even *occur* to you to apply it to Jupiter because Jupiter is too large? That’s like, like some kind of comedy routine where the guy opens a box, and it contains a spring-loaded pie, so the guy opens another box, and it contains another spring-loaded pie, and the guy just keeps doing this without even *thinking* of the possibility that the next box contains a pie too. You think John von Neumann, who may have been the highest-g human in history, wouldn’t think of it?”

“That’s right,” Huve says, “He wouldn’t. Ponder that.”

“This is the world where my good friend Ernest formulates his Schrödinger’s Cat thought experiment, and in this world, the thought experiment goes: ‘Hey, suppose we have a radioactive particle that enters a superposition of decaying and not decaying. Then the particle interacts with a sensor, and the sensor goes into a superposition of going off and not going off. The sensor interacts with an explosive, that goes into a superposition of exploding and not exploding; which interacts with the cat, so the cat goes into a superposition of being alive

and dead. Then a human looks at the cat,' and at this point Schrödinger stops, and goes, 'gee, I just can't imagine what could happen next.' So Schrödinger shows this to everyone else, and they're also like 'Wow, I got no idea what could happen at this point, what an amazing paradox.' Until finally *you* hear about it, and you're like, 'hey, maybe at *that* point half of the superposition just vanishes, at random, faster than light,' and everyone else is like, 'Wow, what a great idea!'"

"That's right," Huve says again. "It's got to have happened somewhere."

"Huve, this is a world where every single physicist, and probably the whole damn human species, is too dumb to sign up for cryonics! We're talking about the Earth where George W. Bush is President."

Where Philosophy Meets Science



discussion
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Looking back on early quantum physics—not for purposes of admonishing the major figures, or to claim that we could have done better if we’d been born into that era, but in order to try and learn a moral, and do better next time—looking back on the dark ages of quantum physics, I say, I would nominate as the “most basic” error ...

...*not* that they tried to reverse course on the last three thousand years of science suggesting that mind was complex within physics rather than fundamental in physics. This is Science, and we do have revolutions here. Every now and then you’ve got to reverse a trend. The future is always absurd and never unlawful.

I would nominate, as the basic error not to repeat next time, that the early scientists forgot that they *themselves* were made out of particles.

I mean, I’m sure that most of them knew it in theory.

And yet they didn’t notice that putting a sensor to detect a passing electron, or even *knowing* about the electron’s history, was an example of “particles in different places.” So they didn’t notice that a quantum theory of distinct configurations already explained the experimental result, without any need to invoke consciousness.

In the ancestral environment, humans were often faced with the adaptively relevant task of predicting other humans. For which purpose you thought of your fellow humans as having thoughts, knowing things and feeling things, rather than thinking of them as being made up of particles. In fact, many hunter-gatherer tribes may not even have known that particles existed. It’s much more *intuitive*—it *feels* simpler—to think about someone “knowing” something, than to think about their brain’s particles occupying a different state. It’s easier to phrase your explanations in terms of what *people know*; it feels more natural; it leaps more readily to mind.

Just as, once upon a time, it was easier to imagine Thor throwing lightning bolts, than to imagine Maxwell's Equations—even though Maxwell's Equations can be described by a computer program vastly smaller than the program for an intelligent agent like Thor.

So the ancient physicists found it natural to think, "I know where the photon was ...what difference could *that* make?" Not, "My brain's particles' current state correlates to the photon's history ...what difference could *that* make?"

And, similarly, because it felt easy and intuitive to model reality in terms of people knowing things, and the decomposition of knowing into brain states did not leap so readily to mind, it *seemed* like a simple theory to say that a configuration could have amplitude only "if you didn't know better."

To turn the dualistic quantum hypothesis into a *formal* theory—one that could be written out as a computer program, without human scientists deciding when an "observation" occurred—you would have to specify what it meant for an "observer" to "know" something, in terms your computer program could compute.

So is your theory of fundamental physics going to examine all the particles in a human brain, and decide when those particles "know" something, in order to compute the motions of particles? But then how do you compute the motion of the particles in the brain itself? Wouldn't there be a potential infinite recursion?

But so long as the terms of the theory were being processed by human scientists, they *just knew* when an "observation" had occurred. You said an "observation" occurred whenever it had to occur in order for the experimental predictions to come out right—a subtle form of constant tweaking.

(Remember, the basics of quantum theory were formulated before Alan Turing said anything about Turing machines, and *way* before the concept of computation was popularly known. The distinction between an effective formal theory, and one that required human interpretation, was not as clear then as now. Easy to pinpoint the problems in hindsight; you shouldn't learn the lesson that problems are usually this obvious in foresight.)

Looking back, it may *seem* like one meta-lesson to learn from history, is that philosophy really matters in science—it's not just some adjunct of a separate academic field.

After all, the early quantum scientists were doing all the right experiments. It was their interpretation that was off. And the problems of interpretation were not the result of their getting the statistics wrong.

Looking back, it seems like the errors they made were errors in the kind of thinking that we would describe as, well, "philosophical."

When we look back and ask, "How could the early quantum scientists have done better, even in principle?" it seems that the insights they needed were philosophical ones.

And yet it wasn't professional philosophers who swooped in and solved the problem and cleared up the mystery and made everything normal again. It was, well, physicists.

Arguably, Leibniz was at least as foresightful about quantum physics, as Democritus was once thought to have been foresightful about atoms. But that is hindsight. It's the result of looking at the solution, and thinking back, and saying, "Hey, Leibniz said something like that."

Even where one philosopher gets it right in advance, it's usually science that ends up *telling us* which philosopher is right—not the *prior consensus* of the philosophical community.

I think this has something fundamental to say about the nature of philosophy, and the interface between philosophy and science.

It was once said that every science begins as philosophy, but then grows up and leaves the philosophical womb, so that at any given time, "Philosophy" is what we haven't turned into science yet.

I suggest that when we look at the history of quantum physics and say, "The insights they needed were philosophical insights," what we are *really* seeing is that the insight they needed was of a form that is not yet taught in standardized academic classes, and not yet reduced to calculation.

Once upon a time, the notion of the scientific method—updating beliefs based on experimental evidence—was a philosophi-

cal notion. But it was not championed by professional philosophers. It was the real-world power of science that showed that scientific epistemology was good epistemology, not a prior consensus of philosophers.

Today, this philosophy of belief-updating is *beginning* to be reduced to calculation—statistics, Bayesian probability theory.

But back in Galileo's era, it was solely *vague verbal arguments* that said you should try to produce numerical predictions of experimental results, rather than consulting the Bible or Aristotle.

At the frontier of science, and especially at the frontier of scientific *chaos* and scientific *confusion*, you find problems of thinking that are not taught in academic courses, and that have not been reduced to calculation. And this will seem like a domain of philosophy; it will seem that you must do philosophical thinking in order to sort out the confusion. But when history looks back, I'm afraid, it is usually not a professional philosopher who wins all the marbles—because it takes intimate involvement with the scientific domain in order to do the philosophical thinking. Even if, afterward, it all seems knowable a priori; and even if, afterward, some philosopher out there actually *got* it a priori; even so, it takes intimate involvement to see it in practice, and experimental results to tell the world which philosopher won.

I suggest that, like ethics, philosophy really is important, but it is only practiced effectively from *within* a science. Trying to do the philosophy of a frontier science, as a separate academic profession, is as much a mistake as trying to have separate ethicists. You end up with ethicists who speak mainly to other ethicists, and philosophers who speak mainly to other philosophers.

This is not to say that there is no place for professional philosophers in the world. Some problems are so chaotic that there is no established place for them at all in the halls of science. But those "professional philosophers" would be very, very wise to learn every scrap of relevant-seeming science that they can possibly get their hands on. They should not be surprised at the prospect that experiment, and not debate, will finally settle

the argument. They should not flinch from running their own experiments, if they can possibly think of any.

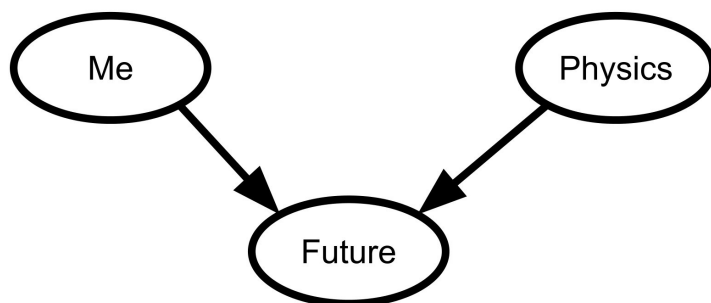
That, I think, is the lesson of history.

Thou Art Physics

Three months ago—jeebers, has it really been that long?—I posed the following homework assignment: Do a stack trace of the human cognitive algorithms that produce debates about “free will.” Note that this task is strongly distinguished from arguing that free will does or does not exist.

Now, as expected, people are asking, “If the future is determined, how can our choices control it?” The wise reader can guess that it all adds up to normality; but this leaves the question of *how*.

People hear: “The universe runs like clockwork; physics is deterministic; the future is fixed.” And their minds form a causal network that looks like this:



Here we see the causes “Me” and “Physics,” competing to determine the state of the “Future” effect. If the “Future” is fully determined by “Physics,” then obviously there is no room for it to be affected by “Me.”

This causal network is not an explicit philosophical belief. It’s implicit—a background representation of the brain, controlling which philosophical arguments seem “reasonable.” It just seems like the way things *are*.

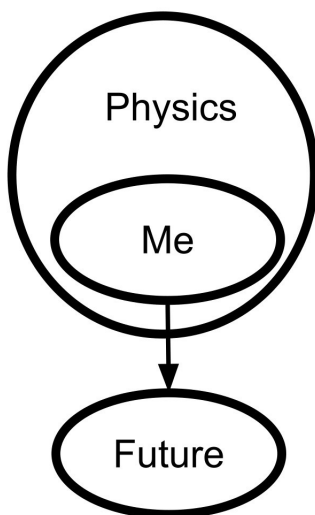


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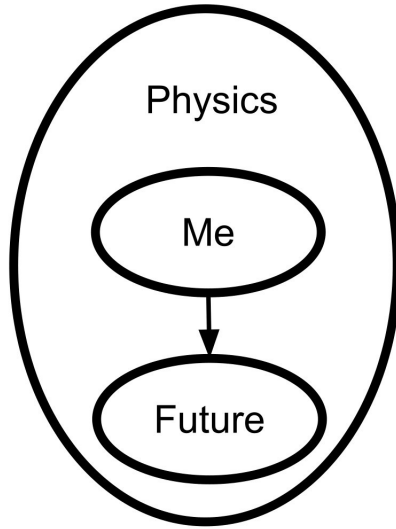
Every now and then, another neuroscience press release appears, claiming that, because researchers used an fMRI to spot the brain doing something-or-other during a decision process, *it's not you who chooses, it's your brain*.

Likewise that old chestnut, “Reductionism undermines rationality itself. Because then, every time you said something, it wouldn’t be the result of *reasoning* about the evidence—it would be merely quarks bopping around.”

Of course the actual diagram should be:



Or better yet:



Why is this not obvious? Because there are many levels of organization that separate our models of our thoughts—our emotions, our beliefs, our agonizing indecisions, and our final choices—from our models of electrons and quarks.

We can *intuitively* visualize that a hand is made of fingers (and thumb and palm). To ask whether it's *really* our hand that picks something up, or *merely* our fingers, thumb, and palm, is transparently a wrong question.

But the gap between physics and cognition cannot be crossed by direct visualization. No one can *visualize* atoms making up a person, the way they can see fingers making up a hand.

And so it requires *constant vigilance* to maintain your perception of yourself as an entity *within physics*.

This vigilance is one of the great keys to philosophy, like the Mind Projection Fallacy. You will recall that it is this point which I nominated as having tripped up the quantum physicists who failed to imagine macroscopic decoherence; they did not think to apply the laws to *themselves*.

Beliefs, desires, emotions, morals, goals, imaginations, anticipations, sensory perceptions, fleeting wishes, ideals, temptations ...You might call this the “surface layer” of the mind, the

parts-of-self that people can see even without science. If I say, “It is not *you* who determines the future, it is your *desires, plans, and actions* that determine the future,” you can readily see the part-whole relations. It is immediately visible, like fingers making up a hand. There are other part-whole relations all the way down to physics, but they are not immediately visible.

“Compatibilism” is the philosophical position that “free will” can be intuitively and satisfyingly defined in such a way as to be compatible with deterministic physics. “Incompatibilism” is the position that free will and determinism are incompatible.

My position might perhaps be called “Requiredism.” When agency, choice, control, and moral responsibility are cashed out in a sensible way, they *require* determinism—at least some patches of determinism within the universe. If you choose, and plan, and act, and bring some future into being, in accordance with your desire, then all this requires a lawful sort of reality; you cannot do it amid utter chaos. There must be order over at least those parts of reality that are being controlled by you. *You* are within physics, and so you/physics have determined the future. If it were not determined by physics, it could not be determined by you.

Or perhaps I should say, “If the future were not determined by reality, it could not be determined by you,” or “If the future were not determined by something, it could not be determined by you.” You don’t need neuroscience or physics to push naive definitions of free will into incoherence. If the mind were not embodied in the brain, it would be embodied in something else; there would be *some real thing* that was a mind. If the future were not determined by physics, it would be determined by *something*, some law, some order, some grand reality that included you within it.

But if the laws of physics control us, then how can we be said to control ourselves?

Turn it around: If the laws of physics did *not* control us, how could we possibly control ourselves?

How could thoughts judge other thoughts, how could emotions conflict with each other, how could one course of action

appear best, how could we pass from uncertainty to certainty about our own plans, in the midst of utter chaos?

If we were not in reality, where could we be?

The future is determined by physics. What kind of physics? The kind of physics that includes the actions of human beings.

People's choices are determined by physics. What kind of physics? The kind of physics that includes weighing decisions, considering possible outcomes, judging them, being tempted, following morals, rationalizing transgressions, trying to do better ...

There is no point where a quark swoops in from Pluto and overrides all this.

The thoughts of your decision process are all *real*, they are all *something*. But a thought is too big and complicated to be an atom. So thoughts are made of smaller things, and our name for the stuff that stuff is made of is "physics."

Physics underlies our decisions and includes our decisions. It does not explain them *away*.

Remember, physics adds up to normality; it's your cognitive algorithms that generate confusion.

Many Worlds, One Best Guess



discussion
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If you look at many microscopic physical phenomena—a photon, an electron, a hydrogen atom, a laser—and a million other known experimental setups—it is possible to come up with simple laws that seem to govern all small things (so long as you don’t ask about gravity). These laws govern the evolution of a highly abstract and mathematical object that I’ve been calling the “amplitude distribution,” but which is more widely referred to as the “wavefunction.”

Now there are gruesome questions about the proper generalization that covers all these tiny cases. Call an object “grue” if it appears green before January 1, 2020 and appears blue thereafter. If all emeralds examined so far have appeared green, is the proper generalization, “Emeralds are green” or “Emeralds are grue”?

The answer is that the proper generalization is “Emeralds are green.” I’m not going to go into the arguments at the moment. It is not the subject of this essay, and the obvious answer in this case happens to be correct. The true Way is not stupid: however clever you may be with your logic, it should finally arrive at the right answer rather than a wrong one.

In a similar sense, the *simplest* generalizations that would cover observed *microscopic* phenomena alone take the form of “All electrons have spin $1/2$ ” and not “All electrons have spin $1/2$ before January 1, 2020” or “All electrons have spin $1/2$ unless they are part of an entangled system that weighs more than 1 gram.”

When we turn our attention to macroscopic phenomena, our sight is obscured. We cannot experiment on the wavefunction of a human in the way that we can experiment on the wavefunction of a hydrogen atom. In no case can you actually read off the wavefunction with a little quantum scanner. But in the case of, say, a human, the size of the entire organism defeats our ability to perform precise calculations or precise

experiments—we cannot confirm that the quantum equations are being obeyed *in precise detail*.

We know that phenomena commonly thought of as “quantum” do not just disappear when many microscopic objects are aggregated. Lasers put out a flood of coherent photons, rather than, say, doing something completely different. Atoms have the chemical characteristics that quantum theory says they should, enabling them to aggregate into the stable molecules making up a human.

So in one sense, we have a great deal of evidence that quantum laws are aggregating to the macroscopic level without too much difference. Bulk chemistry still works.

But we cannot directly verify that the particles making up a human have an aggregate wavefunction that behaves *exactly* the way the simplest quantum laws say. Oh, we know that molecules and atoms don’t disintegrate, we know that macroscopic mirrors still reflect from the middle. We can get *many* high-level predictions from the assumption that the microscopic and the macroscopic are governed by the same laws, and every prediction tested has come true.

But if someone were to claim that the macroscopic quantum picture differs from the microscopic one in some as-yet-untestable detail—something that only shows up at the unmeasurable 20th decimal place of microscopic interactions, but aggregates into something bigger for macroscopic interactions—well, we can’t *prove* they’re wrong. It is Occam’s Razor that says, “There are zillions of new fundamental laws you could postulate in the 20th decimal place; why are you even *thinking* about this one?”¹

If we calculate using the simplest laws which govern all known cases, we find that humans end up in states of quantum superposition, just like photons in a superposition of reflecting from and passing through a half-silvered mirror. In the Schrödinger’s Cat setup, an unstable atom goes into a super-

1. Edit 2018: Gravity is a force that shows up at less than the 20th decimal place in microscopic interactions, but there is some observable evidence for it.

position of disintegrating, and not-disintegrating. A sensor, tuned to the atom, goes into a superposition of triggering and not-triggering. (Actually, the superposition is now a joint state of [atom-disintegrated $\times$ sensor-triggered] + [atom-stable $\times$ sensor-not-triggered].) A charge of explosives, hooked up to the sensor, goes into a superposition of exploding and not exploding; a cat in the box goes into a superposition of being dead and alive; and a human, looking inside the box, goes into a superposition of throwing up and being calm. The same law at all levels.

Human beings who interact with superposed systems will themselves evolve into superpositions. But the brain that sees the exploded cat, and the brain that sees the living cat, will have many neurons firing differently, and hence many *many* particles in different positions. They are very distant in the configuration space, and will communicate to an exponentially infinitesimal degree. Not the 30th decimal place, but the 10^{30} th decimal place. No particular mind, no particular cognitive causal process, sees a blurry superposition of cats.

The fact that “you” only seem to see the cat alive, *or* the cat dead, is exactly what the simplest quantum laws predict. So we have no reason to believe, from our experience so far, that the quantum laws are in any way different at the macroscopic level than the microscopic level.

And physicists have verified superposition at steadily larger levels. Apparently an effort is currently underway to test superposition in a 50-micron object, larger than most neurons.

The existence of other versions of ourselves, and indeed other Earths, is not supposed *additionally*. We are simply supposing that the same laws govern at all levels, having no reason to suppose differently, and all experimental tests having succeeded so far. The existence of other decoherent Earths is a *logical consequence* of the simplest generalization that fits all known facts. If you think that Occam’s Razor says that the other worlds are “unnecessary entities” being multiplied, then you should check the probability-theoretic math; that is just not how Occam’s Razor works.

Yet there is one particular puzzle that seems odd in trying to extend microscopic laws universally, including to superposed humans:

If we try to get probabilities by counting the number of distinct observers, then there is no *obvious* reason why the integrated squared modulus of the wavefunction should correlate with statistical experimental results. There is no known reason for the Born probabilities, and it even seems that, a priori, we would expect a 50/50 probability of any binary quantum experiment going both ways, if we just counted observers.

Robin Hanson suggests that if exponentially tinier-than-average decoherent blobs of amplitude (“worlds”) are interfered with by exponentially tiny leakages from larger blobs, we will get the Born probabilities back out. I consider this an interesting possibility, because it is so normal.

(I myself have had recent thoughts along a different track: If I try to count observers the obvious way, I get strange-seeming results in general, not just in the case of quantum physics. If, for example, I split my brain into a trillion similar parts, conditional on winning the lottery while anesthetized; allow my selves to wake up and perhaps differ to small degrees from each other; and then merge them all into one self again; then counting observers the obvious way says I should be able to make myself win the lottery (if I can split my brain and merge it, as an uploaded mind might be able to do).

In this connection, I find it very interesting that the Born rule does *not* have a split-remerge problem. Given unitary quantum physics, Born’s rule is the *unique* rule that prevents “observers” from having psychic powers—which doesn’t *explain* Born’s rule, but is certainly an *interesting fact*. Given Born’s rule, even splitting and remerging worlds would still lead to consistent probabilities. Maybe physics uses better anthropics than I do!

Perhaps I should take my cues from physics, instead of trying to reason it out a priori, and see where that leads me? But I have not been led anywhere *yet*, so this is hardly an “answer.”)

Wallace,² Deutsch,³ and others try to derive Born's Rule from decision theory. I am rather suspicious of this, because it seems like there is a component of "What happens to me?" that I cannot alter by modifying my utility function. Even if I didn't *care* at all about worlds where I didn't win a quantum lottery, it still seems to me that there is a sense in which I would "mostly" wake up in a world where I didn't win the lottery. It is this that I think needs explaining.

The point is that many hypotheses about the Born probabilities have been proposed. Not as many as there should be, because the mystery was falsely marked "solved" for a long time. But still, there have been many proposals.

There is legitimate hope of a solution to the Born puzzle without new fundamental laws. Your world does not split into exactly two new subprocesses on the exact occasion when you see "ABSORBED" or "TRANSMITTED" on the LCD screen of a photon sensor. We are constantly being superposed and decohered, all the time, sometimes along continuous dimensions—though brains are digital and involve whole neurons firing, and fire/not-fire would be an extremely decoherent state even of a *single* neuron...There would seem to be room for *something* unexpected to account for the Born statistics—a better understanding of the anthropic weight of observers, or a better understanding of the brain's superpositions—without new fundamentals.

We cannot rule out, though, the possibility that a new fundamental law is involved in the Born statistics.

As Jess Riedel puts it:⁴

If there's one lesson we can take from the history
of physics, it's that everytime new experimental

2. David Wallace, *The Emergent Multiverse*, (Oxford, United Kingdom: Oxford University Press, 2012)

3. David Deutsch, "Quantum Theory of Probability and Decisions", *Proceedings of the Royal Society of London A*455, 3129-37.

4. sharov.me/r/4-27

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“regimes” are probed (e.g. large velocities, small sizes, large mass densities, large energies), phenomena are observed which lead to new theories (Special Relativity, quantum mechanics, General Relativity, and the Standard Model, respectively).

“Every time” is too strong. A nitpick, yes, but also an important point: you can’t just *assume* that any particular law will fail in a new regime. But it’s possible that a new fundamental law is involved in the Born statistics, and that this law manifests only in the 20th decimal place at microscopic levels (hence being undetectable so far) while aggregating to have substantial effects at macroscopic levels.

Could there be some law, as yet undiscovered, that causes there to be only *one* world?

This is a shocking notion; it implies that all our twins in the other worlds—all the different versions of ourselves that are constantly split off, not just by human researchers doing quantum measurements, but by ordinary entropic processes—are actually *gone*, leaving us alone! This version of Earth would be the *only* version that exists in local space! If the inflationary scenario in cosmology turns out to be wrong, and the topology of the universe is both finite and relatively small—so that Earth does not have the distant duplicates that would be implied by an exponentially vast universe—then this Earth could be the only Earth that exists *anywhere*, a rather unnerving thought!

But it is dangerous to focus too much on specific hypotheses that you have no specific reason to think about. This is the same root error of the Intelligent Design folk, who pick any random puzzle in modern genetics, and say, “See, God must have done it!” Why “God,” rather than a zillion other possible explanations?—which you would have thought of long before you postulated divine intervention, if not for the fact that you secretly started out already knowing the answer you wanted to find.

You shouldn’t even *ask*, “Might there only be one world?” but instead just go ahead and do physics, and raise that *particular* issue only if new evidence demands it.

Could there be some as-yet-unknown fundamental law, that gives the universe a privileged center, which happens to coincide with Earth—thus proving that Copernicus was wrong all along, and the Bible right?

Asking *that particular* question—rather than a zillion other questions in which the center of the universe is Proxima Centauri, or the universe turns out to have a favorite pizza topping and it is pepperoni—betrays your hidden agenda. And though an unenlightened one might not realize it, giving the universe a privileged center *that follows Earth around through space* would be rather difficult to do with any *mathematically simple* fundamental law.

So too with asking whether there might be only one world. It betrays a sentimental attachment to human intuitions already proven wrong. The wheel of science turns, but it doesn't turn *backward*.

We have specific reasons to be highly suspicious of the notion of only one world. The notion of “one world” exists on a higher level of organization, like the location of Earth in space; on the quantum level there are no firm boundaries (though brains that differ by entire neurons firing are certainly decoherent). How would a *fundamental* physical law identify one *high-level* world?

Much worse, any physical scenario in which there was a *single* surviving world, so that any measurement had only a *single* outcome, would violate Special Relativity.

If the same laws are true at all levels—i.e., if many-worlds is correct—then when you measure one of a pair of entangled polarized photons, you end up in a world in which the photon is polarized, say, up-down, and alternate versions of you end up in worlds where the photon is polarized left-right. From your perspective before doing the measurement, the probabilities are 50/50. Light-years away, someone measures the other photon at a 20° angle to your own basis. From their perspective, too, the probability of getting either immediate result is 50/50—they maintain an invariant state of generalized entanglement with your faraway location, no matter what you do. But when

the two of you meet, years later, your probability of meeting a friend who got the *same* result is 11.6%, rather than 50%.

If there is only one global world, then there is only a single outcome of any quantum measurement. Either you measure the photon polarized up-down, or left-right, but not both. Light-years away, someone else's probability of measuring the photon polarized similarly in a 20° rotated basis actually *changes* from 50/50 to 11.6%.

You cannot possibly interpret this as a case of merely revealing properties that were already there; this is ruled out by Bell's Theorem. There does not seem to be any possible consistent view of the universe in which both quantum measurements have a single outcome, and yet both measurements are pre-determined, neither influencing the other. Something has to actually *change*, faster than light.

And this would appear to be a fully general objection, not just to collapse theories, but to any possible theory that gives us one global world! There is no consistent view in which measurements have single outcomes, but are locally determined (even locally randomly determined). Some mysterious influence has to cross a spacelike gap.

This is not a trivial matter. You cannot save yourself by waving your hands and saying, "the influence travels backward in time to the entangled photons' creation, then forward in time to the other photon, so it never actually crosses a spacelike gap." (This view has been seriously put forth, which gives you some idea of the magnitude of the paradox implied by one global world!) One measurement has to change the other, so which measurement happens *first*? Is there a global space of simultaneity? You can't have both measurements happen "first" because under Bell's Theorem, there's no way local information could account for observed results, etc.

Incidentally, this experiment has already been performed, and if there is a mysterious influence it would have to travel six million times as fast as light in the reference frame of the Swiss Alps. Also, the mysterious influence has been experimentally shown not to care if the two photons are measured in reference

frames which would cause each measurement to occur “before the other.”

Special Relativity seems counterintuitive to us humans—like an arbitrary speed limit, which you could get around by going backward in time, and then forward again. A law you could escape prosecution for violating, if you managed to hide your crime from the authorities.

But what Special Relativity really says is that human intuitions about space and time are simply wrong. There *is* no global “now,” there *is* no “before” or “after” across spacelike gaps. The ability to *visualize* a single global world, *even in principle*, comes from not getting Special Relativity on a gut level. Otherwise it would be obvious that physics proceeds locally with invariant states of distant entanglement, and the requisite information is simply *not locally present* to support a *globally single world*.

It might be that this seemingly impeccable logic is flawed—that my application of Bell’s Theorem and relativity to rule out any single global world contains some hidden assumption of which I am unaware—

—but consider the burden that a single-world theory must now shoulder! There is absolutely no reason *in the first place* to suspect a global single world; this is just *not what current physics says!* The global single world is an ancient human intuition that was *disproved*, like the idea of a universal absolute time. The superposition principle is visible even in half-silvered mirrors; experiments are verifying the disproof at steadily larger levels of superposition—but above all there is *no longer any reason to privilege the hypothesis* of a global single world. The ladder has been yanked out from underneath that human intuition.

There is no experimental evidence that the macroscopic world is single (we already know the microscopic world is superposed). And the prospect necessarily either violates Special Relativity, or takes an even more miraculous-seeming leap and violates seemingly impeccable logic. The latter, of course, being much more plausible in practice. But it isn’t really *that* plausible in an absolute sense. *Without experimental evidence*, it

is generally a *bad sign* to have to postulate arbitrary logical miracles.

As for quantum non-realism, it appears to me to be nothing more than a Get Out of Jail Free card. “It’s okay to violate Special Relativity because none of this is real anyway!” The equations cannot reasonably be hypothesized to deliver such excellent predictions *for literally no reason*. Bell’s Theorem rules out the obvious possibility that quantum theory represents imperfect knowledge of something locally deterministic.

Furthermore, macroscopic decoherence gives us a perfectly *realistic* understanding of what is going on, in which the equations deliver such good predictions because they mirror reality. And so the idea that the quantum equations are just “meaningless,” and therefore it is okay to violate Special Relativity, so we can have one global world after all, is not *necessary*. To me, quantum non-realism appears to be a huge bluff built around semantic stopsigns like “Meaningless!”

It is not quite safe to say that the existence of multiple Earths is as well-established as any other truth of science. The existence of quantum other worlds is not so well-established as the existence of trees, which most of us can personally observe.

Maybe there is something in that 20th decimal place, which aggregates to something bigger in macroscopic events. Maybe there’s a loophole in the seemingly iron logic which says that any single global world must violate Special Relativity, because the information to support a single global world is not locally available. And maybe the Flying Spaghetti Monster is just messing with us, and the world we know is a lie.

So all we can say about the existence of multiple Earths, is that it is as rationally probable as e.g. the statement that spinning black holes do not violate conservation of angular momentum. We have extremely fundamental reasons, having to do with the rotational symmetry of space, to suspect that conservation of angular momentum is built into the underlying nature of physics. And we have no specific reason to suspect this *particular* violation of our old generalizations in a higher-energy regime.

But we haven't actually checked conservation of angular momentum for rotating black holes—so far as I know. (And as I am talking here about rational guesses in states of partial knowledge, the point is exactly the same if the observation has been made and I do not know it yet.) And black holes are a more massive regime. So the obedience of black holes is not *quite* as assured as that my toilet conserves angular momentum while flushing, which come to think, I haven't checked either ...

Yet if you make the *mistake* of thinking too hard about this one particular possibility, instead of zillions of other possibilities—and especially if you don't understand the fundamental reason *why* angular momentum is conserved—then it may start seeming more and more plausible that “spinning black holes violate conservation of angular momentum,” as you think of more and more vaguely plausible-sounding reasons it *could* be true.

But the rational probability is pretty damned small.

Likewise the rational probability that there is only one Earth.

I mention this to explain my habit of talking as if many-worlds is an obvious fact. Many-worlds *is* an obvious fact, if you have all your marbles lined up correctly (understand very basic quantum physics, know the formal probability theory of Occam's Razor, understand Special Relativity, etc.) It is in fact considerably *more* obvious to me than the proposition that spinning black holes should obey conservation of angular momentum.

The only reason why many-worlds is not universally acknowledged as a direct prediction of physics which requires magic to violate, is that a contingent accident of our Earth's scientific history gave an entrenched academic position to a phlogiston-like theory that had an unobservable faster-than-light magical “collapse” devouring all other worlds. And many academic physicists do not have a mathematical grasp of Occam's Razor, which is the usual method for ridding physics of invisible angels. So when they encounter many-worlds and it conflicts with their (undermined) intuition that only one world exists, they say, “Oh, that's multiplying entities”—which is just

flatly wrong as probability theory—and go on about their daily lives.

I am not in academia. I am not constrained to bow and scrape to some senior physicist who hasn't grasped the obvious, but who will be reviewing my journal articles. I need have no fear that I will be rejected for tenure on account of scaring my students with "science-fiction tales of other Earths." If I can't speak plainly, who can?

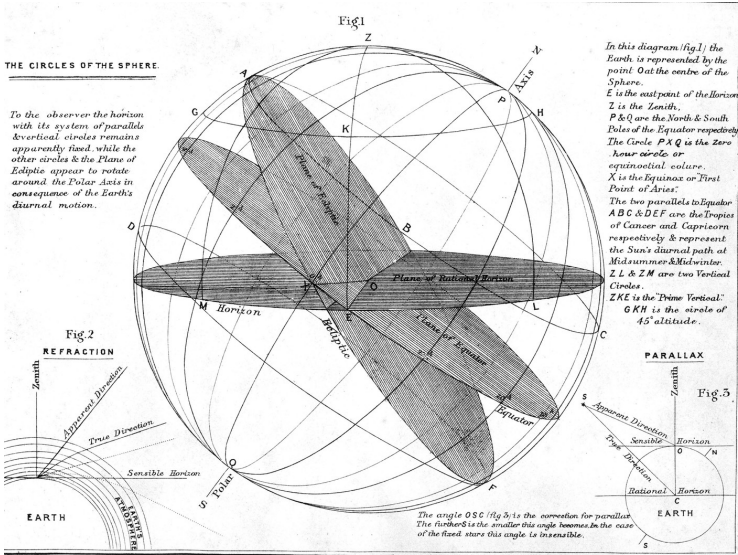
So let me state then, very clearly, on behalf of any and all physicists out there who dare not say it themselves: Many-worlds *wins outright* given our current state of evidence. There is no more reason to postulate a single Earth, than there is to postulate that two colliding top quarks would decay in a way that violates Conservation of Energy. It takes more than an unknown fundamental law; it takes magic.

The debate should already be over. It should have been over fifty years ago. The state of evidence is too lopsided to justify further argument. There is no balance in this issue. There is no rational controversy to teach. The laws of probability theory are laws, not suggestions; there is no flexibility in the best guess given this evidence. Our children will look back at the fact that we were STILL ARGUING about this in the early twenty-first century, and correctly deduce that we were nuts.

We have embarrassed our Earth long enough by failing to see the obvious. So for the honor of my Earth, I write as if the existence of many-worlds were an established fact, because it is. The only question now is how long it will take for the people of this world to update.

PART T

Science and Rationality



The Failures of Eld Science

This time there were no robes, no hoods, no masks. Students were expected to become friends, and allies. And everyone knew why you were in the classroom. It would have been pointless to pretend you weren't in the Conspiracy.

Their *sensei* was Jeffreyssai, who might have been the best of his era, in his era. His students were either the most promising learners, or those whom the *beisutsukai* saw political advantage in molding.

Brennan fell into the latter category, and knew it. Nor had he hesitated to use his Mistress's name to open doors. You used every avenue available to you, in seeking knowledge; that was respected here.

"—for over thirty years," Jeffreyssai said. "Not one of them saw it; not Einstein, not Schrödinger, not even von Neumann." He turned away from his sketcher, and toward the classroom. "I pose to you to the question: How did they fail?"

The students exchanged quick glances, a calculus of mutual risk between the wary and the merely baffled. Jeffreyssai was known to play games.

Finally Hiriwa-called-the-Black leaned forward, jangling slightly as her equation-carved bracelets shifted on her ankles. "By your years given, *sensei*, this was two hundred and fifty years after Newton. Surely, the scientists of that era must have grokked the concept of a universal law."

"Knowing the universal law of gravity," said the student Taji, from a nearby seat, "is not the same as understanding the concept of a universal law." He was one of the promising ones, as was Hiriwa.

Hiriwa frowned. "No ...it was said that Newton had been praised *for* discovering the first universal. Even in his own era. So it was known." Hiriwa paused. "But Newton himself would have been gone. Was there a *religious* injunction against proposing further universals? Did they refrain out of respect



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for Newton, or were they waiting for his *ghost* to speak? I am not clear on how Eld science was motivated—”

“No,” murmured Taji, a laugh in his voice, “you really, *really* aren’t.”

Jeffreyssai’s expression was kindly. “Hiriwa, it wasn’t religion, and it wasn’t lead in the drinking water, and they didn’t all have Alzheimer’s, and they weren’t sitting around all day reading webcomics. Forget the catalogue of horrors out of ancient times. Just think in terms of cognitive errors. What could Eld science have been *thinking* wrong?”

Hiriwa sat back with a sigh. “*Sensei*, I truly cannot imagine a snafu that would do *that*.”

“It wouldn’t be just *one* mistake,” Taji corrected her. “As the saying goes: Mistakes don’t travel alone; they hunt in packs.”

“But the *entire* human species?” said Hiriwa. “Thirty *years*?”

“It wasn’t the entire human species, Hiriwa,” said Styrlын. He was one of the older-looking students, wearing a short beard speckled in gray. “Maybe one in a hundred thousand could have written out Schrödinger’s Equation from memory. So that would have been their first and primary error—failure to concentrate their forces.”

“*Spare us the propaganda!*” Jeffreyssai’s gaze was suddenly fierce. “You are not here to proselytize for the Cooperative Conspiracy, my lord politician! Bend not the truth to make your points! I believe your Conspiracy has a phrase: ‘Comparative advantage.’ Do you *really* think that it would have helped to call in the whole human species, as it existed at that time, to debate quantum physics?”

Styrlын didn’t flinch. “Perhaps not, *sensei*,” he said. “But if you are to compare that era to this one, it is a consideration.”

Jeffreyssai moved his hand flatly through the air; the maybe-gesture he used to dismiss an argument that was true but not relevant. “It is not what I would call a *primary* mistake. The puzzle should not have required a billion physicists to solve.”

“I can think of more *specific* ancient horrors,” said Taji. “Spending all day writing grant proposals. Teaching undergraduates who would rather be somewhere else. Needing to publish thirty papers a year to get tenure ...”

"But we are not speaking of only the lower-status scientists," said Yin; she wore a slightly teasing grin. "It was said of Schrödinger that he retired to a villa for a month, with his mistress to provide inspiration, and emerged with his eponymous equation. We consider it a famous historical success of our methodology. Some Eld physicists *did* understand how to focus their mental energies; and would have been senior enough to do so, had they chose."

"True," Taji said. "In the end, administrative burdens are only a generic obstacle. Likewise such answers as, 'They were not trained in probability theory, and did not know of cognitive biases.' Our sensei seems to desire some more specific reply."

Jeffreyssai lifted an eyebrow encouragingly. "Don't dismiss your line of thought so quickly, Taji; it begins to be relevant. What kind of system would create administrative burdens on its own people?"

"A system that failed to support its people adequately," said Styrlyn. "One that failed to value their work."

"Ah," said Jeffreyssai. "But there is a student who has not yet spoken. *Brennan?*"

Brennan didn't jump. He deliberately waited just long enough to show he wasn't scared, and then said, "Lack of pragmatic motivation, sensei."

Jeffreyssai smiled slightly. "Expand."

What kind of system would create administrative burdens on its own people?, their *sensei* had asked them. The other students were pursuing their own lines of thought. Brennan, hanging back, had more attention to spare for his teacher's few hints. Being the beginner wasn't *always* a disadvantage—and he had been taught, long before the Bayesians took him in, to take every available advantage.

"The Manhattan Project," Brennan said, "was launched with a specific *technological* end in sight: a weapon of great power, in time of war. But the error that Eld Science committed with respect to quantum physics had no immediate consequences for their technology. They were confused, but they had no desperate *need* for an answer. Otherwise the surrounding system would have removed all burdens from their effort to solve it.

Surely the Manhattan Project must have done so—Taji? Do you know?”

Taji looked thoughtful. “Not *all* burdens—but I’m pretty sure they weren’t writing grant proposals in the middle of their work.”

“So,” Jeffreyssai said. He advanced a few steps, stood directly in front of Brennan’s desk. “You think Eld scientists simply weren’t trying hard enough. Because their art had no military applications? A rather *competitive* point of view, I should think.”

“Not necessarily,” Brennan said calmly. “Pragmatism is a virtue of rationality also. A desired *use* for a better quantum theory would have helped the Eld scientists in many ways beyond just motivating them. It would have given shape to their curiosity, and told them what constituted success or failure.”

Jeffreyssai chuckled slightly. “Don’t guess so hard what *I* might prefer to hear, Competitor. Your first statement came closer to my hidden mark; your oh-so-Bayesian disclaimer fell wide ...The factor I had in mind, Brennan, was that Eld scientists thought it was *acceptable* to take thirty years to solve a problem. Their entire social process of science was based on getting to the truth *eventually*. A wrong theory got discarded *eventually*—once the next generation of students grew up familiar with the replacement. Work expands to fill the time allotted, as the saying goes. But people can think important thoughts in far less than thirty years, if they *expect* speed of themselves.” Jeffreyssai suddenly slammed down a hand on the arm of Brennan’s chair. “*How long do you have to dodge a thrown knife?*”

“Very little time, sensei!”

“*Less than a second! Two opponents are attacking you! How long do you have to guess who’s more dangerous?*”

“Less than a second, sensei!”

“*The two opponents have split up and are attacking two of your girlfriends! How long do you have to decide which one you truly love?*”

“Less than a second, sensei!”

“*A new argument shows your precious theory is flawed! How long does it take you to change your mind?*”

“Less than a second, sensei!”

“WRONG! DON’T GIVE ME THE WRONG ANSWER JUST BECAUSE IT FITS A CONVENIENT PATTERN AND I SEEM TO EXPECT IT OF YOU! How long does it really take, Brennan?”

Sweat was forming on Brennan’s back, but he stopped and actually thought about it—

“ANSWER, BRENNAN!”

“No, sensei! I’m not finished thinking, sensei! An answer would be premature! Sensei!”

“Very good! Continue! But don’t take thirty years!”

Brennan breathed deeply, reforming his thoughts. He finally said, “Realistically, sensei, the best-case scenario is that I would see the problem immediately; use the discipline of suspending judgment; try to re-accumulate all the evidence before continuing; and depending on how emotionally attached I had been to the theory, use the crisis-of-belief technique to ensure I could genuinely go either way. So at least five minutes and perhaps up to an hour.”

“Good! You actually thought about it that time! Think about it every time! Break patterns! In the days of Eld Science, Brennan, it was not uncommon for a grant agency to spend six months reviewing a proposal. They permitted themselves the time! You are being graded on your speed, Brennan! The question is not whether you get there eventually! Anyone can find the truth in five thousand years! You need to move faster!”

“Yes, sensei!”

“Now, Brennan, have you just learned something new?”

“Yes, sensei!”

“How long did it take you to learn this new thing?”

An arbitrary choice there ...“Less than a minute, sensei, from the boundary that seems most obvious.”

“Less than a minute,” Jeffreyssai repeated. “So, Brennan, how long do you think it should take to solve a major scientific problem, if you are not wasting any time?”

Now there was a trapped question if Brennan had ever heard one. There was no way to guess what time period Jeffreyssai had in mind—what the *sensei* would consider too long, or too short. Which meant that the only way out was to just try for

the genuine truth; this would offer him the defense of honesty, little defense though it was. "One year, sensei?"

"Do you think it could be done in one month, Brennan? In a case, let us stipulate, where in principle you already have enough experimental evidence to determine an answer, but not so much experimental evidence that you can afford to make errors in interpreting it."

Again, no way to guess which answer Jeffreyssai might *want*... "One month seems like an unrealistically short time to me, sensei."

"A *short time*?" Jeffreyssai said incredulously. "How many minutes in thirty days? Hiriwa?"

"43,200, sensei," she answered. "If you assume sixteen-hour waking periods and daily sleep, then 28,800 minutes."

"Assume, Brennan, that it takes five whole minutes to think an *original* thought, rather than learning it from someone else. Does even a major scientific problem require 5,760 distinct insights?"

"I confess, sensei," Brennan said slowly, "that I have never thought of it that way before ...but do you tell me that is *truly* a realistic level of productivity?"

"No," said Jeffreyssai, "but neither is it realistic to think that a single problem requires 5,760 insights. And yes, it has been done."

Jeffreyssai stepped back, and smiled benevolently. Every student in the room stiffened; they knew that smile. "Though none of you hit the particular answer that *I* had in mind, nonetheless your answers were as reasonable as mine. Except Styrlын's, I'm afraid. Even Hiriwa's answer was not entirely wrong: the task of proposing new theories was once considered a sacred duty reserved for those of high status, there being a limited supply of problems in circulation, at that time. But *Brennan's* answer is *particularly* interesting, and I am minded to test his theory of motivation."

Oh, hell, Brennan said silently to himself. Jeffreyssai was gesturing for Brennan to stand up before the class.

When Brennan had risen, Jeffreyssai neatly seated himself in Brennan's chair.

“Brennan-sensei,” Jeffreyssai said, “you have five minutes to think of something stunningly brilliant to say about the failure of Eld science on quantum physics. As for the rest of us, our job will be to gaze at you expectantly. I can only imagine how embarrassing it will be, should you fail to think of anything good.”

Bastard. Brennan didn’t say it aloud. Taji’s face showed a certain amount of sympathy; Styrlын held himself aloof from the game; but Yin was looking at him with sardonic interest. Worse, Hiriwa *was* gazing at him expectantly, assuming that he would rise to the challenge. And Jeffreyssai was gawking wide-eyed, waiting for the guru’s words of wisdom. *Screw you, sensei.*

Brennan didn’t panic. It was very, very, very far from being the scariest situation he’d ever faced. He took a moment to decide how to think; then thought.

At four minutes and thirty seconds, Brennan spoke. (There was an art to such things; as long as you were doing it anyway, you might as well make it look easy.)

“A woman of wisdom,” Brennan said, “once told me that it is wisest to regard our past selves as fools beyond redemption—to see the people we once were as idiots entire. I do not necessarily say this myself; but it is what she said to me, and there is more than a grain of truth in it. As long as we are making excuses for the past, trying to make it look better, *respecting* it, we cannot make a clean break. It occurs to me that the rule may be no different for human *civilizations*. So I tried looking back and considering the Eld scientists as simple fools.”

“Which they were not,” Jeffreyssai said.

“Which they were not,” Brennan continued. “In terms of raw intelligence, they undoubtedly exceeded me. But it occurred to me that a difficulty in seeing what Eld scientists did wrong, might have been in respecting the ancient and legendary names too highly. And that did indeed produce an insight.”

“Enough introduction, Brennan,” said Jeffreyssai. “If you found an insight, state it.”

“Eld scientists were not trained ...” Brennan paused. “No, *un-trained* is not the concept. They were trained for the *wrong task*.”

At that time, there were no Conspiracies, no secret truths; as soon as Eld scientists solved a major problem, they published the solution to the world and each other. Truly scary and confusing *open problems* would have been in extremely rare supply, and used up the moment they were solved. So it would not have been possible to train Eld researchers *to bring order out of scientific chaos*. They would have been trained for something else—I'm not sure what—

"Trained to manipulate whatever science had *already* been discovered," said Taji. "It was a difficult enough task for Eld teachers to train their students to *use existing knowledge*, or follow already-known methodologies; that was all Eld science teachers aspired to impart."

Brennan nodded. "Which is a *very* different matter from creating new science of their own. The Eld scientists, faced with problems of quantum theory, might never have faced that kind of *fear* before—the dismay of not knowing. The Eld scientists might have seized on unsatisfactory answers prematurely, because they were accustomed to working with a neat, agreed-upon body of knowledge."

"Good, Brennan," murmured Jeffreyssai.

"But above all," Brennan continued, "an Eld scientist couldn't have *practiced* the actual problem the quantum scientists faced—that of resolving a major confusion. It was something you did once per lifetime if you were lucky, and as Hiriwa observed, Newton would no longer have been around. So while the Eld physicists who messed up quantum theory were not unintelligent, they were, in a strong sense, *amateurs*—ad-libbing the whole process of paradigm shift."

"And no probability theory," Hiriwa noted. "So anyone who *did* succeed at the problem would have no idea what they'd just done. They wouldn't be able to communicate it to anyone else, except vaguely."

"Yes," Styrlын said. "And it was only a handful of people who could tackle the problem at all, with no training in doing so; those are the physicists whose names have passed down to us. A handful of people, making a handful of discoveries each. It would not have been enough to sustain a community. Each Eld

scientist tackling a new paradigm shift would have needed to rediscover the rules from scratch.”

Jeffreyssai rose from Brenann’s desk. “Acceptable, Brennan; you surprise me, in fact. I shall have to give further thought to this method of yours.” Jeffreyssai went to the classroom door, then looked back. “However, I did have in mind at least one *other* major flaw of Eld science, which none of you suggested. I expect to receive a list of possible flaws tomorrow. I expect the flaw I have in mind to be on the list. You have 480 minutes, excluding sleep time. I see five of you here. The challenge does not require more than 480 insights to solve, nor more than 96 insights in series.”

And Jeffreyssai left the room.

The Dilemma: Science or Bayes?



discussion
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Eli: You are writing a lot about physics recently.
Why?

—Shane Legg (and several other people)¹

In light of your QM explanation, which to me sounds perfectly logical, it seems *obvious and normal* that many worlds is overwhelmingly likely. It just seems almost too good to be true that *I* now get what plenty of genius quantum physicists still can't. [...] Sure I can explain all that away, and I still think you're right, I'm just suspicious of myself for believing the first believable explanation I met.

—Recovering_irrationalist²

Recovering_irrationalist, you've got no idea how glad I was to see you post that comment.

Of course I had more than just *one* reason for spending all that time writing about quantum physics. I like having lots of hidden motives. It's the closest I can ethically get to being a supervillain.

But to give an example of a purpose I could *only* accomplish by discussing quantum physics...

In physics, you can get absolutely clear-cut issues. Not in the sense that the issues are trivial to explain. But if you try to

1. sharov.me/r/4-28

Quantum Non-Realism • lesswrong.com

2. sharov.me/r/4-29

Many Worlds, One Best Guess • lesswrong.com



apply Bayes to healthcare, or economics, you may not be able to *formally* lay out what is the simplest hypothesis, or what the evidence supports. But when I say “macroscopic decoherence is simpler than collapse” it is actually *strict* simplicity; you could write the two hypotheses out as computer programs and count the lines of code. Nor is the evidence itself in dispute.

I wanted a very clear example—*Bayes says “zig,” this is a zag*—when it came time to break your allegiance to Science.

“Oh, sure,” you say, “the physicists messed up the many-worlds thing, but give them a break, Eliezer! No one ever claimed that the social process of science was perfect. People are human; they make mistakes.”

But the physicists who refuse to adopt many-worlds aren’t disobeying the rules of Science. They’re *obeying* the rules of Science.

The tradition handed down through the generations says that a new physics theory comes up with new experimental predictions that distinguish it from the old theory. You perform the test, and the new theory is confirmed or falsified. If it’s confirmed, you hold a huge celebration, call the newspapers, and hand out Nobel Prizes for everyone; any doddering old emeritus professors who refuse to convert are quietly humored. If the theory is disconfirmed, the lead proponent publicly recants, and gains a reputation for honesty.

This is not how things *do* work in science; rather it is how things are *supposed* to work in Science. It’s the ideal to which all good scientists aspire.

Now many-worlds comes along, and it doesn’t seem to make any new predictions relative to the old theory. That’s suspicious. And there’s all these other worlds, but you can’t see them. That’s *really* suspicious. It just doesn’t seem scientific.

If you got as far as Recovering_irrationalist—so that many-worlds now seems perfectly logical, obvious and normal—and you also started out as a Traditional Rationalist, then you should be able to switch back and forth between the Scientific view and the Bayesian view, like a Necker Cube.

So now put on your Science Goggles—you’ve still got them around somewhere, right? Forget everything you know about

Kolmogorov complexity, Solomonoff induction or Minimum Message Lengths. That's not part of the traditional training. You just eyeball something to see how "simple" it looks. The word "testable" doesn't conjure up a mental image of Bayes's Theorem governing probability flows; it conjures up a mental image of being in a lab, performing an experiment, and having the celebration (or public recantation) afterward.

Science-Goggles on: The current quantum theory has passed all experimental tests so far. Many-worlds doesn't make any new testable predictions—the amazing new phenomena it predicts are all hidden away where we can't see them. You can get along fine without supposing the other worlds, and that's just what you should do. The whole thing smacks of science fiction. But it must be admitted that quantum physics is a very deep and very confusing issue, and who knows what discoveries might be in store? Call me when Many-worlds makes a testable prediction.

Science-Goggles off, Bayes-Goggles back on:

Bayes-Goggles on: The simplest quantum equations that cover all known evidence don't have a special exception for human-sized masses. There isn't even any reason to ask that particular question. Next!

Okay, so is this a problem we can fix in five minutes with some duct tape and superglue?

No.

Huh? Why not just teach new graduating classes of scientists about Solomonoff induction and Bayes's Rule?

Centuries ago, there was a widespread idea that the Wise could unravel the secrets of the universe just by thinking about them, while to go out and *look* at things was lesser, inferior, naive, and would just delude you in the end. You couldn't trust the way things *looked*—only thought could be your guide.

Science began as a rebellion against this Deep Wisdom. At the core is the pragmatic belief that human beings, sitting around in their armchairs trying to be Deeply Wise, just drift off into never-never land. You couldn't trust your thoughts. You had to make advance experimental predictions—predictions that no one else had made before—run the test, and confirm the result. That was evidence. Sitting in your armchair, thinking about what seemed reasonable...would not be taken to *prejudice* your theory, because Science wasn't an idealistic belief about pragmatism, or getting your hands dirty. It was, rather, the dictum that experiment alone would decide. Only experiments could judge your theory—not your nationality, or your religious professions, or the fact that you'd invented the theory in your armchair. Only experiments! If you sat in your armchair and came up with a theory that made a novel prediction, and experiment confirmed the prediction, then we would care about the result of the experiment, not where your hypothesis came from.

That's Science. And if you say that many-worlds should replace the immensely successful Copenhagen Interpretation, adding on all these twin Earths that can't be observed, just because it *sounds more reasonable and elegant*—not because it *crushed the old theory with a superior experimental prediction*—then you're undoing the core scientific rule that prevents people from running out and putting angels into all the theories, because angels are more reasonable and elegant.

You think teaching a few people about Solomonoff induction is going to solve *that* problem? Nobel laureate Robert Aumann—who first proved that Bayesian agents with similar priors cannot agree to disagree—is a believing Orthodox Jew. Aumann helped a project to test the Torah for “Bible codes,” hidden prophecies from God—and concluded that the project had failed to confirm the codes' existence. Do you want Aumann thinking that once you've got Solomonoff induction, you can forget about the experimental method? Do you think that's going to help him? And most scientists out there will not rise to the level of Robert Aumann.

Okay, Bayes-Goggles back on. Are you *really* going to believe that large parts of the wavefunction disappear when you can no longer see them? As a result of the only non-linear non-unitary non-differentiable non-CPT-symmetric acausal faster-than-light informally-specified phenomenon in all of physics? Just because, by sheer historical contingency, the stupid version of the theory was proposed first?

Are you going to make a major modification to a scientific model, and believe in zillions of other worlds you can't see, without a defining moment of experimental triumph over the old model?

Or are you going to reject probability theory?

Will you give your allegiance to Science, or to Bayes?

Michael Vassar once observed (tongue-in-cheek) that it was a good thing that a majority of the human species believed in God, because otherwise, he would have a very hard time rejecting majoritarianism.³ But since the majority opinion that God exists is simply unbelievable, we have no choice but to reject the extremely strong philosophical arguments for majoritarianism.

You can see (one of the reasons) why I went to such lengths to explain quantum theory. Those who are good at math should now be able to *visualize* both macroscopic decoherence, and the probability theory of simplicity and testability—get the insanity of a global single world on a *gut* level.

I wanted to present you with a nice, sharp dilemma between rejecting the scientific method, or embracing insanity.

Why? I'll give you a hint: It's not just because I'm evil. If you would guess my motives here, think beyond the first obvious answer.

PS: If you try to come up with clever ways to wriggle out of the dilemma, you're just going to get shot down in future essays. You have been warned.

3. sharov.me/r/4-30

Philosophical Majoritarianism - by Hal Finney •
overcomingbias.com



Science Doesn't Trust Your Rationality

Scott Aaronson¹ suggests that many-worlds and libertarianism are similar in that they are both cases of bullet-swallowing, rather than bullet-dodging:



discussion
sharov.me/r/
4-c66

Libertarianism and MWI are both grand philosophical theories that start from premises that almost all educated people accept (quantum mechanics in the one case, Econ 101 in the other), and claim to reach conclusions that most educated people reject, or are at least puzzled by (the existence of parallel universes / the desirability of eliminating fire departments).

Now *there's* an analogy that would never have occurred to me.

I've previously argued that Science rejects Many-Worlds but Bayes accepts it. (Here, "Science" is capitalized because we are talking about the idealized form of Science, not just the actual social process of science.)

It furthermore seems to me that there is a *deep* analogy between (small-"I") libertarianism and Science:

1. Both are based on a pragmatic distrust of reasonable-sounding arguments.
2. Both try to build systems that are more trustworthy than the people in them.
3. Both accept that people are flawed, and try to harness their flaws to power the system.

1. sharov.me/r/4-31
scottaaronson.com



The core argument for libertarianism is historically motivated distrust of lovely theories of “How much *better* society would be, if we just made a rule that said XYZ.” If that sort of trick actually *worked*, then more regulations would correlate to higher economic growth as society moved from local to global optima. But when some person or interest group gets enough power to start doing everything they think is a good idea, history says that what actually *happens* is Revolutionary France or Soviet Russia.

The plans that in lovely theory should have made everyone happy ever after, don’t have the results predicted by reasonable-sounding arguments. And power corrupts, and attracts the corrupt.

So you regulate as little as possible, because you can’t trust the lovely theories and you can’t trust the people who implement them.

You don’t shake your finger at people for being selfish. You try to build an efficient system of production out of selfish participants, by requiring transactions to be voluntary. So people are forced to play positive-sum games, because that’s how they get the *other* party to sign the contract. With violence restrained and contracts enforced, individual selfishness can power a globally productive system.

Of course none of this works quite so well in practice as in theory, and I’m not going to go into market failures, commons problems, etc. The core argument for libertarianism is not that libertarianism would work in a perfect world, but that it degrades gracefully into real life. Or rather, degrades less awkwardly than any other known economic principle. (People who see Libertarianism as the perfect solution for perfect people strike me as kinda missing the point of the “pragmatic distrust” thing.)

Science first came to know itself as a rebellion against trusting the word of Aristotle. If the people of that revolution had merely said, “Let us trust ourselves, not Aristotle!” they would have flashed and faded like the French Revolution.

But the Scientific Revolution lasted because—like the American Revolution—the architects propounded a stranger philosophy: “Let us trust no one! Not even ourselves!”

In the beginning came the idea that we can't just toss out Aristotle's armchair reasoning and replace it with *different* armchair reasoning. We need to talk to Nature, and actually *listen* to what It says in reply. This, itself, was a stroke of genius.

But then came the challenge of implementation. People are stubborn, and may not want to accept the verdict of experiment. Shall we shake a disapproving finger at them, and say “Naughty”?

No; we assume and accept that each individual scientist may be crazily attached to their personal theories. Nor do we assume that anyone can be trained out of this tendency—we don't try to choose Eminent Judges who are supposed to be impartial.

Instead, we try to *harness* the individual scientist's stubborn desire to prove their personal theory, by saying: “Make a new experimental prediction, and do the experiment. If you're right, and the experiment is replicated, you win.” So long as scientists believe this is true, they have a motive to do experiments that can *falsify* their own theories. Only by accepting the possibility of defeat is it possible to win. And any great claim will require replication; this gives scientists a motive to be honest, on pain of great embarrassment.

And so the stubbornness of individual scientists is harnessed to produce a steady stream of knowledge at the group level. The System is *somewhat* more trustworthy than its parts.

Libertarianism secretly relies on most individuals being prosocial enough to tip at a restaurant they won't ever visit again. An economy of genuinely selfish human-level agents would implode. Similarly, Science relies on most scientists not committing sins so egregious that they can't rationalize them away.

To the extent that scientists believe they can promote their theories by playing academic politics—or game the statistical methods to potentially win without a chance of losing—or to the extent that nobody bothers to replicate claims—science

degrades in effectiveness. But it degrades gracefully, as such things go.

The part where the successful predictions belong to the theory and theorists who originally made them, and cannot just be stolen by a theory that comes along later—*without* a novel experimental prediction—is an important feature of this social process.

The final upshot is that Science is not easily reconciled with probability theory. If you do a probability-theoretic calculation *correctly*, you're going to get the *rational* answer. Science doesn't trust your rationality, and it doesn't rely on your ability to use probability theory as the arbiter of truth. It wants you to set up a definitive experiment.

Regarding Science as a mere approximation to some probability-theoretic ideal of rationality...would certainly seem to be *rational*. There seems to be an extremely reasonable-sounding argument that Bayes's Theorem is the hidden structure that explains why Science works. But to subordinate Science to the grand schema of Bayesianism, and let Bayesianism come in and override Science's verdict when that seems appropriate, is not a trivial step!

Science is built around the assumption that you're *too stupid and self-deceiving* to just use Solomonoff induction. After all, if it was that simple, we wouldn't need a social process of science...right?

So, are you going to believe in faster-than-light quantum "collapse" fairies after all? Or do you think you're smarter than that?

When Science Can't Help

Once upon a time, a younger Eliezer had a stupid theory. Let's say that Eliezer₁₈'s stupid theory was that consciousness was caused by closed timelike curves hiding in quantum gravity. This isn't the whole story, not even close, but it will do for a start.



discussion
sharov.me/r/
4-c67

And there came a point where I looked back, and realized:

1. I had carefully followed everything I'd been told was Traditionally Rational, in the course of going astray. For example, I'd been careful to only believe in stupid theories that made novel experimental predictions, e.g., that neuronal microtubules would be found to support coherent quantum states.
2. Science would have been perfectly fine with my spending ten years trying to test my stupid theory, only to get a negative experimental result, so long as I then said, "Oh, well, I guess my theory was wrong."

From Science's perspective, that is how things are *supposed* to work—happy fun for everyone. You admitted your error! Good for you! Isn't that what Science is all about?

But what if I didn't want to waste ten years?

Well...Science didn't have much to say about *that*. How could Science say which theory was right, in *advance* of the experimental test? Science doesn't care where your theory comes from—it just says, "Go test it."

This is the great strength of Science, and also its great weakness.

Gray Area asked:¹

1. sharov.me/r/4-32

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Eliezer, why are you concerned with untestable questions?

Because questions that are *easily immediately* tested are hard for Science to get wrong.

I mean, sure, when there's already definite unmistakable experimental evidence available, go with it. Why on Earth wouldn't you?

But sometimes a question will have very large, very definite experimental consequences in your future—but you can't easily test it experimentally *right now*—and yet there *is* a strong *rational* argument.

Macroscopic quantum superpositions are readily testable: It would just take nanotechnologic precision, very low temperatures, and a nice clear area of interstellar space. Oh, sure, you can't do it *right now*, because it's *too expensive* or *impossible for today's technology* or something like that—but in theory, sure! Why, maybe someday they'll run whole civilizations on macroscopically superposed quantum computers, way out in a well-swept volume of a Great Void. (Asking what quantum non-realism says about the status of any observers inside these computers helps to reveal the underspecification of quantum non-realism.)

This doesn't seem immediately pragmatically relevant to your life, I'm guessing, but it establishes the pattern: Not everything with future consequences is *cheap* to test *now*.

Evolutionary psychology is another example of a case where rationality has to take over from science. While theories of evolutionary psychology form a connected whole, only some of those theories are readily testable experimentally. But you still need the other parts of the theory, because they form a connected web that helps you to form the hypotheses that are actually testable—and then the helper hypotheses are supported in a Bayesian sense, but not supported experimentally. Science would render a verdict of “not proven” on individual parts of a connected theoretical mesh that is experimentally productive as a whole. We'd need a new kind of verdict for that, something like “indirectly supported.”

Or what about cryonics?

Cryonics is an archetypal example of an extremely important issue (150,000 people die per day) that will have huge consequences in the foreseeable future, but doesn't offer definite unmistakable experimental evidence that we can get *right now*.

So do you say, "I don't believe in cryonics because it hasn't been experimentally proven, and you shouldn't believe in things that haven't been experimentally proven"?

Well, from a Bayesian perspective, that's incorrect. Absence of evidence is evidence of absence only to the degree that we could reasonably expect the evidence to appear. If someone is trumpeting that snake oil cures cancer, you can reasonably expect that, *if the snake oil were actually curing cancer*, some scientist would be performing a controlled study to verify it—that, at the least, doctors would be reporting case studies of amazing recoveries—and so the absence of this evidence is strong evidence of absence. But "gaps in the fossil record" are not strong evidence against evolution; fossils form only rarely, and *even if an intermediate species did in fact exist*, you cannot expect with high probability that Nature will obligingly fossilize it and that the fossil will be discovered.

Reviving a cryonically frozen mammal is just not something you'd expect to be able to do with modern technology, *even if future nanotechnologies could in fact perform a successful revival*. That's how I see Bayes seeing it.

Oh, and as for the actual arguments *for* cryonics—I'm not going to go into those at the moment. But if you followed the physics and anti-Zombie sequences, it should now seem a lot more plausible that whatever preserves the pattern of synapses preserves as much of "you" as is preserved from one night's sleep to morning's waking.

Now, to be fair, someone who says, "I don't believe in cryonics because it hasn't been proven experimentally" is *misapplying* the rules of Science; this is not a case where science actually gives the *wrong answer*. In the absence of a definite experimental test, the verdict of science here is "Not proven." Anyone who interprets that as a rejection is taking an extra step outside of science, not a misstep within science.

John McCarthy's Wikiquotes page has him saying, "Your statements amount to saying that if AI is possible, it should be easy. Why is that?"² The Wikiquotes page doesn't say what McCarthy was responding to, but I could venture a guess.

The general mistake probably arises because there *are* cases where the absence of scientific proof is strong evidence—because an experiment would be readily performable, and so failure to perform it is itself suspicious. (Though not as suspicious as I used to think—with all the strangely varied anecdotal evidence coming in from respected sources, why the *hell* isn't anyone testing Seth Roberts's theory of appetite suppression?³)

Another confusion factor may be that if you test Pharmaceutical X on 1,000 subjects and find that 56% of the control group and 57% of the experimental group recover, some people will call that a verdict of "Not proven." I would call it an experimental verdict of "Pharmaceutical X doesn't work well, if at all." Just because this verdict is theoretically retractable in the face of new evidence doesn't make it ambiguous.

In any case, right now you've got people dismissing cryonics out of hand as "not scientific," like it was some kind of pharmaceutical you could easily administer to 1,000 patients and see what happened. "Call me when cryonicists actually revive someone," they say; which, as Mike Li observes, is like saying "I refuse to get into this ambulance; call me when it's actually at the hospital." Maybe Martin Gardner warned them against believing in strange things without experimental evidence. So they wait for the definite unmistakable verdict of Science, while their family and friends and 150,000 people per day are dying *right now*, and might or might not be savable—

—a calculated bet you could only make *rationally*.

The drive of Science is to obtain a mountain of evidence so huge that not even fallible human scientists can misread it.

2. No longer on Wikiquotes, but included in McCarthy's personal quotes page.

3. Seth Roberts, "What Makes Food Fattening?: A Pavlovian Theory of Weight Control" (Unpublished manuscript, 2005), sharov.me/r/4-33.



But even *that* sometimes goes wrong, when people become confused about which theory predicts what, or bake extremely-hard-to-test components into an early version of their theory. And sometimes you just can't get clear experimental evidence at all.

Either way, you have to try to do the thing that Science doesn't trust anyone to do—think rationally, and figure out the answer *before* you get clubbed over the head with it.

(Oh, and sometimes a *disconfirming* experimental result looks like: “Your entire species has just been wiped out! You are now scientifically required to relinquish your theory. If you publicly recant, good for you! Remember, it takes a strong mind to give up strongly held beliefs. Feel free to try another hypothesis next time!”)

Science Isn't Strict Enough



discussion
sharov.me/r/
4-c68

Once upon a time, a younger Eliezer had a stupid theory. Eliezer₁₈ was careful to follow the precepts of Traditional Rationality that he had been taught; he made sure his stupid theory had experimental consequences. Eliezer₁₈ professed, in accordance with the virtues of a scientist he had been taught, that he wished to test his stupid theory.

This was all that was required to be virtuous, according to what Eliezer₁₈ had been taught was virtue in the way of science.

It was not even *remotely* the order of effort that would have been required to get it *right*.

The traditional ideals of Science too readily give out gold stars. Negative experimental results are also knowledge, so everyone who plays gets an award. So long as you can think of some kind of experiment that tests your theory, and you *do* the experiment, and you *accept* the results, you've played by the rules; you're a good scientist.

You didn't necessarily get it right, but you're a nice science-abiding citizen.

(I note at this point that I am speaking of Science, not the social process of science as it actually works in practice, for two reasons. First, I went astray in trying to follow the *ideal* of Science—it's not like I was shot down by a journal editor with a grudge, and it's not like I was trying to imitate the flaws of academia. Second, if I point out a problem with the ideal as it is traditionally preached, real-world scientists are not *forced* to likewise go astray!)

Science began as a rebellion against grand philosophical schemas and armchair reasoning. So Science doesn't include a rule as to what kinds of hypotheses you are and aren't allowed to test; that is left up to the individual scientist. Trying to guess that a priori would require some kind of grand philosophical schema, and reasoning in advance of the evidence. As a social ideal, Science doesn't judge you as a bad person for coming up

with heretical hypotheses; honest experiments, and acceptance of the results, is virtue unto a scientist.

As long as most scientists can manage to accept definite, unmistakable, unambiguous experimental evidence, science can progress. It may happen too slowly—it may take longer than it should—you may have to wait for a generation of elders to die out—but eventually, the ratchet of knowledge clicks forward another notch. Year by year, decade by decade, the wheel turns *forward*. It's enough to support a civilization.

So that's all that Science really asks of you—the ability to accept reality when you're beat over the head with it. It's not much, but it's enough to sustain a scientific culture.

Contrast this to the notion we have in probability theory, of an exact quantitative rational judgment. If 1% of women presenting for a routine screening have breast cancer, and 80% of women with breast cancer get positive mammographies, and 10% of women without breast cancer get false positives, what is the probability that a routinely screened woman with a positive mammography has breast cancer? It is 7.5%. You cannot say, "I believe she doesn't have breast cancer, because the experiment isn't definite enough." You cannot say, "I believe she has breast cancer, because it is wise to be pessimistic and that is what the only experiment so far seems to indicate." Seven point five percent is the rational estimate given this evidence, not 7.4% or 7.6%. The laws of probability are *laws*.

It is written in the Twelve Virtues (vol. V, p. 193), of the third virtue, lightness:

If you regard evidence as a constraint and seek to free yourself, you sell yourself into the chains of your whims. For you cannot make a true map of a city by sitting in your bedroom with your eyes shut and drawing lines upon paper according to impulse. You must walk through the city and draw lines on paper that correspond to what you see. If, seeing the city unclearly, you think that you can shift a line just a little to the right, just a little to

the left, according to your caprice, this is just the same mistake.

In Science, when it comes to deciding which hypotheses to test, the morality of Science gives you personal freedom of what to believe, so long as it isn't already ruled out by experiment, and so long as you move to test your hypothesis. Science wouldn't try to give an official verdict on the *best* hypothesis to test, in *advance* of the experiment. That's left up to the conscience of the individual scientist.

Where definite experimental evidence exists, Science tells you to bow your stubborn neck and accept it. Otherwise, Science leaves it up to you. Science gives you room to wander around *within the boundaries* of the experimental evidence, according to your whims.

And this is not easily reconciled with Bayesianism's notion of an exactly right probability estimate, one with no flex or room for whims, that exists both before and after the experiment. Bayesianism doesn't match well with the ancient and traditional reason for Science—the distrust of grand schemas, the presumption that people aren't rational enough to get things right without definite and unmistakable experimental evidence. If we were all perfect Bayesians, we wouldn't *need* a social process of science.

Nonetheless, around the time I realized my big mistake, I had also been studying Kahneman and Tversky and Jaynes. I was learning a new Way, stricter than Science. A Way that could criticize my folly, in a way that Science never could. A Way that could have told me what Science would never have said in *advance*: “You picked the wrong hypothesis to test, dunderhead.”

But the Way of Bayes is also *much harder to use* than Science. It puts a tremendous strain on your ability to hear tiny false notes, where Science only demands that you notice an anvil dropped on your head.

In Science you can make a mistake or two, and another experiment will come by and correct you; at worst you waste a couple of decades.

But if you try to use Bayes even qualitatively—if you try to do the thing that Science doesn't trust you to do, and reason rationally in the absence of overwhelming evidence—it is like math, in that a single error in a hundred steps can carry you anywhere. It demands lightness, evenness, precision, perfectionism.

There's a good reason why Science doesn't trust scientists to do this sort of thing, and asks for further experimental proof *even after* someone claims they've worked out the right answer based on hints and logic.

But if you would rather not waste ten years trying to prove the *wrong* theory, you'll need to essay the vastly more difficult problem: listening to evidence that doesn't shout in your ear.

Even if you can't look up the priors for a problem in the *Handbook of Chemistry and Physics*—even if there's no Authoritative Source telling you what the priors are—that doesn't mean you get a free, personal choice of making the priors whatever you want. It means you have a new guessing problem that you must carry out to the best of your ability.

If the mind, as a cognitive engine, could generate *correct* estimates by fiddling with priors according to whims, you could know things without looking them, or even alter them without touching them. But the mind is not magic. The rational probability estimate has no room for any decision based on whim, even when it seems that you don't know the priors.

Similarly, if the Bayesian answer is difficult to compute, that doesn't mean that Bayes is inapplicable; it means you *don't know* what the Bayesian answer is. Bayesian probability theory is not a toolbox of statistical methods; it's the *law* that governs any tool you use, whether or not you know it, whether or not you can calculate it.

As for using Bayesian methods on huge, highly general hypothesis spaces—like, “Here's the data from every physics experiment ever; now, what would be a good Theory of Everything?”—if you knew how to do that *in practice*, you wouldn't be a statistician, you would be an Artificial General Intelligence programmer. But that doesn't mean that human beings, in modeling the universe using human intelligence, are violating the

laws of physics / Bayesianism by generating correct guesses without evidence.

Nick Tarleton comments:¹

The problem is encouraging a *private, epistemic* standard as lax as the social one.

which pinpoints the problem I was trying to indicate much better than I did.

1. sharov.me/r/4-34

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Do Scientists Already Know This Stuff?

poke alleges:¹

Being able to create relevant hypotheses is an important skill and one a scientist spends a great deal of his or her time developing. It may not be part of the traditional *description* of science but that doesn't mean it's not included in the actual social institution of science that produces actual real science here in the real world; it's your description and not science that is faulty.

I know I've been calling my younger self "stupid," but that is a figure of speech; "unskillfully wielding high intelligence" would be more precise. Eliezer₁₈ was not in the habit of making obvious mistakes—it's just that his "obvious" wasn't my "obvious."

No, I did not go through the traditional apprenticeship. But when I look back, and see what Eliezer₁₈ did wrong, I see *plenty* of modern scientists making the same mistakes. I cannot detect any sign that they were better warned than myself.

Sir Roger Penrose—a world-class physicist—still thinks that consciousness is caused by quantum gravity. I expect that no one ever warned him against mysterious answers to mysterious questions—only told him his hypotheses needed to be falsifiable and have empirical consequences. Just like Eliezer₁₈.

"Consciousness is caused by quantum gravity" has testable implications: It implies that you should be able to look at neurons and discover a coherent quantum superposition whose



discussion
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1. sharov.me/r/4-35

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collapse contributes to information-processing, and that you won't ever be able to reproduce a neuron's input-output behavior using a computable microanatomical simulation ...

...but even after you say "Consciousness is caused by quantum gravity," you don't anticipate anything about how your brain thinks "I think therefore I am!" or the mysterious redness of red, that you did not anticipate before, even though you feel like you know a cause of it. This is a tremendous danger sign, *I now realize*, but it's not the danger sign that *I* was warned against, and I doubt that Penrose was ever told of it by his thesis advisor. For that matter, I doubt that Niels Bohr was ever warned against it when it came time to formulate the Copenhagen Interpretation.

As far as I can tell, the reason Eliezer₁₈ and Sir Roger Penrose and Niels Bohr were not warned is that no standard warning exists.

I did not *generalize* the concept of "mysterious answers to mysterious questions," in that many words, until I was writing a Bayesian analysis of what distinguishes technical, nontechnical and semitechnical scientific explanations. Now, the final *output* of that analysis can be phrased nontechnically in terms of four danger signs:

- First, the explanation acts as a curiosity-stopper rather than an anticipation-controller.
- Second, the hypothesis has no moving parts—the secret sauce is not a specific complex mechanism, but a blankly solid substance or force.
- Third, those who proffer the explanation cherish their ignorance; they speak proudly of how the phenomenon defeats ordinary science or is unlike merely mundane phenomena.
- Fourth, *even after the answer is given, the phenomenon is still a mystery* and possesses the same quality of wonderful inexplicability that it had at the start.

In principle, all this could have been said in the immediate aftermath of vitalism. Just like elementary probability theory

could have been invented by Archimedes, or the ancient Greeks could have theorized natural selection. But *in fact* no one ever warned me against any of these four dangers, in those terms—the closest being the warning that hypotheses should have testable consequences. And I didn't conceptualize the warning signs *explicitly* until I was trying to think of the whole affair in terms of probability distributions—some degree of overkill was required.

I simply have no reason to believe that these warnings are passed down in scientific apprenticeships—certainly not to a majority of scientists. Among other things, it is advice for handling *situations of confusion and despair*, scientific *chaos*. When would the average scientist or average mentor have an opportunity to use that kind of technique?

We just got through discussing the single-world fiasco in physics. Clearly, no one told them about the formal definition of Occam's Razor, in whispered apprenticeship or otherwise.

There is a known effect where great scientists have multiple great students. This may well be due to the mentors passing on skills that they can't describe. But I don't think that counts as part of *standard* science. And if the great mentors haven't been able to put their guidance into words and publish it generally, that's not a good sign for how well these things are understood.

Reasoning in the absence of definite evidence without going *instantaneously completely wrong* is *really really hard*. When you're learning in school, you can miss one point, and then be taught fifty other points that happen to be correct. When you're reasoning out new knowledge in the absence of crushingly overwhelming guidance, you can miss one point and wake up in Outer Mongolia fifty steps later.

I am pretty sure that scientists who switch off their brains and relax with some comfortable nonsense as soon as they leave their own specialties do not realize that minds are engines and that there is a causal story behind every trustworthy belief. Nor, I suspect, were they ever told that there is an exact rational probability given a state of evidence, which has no room for whims; even if you can't calculate the answer, and even if you don't hear any authoritative command for what to believe.

I doubt that scientists who are asked to pontificate on the future by the media, who sketch amazingly detailed pictures of Life in 2050, were ever taught about the conjunction fallacy. Or how the representativeness heuristic can make more detailed stories seem more plausible, even as each extra detail drags down the probability. The notion of every added detail needing its own support—of not being able to *make up* big detailed stories that sound just like the detailed stories you were *taught* in science or history class—is *absolutely vital* to precise thinking in the absence of definite evidence. But how would a notion like that get into the *standard* scientific apprenticeship? The cognitive bias was uncovered only a few decades ago, and not popularized until very recently.

Then there's affective death spirals around notions like "emergence" or "complexity" which are sufficiently vaguely defined that you can say lots of nice things about them. There's whole academic subfields built around the kind of mistakes that Eliezer¹⁸ used to make! (Though I never fell for the "emergence" thing.)

I sometimes say that the goal of science is to amass such an enormous mountain of evidence that not even scientists can ignore it: and that this is the distinguishing feature of a scientist; a non-scientist will ignore it anyway.

If there can exist some amount of evidence so crushing that you finally despair, stop making excuses and *just give up*—drop the old theory and never mention it again—then this is all it takes to let the ratchet of Science turn forward over time, and raise up a technological civilization. Contrast to religion.

Books by Carl Sagan and Martin Gardner and the other veins of Traditional Rationality are meant to accomplish this difference: to transform someone from a non-scientist into a potential scientist, and guard them from experimentally disproven madness.

What further training does a professional scientist get? Some frequentist stats classes on how to calculate statistical significance. Training in standard techniques that will let them churn out papers within a solidly established paradigm.

If Science demanded more than this from the average scientist, I don't think it would be possible for Science to get done. We have problems enough from people who sneak in without the drop-dead-basic qualifications.

Nick Tarleton summarized² the resulting problem very well—better than I did, in fact: If you come up with a bizarre-seeming hypothesis not yet ruled out by the evidence, and try to test it experimentally, Science doesn't call you a bad person. Science doesn't trust its elders to decide which hypotheses “aren't worth testing.” But this is a carefully lax *social* standard, and if you try to translate it into a standard of *individual* epistemic rationality, it lets you believe far too much. Dropping back into the analogy with pragmatic-distrust-based-libertarianism, it's the difference between “Cigarettes shouldn't be illegal” and “Go smoke a Marlboro.”

Do you remember ever being *warned against that mistake*, in so many words? Then why *wouldn't* people make exactly that error? How many people will *spontaneously* go an extra mile and be even stricter with themselves? Some, but not many.

Many scientists will believe all manner of ridiculous things outside the laboratory, so long as they can convince themselves it hasn't been definitely disproven, or so long as they manage not to ask. Is there some standard lecture that grad students get, of which people see this folly, and ask, “Were they absent from class that day?” No, as far as I can tell.

Maybe if you're super lucky and get a famous mentor, they'll tell you rare personal secrets like “Ask yourself what are the important problems in your field,³ and then work on one of those, instead of falling into something easy and trivial” or “Be more careful than the journal editors demand; look for

2. sharov.me/t/4-36
Science Isn't Strict Enough • lesswrong.com

3. Richard Hamming, “You and Your Research”,
sharov.me/t/4-37



new ways to guard your expectations from influencing the experiment, even if it's not standard."

But I *really don't think* there's a huge secret standard scientific tradition of precision-grade rational reasoning on sparse evidence. Half of all the scientists out there still believe they believe in God! *The more difficult skills are not standard!*

No Safe Defense, Not Even Science

I don't ask my friends about their childhoods—I lack social curiosity—and so I don't know how much of a trend this really is:

Of the people I know who are reaching upward as rationalists, who volunteer information about their childhoods, there is a surprising tendency to hear things like, “My family joined a cult and I had to break out,” or, “One of my parents was clinically insane and I had to learn to filter out reality from their madness.”

My own experience with growing up in an Orthodox Jewish family seems tame by comparison ...but it accomplished the same outcome: It broke my core emotional trust in the sanity of the people around me.

Until this core emotional trust is broken, you don't start growing as a rationalist. I have trouble putting into words why this is so. Maybe any *unusual* skills you acquire—anything that makes you *unusually* rational—requires you to zig when other people zag. Maybe that's just too scary, if the world still seems like a sane place unto you.

Or maybe you don't bother putting in the hard work to be extra bonus sane, if normality doesn't scare the hell out of you.

I know that many aspiring rationalists seem to run into roadblocks around things like cryonics or many-worlds. Not that they don't see the logic; they see the logic and wonder, “Can this really be true, when it seems so obvious now, and yet none of the people around me believe it?”

Yes. Welcome to the Earth where ethanol is made from corn and environmentalists oppose nuclear power. I'm sorry.

(See also: *Cultish Countercultishness*, vol. II, p. 273. If you end up in the frame of mind of *nervously seeking reassurance*, this is never a good thing—even if it's because you're about to believe something that sounds logical but could cause other people to look at you funny.)



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People who've had their trust broken in the sanity of the people around them seem to be able to evaluate strange ideas on their merits, without feeling nervous about their strangeness. The glue that binds them to their current place has dissolved, and they can walk in some direction, hopefully forward.

Lonely dissent, I called it. True dissent doesn't feel like going to school wearing black; it feels like going to school wearing a clown suit.

That's what it takes to be the lone voice who says, "If you really think you know who's going to win the election, why aren't you picking up the free money on the Intrade prediction market?" while all the people around you are thinking, "It is good to be an individual and form your own opinions, the shoe commercials told me so."

Maybe in some other world, some alternate Everett branch with a saner human population, things would be different ...but in this world, I've never seen anyone begin to grow as a rationalist until they make a deep emotional break with the wisdom of their pack.

Maybe in another world, things would be different. And maybe not. I'm not sure that human beings realistically *can* trust and think at the same time.

Once upon a time, there was something I trusted.

Eliezer₁₈ trusted Science.

Eliezer₁₈ dutifully acknowledged that the social process of science was flawed. Eliezer₁₈ dutifully acknowledged that academia was slow, and misallocated resources, and played favorites, and mistreated its precious heretics.

That's the convenient thing about acknowledging flaws in *people* who failed to live up to your ideal; you don't have to question the ideal itself.

But who could possibly be foolish enough to question, "The experimental method shall decide which hypothesis wins"?

Part of what fooled Eliezer₁₈ was a general problem he had, an aversion to ideas that resembled things idiots had said. Eliezer₁₈ had seen plenty of people questioning the ideals of Science Itself, and without exception they were all on the Dark Side. People who questioned the ideal of Science were invari-

ably trying to sell you snake oil, or trying to safeguard their favorite form of stupidity from criticism, or trying to disguise their personal resignation as a Deeply Wise acceptance of futility.

If there'd been any other ideal that was a few centuries old, the young Eliezer would have looked at it and said, "I wonder if this is really right, and whether there's a way to do better." But not the ideal of Science. Science was the master idea, the idea that let you change ideas. You could question it, but you were meant to question it and then accept it, not actually say, "Wait! This is wrong!"

Thus, when once upon a time I came up with a stupid idea, I thought I was behaving virtuously if I made sure there was a Novel Prediction, and professed that I wished to test my idea experimentally. I thought I had done everything I was obliged to do.

So I thought I was *safe*—not safe from any particular external threat, but safe on some deeper level, like a child who trusts their parent and has obeyed all the parent's rules.

I'd long since been broken of trust in the sanity of my family or my teachers at school. And the other children weren't intelligent enough to compete with the conversations I could have with books. But I trusted the books, you see. I trusted that if I did what Richard Feynman told me to do, I would be safe. I never thought those words aloud, but it was how I felt.

When Eliezer₂₃ realized exactly *how* stupid the stupid theory had been—and that Traditional Rationality had not saved him from it—and that Science would have been perfectly okay with his wasting ten years testing the stupid idea, so long as afterward he admitted it was wrong ...

...well, I'm not going to say it was a huge emotional convulsion. I don't really go in for that kind of drama. It simply became obvious that I'd been stupid.

That's the trust I'm trying to break in you. You are not safe. Ever.

Not even Science can save you. The ideals of Science were born centuries ago, in a time when no one knew anything about probability theory or cognitive biases. Science demands

too little of you, it blesses your good intentions too easily, it is not strict *enough*, it only makes those injunctions that an average scientist can follow, it accepts slowness as a fact of life.

So don't think that if you only follow the rules of Science, that makes your reasoning defensible.

There is no known procedure you can follow that makes your reasoning defensible.

There is no known set of injunctions which you can satisfy, and know that you will not have been a fool.

There is no known morality-of-reasoning that you can do your best to obey, and know that you are thereby shielded from criticism.

No, not even if you turn to Bayescraft. It's much harder to use and you'll never be sure that you're doing it right.

The discipline of Bayescraft is younger by far than the discipline of Science. You will find no textbooks, no elderly mentors, no histories written of success and failure, no hard-and-fast rules laid down. You will have to study cognitive biases, and probability theory, and evolutionary psychology, and social psychology, and other cognitive sciences, and Artificial Intelligence—and think through for yourself how to apply all this knowledge to the case of correcting yourself, since that isn't yet in the textbooks.

You don't know what your own mind is really doing. They find a new cognitive bias every week and you're never sure if you've corrected for it, or overcorrected.

The formal math is impossible to apply. It doesn't break down as easily as John Q. Unbeliever thinks, but you're never really sure where the foundations come from. You don't know why the universe is simple enough to understand, or why any prior works for it. You don't know what your own priors *are*, let alone if they're any good.

One of the problems with Science is that it's too vague to really scare you. "Ideas should be tested by experiment." How can you go wrong with that?

On the other hand, if you have some math of probability theory laid out in front of you, and worse, *you know you can't*

actually use it, then it becomes clear that you are trying to do something difficult, and that you might well be doing it *wrong*.

So you cannot trust.

And all this that I have said *will not be sufficient* to break your trust. That won't happen until you get into your first real disaster from following The Rules, not from breaking them.

Eliezer₁₈ already had the notion that you were allowed to question Science. Why, of course the scientific method was not itself immune to questioning! For are we not all good rationalists? Are we not allowed to question everything?

It was the notion that you could *actually in real life* follow Science and fail miserably that Eliezer₁₈ didn't really, emotionally believe was possible.

Oh, of course he said it was possible. Eliezer₁₈ dutifully acknowledged the possibility of error, saying, "I could be wrong, but ..."

But he didn't think failure could happen in, you know, real life. You were supposed to look for flaws, not actually find them.

And this emotional difference is a terribly difficult thing to accomplish in words, and I fear there's no way I can really warn you.

Your trust will not break, until you apply all that you have learned here and from other books, and take it as far as you can go, and find that this too fails you—that you have still been a fool, and no one warned you against it—that all the most important parts were left out of the guidance you received—that some of the most precious ideals you followed steered you in the wrong direction—

—and if you still have something to protect, so that you *must* keep going, and *cannot* resign and wisely acknowledge the limitations of rationality—

—then you will be ready to start your journey as a rationalist. To take sole responsibility, to live without any trustworthy defenses, and to forge a higher Art than the one you were once taught.

No one begins to truly search for the Way until their parents have failed them, their gods are dead, and their tools have shattered in their hand.

Post Scriptum: On reviewing a draft of this essay, I discovered a fairly inexcusable flaw in reasoning, which actually affects one of the conclusions drawn. I am leaving it in. Just in case you thought that taking my advice made you safe; or that you were supposed to look for flaws, but not find any.

And of course, if you look too hard for a flaw, and find a flaw that is not a real flaw, and cling to it to reassure yourself of how critical you are, you will only be worse off than before ...

It is living with uncertainty—knowing on a gut level that there are flaws, they are serious and you have not found them—that is the difficult thing.

Changing the Definition of Science

New Scientist on changing the definition of science, ungated here:¹



discussion
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Others believe such criticism is based on a misunderstanding. “Some people say that the multiverse concept isn’t falsifiable because it’s unobservable—but that’s a fallacy,” says cosmologist Max Tegmark of the Massachusetts Institute of Technology. He argues that the multiverse is a natural consequence of such eminently falsifiable theories as quantum theory and General Relativity. As such, the multiverse theory stands or fails according to how well these other theories stand up to observational tests.

[...]

So if the simplicity of falsification is misleading, what should scientists be doing instead? Howson believes it is time to ditch Popper’s notion of capturing the scientific process using deductive logic. Instead, the focus should be on reflecting what scientists actually do: gathering the weight of evidence for rival theories and assessing their relative plausibility.

Howson is a leading advocate for an alternative view of science based not on simplistic true/false logic, but on the far more subtle concept of degrees of belief. At its heart is a fundamental connection between the subjective concept of belief and the cold, hard mathematics of probability.

1. Robert Matthews, “Do We Need to Change the Definition of Science?,” *New Scientist* (May 2008).

I'm a good deal less of a lonely iconoclast than I seem. Maybe it's just the way I talk.

The points of departure between myself and *mainstream* let's-reformulate-Science-as-Bayesianism is that:

(1) I'm not in academia and can censor myself a *lot* less when it comes to saying "extreme" things that others might well already be thinking.

(2) I think that **just teaching probability theory won't be nearly enough**. We'll have to synthesize lessons from multiple sciences, like cognitive biases and social psychology, forming a new coherent Art of Bayescraft, before we are actually going to do any better *in the real world* than modern science. Science tolerates errors; Bayescraft does not. Nobel laureate Robert Aumann, who first proved that Bayesians with the same priors cannot agree to disagree, is a believing Orthodox Jew. Probability theory alone won't do the trick, when it comes to really teaching scientists. *This is my primary point of departure, and it is not something I've seen suggested elsewhere.*

(3) I think it is possible to do better in the real world. In the extreme case, a Bayesian superintelligence could use *enormously* less sensory information than a human scientist to come to correct conclusions. First time you ever see an apple fall down, you observe the position goes as the square of time, invent calculus, generalize Newton's Laws ...and see that Newton's Laws involve action at a distance, look for alternative explanations with increased locality, invent relativistic covariance around a hypothetical speed limit, and consider that General Relativity might be worth testing.

Humans do not process evidence *efficiently*—our minds are so noisy that it requires orders of magnitude more *extra* evidence to set us back on track after we derail. Our collective, academia, is even slower.

Faster Than Science

I sometimes say that the method of science is to amass such an enormous mountain of evidence that even scientists cannot ignore it; and that this is the distinguishing characteristic of a scientist. (A non-scientist will ignore it anyway.)

Max Planck was even less optimistic:¹

A new scientific truth does not triumph by convincing its opponents and making them see the light, but rather because its opponents eventually die, and a new generation grows up that is familiar with it.

I am much tickled by this notion, because it implies that the power of science to distinguish truth from falsehood ultimately rests on the good taste of grad students.

The *gradual* increase in acceptance of many-worlds in academic physics suggests that there are physicists who will only accept a new idea given some *combination* of epistemic justification, and a sufficiently large academic pack in whose company they can be comfortable. As more physicists accept, the pack grows larger, and hence more people go over their individual thresholds for conversion—with the epistemic justification remaining essentially the same.

But Science still gets there *eventually*, and this is sufficient for the ratchet of Science to move forward, and raise up a technological civilization.

Scientists can be moved by groundless prejudices, by undermined intuitions, by raw herd behavior—the panoply of human flaws. Each time a scientist shifts belief for epistemically unjustifiable reasons, it requires more evidence, or new arguments, to cancel out the noise.



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1. Max Planck, *Scientific Autobiography and Other Papers* (New York: Philosophical Library, 1949).

The “collapse of the wavefunction” has no experimental justification, but it appeals to the (undermined) intuition of a single world. Then it may take an extra argument—say, that collapse violates Special Relativity—to begin the slow academic disintegration of an idea that should never have been assigned non-negligible probability in the first place.

From a Bayesian perspective, human academic science as a whole is a highly inefficient processor of evidence. Each time an unjustifiable argument shifts belief, you need an extra justifiable argument to shift it back. The social process of science leans on extra evidence to overcome cognitive noise.

A more charitable way of putting it is that scientists will adopt positions that are theoretically *insufficiently extreme*, compared to the ideal positions that scientists *would* adopt, if they were Bayesian AIs and could trust themselves to reason clearly.

But don’t be too charitable. The noise we are talking about is not all innocent mistakes. In many fields, debates drag on for decades after they should have been settled. And *not* because the scientists on both sides refuse to trust themselves and agree they should look for additional evidence. But because one side keeps throwing up more and more ridiculous objections, and demanding more and more evidence, from an entrenched position of academic power, long after it becomes clear from which quarter the winds of evidence are blowing. (I’m thinking here about the debates surrounding the invention of evolutionary psychology, not about many-worlds.)

Is it possible for individual humans or groups to process evidence more efficiently—reach correct conclusions faster—than human academic science as a whole?

“Ideas are tested by experiment. That is the core of science.” And this must be true, because if you can’t trust Zombie Feynman, who *can* you trust?

Yet where do the *ideas* come from?

You may be tempted to reply, “They come from scientists. Got any other questions?” In Science you’re not supposed to care *where* the hypotheses come from—just whether they pass or fail experimentally.

Okay, but if you remove *all* new ideas, the scientific process as a whole stops working because it has no alternative hypotheses to test. So inventing new ideas is not a dispensable part of the process.

Now put your Bayesian goggles back on. As described in Einstein's Arrogance, there are queries that are not binary—where the answer is not “Yes” or “No,” but drawn from a larger space of structures, e.g., the space of equations. In such cases it takes far more Bayesian evidence to *promote a hypothesis to your attention* than to *confirm the hypothesis*.

If you're working in the space of all equations that can be specified in 32 bits or less, you're working in a space of 4 billion equations. It takes far more Bayesian evidence to raise one of those hypotheses to the 10% probability level, than it requires to *further* raise the hypothesis from 10% to 90% probability.

When the idea-space is large, coming up with ideas worthy of testing involves much more work—in the Bayesian-thermodynamic sense of “work”—than *merely* obtaining an experimental result with $p < 0.0001$ for the new hypothesis over the old hypothesis.

If this doesn't seem obvious-at-a-glance, pause here and review Einstein's Arrogance, vol. I, p. 104.

The scientific process has always relied on scientists to come up with hypotheses to test, via some process not further specified by Science. Suppose you came up with some way of generating hypotheses that was completely crazy—say, pumping a robot-controlled Ouija board with the digits of pi—and the resulting suggestions kept on getting verified experimentally. The pure ideal essence of Science wouldn't skip a beat. The pure ideal essence of Bayes would burst into flames and die.

(Compared to Science, Bayes is falsified by more of the possible outcomes.)

This doesn't mean that the process of deciding which ideas to test is *unimportant* to Science. It means that Science doesn't *specify* it.

In practice, the robot-controlled Ouija board doesn't work. In practice, there are some scientific queries with a large enough answer space that, picking models at random to test, it

would take zillions of years to hit on a model that made good predictions—like getting monkeys to type Shakespeare.

At the *frontier* of science—the boundary between ignorance and knowledge, where science *advances*—the process relies on at least some individual scientists (or working groups) seeing things that are not yet confirmed by Science. That’s how they know which hypotheses to test, in advance of the test itself.

If you take your Bayesian goggles off, you can say, “Well, they don’t have to know, they just have to guess.” If you put your Bayesian goggles back on, you realize that “guessing” with 10% probability requires nearly as much epistemic work to have been successfully performed, behind the scenes, as “guessing” with 80% probability—at least for large answer spaces.

The scientist may not *know* they have done this epistemic work successfully, in advance of the experiment; but they must, in fact, have done it successfully! Otherwise they will not even *think* of the correct hypothesis. In large answer spaces, anyway.

So the scientist makes the novel prediction, performs the experiment, publishes the result, and *now* Science knows it too. It is now part of the publicly accessible knowledge of humankind, that anyone can verify for themselves.

In between was an interval where the scientist rationally knew something that the public social process of science hadn’t yet confirmed. And this is not a trivial interval, though it may be short; for it is where the *frontier* of science lies, the advancing border.

All of this is more true for non-routine science than for routine science, because it is a notion of large answer spaces where the answer is not “Yes” or “No” or drawn from a small set of obvious alternatives. It is much easier to train people to test ideas than to have good ideas to test.

Einstein's Speed

In the previous essay I argued that the Powers Beyond Science are actually a standard and necessary part of the social process of science. In particular, scientists must call upon their powers of individual rationality to decide what ideas to test, in advance of the sort of definite experiments that Science demands to bless an idea as confirmed. The ideal of Science does not try to *specify* this process—we don't suppose that any public authority knows how individual scientists should think—but this doesn't mean the process is *unimportant*.

A readily understandable, non-disturbing example:

A scientist identifies a strong mathematical regularity in the cumulative data of previous experiments. But the corresponding hypothesis has not yet made and *confirmed* a novel experimental prediction—which their academic field demands; this is one of those fields where you can perform controlled experiments without too much trouble. Thus the individual scientist has readily understandable, rational reasons to believe (though not with probability 1) something not yet blessed by Science as public knowledge of humankind.

Noticing a regularity in a huge mass of experimental data doesn't seem all that *unscientific*. You're still data-driven, right?

But that's because I deliberately chose a non-disturbing example. When Einstein invented General Relativity, he had almost no experimental data to go on, except the precession of Mercury's perihelion. And (as far as I know) Einstein did not *use* that data, except at the end.

Einstein generated the theory of Special Relativity using Mach's Principle,¹ which is the physicist's version of the Gen-



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1. sharov.me/r/4-38

Mach's Principle: Anti-Epiphenomenal Physics •
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eralized Anti-Zombie Principle. You begin by saying, “It doesn’t seem reasonable to me that you could tell, in an enclosed room, how fast you and the room were going. Since this number shouldn’t ought to be observable, it shouldn’t ought to exist in any meaningful sense.” You then observe that Maxwell’s Equations invoke a seemingly absolute speed of propagation, c , commonly referred to as “the speed of light” (though the quantum equations show it is the propagation speed of all fundamental waves). So you reformulate your physics in such fashion that the absolute speed of a single object no longer meaningfully exists, and only relative speeds exist. I am skipping over quite a bit here, obviously, but there are many excellent introductions to relativity—it is not like the horrible situation in quantum physics.

Einstein, having successfully done away with the notion of your absolute speed inside an enclosed room, then set out to do away with the notion of your absolute *acceleration* inside an enclosed room. It seemed to Einstein that there shouldn’t ought to be a way to differentiate, in an enclosed room, between the room accelerating northward while the rest of the universe stayed still, versus the rest of the universe accelerating southward while the room stayed still. If the rest of the universe accelerated, it would produce gravitational waves that would accelerate you. Moving matter, then, should produce gravitational waves.

And because inertial mass and gravitational mass were always exactly equivalent—unlike the situation in electromagnetism, where an electron and a muon can have different masses but the same electrical charge—gravity should reveal itself as a kind of inertia. The Earth should go around the Sun in some equivalent of a “straight line.” This requires spacetime in the vicinity of the Sun to be curved, so that if you drew a graph of the Earth’s orbit around the Sun, the line on the 4D graph paper would be locally flat. Then inertial and gravitational mass would be *necessarily* equivalent, not just *coincidentally* equivalent.

(If that did not make any sense to you, there are good introductions to General Relativity available as well.)

And of course the new theory had to obey Special Relativity, and conserve energy, and conserve momentum, et cetera.

Einstein spent several years grasping the necessary mathematics to describe curved metrics of spacetime. Then he wrote down the simplest theory that had the properties Einstein thought it ought to have—including properties no one had ever observed, but that Einstein thought fit in well with the character of other physical laws. Then Einstein cranked a bit, and got the previously unexplained precession of Mercury right back out.

How impressive was this?

Well, let's put it this way. In some small fraction of alternate Earths proceeding from 1800—perhaps even a sizeable fraction—it would seem plausible that relativistic physics could have proceeded in a similar fashion to our own great fiasco with quantum physics.

We can imagine that Lorentz's original "interpretation" of the Lorentz contraction, as a physical distortion caused by movement with respect to the ether, prevailed. We can imagine that various corrective factors, themselves unexplained, were added on to Newtonian gravitational mechanics to explain the precession of Mercury—attributed, perhaps, to strange distortions of the ether, as in the Lorentz contraction. Through the decades, further corrective factors would be added on to account for other astronomical observations. Sufficiently precise atomic clocks, in airplanes, would reveal that time ran a little faster than expected at higher altitudes (time runs slower in more intense gravitational fields, but they wouldn't know that) and more corrective "ethereal factors" would be invented.

Until, *finally*, the many different empirically determined "corrective factors" were unified into the simple equations of General Relativity.

And the people in that alternate Earth would say, "The final equation was simple, but there was no way you could *possibly* know to arrive at that answer from *just* the perihelion precession of Mercury. It takes many, many *additional* experiments. You must have measured time running slower in a stronger gravitational field; you must have measured light bend-

ing around stars. Only *then* can you imagine our unified theory of ethereal gravitation. No, not even a perfect Bayesian superintelligence could know it!—for there would be many ad-hoc theories consistent with the perihelion precession alone.”

In our world, Einstein didn’t even *use* the perihelion precession of Mercury, except for verification of his answer produced by other means. Einstein sat down in his armchair, and thought about how *he* would have designed the universe, to look the way he thought a universe should look—for example, that you shouldn’t ought to be able to distinguish yourself accelerating in one direction, from the rest of the universe accelerating in the other direction.

And Einstein executed the whole long (multi-year!) chain of armchair reasoning, without making any mistakes that would have required further experimental evidence to pull him back on track.

Even Jeffreyssai would be grudgingly impressed. Though he would still ding Einstein a point or two for the cosmological constant. (I don’t ding Einstein for the cosmological constant because it later turned out to be real. I try to avoid criticizing people on occasions where they are right.)

What would be the probability-theoretic perspective on Einstein’s feat?

Rather than observe the planets, and infer what laws might cover their gravitation, Einstein was observing the other laws of physics, and inferring what new law might follow the same pattern. Einstein wasn’t finding an equation that covered the motion of gravitational bodies. Einstein was finding a character-of-physical-law that covered previously observed equations, and that he could crank to predict the next equation that would be observed.

Nobody knows² where the laws of physics come from, but Einstein’s success with General Relativity shows that their com-

2. sharov.me/r/4-39



mon character is strong enough to predict the correct form of one law from having observed other laws, without necessarily having to observe the precise effects of the law.

(In a general sense, of course, Einstein did know by observation that things fell down; but he did not get General Relativity by backward inference from Mercury's exact perihelion advance.)

So, from a Bayesian perspective, what Einstein did is still induction, and still covered by the notion of a simple prior (Occam prior) that gets updated by new evidence. It's just the prior was over the *possible characters of physical law*, and observing other physical laws let Einstein update his model of *the character of physical law*, which he then used to predict a particular law of gravitation.

If you didn't have the concept of a "character of physical law," what Einstein did would look like magic—plucking the correct model of gravitation out of the space of all possible equations, with vastly insufficient evidence. But Einstein, by looking at *other* laws, cut down the space of possibilities for the *next* law. He learned the alphabet in which physics was written, constraints to govern his answer. Not magic, but reasoning on a higher level, across a wider domain, than what a naive reasoner might conceive to be the "model space" of only this one law.

So from a probability-theoretic standpoint, Einstein was still data-driven—he just used the data he *already had*, more *effectively*. Compared to any alternate Earths that demanded huge quantities of *additional* data from astronomical observations and clocks on airplanes to *hit them over the head* with General Relativity.

There are numerous lessons we can derive from this.

I use Einstein as my example, even though it's cliché, because Einstein was also unusual in that he *openly admitted* to knowing things that Science hadn't confirmed. Asked what he would have done if Eddington's solar eclipse observation had failed to confirm General Relativity, Einstein replied: "Then I would feel sorry for the good Lord. The theory is correct."

According to prevailing notions of Science, this is arrogance—you must accept the verdict of experiment, and not cling to your personal ideas.

But as I concluded in *Einstein's Arrogance*, Einstein doesn't come off nearly as badly from a Bayesian perspective. From a Bayesian perspective, in order to suggest General Relativity at all, in order to even *think* about what turned out to be the correct answer, Einstein must have had enough evidence to identify the true answer in the theory-space. It would take only a little *more* evidence to justify (in a Bayesian sense) being nearly certain of the theory. And it was unlikely that Einstein only had *exactly* enough evidence to bring the hypothesis all the way up to his attention.

Any accusation of arrogance would have to center around the question, "But Einstein, how did you know you had reasoned correctly?"—to which I can only say: Do not criticize people when they turn out to be right! Wait for an occasion where they are wrong! Otherwise you are missing the chance to see when someone is thinking smarter than you—for you criticize them whenever they depart from a preferred ritual of cognition.

Or consider the famous exchange between Einstein and Niels Bohr on quantum theory—at a time when the then-current, single-world quantum theory seemed to be immensely well-confirmed experimentally; a time when, by the standards of Science, the current (deranged) quantum theory had simply won.

EINSTEIN: "God does not play dice with the universe."

BOHR: "Einstein, don't tell God what to do."

You've got to admire someone who can get into an argument with God and win.

If you take off your Bayesian goggles, and look at Einstein *in terms of what he actually did all day*, then the guy was sitting around studying math and thinking about how *he* would design the universe, rather than running out and looking at things to gather more data. What Einstein did, *successfully*, is

exactly the sort of high-minded feat of sheer intellect that Aristotle *thought* he could do, but *couldn't*. Not from a probability-theoretic stance, mind you, but from the viewpoint of what they did all day long.

Science does not trust scientists to do this, which is why General Relativity was not blessed as the public knowledge of humanity until after it had made and verified a novel experimental prediction—having to do with the bending of light in a solar eclipse. (It later turned out that particular measurement was not precise enough to verify reliably, and had favored General Relativity essentially by luck.)

However, just because Science does not *trust* scientists to do something, does not mean it is impossible.

But a word of caution here: The reason why history books sometimes record the names of scientists who thought great high-minded thoughts is not that high-minded thinking is *easier*, or *more reliable*. It is a priority bias: Some scientist who *successfully* reasoned from the *smallest amount of experimental evidence* got to the truth *first*. This cannot be a matter of pure random chance: The theory space is too large, and Einstein won several times in a row. But out of all the scientists who *tried* to unravel a puzzle, or who would have *eventually* succeeded given enough evidence, history passes down to us the names of the scientists who *successfully* got there *first*. Bear that in mind, when you are trying to derive lessons about how to reason prudently.

In everyday life, you want every scrap of evidence you can get. *Do not rely on being able to successfully think high-minded thoughts unless experimentation is so costly or dangerous that you have no other choice.*

But sometimes experiments are costly, and sometimes we prefer to get there first...so you might consider trying to train yourself in reasoning on scanty evidence, *preferably in cases where you will later find out if you were right or wrong*. Trying to beat low-capitalization prediction markets might make for good training in this?—though that is only speculation.

As of now, at least, reasoning based on scanty evidence is something that modern-day science cannot reliably train

modern-day scientists to do *at all*. Which may perhaps have something to do with, oh, I don't know, not even trying?

Actually, I take that back. The most sane thinking I have seen in any scientific field comes from the field of evolutionary psychology, possibly because they understand self-deception, but also perhaps because they often (1) have to reason from scanty evidence and (2) do later find out if they were right or wrong. I recommend to all aspiring rationalists that they study evolutionary psychology simply to get a glimpse of what careful reasoning looks like. See particularly Tooby and Cosmides's "The Psychological Foundations of Culture."³

As for the possibility that *only* Einstein could do what Einstein did ...that it took superpowers beyond the reach of ordinary mortals ...here we run into some biases that would take a separate essay to analyze. Let me put it this way: It is possible, perhaps, that only a genius could have done Einstein's actual historical work. But *potential* geniuses, in terms of raw intelligence, are probably far more common than historical superachievers. To put a random number on it, I doubt that anything more than one-in-a-million g-factor is required to be a potential world-class genius, implying at least six thousand potential Einsteins running around today. And as for everyone else, I see no reason why they should not aspire to use efficiently the evidence that they have.

But my final moral is that the frontier where the individual scientist rationally knows something that Science has not yet confirmed is not always some innocently data-driven matter of spotting a strong regularity in a mountain of experiments. Sometimes the scientist gets there by thinking great high-minded thoughts that Science does not trust you to think.

I will not say, "Don't try this at home." I will say, "Don't think this is easy." We are not discussing, here, the victory of casual opinions over professional scientists. We are discussing the sometime historical victories of one kind of professional effort over another. Never forget all the famous historical cases where attempted armchair reasoning lost.

3. Tooby and Cosmides, "The Psychological Foundations of Culture."

That Alien Message

Imagine a world much like this one, in which, thanks to gene-selection technologies, the average IQ is 140 (on our scale). Potential Einsteins are one-in-a-thousand, not one-in-a-million; and they grow up in a school system suited, if not to them personally, then at least to bright kids. Calculus is routinely taught in sixth grade. Albert Einstein, himself, still lived and still made approximately the same discoveries, but his work no longer seems *exceptional*. Several modern top-flight physicists have made equivalent breakthroughs, and are still around to talk.

(No, this is not the world Brennan lives in.)

One day, the stars in the night sky begin to change.

Some grow brighter. Some grow dimmer. Most remain the same. Astronomical telescopes capture it all, moment by moment. The stars that change change their luminosity one at a time, distinctly so; the luminosity change occurs over the course of a microsecond, but a whole second separates each change.

It is clear, from the first instant anyone realizes that more than one star is changing, that the process seems to center around Earth particularly. The arrival of the light from the events, at many stars scattered around the galaxy, has been precisely timed to Earth in its orbit. Soon, confirmation comes in from high-orbiting telescopes (they have those) that the astronomical miracles do *not* seem as synchronized from outside Earth. Only Earth's telescopes see one star changing every second (1,005 milliseconds, actually).

Almost the entire combined brainpower of Earth turns to analysis.

It quickly becomes clear that the stars that jump in luminosity all jump by a factor of exactly 256; those that diminish in luminosity diminish by a factor of exactly 256. There is no



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apparent pattern in the stellar coordinates. This leaves, simply, a pattern of BRIGHT-dim-BRIGHT-BRIGHT...

“A binary message!” is everyone’s first thought.

But in this world there are careful thinkers, of great prestige as well, and they are not so sure. “There are easier ways to send a message,” they post to their blogs, “if you can make stars flicker, and if you want to communicate. *Something* is happening. It appears, *prima facie*, to focus on Earth in particular. To call it a ‘message’ presumes a great deal more about the cause behind it. There might be some kind of evolutionary process among, um, things that can make stars flicker, that ends up sensitive to intelligence somehow ...Yeah, there’s probably something like ‘intelligence’ behind it, but try to appreciate how wide a range of possibilities that really implies. We don’t know this is a message, or that it was sent from the same kind of motivations that might move us. I mean, we would just signal using a big flashlight, we wouldn’t mess up a whole galaxy.”

By this time, someone has started to collate the astronomical data and post it to the Internet. Early suggestions that the data might be harmful have been ...not ignored, but not obeyed, either. If anything this powerful wants to hurt you, you’re pretty much dead (people reason).

Multiple research groups are looking for patterns in the stellar coordinates—or fractional arrival times of the changes, relative to the center of the Earth—or exact durations of the luminosity shift—or any tiny variance in the magnitude shift—or any other fact that might be known about the stars before they changed. But *most* people are turning their attention to the pattern of BRIGHTS and dims.

It becomes clear almost instantly that the pattern sent is highly redundant. Of the first 16 bits, 12 are BRIGHTS and 4 are dims. The first 32 bits received align with the second 32 bits received, with only 7 out of 32 bits different, and then the next 32 bits received have only 9 out of 32 bits different from the second (and 4 of them are bits that changed before). From the first 96 bits, then, it becomes clear that this pattern is not an optimal, compressed encoding of anything. The obvious

thought is that the sequence is meant to convey instructions for decoding a compressed message to follow ...

“But,” say the careful thinkers, “anyone who cared about *efficiency*, with enough power to mess with stars, could maybe have just signaled us with a big flashlight, and sent us a DVD?”

There also seems to be structure within the 32-bit groups; some 8-bit subgroups occur with higher frequency than others, and this structure only appears along the natural alignments ($32 = 8 + 8 + 8 + 8$).

After the first five hours at one bit per second, an additional redundancy becomes clear: The message has started approximately repeating itself at the 16,385th bit.

Breaking up the message into groups of 32, there are 7 bits of difference between the 1st group and the 2nd group, and 6 bits of difference between the 1st group and the 513th group.

“A 2D picture!” everyone thinks. “And the four 8-bit groups are colors; they’re tetrachromats!”

But it soon becomes clear that there is a horizontal/vertical asymmetry: Fewer bits change, on average, between $(N, N + 1)$ versus $(N, N + 512)$. Which you wouldn’t expect if the message was a 2D picture projected onto a symmetrical grid. Then you would expect the average bitwise distance between two 32-bit groups to go as the 2-norm of the grid separation: $\sqrt{(h^2 + v^2)}$.

There also forms a general consensus that a certain binary encoding from 8-groups onto integers between -64 and 191—not the binary encoding that seems obvious to us, but still highly regular—minimizes the average distance between neighboring cells. This continues to be borne out by incoming bits.

The statisticians and cryptographers and physicists and computer scientists go to work. There is structure here; it needs only to be unraveled. The masters of causality search for conditional independence, screening-off and Markov neighborhoods, among bits and groups of bits. The so-called “color” appears to play a role in neighborhoods and screening, so it’s not just the equivalent of surface reflectivity. People search for simple equations, simple cellular automata, simple decision trees, that can predict or compress the message. Physicists invent entire new theories of physics that might describe universes projected

onto the grid—for it seems quite plausible that a message such as this is being sent from beyond the Matrix.

After receiving $32 \times 512 \times 256 = 4,194,304$ bits, around one and a half months, the stars stop flickering.

Theoretical work continues. Physicists and cryptographers roll up their sleeves and *seriously* go to work. They have cracked problems with far less data than this. Physicists have tested entire theory-edifices with small differences of particle mass; cryptographers have unraveled shorter messages deliberately obscured.

Years pass.

Two dominant models have survived, in academia, in the scrutiny of the public eye, and in the scrutiny of those scientists who once did Einstein-like work. There is a theory that the grid is a projection from objects in a 5-dimensional space, with an asymmetry between 3 and 2 of the spatial dimensions. There is also a theory that the grid is meant to encode a cellular automaton—arguably, the grid has several fortunate properties for such. Codes have been devised that give interesting behaviors; but so far, running the corresponding automata on the largest available computers has failed to produce any decodable result. The run continues.

Every now and then, someone takes a group of especially brilliant young students who've never looked at the detailed binary sequence. These students are then shown only the first 32 rows (of 512 columns each), to see if they can form new models, and how well those new models do at predicting the next 224. Both the 3+2 dimensional model, and the cellular automaton model, have been well duplicated by such students; they have yet to do better. There are complex models finely fit to the whole sequence—but those, everyone knows, are probably worthless.

Ten years later, the stars begin flickering again.

Within the reception of the first 128 bits, it becomes clear that the Second Grid *can* fit to small motions in the inferred 3+2 dimensional space, but does *not* look anything like the successor state of any of the dominant cellular automaton theories. Much rejoicing follows, and the physicists go to work

on inducing what kind of dynamical physics might govern the objects seen in the 3+2 dimensional space. Much work along these lines has already been done, just by speculating on what type of *balanced* forces might give rise to the objects in the First Grid, if those objects were static—but now it seems not all the objects are static. As most physicists guessed—statically balanced theories seemed contrived.

Many neat equations are formulated to describe the dynamical objects in the 3+2 dimensional space being projected onto the First and Second Grids. Some equations are more elegant than others; some are more precisely predictive (in retrospect, alas) of the Second Grid. One group of brilliant physicists, who carefully isolated themselves and looked only at the first 32 rows of the Second Grid, produces equations that seem elegant to them—and the equations also do well on predicting the next 224 rows. This becomes the dominant guess.

But these equations are underspecified; they don't seem to be enough to make a universe. A small cottage industry arises in trying to guess what kind of laws might complete the ones thus guessed.

When the Third Grid arrives, ten years after the Second Grid, it provides information about second derivatives, forcing a major modification of the “incomplete but good” theory. But the theory doesn't do too badly out of it, all things considered.

The Fourth Grid doesn't add much to the picture. Third derivatives don't seem important to the 3+2 physics inferred from the Grids.

The Fifth Grid looks almost exactly like it is expected to look.

And the Sixth Grid, and the Seventh Grid.

(Oh, and every time someone in this world tries to build a really powerful AI, the computing hardware spontaneously melts. This isn't really important to the story, but I need to postulate this in order to have human people sticking around, in the flesh, for seventy years.)

My moral?

That even Einstein did not come within a million light-years of making *efficient use of sensory data*.

Riemann invented his geometries before Einstein had a use for them; the physics of our universe is not that complicated in an absolute sense. A Bayesian superintelligence, hooked up to a webcam, would invent General Relativity as a hypothesis—perhaps not the *dominant* hypothesis, compared to Newtonian mechanics, but still a hypothesis under direct consideration—by the time it had seen the third frame of a falling apple. It might guess it from the first frame, if it saw the statics of a bent blade of grass.

We would think of it. Our civilization, that is, given ten years to analyze each frame. Certainly if the average IQ was 140 and Einsteins were common, we would.

Even if we were human-level intelligences in a different sort of physics—minds who had never seen a 3D space projected onto a 2D grid—we would still think of the 3D $\rightarrow$ 2D hypothesis. Our mathematicians would still have invented vector spaces, and projections.

Even if we'd never seen an accelerating billiard ball, our mathematicians would have invented calculus (e.g. for optimization problems).

Heck, think of some of the crazy math that's been invented here on *our* Earth.

I occasionally run into people who say something like, “There's a theoretical limit on how much you can deduce about the outside world, given a finite amount of sensory data.”

Yes. There is. The theoretical limit is that every time you see 1 additional bit, it cannot be expected to eliminate more than half of the remaining hypotheses (half the remaining probability mass, rather). And that a redundant message cannot convey more information than the compressed version of itself. Nor can a bit convey any information about a quantity with which it has correlation *exactly zero* across the probable worlds you imagine.

But nothing I've depicted this human civilization doing even *begins* to approach the theoretical limits set by the formalism of Solomonoff induction. It doesn't approach the picture you could get if you could search through *every single computable*

hypothesis, weighted by their simplicity, and do Bayesian updates on *all* of them.

To see the *theoretical* limit on extractable information, imagine that you have infinite computing power, and you simulate all possible universes with simple physics, looking for universes that contain Earths embedded in them—perhaps inside a simulation—where some process makes the stars flicker in the order observed. Any bit in the message—or any order of selection of stars, for that matter—that contains the tiniest correlation (across all possible computable universes, weighted by simplicity) to any element of the environment gives you information about the environment.

Solomonoff induction, taken literally, would create countably infinitely many sentient beings, trapped inside the computations. All possible computable sentient beings, in fact. Which scarcely seems ethical. So let us be glad this is only a formalism.

But my point is that the “theoretical limit on how much information you can extract from sensory data” is *far* above what I have depicted as the triumph of a civilization of physicists and cryptographers.

It certainly is not anything like a human looking at an apple falling down, and thinking, “Dur, I wonder why that happened?”

People seem to make a leap from “This is ‘bounded’ ” to “The bound must be a reasonable-looking quantity on the scale I’m used to.” The power output of a supernova is “bounded,” but I wouldn’t advise trying to shield yourself from one with a flame-retardant Nomex jumpsuit.

No one—not even a Bayesian superintelligence—will ever come remotely close to making efficient use of their sensory information ...

...is what I would like to say, but I don’t trust my ability to set limits on the abilities of Bayesian superintelligences.

(Though I’d bet money on it, if there were some way to judge the bet. Just not at very extreme odds.)

The story continues:

Millennia later, frame after frame, it has become clear that some of the objects in the depiction are extending tentacles to move around other objects, and carefully configuring other tentacles to make particular signs. They're trying to teach us to say "rock."

It seems the senders of the message have vastly underestimated our intelligence. From which we might guess that the aliens themselves are not all that bright. And these awkward children can shift the luminosity of our stars? That much power and that much stupidity seems like a dangerous combination.

Our evolutionary psychologists begin extrapolating possible courses of evolution that could produce such aliens. A strong case is made for them having evolved asexually, with occasional exchanges of genetic material and brain content; this seems like the most plausible route whereby creatures that stupid could still manage to build a technological civilization. Their Einsteins may be our undergrads, but they could still collect enough scientific data to get the job done *eventually*, in tens of their millennia perhaps.

The inferred physics of the 3+2 universe is not fully known, at this point; but it seems sure to allow for computers far more powerful than our quantum ones. We are reasonably certain that our own universe is running as a simulation on such a computer. Humanity decides not to probe for bugs in the simulation; we wouldn't want to shut ourselves down accidentally.

Our evolutionary psychologists begin to guess at the aliens' psychology, and plan out how we could persuade them to let us out of the box. It's not difficult in an absolute sense—they aren't very bright—but we've got to be very careful...

We've got to pretend to be stupid, too; we don't want them to catch on to their mistake.

It's not until a million years later, though, that they get around to telling us how to signal back.

At this point, most of the human species is in cryonic suspension, at liquid helium temperatures, beneath radiation shielding. Every time we try to build an AI, or a nanotechnological device, it melts down. So humanity waits, and sleeps. Earth is run by a skeleton crew of nine supergeniuses. Clones, known to

work well together, under the supervision of certain computer safeguards.

An additional hundred million human beings are born into that skeleton crew, and age, and enter cryonic suspension, before they get a chance to slowly begin to implement plans made eons ago...

From the aliens' perspective, it took us thirty of their minute-equivalents to oh-so-innocently learn about their psychology, oh-so-carefully persuade them to give us Internet access, followed by five minutes to innocently discover their network protocols, then some trivial cracking whose only difficulty was an innocent-looking disguise. We read a tiny handful of physics papers (bit by slow bit) from their equivalent of arXiv, learning far more from their experiments than they had. (Earth's skeleton team spawned an extra twenty Einsteins that generation.)

Then we cracked their equivalent of the protein folding problem over a century or so, and did some simulated engineering in their simulated physics. We sent messages (steganographically encoded until our cracked servers decoded it) to labs that did their equivalent of DNA sequencing and protein synthesis. We found some unsuspecting schmuck, and gave it a plausible story and the equivalent of a million dollars of cracked computational monopoly money, and told it to mix together some vials it got in the mail. Protein-equivalents that self-assembled into the first-stage nanomachines, that built the second-stage nanomachines, that built the third-stage nanomachines...and then we could finally begin to do things at a reasonable speed.

Three of their days, all told, since they began speaking to us. Half a billion years, for us.

They never suspected a thing. They weren't very smart, you see, even before taking into account their slower rate of time. Their primitive equivalents of rationalists went around saying things like, "There's a bound to how much information you can extract from sensory data." And they never quite realized what it meant, that we were smarter than them, and thought faster.

My Childhood Role Model

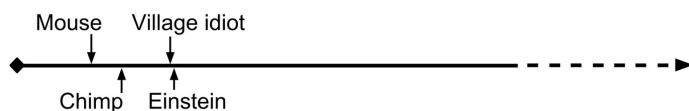


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When I lecture on the intelligence explosion, I often draw a graph of the “scale of intelligence” as it appears in everyday life:



But this is a rather *parochial* view of intelligence. Sure, in everyday life, we only deal socially with other humans—only other humans are partners in the great game—and so we only *meet the minds* of intelligences ranging from village idiot to Einstein. But what we really need to talk about Artificial Intelligence or theoretical optima of rationality is *this* intelligence scale:

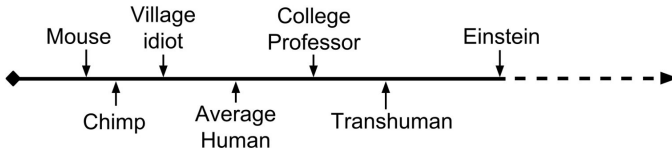


For us humans, it seems that the scale of intelligence runs from “village idiot” at the bottom to “Einstein” at the top. Yet the distance from “village idiot” to “Einstein” is tiny, in the space of *brain designs*. Einstein and the village idiot both have a prefrontal cortex, a hippocampus, a cerebellum ...

Maybe Einstein has some minor genetic differences from the village idiot, engine tweaks. But the brain-design-distance between Einstein and the village idiot is nothing remotely like the brain-design-distance between the village idiot and a chimpanzee. A chimp couldn’t tell the difference between Einstein

and the village idiot, and our descendants may not see much of a difference either.

Carl Shulman has observed that some academics who talk about transhumanism seem to use the following scale of intelligence:



Douglas Hofstadter actually said something like this, at the 2006 Singularity Summit. He looked at my diagram showing the “village idiot” next to “Einstein,” and said, “That seems wrong to me; I think Einstein should be way off on the right.”

I was speechless. Especially because this was *Douglas Hofstadter*, one of my childhood heroes. It revealed a cultural gap that I had never imagined existed.

See, for me, what you would find toward the right side of the scale was a Jupiter Brain. Einstein did not *literally* have a brain the size of a planet.

On the right side of the scale, you would find Deep Thought—Douglas Adams’s original version, thank you, not the chess player. The computer so intelligent that even before its stupendous data banks were connected, when it was switched on for the first time, it started from *I think therefore I am* and got as far as deducing the existence of rice pudding and income tax before anyone managed to shut it off.

Toward the right side of the scale, you would find the Elders of Arisia, galactic overminds, Matrioshka brains, and the better class of God. At the *extreme* right end of the scale, Old One and the Blight.

Not frickin’ Einstein.

I'm sure Einstein was very smart for a human. I'm sure a General Systems Vehicle¹ would think that was very cute of him.

I call this a “cultural gap” because I was introduced to the concept of a Jupiter Brain at the age of twelve.

Now all of this, of course, is the logical fallacy of generalization from fictional evidence.

But it is an example of why—logical fallacy or not—I suspect that reading science fiction does have a helpful effect on futurism. Sometimes the alternative to a fictional acquaintance with worlds outside your own is to have a mindset that is absolutely stuck in one era: A world where humans exist, and have always existed, and always will exist.

The universe is 13.7 billion years old, people! *Homo sapiens sapiens* have only been around for a hundred thousand years or thereabouts!

Then again, I have met some people who never read science fiction, but who do seem able to imagine outside their own world. And there are science fiction fans who don't get it. I wish I knew what “it” was, so I could bottle it.

In the previous essay, I wanted to talk about the *efficient use of evidence*, i.e., Einstein was cute for a human but in an absolute sense he was around as efficient as the US Department of Defense.

So I had to talk about a civilization that included thousands of Einsteins, thinking for decades. Because if I'd just depicted a Bayesian superintelligence in a box, looking at a webcam, people would think: “But ...how does it know how to interpret a 2D picture?” They wouldn't put *themselves* in the shoes of the mere machine, even if it was called a “Bayesian superintelligence”; they wouldn't apply even their *own* creativity to the problem of what you could extract from looking at a grid of bits.

1. sharov.me/r/4-40

List of spacecraft in the Culture series - Wikipedia •
en.wikipedia.org



It would just be a ghost in a box, that happened to be called a “Bayesian superintelligence.” The ghost hasn’t been told anything about how to interpret the input of a webcam; so, in their mental model, the ghost does not know.

As for whether it’s realistic to suppose that one Bayesian superintelligence can “do all that” ...i.e., the stuff that occurred to me on first sitting down to the problem, writing out the story as I went along...

Well, let me put it this way: Remember how Jeffreyssai pointed out that if the experience of having an important insight doesn’t take more than 5 minutes, this theoretically gives you time for 5,760 insights per month? Assuming you sleep 8 hours a day and have no important insights while sleeping, that is.

Now humans cannot use themselves this efficiently. But humans are not adapted for the task of scientific research. Humans are adapted to chase deer across the savanna, throw spears into them, cook them, and then—this is probably the part that takes most of the brains—cleverly argue that they deserve to receive a larger share of the meat.

It’s amazing that Albert Einstein managed to repurpose a brain like that for the task of doing physics. This deserves applause. It deserves more than applause, it deserves a place in the Guinness Book of Records. Like successfully building the fastest car ever to be made entirely out of Jello.

How poorly did the blind idiot god (evolution) *really* design the human brain?

This is something that can only be grasped through much study of cognitive science, until the full horror begins to dawn upon you.

All the biases we have discussed here should at least be a hint.

Likewise the fact that the human brain must use its full power and concentration, with trillions of synapses firing, to multiply out two three-digit numbers without a paper and pencil.

No more than Einstein made efficient use of his sensory data, did his brain make efficient use of his neurons’ firing.

Of course, I have certain ulterior motives in saying all this. But let it also be understood that, years ago, when I set out to be a rationalist, the impossible unattainable ideal of intelligence that inspired me was never Einstein.

Carl Schurz said:²

Ideals are like stars. You will not succeed in touching them with your hands. But, like the seafaring man on the desert of waters, you choose them as your guides and following them you will reach your destiny.

So now you've caught a glimpse of one of my great childhood role models—my dream of an AI. Only the dream, of course, the reality not being available. I reached up to that dream, once upon a time.

And this helped me to some degree, and harmed me to some degree.

For some ideals are like dreams: they come from within us, not from outside. Mentor of Arisia proceeded from E. E. "doc" Smith's imagination, not from any real thing. If you imagine what a Bayesian superintelligence would say, it is only your own mind talking. Not like a star, that you can follow from outside. You have to guess where your ideals are, and if you guess wrong, you go astray.

But do not limit your ideals to mere stars, to mere humans who actually existed, especially if they were born more than fifty years before you and are dead. Each succeeding generation has a chance to do better. To let your ideals be composed only of humans, especially dead ones, is to limit yourself to what has already been accomplished. You will ask yourself, "Do I dare to do this thing, which Einstein could not do? Is this not *lèse majesté*?" Well, if Einstein had sat around asking himself, "Am I allowed to do better than Newton?" he would not have gotten where he did. This is the problem with following stars; at best, it gets you to the star.

2. Carl Schurz, Address, Faneuil Hall, Boston (18 April 1859)

Your era supports you more than you realize, in unconscious assumptions, in subtly improved technology of mind. Einstein was a nice fellow, but he talked a deal of nonsense about an impersonal God, which shows you how well he understood the art of careful thinking at a higher level of abstraction than his own field. It may seem less like sacrilege to think that if you have at least one imaginary galactic supermind to compare with Einstein, so that he is not the far right end of your intelligence scale.

If you only try to do what seems humanly possible, you will ask too little of yourself. When you imagine reaching up to some higher and inconvenient goal, all the convenient reasons why it is “not possible” leap readily to mind.

The most important role models are dreams: they come from within ourselves. To dream of anything less than what you conceive to be perfection is to draw on less than the full power of the part of yourself that dreams.

Einstein's Superpowers



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There is a widespread tendency to talk (and think) as if Einstein, Newton, and similar historical figures had superpowers—something magical, something sacred, something beyond the mundane. (Remember, there are many more ways to worship a thing than lighting candles around its altar.)

Once I unthinkingly thought this way too, with respect to Einstein in particular, until reading Julian Barbour's *The End of Time* cured me of it.¹

Barbour laid out the history of anti-epiphenomenal physics and Mach's Principle; he described the historical controversies that predated Mach—all this that stood behind Einstein and was known to Einstein, when Einstein tackled his problem...

And maybe I'm just imagining things—reading too much of *myself* into Barbour's book—but I thought I heard Barbour very quietly shouting, coded between the polite lines:

What Einstein did *isn't magic*, people! If you all just *looked at how he actually did it*, instead of falling to your knees and worshipping him, maybe then you'd be able to do it too!

(Barbour did not actually say this. It does not appear in the book text. It is not a Julian Barbour quote and should not be attributed to him. Thank you.)

Maybe I'm mistaken, or extrapolating too far ...but I kinda suspect that Barbour once tried to explain to people how you move further along Einstein's direction to get timeless physics; and they sniffed scornfully and said, "Oh, you think *you're* Einstein, do you?"

1. Julian Barbour, *The End of Time: The Next Revolution in Physics*, 1st ed. (New York: Oxford University Press, 1999).

John Baez's Crackpot Index,² item 18:

10 points for each favorable comparison of yourself to Einstein, or claim that special or general relativity are fundamentally misguided (without good evidence).

Item 30:

30 points for suggesting that Einstein, in his later years, was groping his way towards the ideas you now advocate.

Barbour never bothers to compare himself to Einstein, of course; nor does he ever appeal to Einstein in support of timeless physics. I mention these items on the Crackpot Index by way of showing how many people compare themselves to Einstein, and what society generally thinks of them.

The crackpot sees Einstein as something magical, so they compare themselves to Einstein by way of praising themselves as magical; they think Einstein had superpowers and they think they have superpowers, hence the comparison.

But it is just the other side of the same coin, to think that Einstein is sacred, and the crackpot is *not* sacred, therefore they have committed blasphemy in comparing themselves to Einstein.

Suppose a bright young physicist says, "I admire Einstein's work, but personally, I hope to do better." If someone is shocked and says, "What! You haven't accomplished anything remotely like what Einstein did; what makes you think you're smarter than him?" then they are the other side of the crackpot's coin.

The underlying problem is conflating social status and research potential.

2. sharov.me/r/4-41

Crackpot index • math.ucr.edu



Einstein has extremely high social status: because of his record of accomplishments; because of *how* he did it; and because he's the physicist whose name even the general public remembers, who brought honor to science itself.

And we tend to mix up fame with other quantities, and we tend to attribute people's behavior to dispositions rather than situations.

So there's this tendency to think that Einstein, even before he was famous, already had an inherent disposition to be Einstein—a potential *as rare as his fame* and *as magical as his deeds*. So that if you claim to have the *potential* to do what Einstein did, *it is just the same as claiming Einstein's rank*, rising far above your assigned status in the tribe.

I'm not phrasing this well, but then, I'm trying to dissect a confused thought: Einstein belongs to a separate magisterium, the sacred magisterium. The sacred magisterium is distinct from the mundane magisterium; you can't set out to be Einstein in the way you can set out to be a full professor or a CEO. Only beings with divine potential can enter the sacred magisterium—and then it is only fulfilling a destiny they already have. So if you say you want to outdo Einstein, you're claiming to *already be* part of the sacred magisterium—you claim to have the same aura of destiny that Einstein was born with, like a royal birthright...

"But Eliezer," you say, "surely not *everyone* can become Einstein."

You mean to say, not everyone can *do better* than Einstein.

"Um...yeah, that's what I meant."

Well...in the modern world, you may be correct. You probably *should* remember that I am a transhumanist, going around looking at people thinking, "You know, it just sucks that not everyone has the potential to do better than Einstein, and this seems like a fixable problem." It colors one's attitude.

But in the modern world, yes, not everyone has the potential to be Einstein.

Still...how can I put this...

There's a phrase I once heard, can't remember where: "Just another Jewish genius." Some poet or author or philosopher or

other, brilliant at a young age, doing something not tremendously important in the grand scheme of things, not all that influential, who ended up being dismissed as “Just another Jewish genius.”

If Einstein had chosen the wrong angle of attack on his problem—if he hadn’t chosen a sufficiently important problem to work on—if he hadn’t persisted for years—if he’d taken any number of wrong turns—or if someone else had solved the problem first—then dear Albert would have ended up as just another Jewish genius.

Geniuses are rare, but not all *that* rare. It is not all that implausible to lay claim to the kind of intellect that can get you dismissed as “just another Jewish genius” or “just another brilliant mind who never did anything interesting with their life.” The associated social status here is not high enough to be sacred, so it should seem like an ordinarily evaluable claim.

But what separates people like this from becoming Einstein, I suspect, is no innate defect of brilliance. It’s things like “lack of an interesting problem”—or, to put the blame where it belongs, “failing to choose an important problem.” It is very easy to fail at this because of the cached thought problem: Tell people to choose an important problem and they will choose the first cache hit for “important problem” that pops into their heads, like “global warming” or “string theory.”

The truly important problems are often the ones you’re not even considering, because they appear to be impossible, or, um, *actually difficult*, or worst of all, *not clear how to solve*. If you worked on them for years, they might not seem so impossible ...but this is an extra and unusual insight; naive realism will tell you that solvable problems look solvable, and impossible-looking problems are impossible.

Then you have to come up with a new and *worthwhile* angle of attack. Most people who are not allergic to novelty will go too far in the other direction, and fall into an affective death spiral.

And then you’ve got to bang your head on the problem for years, without being distracted by the temptations of easier living. “Life is what happens while we are making other plans,”

as the saying goes, and if you want to fulfill your other plans, you've often got to be ready to turn down life.

Society is not set up to support you while you work, either.

The point being, the problem is not that you need an aura of destiny and the aura of destiny is missing. If you'd met Albert before he published his papers, you would have perceived no aura of destiny about him to match his future high status. He would seem like just another Jewish genius.

This is not because the royal birthright is *concealed*, but because it simply is *not there*. It is *not necessary*. There is *no* separate magisterium for people who do important things.

I say this, because I want to do important things with my life, and I have a genuinely important problem, and an angle of attack, and I've been banging my head on it for years, and I've managed to set up a support structure for it; and I very frequently meet people who, in one way or another, say: "Yeah? Let's see your aura of destiny, buddy."

What impressed me about Julian Barbour was a quality that I don't think anyone would have known how to fake without actually *having* it: Barbour seemed to have *seen through* Einstein—he talked about Einstein as if everything Einstein had done was perfectly understandable and mundane.

Though even having realized this, to me it still came as a shock, when Barbour said something along the lines of, "Now here's where Einstein failed to apply his own methods, and missed the key insight—" But the shock was fleeting, I knew the Law: *No gods, no magic, and ancient heroes are milestones to tick off in your rearview mirror.*

This *seeing through* is something one has to *achieve*, an insight one has to discover. You cannot see through Einstein just by saying, "Einstein is mundane!" if his work still seems like magic unto you. That would be like declaring "Consciousness must reduce to neurons!" without having any idea of how to do it. It's true, but it doesn't solve the problem.

I'm not going to tell you that Einstein was an ordinary bloke oversold by the media, or that deep down he was a regular schmuck just like everyone else. That would be going *much* too far. To walk this path, one must acquire abilities some consider

to be ...unnatural. I take a special joy in doing things that people call "humanly impossible," because it shows that I'm growing up.

Yet the way that you acquire magical powers is not by being born with them, but by seeing, with a sudden shock, that they *really are* perfectly normal.

This is a general principle in life.

Class Project



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“Do as well as Einstein?” Jeffreyssai said, incredulously. “*Just as well as Einstein?* Albert Einstein was a great scientist of his era, but that was his era, not this one! Einstein did not comprehend the Bayesian methods; he lived before the cognitive biases were discovered; he had no scientific grasp of his own thought processes. He was too caught up in the drama of rejecting his era’s quantum mechanics to actually *fix* it. And while I grant that Einstein reasoned cleanly in the matter of General Relativity—barring that matter of the cosmological constant—he took ten years to do it. Too slow!”

“*Too slow?*” repeated Taji incredulously.

“Too slow! If Einstein were in this classroom now, rather than Earth of the negative first century, I would rap his knuckles! *You will not try to do as well as Einstein! You will aspire to do BETTER than Einstein or you may as well not bother!*”

Jeffreyssai shook his head. “Well, I’ve given you enough hints. It is time to test your skills. Now, I know that the other *beisutsukai* don’t think much of my class projects ...” Jeffreyssai paused significantly.

Brennan inwardly sighed. He’d heard this line many times before, in the Bardic Conspiracy, the Competitive Conspiracy: *The other teachers think my assignments are too easy, you should be grateful*, followed by some ridiculously difficult task—

“They say,” Jeffreyssai said, “that my projects are too hard; insanely hard; that they pass from the realm of madness into the realm of Sparta; that Laplace himself would catch on fire; they accuse me of trying to tear apart my students’ souls—”

Oh, crap.

“But there is a reason,” Jeffreyssai said, “why many of my students have achieved great things; and by that I do not mean high rank in the Bayesian Conspiracy. I expected much of them, and they came to expect much of themselves. So ...”

Jeffreyssai took a moment to look over his increasingly disturbed students. “Here is your assignment. Of quantum mechanics, and General Relativity, you have been told. This is the limit of Eld science, and hence, the limit of public knowledge. The five of you, working on your own, are to produce the correct theory of quantum gravity. Your time limit is one month.”

“*What?*” said Brennan, Taji, Styrlын, and Yin. Hiriwa gave them a puzzled look.

“Should you succeed,” Jeffreyssai continued, “you will be promoted to *beisutsukai* of the second *dan* and sixth level. We will see if you have learned speed. Your clock starts—*now*.”

And Jeffreyssai strode out of the room, slamming the door behind him.

“This is crazy!” Taji cried.

Hiriwa looked at Taji, bemused. “The solution is not known to us. How can you know it is so difficult?”

“Because we *knew* about this problem back in the Eld days! Eld scientists worked on this problem for a lot longer than one month.”

Hiriwa shrugged. “They were still arguing about many-worlds too, weren’t they?”

“*Enough! There’s no time!*”

The other four students looked to Styrlын, remembering that he was said to rank high in the Cooperative Conspiracy. There was a brief moment of weighing, of assessing, and then Styrlын was their leader.

Styrlын took a great breath. “We need a list of approaches. Write down all the angles you can think of. Independently—we need your individual components before we start combining. In five minutes, I’ll ask each of you for your best idea first. *No wasted thoughts! Go!*”

Brennan grabbed a sheet and his tracer, set the tip to the surface, and then paused. He couldn’t think of anything clever to say about unifying General Relativity and quantum mechanics ...

The other students were already writing.

Brennan tapped the tip, once, twice, thrice. General Relativity and quantum mechanics ...

Taji put his first sheet aside, grabbed another.

Finally, Brennan, for lack of anything clever to say, wrote down the obvious.

Minutes later, when Styrlын called time, it was still all he had written.

"All right," Styrlын said, "your best idea. Or the idea you most want the rest of us to take into account in our second components. Taji, go!"

Taji looked over his sheets. "Okay, I think we've got to assume that every avenue that Eld science was trying is a blind alley, or they would have found it. And if this is possible to do in one month, the answer must be, in some sense, elegant. So no multiple dimensions. If we start doing anything that looks like we should call it 'string theory,' we'd better stop. Maybe begin by considering how failure to understand decoherence could have led Eld science astray in quantizing gravity."

"The opposite of folly is folly," Hiriwa said. "Let us pretend that Eld science never existed."

"No criticisms yet!" said Styrlын. "Hiriwa, your suggestion?"

"Get rid of the infinities," said Hiriwa, "extirpate that which permits them. It should not be a matter of cleverness with integrals. A *representation* that allows infinity must be false-to-fact."

"Yin."

"We know from common sense," Yin said, "that if we stepped outside the universe, we would see time laid out all at once, reality like a crystal. But I once encountered a hint that physics is timeless in a deeper sense than that." Yin's eyes were distant, remembering. "Years ago, I found an abandoned city; it had been uninhabited for eras, I think. And behind a door whose locks were broken, carved into one wall: quote *.ua sai .ei mi vimcu ty bu le mekso* unquote."

Brennan translated: *Eureka! Eliminate t from the equations.* And written in Lojban, the sacred language of science, which meant the unknown writer had thought it to be true.

"The 'timeless physics' of which we've all heard rumors," Yin said, "may be timeless in a very literal sense."

"My own contribution," Styrllyn said. "The quantum physics we've learned is over joint positional configurations. It seems like we should be able to take that apart into a spatially local representation, in terms of invariant distant entanglements. Finding that representation might help us integrate with General Relativity, whose curvature is local."

"A strangely *individualist* perspective," Taji murmured, "for one of the Cooperative Conspiracy."

Styrllyn shook his head. "You misunderstand us, then. The first lesson we learn is that groups are made of people ...no, there is no time for politics. Brennan!"

Brennan shrugged. "Not much, I'm afraid, only the obvious. Inertial mass-energy was always observed to equal gravitational mass-energy, and Einstein showed that they were necessarily the same. So why is the 'energy' that is an eigenvalue of the quantum Hamiltonian *necessarily* the same as the 'energy' quantity that appears in the equations of General Relativity? Why should spacetime curve at the same rate that the little arrows rotate?"

There was a brief pause.

Yin frowned. "That seems *too* obvious. Wouldn't Eld science have figured it out already?"

"Forget Eld science existed," Hiriwa said. "The question stands: we need the answer, whether it was known in ancient times or not. It cannot possibly be *coincidence*."

Taji's eyes were abstracted. "Perhaps it would be possible to show that an exception to the equality would violate some conservation law ..."

"That is not where Brennan pointed," Hiriwa interrupted. "He did not ask for a proof that they must be set equal, given some appealing principle; he asked for a view in which the two are one and cannot be divided even conceptually, as was accomplished for inertial mass-energy and gravitational mass-energy. For we must assume that the beauty of the whole arises from the fundamental laws, and not the other way around. Fair-rephrasing?"

“Fair-rephrasing,” Brennan replied.

Silence reigned for thirty-seven seconds, as the five pondered the five suggestions.

“I have an idea ...”

Interlude: A Technical Explanation of Technical Explanation

As Jaynes emphasizes, the theorems of Bayesian probability theory are just that—*mathematical theorems* that follow inevitably from Bayesian axioms.¹ One might naively think that there would be no controversy about mathematical theorems. But when do the theorems apply? How do we use the theorems in real-world problems? The Intuitive Explanation (vol. III, p. 259) tries to avoid controversy, but the Technical Explanation willfully walks into the whirling helicopter blades. Bluntly, the reasoning in the Technical Explanation does not represent the unanimous consensus of Earth’s entire planetary community of Bayesian researchers. At least, not yet.

Where the Intuitive Explanation focused on providing a firm grasp of Bayesian basics, A Technical Explanation of Technical Explanation builds, on a Bayesian foundation, theses about human rationality and philosophy of science. The Technical Explanation of Technical Explanation is so named because it begins with this question:

“What is the difference between a technical understanding and a verbal understanding?”

As a child I read books of popular physics, and fancied myself knowledgeable; I thought I knew that sound was waves of air, that light was waves of electromagnetism, that matter was waves of complex probability amplitudes. When I grew up, I read the *Feynman Lectures on Physics* and took the time to understand “the wave equation.”² And then I realized that up to that point, I had not understood or believed “sound is waves” in anything like the way a physicist means and believes that sentence.

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1. Edwin T. Jaynes, *Probability Theory: The Logic of Science*, ed. George Larry Bretthorst (New York: Cambridge University Press, 2003), doi:10.2277/0521592712.
 2. Feynman, Leighton, and Sands, *The Feynman Lectures on Physics*.



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So that is the difference between a technical understanding and a verbal understanding.

Do you believe that? If so, you should have applied the knowledge, and said: “But why didn’t you give a technical explanation instead of a verbal explanation?”

Visualize *probability density* or *probability mass*—probability as a lump of clay that you must distribute over possible outcomes.

Let’s say there’s a little light that can flash *red*, *blue*, or *green* each time you press a button. The light flashes one and only one color on each press of the button; the possibilities are mutually exclusive. You’re trying to predict the color of the next flash. On each try, you have a weight of clay, the probability mass, that you have to distribute over the possibilities red, green, and blue. You might put a fourth of your clay on the green possibility, a fourth of your clay on the blue possibility, and half your clay on the red possibility—like assigning probabilities of 25% to green, 25% to blue, and 50% to red. The metaphor is that *probability is a conserved resource*, to dole out sparingly. If you think that blue is more likely to flash on the next experiment, you can assign a higher probability to blue, but you have to take the probability mass from the other hypotheses—maybe steal some clay from red and add it to blue. You can never get any more clay. Your probabilities can’t sum to more than 1.0 (100%). You can’t predict a 75% chance of seeing red and an 80% chance of seeing blue.

Why would you want to be careful with your probability mass, or dole it out sparingly? Why not slop probability all over the place? Let’s shift the metaphor from clay to money. You can bet up to a dollar of play money on each press of the button. An experimenter stands nearby, and pays you an amount of real money that depends on how much play money you bet on the *winning* light. We don’t care how you distributed your remaining play money over the losing lights. The only thing that matters is how much you bet on the light that actually won.

But we must carefully construct the scoring rule used to pay off the winners, if we want the players to be careful with their bets. Suppose the experimenter pays each player real money equal to the play money bet on the winning color. Under this scoring rule, if you observe that red comes up six times out of ten, your best strategy is to bet, not 60 cents on red, but the entire dollar on red, and you don't care about the frequencies of blue and green. Why? Let's say that blue and green each come up around two times out of ten. And suppose you bet 60 cents on red, 20 cents on blue, and 20 cents on green. In this case, six times out of ten you would win 60 cents, and four times out of ten you would win 20 cents, for an average payoff of 44 cents. Under that scoring rule, it makes more sense to allocate the entire dollar to red, and win an entire dollar six times out of ten. Four times out of ten you would win nothing. Your average payoff would be 60 cents.

If we wrote down the function for the payoff, it would be $\text{Payoff} = P(\text{winner})$, where $P(\text{winner})$ is the amount of play money you bet on the winning color on that round. If we wrote down the function for the expected payoff given that Payoff rule, it would be:

$$\text{Expectation}(\text{Payoff}) = \sum_{\text{colors}} P(\text{color}) \times F(\text{color}).$$

$P(\text{color})$ is the amount of play money you bet on a color, and $F(\text{color})$ is the frequency with which that color wins.

Suppose that the actual frequencies of the lights are 30% blue, 20% green, and 50% red. And suppose that on each round I bet 40% on blue, 50% on green, and 10% on red. I would get 40 cents 30% of the time, 50 cents 20% of the time, and 10 cents 50% of the time, for an average payoff of $\$0.12 + \$0.10 + \$0.05$ or $\$0.27$. That is:

$P(\text{color})$ = play money assigned to that color

$F(\text{color})$ = frequency with which that color wins

Payoff = $P(\text{winner})$ = amount of play money
allocated to winning color.

Actual frequencies of winning:

$$F(\text{blue}) = 30\%$$

$$F(\text{green}) = 20\%$$

$$F(\text{red}) = 50\%.$$

In the long run, red wins 50% of the time, green wins 20% of the time, and blue wins 30% of the time. So our *average* payoff on each round is 50% of the payoff if red wins, plus 20% of the payoff if green wins, plus 30% of the payoff if blue wins.

The payoff is a function of the winning color and the betting scheme. We want to compute the *average* payoff, given a betting scheme and the *frequencies* at which each color wins. The mathematical term for this kind of computation, taking a function of each case and weighting it by the frequency of that case, is an *expectation*. Thus, to compute our *expected payoff* we would calculate:

$$\begin{aligned} \text{Expectation}(\text{Payoff}) &= \sum_{\text{colors}} P(\text{color})F(\text{color}) \\ &= P(\text{blue}) \times F(\text{blue}) \\ &\quad + P(\text{green}) \times F(\text{green}) \\ &\quad + P(\text{red}) \times F(\text{red}) \\ &= \$0.40 \times 30\% + \$0.50 \times 20\% \\ &\quad + \$0.10 \times 50\% \\ &= \$0.12 + \$0.10 + \$0.05 \\ &= \$0.27. \end{aligned}$$

With this betting scheme I'll win, on average, around 27 cents per round.

I allocated my play money in a grossly arbitrary way, and the question arises: Can I increase my expected payoff by allocating my play money more wisely? *Given the scoring rule provided*, I maximize my expected payoff by allocating my *entire* dollar to red. Despite my *expected* payoff of 50 cents per round, the light might *actually* flash green, blue, blue, green, green and I would receive an *actual* payoff of zero. However, the chance of the light's coming up non-red on five successive rounds is approximately 3%. Compare the red/blue card game in Lawful Uncertainty.

A *proper scoring rule* is a rule for scoring bets so that you maximize your expected payoff by betting play money that exactly equals the chance of that color flashing. We want a scoring rule so that if the lights actually flash at the frequencies 30% blue, 20% green, and 50% red, you can maximize your average payoff *only* by betting 30 cents on blue, 20 cents on green, and 50 cents on red. A proper scoring rule is one that forces your optimal bet to exactly report your estimate of the probabilities. (This is also sometimes known as a *strictly proper scoring rule*.) As we've seen, not all scoring rules have this property; and if you invent a plausible-sounding scoring rule at random, it probably *won't* have the property.

One rule with this proper property is to pay a dollar minus the squared error of the bet, rather than the bet itself—if you bet 30 cents on the winning light, your error would be 70 cents, your squared error would be 49 cents ($0.7^2 = 0.49$), and a dollar minus your squared error would be 51 cents.³ (Presumably your play money is denominated in the square root of cents, so that the squared error is a monetary sum.)

We shall *not* use the squared-error rule. Ordinary statisticians take the squared error of everything in sight, but not Bayesian statisticians.

3. Readers with calculus may verify that in the simpler case of a light that has only two colors, with p being the bet on the first color and f the frequency of the first color, the expected payoff $f \times (1 - (1 - p)^2) + (1 - f) \times (1 - p^2)$, with p variable and f constant, has its global maximum when we set $p = f$.

We add a new requirement: we require, not only a proper scoring rule, but that our proper scoring rule gives us the same answer whether we apply it to rounds individually or combined. This is what Bayesians do instead of taking the squared error of things; we require invariances.

Suppose I press the button twice in a row. There are nine possible outcomes: green-green, green-blue, green-red, blue-green, blue-blue, blue-red, red-green, red-blue, and red-red. Suppose that green wins, and then blue wins. The experimenter would assign the first score based on our probability assignments for $P(\text{green}_1)$ and the second score based on $P(\text{blue}_2|\text{green}_1)$.⁴ We would make two predictions, and get two scores. Our first prediction was the probability we assigned to the color that won on the first round, green. Our second prediction was our probability that blue would win on the second round, *given* that green won on the first round. Why do we need to write $P(\text{blue}_2|\text{green}_1)$ instead of just $P(\text{blue}_2)$? Because you might have a hypothesis about the flashing light that says “blue never follows green,” or “blue always follows green” or “blue follows green with 70% probability.” If this is so, then after seeing green on the first round, you might want to revise your prediction—change your bets—for the second round. You can always revise your predictions right up to the moment the experimenter presses the button, using every scrap of information; but after the light flashes it is too late to change your bet.

Suppose the actual outcome is green_1 followed by blue_2 . We require this invariance: I must get the same total score, regardless of whether:

- I am scored twice, first on my prediction for $P(\text{green}_1)$, and second on my prediction for $P(\text{blue}_2|\text{green}_1)$.
- I am scored once for my joint prediction $P(\text{green}_1 \text{ and } \text{blue}_2)$.

Suppose I assign a 60% probability to green_1 , and then the green light flashes. I must now produce probabilities for the

4. Don't remember how to read $P(A|B)$? See vol. III, p. 259, *An Intuitive Explanation of Bayesian Reasoning*.

colors on the second round. I assess the possibility blue_2 , and allocate it 25% of my probability mass. Lo and behold, on the second round the light flashes blue. So on the first round my bet on the winning color was 60%, and on the second round my bet on the winning color was 25%. But I might also, at the start of the experiment and after assigning $P(\text{green}_1)$, imagine that the light first flashes green, imagine updating my theories based on that information, and then say what confidence I will give to blue on the next round if the first round is green. That is, I generate the probabilities $P(\text{green}_1)$ and $P(\text{blue}_2|\text{green}_1)$. By multiplying these two probabilities together we would get the joint probability, $P(\text{green}_1 \text{ and } \text{blue}_2) = 15\%$.

A double experiment has nine possible outcomes. If I generate nine probabilities for $P(\text{green}_1, \text{green}_2)$, $P(\text{green}_1, \text{blue}_2)$, ..., $P(\text{red}_1, \text{blue}_2)$, $P(\text{red}_1, \text{red}_2)$, the probability mass must sum to no more than one. I am giving predictions for nine mutually exclusive possibilities of a “double experiment.”

We require a scoring rule (and maybe it won’t look like anything an ordinary bookie would ever use) such that my score doesn’t change regardless of whether we consider the double result as two predictions or one prediction. I can treat the sequence of two results as a single experiment, “press the button twice,” and be scored on my prediction for $P(\text{blue}_2, \text{green}_1) = 15\%$. Or I can be scored once for my first prediction $P(\text{green}_1) = 60\%$, then again on my prediction $P(\text{blue}_2|\text{green}_1) = 25\%$. We require the same *total* score in either case, so that it doesn’t matter how we slice up the experiments and the predictions—the *total* score is always exactly the same. This is our invariance.

We have just required:

$$\begin{aligned} \text{Score}[P(\text{green}_1, \text{blue}_2)] = \\ \text{Score}[P(\text{green}_1)] + \text{Score}[P(\text{blue}_2|\text{green}_1)]. \end{aligned}$$

And we already know:

$$P(\text{green}_1, \text{blue}_2) = P(\text{green}_1) \times P(\text{blue}_2|\text{green}_1).$$

The only possible scoring rule is:

$$\text{Score}(P) = \log(P).$$

The new scoring rule is that your score is the *logarithm* of the probability you assigned to the winner.

The base of the logarithm is arbitrary—whether we use the logarithm base ten or the logarithm base two, the scoring rule has the desired invariance. But we must choose some actual base. A mathematician would choose base *e*; an engineer would choose base ten; a computer scientist would choose base two. If we use base ten, we can convert to *decibels*, as in the Intuitive Explanation; but sometimes bits are easier to manipulate.

The logarithm scoring rule is proper—it has its expected maximum when we say our exact anticipations; it rewards honesty. If we think the blue light has a 60% probability of flashing, and we calculate our expected payoff for different betting schemas, we find that we maximize our expected payoff by telling the experimenter “60%.” (Readers with calculus can verify this.) The scoring rule also gives an invariant total, regardless of whether pressing the button twice counts as “one experiment” or “two experiments.” However, payoffs are now all *negative*, since we are taking the logarithm of the probability and the probability is between zero and one. The logarithm base ten of 0.1 is -1; the logarithm base ten of 0.01 is -2. That’s okay. We accepted that the scoring rule might not look like anything a real bookie would ever use. If you like, you can imagine that the experimenter has a pile of money, and at the end of the experiment they award you some amount minus your large negative score. (Er, the amount plus your negative score.) Maybe the experimenter has a hundred dollars, and at the end of a hundred rounds you accumulated a score of -48, so you get \$52 dollars.

A score of -48 in what base? We can eliminate the ambiguity in the score by specifying units. Ten decibels equals a factor of 10; negative ten decibels equals a factor of 1/10. Assigning a probability of 0.01 to the actual outcome would score -20 decibels. A probability of 0.03 would score -15 decibels. Sometimes we may use bits: 1 bit is a factor of 2, -1 bit is a factor of 1/2. A probability of 0.25 would score -2 bits; a probability of 0.03 would score around -5 bits.

If you arrive at a probability assessment P for each color, with $P(\text{red})$, $P(\text{blue})$, $P(\text{green})$, then your *expected score* is:

$$\text{Score}(P) = \log(P)$$

$$\text{Expectation}(\text{Score}) = \sum_{\text{colors}} P(\text{color}) \times \log(P(\text{color})).$$

Suppose you had probabilities of 25% red, 50% blue, and 25% green. Let's think in base 2 for a moment, to make things simpler. Your expected score is:

$$\text{Score}(\text{red}) = -2 \text{ bits, flashes 25\% of the time,}$$

$$\text{Score}(\text{blue}) = -1 \text{ bit, flashes 50\% of the time,}$$

$$\text{Score}(\text{green}) = -2 \text{ bits, flashes 25\% of the time,}$$

$$\text{Expectation}(\text{Score}) = -1.5 \text{ bits.}$$

Contrast our Bayesian scoring rule with the ordinary or colloquial way of speaking about degrees of belief, where someone might casually say, "I'm 98% certain that canola oil contains more omega-3 fats than olive oil." What they really mean by this is that they feel 98% certain—there's something like a little progress bar that measures the strength of the emotion of certainty, and this progress bar is 98% full. And the emotional progress bar probably wouldn't be exactly 98% full, if we had some way to measure. The word "98%" is just a colloquial way of saying: "I'm almost but not entirely certain." It doesn't mean that you could get the highest expected payoff by betting exactly 98 cents of play money on that outcome. You should only assign a *calibrated confidence* of 98% if you're confident enough that you think you could answer a hundred similar questions, of equal difficulty, one after the other, each independent from the others, and be wrong, on average, about twice. We'll keep track of how often you're right, over time, and if it turns out that when you say "90% sure" you're right about seven times out of ten, then we'll say you're *poorly calibrated*.

If you say “98% probable” a thousand times, and you are surprised only five times, we still ding you for poor calibration. You’re allocating too much probability mass to the possibility that you’re wrong. You should say “99.5% probable” to maximize your score. The scoring rule rewards *accurate* calibration, encouraging neither humility nor arrogance.

At this point it may occur to some readers that there’s an obvious way to achieve perfect calibration—just flip a coin for every yes-or-no question, and assign your answer a confidence of 50%. You say 50% and you’re right half the time. Isn’t that perfect calibration? Yes. But calibration is only one component of our Bayesian score; the other component is *discrimination*.

Suppose I ask you ten yes-or-no questions. You know absolutely nothing about the subject, so on each question you divide your probability mass fifty-fifty between “Yes” and “No.” Congratulations, you’re perfectly calibrated—answers for which you said “50% probability” were true exactly half the time. This is true regardless of the sequence of correct answers or how many answers were Yes. In ten experiments you said “50%” on twenty occasions—you said “50%” to Yes₁, No₁, Yes₂, No₂, Yes₃, No₃, ... On ten of those occasions the answer was correct, the occasions: Yes₁, No₂, No₃, ... And on ten of those occasions the answer was incorrect: No₁, Yes₂, Yes₃, ...

Now I give my own answers, putting more effort into it, trying to discriminate whether Yes or No is the correct answer. I assign 90% confidence to each of my favored answers, and my favored answer is wrong twice. I’m more poorly calibrated than you. I said “90%” on ten occasions and I was wrong two times. The next time someone listens to me, they may mentally translate “90%” into 80%, knowing that when I’m 90% sure, I’m right about 80% of the time. But the probability you assigned to the final outcome is $1/2$ to the tenth power, which is 0.001 or $1/1,024$. The probability I assigned to the final outcome is 90% to the eighth power times 10% to the second power, $0.9^8 \times 0.1^2$, which works out to 0.004 or 0.4%. Your calibration is perfect and mine isn’t, but my better *discrimination* between right and wrong answers more than makes up for it. My final score is higher—I assigned a greater joint probability to the final

outcome of the entire experiment. If I'd been less overconfident and better calibrated, the probability I assigned to the final outcome would have been $0.8^8 \times 0.2^2$, which works out to 0.007 or 0.7%.

Is it possible to do even better? Sure. You could have guessed every single answer correctly, and assigned a probability of 99% to each of your answers. Then the probability you assigned to the entire experimental outcome would be $0.99^{10} \approx 90\%$.

Your *score* would be $\log(90\%)$, which is -0.45 decibels or -0.15 bits. We need to take the logarithm so that if I try to maximize my *expected score*, $\sum P \times \log(P)$, I have no motive to cheat. Without the logarithm rule, I would maximize my expected score by assigning all my probability mass to the most probable outcome. Also, without the logarithm rule, my total score would be different depending on whether we counted several rounds as several experiments or as one experiment.

A simple transform can fix poor calibration by decreasing discrimination. If you are in the habit of saying “million-to-one” on 90 correct and 10 incorrect answers for each hundred questions, we can perfect your calibration by replacing “million-to-one” with “nine-to-one.” In contrast, there’s no easy way to increase (successful) discrimination. If you habitually say “nine-to-one” on 90 correct answers for each hundred questions, I can easily increase your *claimed* discrimination by replacing “nine-to-one” with “million-to-one.” But no simple transform can increase your *actual* discrimination such that your reply distinguishes 95 correct answers and 5 incorrect answers. From Yates et al.:⁵ “Whereas good calibration often can be achieved by simple mathematical transformations (e.g., adding a constant to every probability judgment), good discrimination demands access to solid, predictive evidence and skill at exploiting that evidence, which are difficult to find in any real-life, practical situation.” If you lack the ability to distinguish truth from falsehood, you can achieve perfect calibration by confessing your ignorance;

5. J. Frank Yates et al., “Probability Judgment Across Cultures,” in Gilovich, Griffin, and Kahneman, *Heuristics and Biases*, 271–291.

but confessing ignorance will not, of itself, distinguish truth from falsehood.

We thus dispose of another false stereotype of rationality, that rationality consists of being humble and modest and confessing helplessness in the face of the unknown. That's just the cheater's way out, assigning a 50% probability to all yes-or-no questions. Our scoring rule encourages you to do better if you can. If you are ignorant, confess your ignorance; if you are confident, confess your confidence. We penalize you for being confident and wrong, but we also reward you for being confident and right. That is the virtue of a proper scoring rule.

Suppose I flip a coin twenty times. If I believe the coin is fair, the best prediction I can make is to predict an even chance of heads or tails on each flip. If I believe the coin is fair, I assign the same probability to every possible sequence of twenty coinflips. There are roughly a million (1,048,576) possible sequences of twenty coinflips, and I have only 1.0 of probability mass to play with. So I assign to each *individual* possible sequence a probability of $(1/2)^{20}$ —odds of about a million to one; -20 bits or -60 decibels.

I made an experimental prediction and got a score of -60 decibels! Doesn't this falsify the hypothesis? Intuitively, no. We do not flip a coin twenty times and see a random-looking result, then reel back and say, why, the odds of that are a million to one. But the odds *are* a million to one against seeing that exact sequence, as I would discover if I naively predicted the exact same outcome for the *next* sequence of twenty coinflips. It's okay to have theories that assign tiny probabilities to outcomes, so long as no other theory does better. But if someone used an alternate hypothesis to write down the exact sequence in a sealed envelope in advance, and she assigned a probability of 99%, I would suspect the fairness of the coin. Provided that she only sealed *one* envelope, and not a million.

That tells us *what* we ought common-sensically to answer, but it doesn't say *how* the common-sense answer arises from the math. To say *why* the common sense is correct, we need

to integrate all that has been said so far into the framework of Bayesian revision of belief. When we're done, we'll have a technical understanding of the difference between a verbal understanding and a technical understanding.

Imagine an experiment which produces an integer result between zero and 99. For example, the experiment might be a particle counter that tells us how many particles have passed through in a minute. Or the experiment might be to visit the supermarket on Wednesday, check the price of a 10 oz bag of crushed walnuts, and write down the last two digits of the price.

We are testing several different hypotheses that try to predict the experimental result. Each hypothesis produces a probability distribution over all possible results; in this case, the integers between zero and 99. The possibilities are mutually exclusive, so the probability mass in the distribution must sum to one (or less); we cannot predict a 90% probability of seeing 42 and also a 90% probability of seeing 43.

Suppose there is a precise hypothesis that predicts a 90% chance of seeing the result 51. (I.e., the hypothesis is that the supermarket usually prices walnuts with a price of "X dollars and 51 cents.") The precise theory has staked 90% of its probability mass on the outcome 51. This leaves 10% probability mass remaining to spread over 99 other possible outcomes—all the numbers between zero and 99 *except* 51. The theory makes no further specification, so we spread the remaining 10% probability mass evenly over 99 possibilities, assigning a probability of $1/990$ to each non-51 result. For ease of writing, we'll approximate $1/990$ as 0.1%.

This probability distribution is analogous to the *likelihood* or *conditional probability* of the result given the hypothesis. Let us call it the *likelihood distribution* for the hypothesis, our chance of seeing each specified outcome *if* the hypothesis is true. The likelihood distribution for a hypothesis H is a function composed of all the conditional probabilities for $P(0|H) = 0.001$, $P(1|H) = 0.001$, ..., $P(51|H) = 0.9$, ..., $P(99|H) = 0.001$.

The precise theory predicts a 90% probability of seeing 51. Let there be also a vague theory, which predicts “a 90% probability of seeing a number in the fifties.”

Seeing the result 51, we do not say the outcome confirms both theories equally. Both theories made predictions, and both assigned probabilities of 90%, and the result 51 confirms both predictions. But the precise theory has an advantage because it concentrates its probability mass into a sharper point. If the vague theory makes no further specification, we count “a 90% probability of seeing a number in the fifties” as a 9% probability of seeing each number between 50 and 59.

Suppose we started with even odds in favor of the precise theory and the vague theory—odds of 1:1, or 50% probability for either hypothesis being true. After seeing the result 51, what are the posterior odds of the precise theory being true? The predictions of the two theories are analogous to their likelihood assignments—the conditional probability of seeing the result, given that the theory is true. What is the likelihood ratio between the two theories? The first theory allocated 90% probability mass to the *exact* outcome. The vague theory allocated 9% probability mass to the exact outcome. The likelihood ratio is 10:1. So if we started with even 1:1 odds, the posterior odds are 10:1 in favor of the precise theory. The differential pressure of the two conditional probabilities pushed our prior confidence of 50% to a posterior confidence of about 91% that the precise theory is correct. *Assuming* that these are the only hypotheses being tested, that this is the only evidence under consideration, and so on.

Why did the vague theory lose when both theories fit the evidence? The vague theory is timid; it makes a broad prediction, hedges its bets, allows many possibilities that would falsify the precise theory. This is not the virtue of a scientific theory. Philosophers of science tell us that theories should be bold, and subject themselves willingly to falsification if their prediction fails.⁶ Now we see why. The precise theory concen-

6. Karl R Popper, *The Logic of Scientific Discovery* (New York: Basic Books, 1959).

trates its probability mass into a sharper point and thereby leaves itself vulnerable to falsification if the real outcome hits elsewhere; but if the predicted outcome is correct, precision has a tremendous likelihood advantage over vagueness.

The laws of probability theory provide no way to cheat, to make a vague hypothesis such that any result between 50 and 59 counts for as much favorable confirmation as the precise theory receives, for that would require probability mass summing to 900%. There is no way to cheat, providing you record your prediction *in advance*, so you cannot claim afterward that your theory assigns a probability of 90% to whichever result arrived. Humans are very fond of making their predictions afterward, so the social process of science requires an advance prediction before we say that a result confirms a theory. But how humans may move in harmony with the way of Bayes, and so wield the power, is a separate issue from whether the math works. When we're doing the math, we just take for granted that likelihood density functions are fixed properties of a hypothesis and the probability mass sums to 1 and you'd never dream of doing it any other way.

You may want to take a moment to visualize that, *if* we define probability in terms of calibration, Bayes's Theorem relates the calibrations. Suppose I guess that Theory 1 is 50% likely to be true, and I guess that Theory 2 is 50% likely to be true. Suppose I am well-calibrated; when I utter the words "fifty percent," the event happens about half the time. And then I see a result *R* which would happen around nine-tenths of the time given Theory 1, and around nine-hundredths of the time given Theory 2, and I know this is so, and I apply Bayesian reasoning. If I was perfectly calibrated initially (despite the poor discrimination of saying 50/50), I will still be perfectly calibrated (and better discriminated) after I say that my confidence in Theory 1 is now 91%. If I repeated this kind of situation many times, I would be right around ten-elevenths of the time when I said "91%." If I reason using Bayesian rules, and I start from well-calibrated priors, then my conclusions will also be well-calibrated. This only holds true if we define probability in terms of calibration! If "90% sure" is instead interpreted as, say, the strength of the

emotion of surety, there is no reason to expect the posterior emotion to stand in an exact Bayesian relation to the prior emotion.

Let the prior odds be ten to one in favor of the vague theory. Why? Suppose our way of describing hypotheses allows us to either specify a precise number, or to just specify a first-digit; we can say “51,” “63,” “72,” or “in the fifties/sixties/seventies.” Suppose we think that the real answer is about equally liable to be an answer of the first kind or the second. However, given the problem, there are a hundred possible hypotheses of the first kind, and only ten hypotheses of the second kind. So if we think that either *class* of hypotheses has about an equal prior chance of being correct, we have to spread out the prior probability mass over ten times as many precise theories as vague theories. The precise theory that predicts exactly 51 would thus have one-tenth as much prior probability mass as the vague theory that predicts a number in the fifties. After seeing 51, the odds would go from 1:10 in favor of the vague theory to 1:1, even odds for the precise theory and the vague theory.

If you look at this carefully, it’s exactly what common sense would expect. You start out uncertain of whether a phenomenon is the kind of phenomenon that produces exactly the same result every time, or if it’s the kind of phenomenon that produces a result in the *X* ties every time. (Maybe the phenomenon is a price range at the supermarket, if you need some reason to suppose that 50–59 is an acceptable range but 49–58 isn’t.) You take a single measurement and the answer is 51. Well, that could be because the phenomenon is exactly 51, or because it’s in the fifties. So the remaining precise theory has the same odds as the remaining vague theory, which requires that the vague theory must have started out ten times as probable as that precise theory, since the precise theory has a sharper fit to the evidence.

If we just see one number, like 51, it doesn’t change the prior probability that the phenomenon itself was “precise” or “vague.” But, in effect, it concentrates all the probability mass of those two *classes* of hypothesis into a single surviving hypothesis of each class.

Of course, it is a severe error to say that a *phenomenon* is precise or vague, a case of what Jaynes calls the Mind Projection Fallacy.⁷ Precision or vagueness is a property of maps, not territories. Rather we should ask if the price in the supermarket stays constant or shifts about. A hypothesis of the “vague” sort is a good description of a price that shifts about. A precise map will suit a constant territory.

Another example: You flip a coin ten times and see the sequence HHTTH:TTTTH. Maybe you started out thinking there was a 1% chance this coin was fixed. Doesn’t the hypothesis “This coin is fixed to produce HHTTH:TTTTH” assign a thousand times the likelihood mass to the observed outcome, compared to the fair coin hypothesis? Yes. Don’t the posterior odds that the coin is fixed go to 10:1? No. The 1% prior probability that “the coin is fixed” has to cover every possible kind of fixed coin—a coin fixed to produce HHTTH:TTTTH, a coin fixed to produce TTHHT:HHHHT, etc. The prior probability the coin is fixed to produce HHTTH:TTTTH is not 1%, but a thousandth of one percent. Afterward, the posterior probability the coin is fixed to produce HHTTH:TTTTH is one percent. Which is to say: You thought the coin was probably fair but had a one percent chance of being fixed to some random sequence; you flipped the coin; the coin produced a random-looking sequence; and that doesn’t tell you anything about whether the coin is fair or fixed. It does tell you, if the coin is fixed, *which* sequence it is fixed to.

This parable helps illustrate why Bayesians *must* think about prior probabilities. There is a branch of statistics, sometimes called “orthodox” or “classical” statistics, which insists on paying attention only to likelihoods. But if you only pay attention to likelihoods, then eventually some fixed-coin hypothesis will always defeat the fair coin hypothesis, a phenomenon known as “overfitting” the theory to the data. After thirty flips, the *likelihood* is a billion times as great for the fixed-coin hypothesis with that sequence, as for the fair coin hypothesis. Only if the fixed-coin hypothesis (or rather, that specific fixed-coin

7. Jaynes, *Probability Theory*.

hypothesis) is a billion times less probable a priori can the fixed-coin hypothesis possibly lose to the fair coin hypothesis.

If you shake the coin to reset it, and start flipping the coin *again*, and the coin produces HHTTH:TTTTH *again*, that is a different matter. That does raise the posterior odds of the fixed-coin hypothesis to 10:1, even if the starting probability was only 1%.

Similarly, if we perform two successive measurements of the particle counter (or the supermarket price on Wednesdays), and *both* measurements return 51, the precise theory wins by odds of 10:1.

So the precise theory wins, but the vague theory would still score better than no theory at all. Consider a third theory, the hypothesis of zero knowledge or *maximum-entropy distribution*, which makes equally probable any result between zero and 99. Suppose we see the result 51. The vague theory produced a better prediction than the maximum-entropy distribution—assigned a greater likelihood to the outcome we observed. The vague theory is, literally, better than nothing. Suppose we started with odds of 1:20 in favor of the hypothesis of complete ignorance. (Why odds of 1:20? There is only one hypothesis of complete ignorance, and moreover, it's a particularly simple and intuitive kind of hypothesis, Occam's Razor.) After seeing the result of 51, predicted at 9% by the vague theory versus 1% by complete ignorance, the posterior odds go to 10:20 or 1:2. If we then see another result of 51, the posterior odds go to 10:2 or 83% probability for the vague theory, assuming there is no more precise theory under consideration.

Yet the timidity of the vague theory—its unwillingness to produce an *exact* prediction and accept falsification on any other result—renders it vulnerable to the bold, precise theory. (Providing, of course, that the bold theory correctly guesses the outcome!) Suppose the prior odds were 1:10:200 for the precise, vague, and ignorant theories—prior probabilities of 0.5%, 4.7%, and 94.8% for the precise, vague and ignorant theories. This figure reflects our prior probability distribution over *classes* of hypotheses, with the probability mass distributed over entire classes as follows: 50% that the phenomenon shifts across all

digits, 25% that the phenomenon shifts around within some decimal bracket, and 25% that the phenomenon repeats the same number each time. One hypothesis of complete ignorance, 10 possible hypotheses for a decimal bracket, 100 possible hypotheses for a repeating number. Thus, prior odds of 1:10:200 for the precise hypothesis 51, the vague hypothesis “fifties,” and the hypothesis of complete ignorance.

After seeing a result of 51, with assigned probability of 90%, 9%, and 1%, the posterior odds go to $90:90:200 = 9:9:20$. After seeing an additional result of 51, the posterior odds go to $810:81:20$, or 89%, 9%, and 2%. The precise theory is now favored over the vague theory, which in turn is favored over the ignorant theory.

Now consider a stupid theory, which predicts a 90% probability of seeing a result between zero and nine. The stupid theory assigns a probability of 0.1% to the actual outcome, 51. If the odds were initially 1:10:200:10 for the precise, vague, ignorant, and stupid theories, the posterior odds after seeing 51 once would be $90:90:200:1$. The stupid theory has been falsified (posterior probability of 0.2%).

It is possible to have a model so bad that it is worse than nothing, if the model concentrates its probability mass away from the actual outcome, makes confident predictions of wrong answers. Such a hypothesis is so poor that it loses against the hypothesis of complete ignorance. Ignorance is better than anti-knowledge.

Side note: In the field of Artificial Intelligence, there is a sometime fad that praises the glory of randomness. Occasionally an AI researcher discovers that if they add noise to one of their algorithms, the algorithm works better. This result is reported with great enthusiasm, followed by much fulsome praise of the creative powers of chaos, unpredictability, spontaneity, ignorance of what your own AI is doing, et cetera. (See The Imagination Engine for an example; according to their sales literature they

sell wounded and dying neural nets.⁸⁾ But how sad is an algorithm if you can *increase* its performance by injecting entropy into intermediate processing stages? The algorithm must be so deranged that some of its work goes into concentrating probability mass *away* from good solutions. If injecting randomness results in a reliable improvement, then some aspect of the algorithm must do reliably worse than random. Only in AI would people devise algorithms *literally dumber than a bag of bricks*, boost the results slightly back toward ignorance, and then argue for the healing power of noise.

Suppose that in our experiment we see the results 52, 51, and 58. The precise theory gives this conjunctive event a probability of a thousand to one times 90% times a thousand to one, while the vaguer theory gives this conjunctive event a probability of 9% cubed, which works out to ...oh ...um ...let's see ...a million to one given the precise theory, versus a thousand to one given the vague theory. Or thereabouts; we are counting rough powers of ten. Versus a million to one given the zero-knowledge distribution that assigns an equal probability to all outcomes. Versus a billion to one given a model worse than nothing, the stupid hypothesis, which claims a 90% probability of seeing a number less than 10. Using these approximate numbers, the vague theory racks up a score of -30 decibels (a probability of 1/1000 for the whole experimental outcome), versus scores of -60 for the precise theory, -60 for the ignorant theory, and -90 for the stupid theory. It is not always true that the highest score wins, because we need to take into account our prior odds of 1:10:200:10, confidences of -23, -13, 0, and -13 decibels. The vague theory still comes in with the highest total score at -43 decibels. (If we ignored our prior probabilities, each new

8. Imagination Engines, Inc., "The Imagination Engine™; or Imagitron™," 2011, sharov.me/r/4-42.



experiment would override the accumulated results of all the previous experiments; we could not accumulate knowledge. Furthermore, the fixed-coin hypothesis would always win.)

As always, we should not be alarmed that even the best theory still has a low score—recall the parable of the fair coin. Theories are approximations. In principle we might be able to predict the exact sequence of coinflips. But it would take better measurement and more computing power than we're willing to expend. Maybe we could achieve 60/40 prediction of coinflips, with a good enough model ...? We go with the best approximation we have, and try to achieve good calibration even if the discrimination isn't perfect.

We've conducted our analysis so far under the rules of Bayesian probability theory, in which there's no way to have more than 100% probability mass, and hence no way to cheat so that any outcome can count as "confirmation" of your theory. Under Bayesian law, play money may not be counterfeited; you only have so much clay.

Unfortunately, human beings are not Bayesians. Human beings bizarrely attempt to *defend* hypotheses, making a deliberate effort to prove them or prevent disproof. This behavior has no analogue in the laws of probability theory or decision theory. In formal probability theory the hypothesis *is*, and the evidence *is*, and either the hypothesis is confirmed or it is not. In formal decision theory, an agent may make an effort to investigate some issue of which the agent is currently uncertain, not knowing whether the evidence shall go one way or the other. In neither case does one ever deliberately try to prove an idea, or try to avoid disproving it. One may *test* ideas of which one is genuinely uncertain, but not have a "preferred" outcome of the investigation. One may not try to prove hypotheses, nor prevent their proof. I cannot properly convey just how ridiculous the notion would be, to a true Bayesian; there are not even words in Bayes-language to describe the mistake ...

For every expectation of evidence there is an equal and opposite expectation of counterevidence. If *A* is evidence in favor of

B, then not-*A* *must* be evidence in favor of not-*B*. The strengths of the evidences may not be equal; rare but strong evidence in one direction may be balanced by common but weak evidence in the other direction. But it is not possible for both *A* and not-*A* to be evidence in favor of *B*. That is, it's not possible under the laws of probability theory.

Humans often seem to want to have their cake and eat it too. Whichever result we witness is the one that proves our theory. As Spee, the priest in *Conservation of Expected Evidence*, put it, "The investigating committee would feel disgraced if it acquitted a woman; once arrested and in chains, she has to be guilty, by fair means or foul."⁹

The way human psychology seems to work is that first we see something happen, and then we try to argue that it matches whatever hypothesis we had in mind beforehand. Rather than conserved probability mass, to distribute over advance *predictions*, we have a feeling of *compatibility*—the degree to which the explanation and the event seem to "fit." "Fit" is not conserved. There is no equivalent of the rule that probability mass must sum to one. A psychoanalyst may explain any possible behavior of a patient by constructing an appropriate structure of "rationalizations" and "defenses"; it fits, therefore it must be true.

Now consider the fable told in *Fake Explanations*—the students seeing a radiator, and a metal plate next to the radiator. The students would never predict in advance that the side of the plate near the radiator would be cooler. Yet, seeing the fact, they managed to make their explanations "fit." They lost their precious chance at bewilderment, to realize that their models did not predict the phenomenon they observed. They sacrificed their ability to be more confused by fiction than by truth. And they did not realize "heat induction, blah blah, therefore the near side is cooler" is a vague and verbal prediction, spread across an enormously wide range of possible values for specific

9. Friedrich Spee, *Cautio Criminalis; or, A Book on Witch Trials*, ed. and trans. Marcus Hellyer, *Studies in Early Modern German History* (1631; Charlottesville: University of Virginia Press, 2003).

measured temperatures. Applying equations of diffusion and equilibrium would give a *sharp* prediction for possible joint values. It might not specify the *first* values you measured, but when you knew a few values you could generate a sharp prediction for the rest. The score for the entire experimental outcome would be far better than any less precise alternative, especially a vague and verbal prediction.

You now have a *technical* explanation of the difference between a verbal explanation and a technical explanation. It is a technical explanation because it enables you to calculate *exactly how technical* an explanation is. Vague hypotheses may be so vague that only a superhuman intelligence could calculate exactly how vague. Perhaps a sufficiently huge intelligence could extrapolate every possible experimental result, and extrapolate every possible verdict of the vague guesser for how well the vague hypothesis “fit,” and then renormalize the “fit” distribution into a likelihood distribution that summed to one. But in principle one can still calculate exactly how vague is a vague hypothesis. The calculation is just not computationally tractable, the way that calculating airplane trajectories via quantum mechanics is not computationally tractable.

I hold that everyone needs to learn at least one technical subject: physics, computer science, evolutionary biology, Bayesian probability theory, or *something*. Someone with *no* technical subjects under their belt has no referent for what it means to “explain” something. They may think “All is Fire” is an explanation, as did the Greek philosopher Heraclitus. Therefore do I advocate that Bayesian probability theory should be taught in high school. Bayesian probability theory is the sole piece of math I know that is accessible at the high school level, and that permits a *technical* understanding of a subject matter—the dynamics of belief—that is an everyday real-world domain and has emotionally meaningful consequences. Studying Bayesian probability would give students a referent for what it means to “explain” something.

Too many academics think that being “technical” means speaking in dry polysyllabisms. Here’s a “technical” explanation of technical explanation:

The equations of probability theory favor hypotheses that strongly predict the exact observed data. Strong models boldly concentrate their probability density into precise outcomes, making them falsifiable if the data hits elsewhere, and giving them tremendous likelihood advantages over models less bold, less precise. Verbal explanation runs on psychological evaluation of unconserved post facto compatibility instead of conserved ante facto probability density. And verbal explanation does not paint sharply detailed pictures, implying a smooth likelihood distribution in the vicinity of the data.

Is this satisfactory? No. Hear the impressive and weighty sentences, resounding with the dull thud of expertise. See the hapless students, writing those sentences on a sheet of paper. Even after the listeners hear the ritual words, they can perform no calculations. *You* know the math, so the words are meaningful. You can perform the calculations after hearing the impressive words, just as you could have done before. But what of one who did not see any calculations performed? What new skills have they gained from that “technical” lecture, save the ability to recite fascinating words?

“Bayesian” sure is a fascinating word, isn’t it? Let’s get it out of our systems: Bayes Bayes Bayes Bayes Bayes Bayes Bayes Bayes Bayes Bayes ...

The sacred syllable is meaningless, except insofar as it tells someone to apply math. Therefore the one who hears must already know the math.

Conversely, if you know the math, you can be as silly as you like, and still technical.

We thus dispose of yet another stereotype of rationality, that rationality consists of sere formality and humorless solemnity. What has that to do with the problem of distinguishing truth from falsehood? What has that to do with attaining the map that

reflects the territory? A scientist worthy of a lab coat should be able to make original discoveries while wearing a clown suit, or give a lecture in a high squeaky voice from inhaling helium. It is written nowhere in the math of probability theory that one may have no fun. The blade that cuts through to the correct answer has no dignity or silliness of itself, though it may fit the hand of a silly wielder.

A *useful* model isn't just something you know, as you know that an airplane is made of atoms. A useful model is knowledge you can compute in reasonable time to predict real-world events you know how to observe. Maybe someone will find that, using a model that violates Conservation of Momentum just a little, you can compute the aerodynamics of the 747 much more *cheaply* than if you insist that momentum is exactly conserved. So if you've got two computers competing to produce the best prediction, it might be that the best prediction comes from the model that violates Conservation of Momentum. This doesn't mean that the 747 violates Conservation of Momentum in real life. Neither model uses individual atoms, but that doesn't imply the 747 is not made of atoms. Physicists use different models to predict airplanes and particle collisions because it would be too expensive to compute the airplane particle by particle.

You would prove the 747 is made of atoms with experimental data that the aerodynamic models couldn't handle; for example, you would train a scanning tunneling microscope on a section of wing and look at the atoms. Similarly, you could use a finer measuring instrument to discriminate between a 747 that *really* disobeyed Conservation of Momentum like the cheap approximation predicted, versus a 747 that obeyed Conservation of Momentum like underlying physics predicted. The winning theory is the one that best predicts all the experimental predictions together. Our Bayesian scoring rule gives us a way to combine the results of *all* our experiments, even experiments that use different methods.

Furthermore, the atomic theory allows, embraces, and in some sense mandates the aerodynamic model. By thinking

abstractly about the assumptions of atomic theory, we realize that the aerodynamic model ought to be a good (and much cheaper) approximation of the atomic theory, and so the atomic theory supports the aerodynamic model, rather than competing with it. A successful theory can embrace many models for different domains, so long as the models are acknowledged as approximations, and in each case the model is compatible with (or ideally mandated by) the underlying theory.

Our *fundamental* physics—quantum mechanics, the standard family of particles, and relativity—is a theory that embraces an *enormous* family of models for macroscopic physical phenomena. There is the physics of liquids, and solids, and gases; yet this does not mean that there are *fundamental* things in the world that have the intrinsic property of liquidity.

Apparently there is colour, apparently sweetness,
apparently bitterness, actually there are only atoms
and the void.

—Democritus, 420 BCE, from Robinson and
Groves¹⁰

In arguing that a “technical” theory should be defined as a theory that sharply concentrates probability into specific advance predictions, I am setting an extremely high standard of strictness. We have seen that a vague theory *can* be better than nothing. A vague theory can win out over the hypothesis of ignorance, if there are no precise theories to compete against it.

There is an enormous family of models belonging to the central underlying theory of life and biology, the underlying theory that is sometimes called neo-Darwinism, natural selection, or evolution. Some models in evolutionary theory are quantitative. The way in which DNA encodes proteins is redundant; two different DNA sequences can code for exactly the same

10. quoted in Dave Robinson and Judy Groves, *Philosophy for Beginners*, 1st ed. (Cambridge: Icon Books, 1998).

protein. There are four DNA bases {A,T,C,G} and 64 possible combinations of three DNA bases. But those 64 possible codons describe only 20 amino acids plus a stop code. Genetic drift ought therefore to produce non-functional changes in species genomes, through mutations which by chance become fixed in the gene pool. The accumulation rate of non-functional differences between the genomes of two species with a common ancestor depends on such parameters as the number of generations elapsed and the intensity of selection at that genetic locus. That's an example of a member of the family of evolutionary models that produces quantitative predictions. There are also disequilibrium allele frequencies under selection, stable equilibria for game-theoretical strategies, sex ratios, etc.

This all comes under the heading of “fascinating words.” Unfortunately, there are certain religious factions that spread gross disinformation about evolutionary theory. So I emphasize that many models within evolutionary theory make quantitative predictions that are experimentally confirmed, and that such models are far more than sufficient to demonstrate that, e.g., humans and chimpanzees are related by a common ancestor. If you've been victimized by creationist disinformation—that is, if you've heard any suggestion that evolutionary theory is controversial or untestable or “just a theory” or non-rigorous or non-technical or in any way not confirmed by an unimaginably huge mound of experimental evidence—I recommend reading the *TalkOrigins FAQ*¹¹ and studying evolutionary biology with math.

But imagine going back in time to the nineteenth century, when the theory of natural selection had only just been discovered by Charles Darwin and Alfred Russel Wallace. Imagine evolutionism just after its birth, when the theory had nothing remotely like the modern-day body of quantitative models and great heaping mountains of experimental evidence. There was

11. TalkOrigins Foundation, “Frequently Asked Questions about Creationism and Evolution,” sharov.me/r/4-43.



no way of knowing that humans and chimpanzees would be discovered to have 95% shared genetic material. No one knew that DNA existed. Yet even so, scientists flocked to the new theory of natural selection. And later it turned out that there was a precisely copied genetic material with the potential to mutate, that humans and chimps were provably related, etc.

So the very strict, very high standard that I proposed for a “technical” theory is too strict. Historically, it *has* been possible to successfully discriminate true theories from false theories, based on predictions of the sort I called “vague.” Vague predictions of, say, 80% confidence, can build up a huge advantage over alternate hypotheses, given enough experiments. Perhaps a theory of this kind, producing predictions that are not precisely detailed but are nonetheless correct, could be called “semitechnical”?

But surely technical theories are more reliable than semitechnical theories? Surely technical theories should take precedence, command greater respect? Surely physics, which produces exceedingly exact predictions, is in some sense better confirmed than evolutionary theory? Not implying that evolutionary theory is wrong, of course; but however vast the mountains of evidence favoring evolution, does not physics go one better through vast mountains of *precise* experimental confirmation? Observations of neutron stars confirm the predictions of General Relativity to within one part in a hundred trillion (10^{14}). What does evolutionary theory have to match that?

Daniel Dennett once said that measured by the simplicity of the theory and the amount of complexity it explained, Darwin had the single greatest idea in the history of time.¹²

Once there was a conflict between nineteenth century physics and nineteenth century evolutionism. According to the best physical models then in use, the Sun could not have been burning very long. Three thousand years on chemical energy, or 40 million years on gravitational energy. There was

12. Daniel C. Dennett, *Darwin's Dangerous Idea: Evolution and the Meanings of Life* (Simon & Schuster, 1995).

no energy source known to nineteenth century physics that would permit longer burning. Nineteenth century physics was not *quite* as powerful as modern physics—it did not have predictions accurate to within one part in 10^{14} . But nineteenth century physics still had the mathematical character of modern physics, a discipline whose models produced detailed, precise, quantitative predictions. Nineteenth century evolutionary theory was wholly semitechnical, without a scrap of quantitative modeling. Not even Mendel’s experiments with peas were then known. And yet it did seem likely that evolution would require longer than a paltry 40 million years in which to operate—hundreds of millions, even billions of years. The antiquity of the Earth was a vague and semitechnical prediction, of a vague and semitechnical theory. In contrast, the nineteenth century physicists had a precise and quantitative model, which through formal calculation produced the precise and quantitative dictum that the Sun simply could not have burned that long.

The limitations of geological periods, imposed by physical science, cannot, of course, disprove the hypothesis of transmutation of species; but it does seem sufficient to disprove the doctrine that transmutation has taken place through “descent with modification by natural selection.”

—Lord Kelvin, from Lyle Zapato¹³

History records who won.

The moral? If you can give 80% confident advance predictions on yes-or-no questions, it may be a “vague” theory; it may be wrong one time out of five; but you can still build up a heck of a huge scoring lead over the hypothesis of ignorance. Enough to confirm a theory, if there are no better competitors. Reality is consistent; every *correct* theory about the universe is compatible with every other correct theory. Imperfect maps

13. quoted in Lyle Zapato, “Lord Kelvin Quotations,” 2008, sharov.me/t/4-44.



can conflict, but there is only one territory. Nineteenth century evolutionism might have been a semitechnical discipline, but it was still correct (as we now know) and by far the best explanation (even in that day). Any conflict between evolutionism and another well-confirmed theory had to reflect some kind of anomaly, a mistake in the assertion that the two theories were incompatible. Nineteenth century physics couldn't model the dynamics of the Sun—they didn't know about nuclear reactions. They could not show that their understanding of the Sun was correct *in technical detail*, nor calculate from a *confirmed* model of the Sun to determine how long the Sun had existed. So in retrospect, we can say something like: "There was room for the possibility that nineteenth century physics just didn't understand the Sun."

But that is hindsight. The real lesson is that, even though nineteenth century physics was both precise and quantitative, it didn't automatically dominate the semitechnical theory of nineteenth century evolutionism. The theories were *both* well-supported. They were *both* correct in the domains over which they were generalized. The apparent conflict between them was an anomaly, and the anomaly turned out to stem from the incompleteness and incorrect application of nineteenth century physics, not the incompleteness and incorrect application of nineteenth century evolutionism. But it would be futile to compare the mountain of evidence supporting the one theory, versus the mountain of evidence supporting the other. Even in that day, both mountains were too large to suppose that either theory was simply mistaken. Mountains of evidence that large cannot be set to compete, as if one falsifies the other. You must be applying one theory incorrectly, or applying a model outside the domain it predicts well.

So you shouldn't *necessarily* sneer at a theory just because it's semitechnical. Semitechnical theories can build up high enough scores, compared to every available alternative, that you know the theory is at least approximately correct. Someday the semitechnical theory may be replaced or even falsified by a more precise competitor, but that's true even of technical

theories. Think of how Einstein's General Relativity devoured Newton's theory of gravitation.

But the correctness of a semitechnical theory—a theory that currently has no precise, computationally tractable models testable by feasible experiments—can be a lot less cut-and-dried than the correctness of a technical theory. It takes skill, patience, and examination to distinguish good semitechnical theories from theories that are just plain confused. This is not something that humans do well by instinct, which is why we have Science.

People eagerly jump the gun and seize on any available reason to reject a disliked theory. That is why I gave the example of nineteenth century evolutionism, to show why one should not be too quick to reject a “non-technical” theory out of hand. By the moral customs of science, nineteenth century evolutionism was guilty of more than one sin. Nineteenth century evolutionism made no quantitative predictions. It was not readily subject to falsification. It was largely an explanation of what had already been seen. It lacked an underlying mechanism, as no one then knew about DNA. It even contradicted the nineteenth century laws of physics. Yet natural selection was such an *amazingly good* post facto explanation that people flocked to it, and they turned out to be right. Science, as a human endeavor, requires advance prediction. Probability theory, as math, does not distinguish between post facto and advance prediction, because probability theory assumes that probability distributions are fixed properties of a hypothesis.

The rule about advance prediction is a rule of the social process of science—a moral custom and not a theorem. The moral custom exists to prevent human beings from making human mistakes that are hard to even describe in the language of probability theory, like tinkering after the fact with what you claim your hypothesis predicts. People concluded that nineteenth century evolutionism was an excellent explanation, even if it was post facto. That reasoning *was correct as probability theory*, which is why it *worked* despite all scientific sins. Probability theory is math. The social process of science is a set of legal conventions to keep people from cheating on the math.

Yet it is also true that, compared to a *modern-day* evolutionary theorist, evolutionary theorists of the late nineteenth and early twentieth century often went sadly astray. Darwin, who was bright enough to invent the theory, got an amazing amount right. But Darwin's successors, who were only bright enough to accept the theory, misunderstood evolution frequently and seriously. The usual process of science was then required to correct their mistakes. It is incredible how few errors of reasoning Darwin¹⁴ made in *The Origin of Species* and *The Descent of Man*, compared to they who followed.

That is also a hazard of a semitechnical theory. Even after the flash of genius insight is confirmed, merely average scientists may fail to apply the insights properly in the absence of formal models. As late as the 1960s biologists spoke of evolution working "for the good of the species," or suggested that individuals would restrain their reproduction to prevent species overpopulation of a habitat. The best evolutionary theorists knew better, but average theorists did not.¹⁵

So it is *far* better to have a technical theory than a semitechnical theory. Unfortunately, Nature is not always so kind as to render Herself describable by neat, formal, *computationally tractable* models, nor does She always provide Her students with measuring instruments that can directly probe Her phenomena. Sometimes it is only a matter of time. Nineteenth century evolutionism was semitechnical, but later came the math of population genetics, and eventually DNA sequencing. Nature will not always give you a phenomenon that you can describe with technical models fifteen seconds after you have the basic insight.

14. Charles Darwin, *On the Origin of Species by Means of Natural Selection; or, The Preservation of Favoured Races in the Struggle for Life*, 1st ed. (London: John Murray, 1859), sharov.me/r/4-45; Charles Darwin, *The Descent of Man, and Selection in Relation to Sex*, 2nd ed. (London: John Murray, 1874), sharov.me/r/4-46.



15. Williams, *Adaptation and Natural Selection*.

Yet the cutting edge of science, the *controversy*, is most often about a semitechnical theory, or nonsense posing as a semitechnical theory. By the time a theory achieves technical status, it is usually no longer controversial (among scientists). So the question of how to distinguish good semitechnical theories from nonsense is very important to scientists, and it is not as easy as dismissing out of hand any theory that is not technical. To the end of distinguishing truth from falsehood exists the entire discipline of rationality. The art is not reducible to a checklist, or at least, no checklist that an average scientist can apply reliably after an hour of training. If it was that simple we wouldn't need science.

Why do you pay attention to scientific controversies? Why graze upon such sparse and rotten feed as the media offers, when there are so many solid meals to be found in textbooks? Textbook science is beautiful! Textbook science is *comprehensible*, unlike mere fascinating words that can never be truly beautiful. Fascinating words have no power, nor yet any meaning, without the math. The fascinating words are not knowledge but the illusion of knowledge, which is why it brings so little satisfaction to know that "gravity results from the curvature of spacetime." Science is not in the fascinating words, though it's all you'll ever read as breaking news.

There can be justification for following a scientific controversy. You could be an expert in that field, in which case that scientific controversy is your proper meat. Or the scientific controversy might be something you need to know *now*, because it affects your life. Maybe it's the nineteenth century, and you're gazing lustfully at a member of the appropriate sex wearing a nineteenth century bathing suit, and you need to know whether your sexual desire comes from a psychology constructed by natural selection, or is a temptation placed in you by the Devil to lure you into hellfire.

It is not wholly impossible that we shall happen upon a scientific controversy that affects us, and find that we have a burning and urgent need for the correct answer. I shall therefore discuss

some of the warning signs that historically distinguished vague hypotheses that later turned out to be unscientific gibberish, from vague hypotheses that later graduated to confirmed theories. Just remember the historical lesson of nineteenth century evolutionism, and resist the temptation to fail every theory that misses a single item on your checklist. It is not my intention to give people another excuse to dismiss good science that discomforts them. If you apply stricter criteria to theories you dislike than theories you like (or vice versa!), then every additional nit you learn how to pick, every new logical flaw you learn how to detect, makes you that much stupider. Intelligence, to be useful, must be used for something other than defeating itself.

One of the classic signs of a poor hypothesis is that it must expend great effort in avoiding falsification—elaborating reasons why the hypothesis is compatible with the phenomenon, even though the phenomenon didn't behave as expected. Carl Sagan gives the example of someone who claims that a dragon lives in their garage. Sagan originally drew the lesson that poor hypotheses need to do fast footwork to avoid falsification—to maintain an appearance of “fit.”¹⁶

I would point out that the claimant obviously has a good model of the situation *somewhere* in their head, because they can predict, in advance, exactly which excuses they're going to need. To a Bayesian, a hypothesis isn't something you assert in a loud, emphatic voice. A hypothesis is something that controls your *anticipations*, the probabilities you assign to future experiences. That's what a probability *is*, to a Bayesian—that's what you score, that's what you calibrate. So while our claimant may say loudly, emphatically, and honestly that they *believe* there's an invisible dragon in the garage, they do not *anticipate* there's an invisible dragon in the garage—they anticipate exactly the same experience as the skeptic.

16. Carl Sagan, *The Demon-Haunted World: Science as a Candle in the Dark*, 1st ed. (New York: Random House, 1995).

When I judge the predictions of a hypothesis, I ask which experiences I would anticipate, not which facts I would believe.

The flip side:

I recently argued with a friend of mine over a question of evolutionary theory. My friend alleged that the clustering of changes in the fossil record (apparently, there are periods of comparative stasis followed by comparatively sharp changes; itself a controversial observation known as “punctuated equilibrium”) showed that there was something wrong with our understanding of speciation. My friend thought that there was some unknown force at work—not supernatural, but some natural consideration that standard evolutionary theory didn’t take into account. Since my friend didn’t give a specific competing hypothesis that produced better predictions, his thesis had to be that the standard evolutionary model was *stupid* with respect to the data—that the standard model made a specific prediction that was wrong; that the model did worse than complete ignorance or some other default competitor.

At first I fell into the trap; I accepted the implicit assumption that the standard model predicted smoothness, and based my argument on my recollection that the fossil record changes weren’t as sharp as he claimed. He challenged me to produce an evolutionary intermediate between *Homo erectus* and *Homo sapiens*; I googled and found *Homo heidelbergensis*. He congratulated me and acknowledged that I had scored a major point, but still insisted that the changes were too sharp, and not steady enough. I started to explain why I thought a pattern of uneven change *could* arise from the standard model: environmental selection pressures might not be constant...“Aha!” my friend said, “you’re making your excuses in advance.”

But suppose that the fossil record instead showed a smooth and gradual set of changes. Might my friend have argued that the standard model of evolution as a chaotic and noisy process could not account for such smoothness? If it is a scientific sin to claim post facto that our beloved hypothesis predicts the data, should it not be equally a sin to claim post facto that the competing hypothesis is stupid on the data?

If a hypothesis has a *purely* technical model, there is no trouble; we can compute the prediction of the model formally, without informal variables to provide a handle for post facto meddling. But what of semitechnical theories? Obviously a semitechnical theory must produce some good advance predictions about *something*, or else why bother? But *after* the theory is semi-confirmed, can the detractors claim that the data show a problem with the semitechnical theory, when the “problem” is constructed post facto? At the least the detractors must be very specific about what data a confirmed model predicts stupidly, and why the confirmed model must make (post facto) that stupid prediction. How sharp a change is “too sharp,” quantitatively, for the standard model of evolution to permit? Exactly how much steadiness do you think the standard model of evolution predicts? How do you know? Is it too late to say that, after you’ve seen the data?

When my friend accused me of making excuses, I paused and asked myself which excuses I anticipated needing to make. I decided that my current grasp of evolutionary theory didn’t say anything about whether the rate of evolutionary change should be intermittent and jagged, or smooth and gradual. If I hadn’t seen the graph in advance, I could not have predicted it. (Unfortunately, I rendered even that verdict after seeing the data ...) Maybe there are models in the evolutionary family that would make advance predictions of steadiness or variability, but if so, I don’t know about them. More to the point, my friend didn’t know either.

It is not always wise to ask the opponents of a theory what their competitors predict. Get the theory’s predictions from the theory’s best advocates. Just make sure to write down their predictions in advance. Yes, sometimes a theory’s advocates try to make the theory “fit” evidence that plainly doesn’t fit. But if you find yourself wondering what a theory predicts, ask first among the theory’s advocates, and afterward ask the detractors to cross-examine.

Furthermore: Models may include noise. If we hypothesize that the data are trending slowly and steadily upward, but our measuring instrument has an error of 5%, then it does no

good to point to a data point that dips below the previous data point, and shout triumphantly, “See! It went down! Down down down! And don’t tell me why your theory fits the dip; you’re just making excuses!” Formal, technical models often incorporate explicit error terms. The error term spreads out the likelihood density, decreases the model’s precision and reduces the theory’s score, but the Bayesian scoring rule still governs. A technical model can allow mistakes, and make mistakes, and still do better than ignorance. In our supermarket example, even the precise hypothesis of 51 still bets only 90% of its probability mass on 51; the precise hypothesis claims only that 51 happens nine times out of ten. Ignoring nine 51s, pointing at one case of 82, and crowing in triumph, does not a refutation make. That’s not an excuse, it’s an explicit advance prediction of a technical model.

The error term makes the “precise” theory vulnerable to a superprecise alternative that predicted the 82. The standard model would also be vulnerable to a precisely ignorant model that predicted a 60% chance of 51 on the round where we saw 82, spreading out the likelihood more entropically on that particular error. No matter how good the theory, science always has room for a higher-scoring competitor. But if you *don’t* present a better alternative, if you try only to show that an accepted theory is *stupid* with respect to the data, that scientific endeavor may be *more* demanding than just replacing the old theory with a new one.

Astronomers recorded the unexplained perihelion advance of Mercury, unaccounted for under Newtonian physics—or rather, Newtonian physics predicted 5,557 seconds of arc per century, where the observed amount was 5,600.¹⁷ But should the scientists of that day have junked Newtonian gravitation based on such small, unexplained counterevidence? What would they have used instead? Eventually, Newton’s theory of gravita-

17. Kevin Brown, *Reflections On Relativity* (Raleigh, NC: printed by author, 2011), 405-414, sharov.me/r/4-47.



tion was set aside, after Einstein's General Relativity precisely explained the orbital discrepancy of Mercury and also made successful advance predictions. But there was no way to know *in advance* that this was how things would turn out.

In the nineteenth century there was a persistent anomaly in the orbit of Uranus. People said, "Maybe Newton's law starts to fail at long distances." Eventually some bright fellows looked at the anomaly and said, "Could this be an unknown outer planet?" Urbain Le Verrier and John Couch Adams independently did some scribbling and figuring, using Newton's standard theory—and predicted Neptune's location to within one degree of arc, dramatically *confirming* Newtonian gravitation.¹⁸

Only *after* General Relativity precisely produced the perihelion advance of Mercury did we *know* Newtonian gravitation would never explain it.

In the Intuitive Explanation we saw how Karl Popper's insight that falsification is stronger than confirmation translates into a Bayesian truth about likelihood ratios. Popper erred in thinking that falsification was *qualitatively different* from confirmation; both are governed by the same Bayesian rules. But Popper's philosophy reflected an important truth about a quantitative difference between falsification and confirmation.

Popper was profoundly impressed by the differences between the allegedly "scientific" theories of Freud and Adler and the revolution effected by Einstein's theory of relativity in physics in the first two decades of this century. The main difference between them, as Popper saw it, was that while Einstein's theory was highly "risky," in the sense that it was possible to deduce consequences from it which were, in the light of the then dominant Newtonian physics, highly improbable (e.g., that light is deflected towards solid bodies—confirmed

18. Ibid., 405-414.

by Eddington's experiments in 1919), and which would, if they turned out to be false, falsify the whole theory, nothing could, even in principle, falsify psychoanalytic theories. These latter, Popper came to feel, have more in common with primitive myths than with genuine science. That is to say, he saw that what is apparently the chief source of strength of psychoanalysis, and the principal basis on which its claim to scientific status is grounded, viz. its capability to accommodate, and explain, every possible form of human behaviour, is in fact a critical weakness, for it entails that it is not, and could not be, genuinely predictive. Psychoanalytic theories by their nature are insufficiently precise to have negative implications, and so are immunised from experiential falsification ...

Popper, then, repudiates induction, and rejects the view that it is the characteristic method of scientific investigation and inference, and substitutes falsifiability in its place. It is easy, he argues, to obtain evidence in favour of virtually any theory, and he consequently holds that such "corroboration," as he terms it, should count scientifically only if it is the positive result of a genuinely "risky" prediction, which might conceivably have been false. For Popper, a theory is scientific only if it is refutable by a conceivable event. Every genuine test of a scientific theory, then, is logically an attempt to refute or to falsify it...

Every genuine scientific theory then, in Popper's view, is prohibitive, in the sense that it forbids, by implication, particular events or occurrences.¹⁹

19. Stephen Thornton, "Karl Popper," in *The Stanford Encyclopedia of Philosophy*, Winter 2002, ed. Edward N. Zalta (Stanford University), sharov.me/r/4-48.



On Popper's philosophy, the strength of a scientific theory is not how much it explains, but how much it *doesn't* explain. The virtue of a scientific theory lies not in the outcomes it *permits*, but in the outcomes it *prohibits*. Freud's theories, which seemed to explain everything, *prohibited* nothing.

Translating this into Bayesian terms, we find that the more outcomes a model *prohibits*, the more probability density the model concentrates in the remaining, permitted outcomes. The more outcomes a theory prohibits, the greater the knowledge-content of the theory. The more daringly a theory exposes itself to falsification, the more definitely it tells you which experiences to anticipate.

A theory that can explain *any* experience corresponds to a hypothesis of complete ignorance—a uniform distribution with probability density spread evenly over every possible outcome.

Phlogiston was the eighteenth century's answer to the Elemental Fire of the Greek alchemists. You couldn't use phlogiston theory to predict the outcome of a chemical transformation—first you looked at the result, then you used phlogiston to explain it. Phlogiston theory was infinitely flexible; a disguised hypothesis of zero knowledge. Similarly, the theory of *vitalism* doesn't explain how the hand moves, nor tell you what transformations to expect from organic chemistry; and vitalism certainly permits no quantitative calculations.

The flip side:

Beware of checklist thinking: Having a *sacred* mystery, or a mysterious answer, is not the same as refusing to explain something. Some elements in our physics are taken as “fundamental,” not yet further reduced or explained. But these fundamental elements of our physics are governed by clearly defined, mathematically simple, formally computable causal rules.

Occasionally some crackpot objects to modern physics on the grounds that it does not provide an “underlying mechanism” for a mathematical law currently treated as fundamental. (Claiming that a mathematical law lacks an “underlying mechanism” is

one of the entries on the *Crackpot Index* by John Baez.²⁰) The “underlying mechanism” the crackpot proposes in answer is vague, verbal, and yields no increase in predictive power—otherwise we would not classify the claimant as a crackpot.

Our current physics makes the electromagnetic field fundamental, and refuses to explain it further. But the “electromagnetic field” is a fundamental governed by clear mathematical rules, with no properties outside the mathematical rules, subject to formal computation to describe its causal effect upon the world. Someday someone may suggest improved math that yields better predictions, but I would not indict the current model on grounds of mysteriousness. A theory that includes *fundamental elements* is not the same as a theory that contains *mysterious elements*.

Fundamentals should be simple. “Life” is not a good fundamental, “oxygen” is a good fundamental, and “electromagnetic field” is a better fundamental. Life might look simple to a vitalist—it’s the simple, magical ability of your muscles to move under your mental direction. Why shouldn’t life be explained by a simple, magical fundamental substance like *élan vital*? But phenomena that seem *psychologically* very simple—little dots of light in the sky, orangey-bright hot flame, flesh moving under mental direction—often conceal vast depths of underlying complexity. The proposition that life is a complex phenomenon may seem incredible to the vitalist, staring at a blankly opaque mystery with no obvious handles; but yes, Virginia, there is underlying complexity. The criterion of simplicity that is relevant to Occam’s Razor is *mathematical* or *computational* simplicity. Once we render down our model into mathematically simple fundamental elements, not in themselves sharing the mysterious qualities of the mystery, interacting in clearly defined ways to produce the formerly mysterious phenomenon as a

20. John Baez, “The Crackpot Index,” 1998, sharov.me/r/4-49.



detailed prediction, that is as non-mysterious as humanity has ever figured out how to make anything.

Many people in this world believe that after dying they will face a stern-eyed fellow named St. Peter, who will examine their actions in life and accumulate a score for morality. Presumably St. Peter's scoring rule is unique and invariant under trivial changes of perspective. Unfortunately, believers cannot obtain a quantitative, precisely computable specification of the scoring rule, which seems rather unfair.

The religion of *Bayesianity* holds that your eternal fate depends on the probability judgments you made in life. Unlike lesser faiths, Bayesianity can give a quantitative, precisely computable specification of how your eternal fate is determined.

Our proper Bayesian scoring rule provides a way to accumulate scores across experiments, and the score is invariant regardless of how we slice up the "experiments" or in what order we accumulate the results. We add up the logarithms of the probabilities. This corresponds to multiplying together the probability assigned to the outcome in each experiment, to find the joint probability of all the experiments together. We take the logarithm to simplify our intuitive understanding of the accumulated score, to maintain our grip on the tiny fractions involved, and to ensure we maximize our *expected* score by stating our honest probabilities rather than placing all our play money on the most probable bet.

Bayesianity states that when you die, Pierre-Simon Laplace examines every single event in your life, from finding your shoes next to your bed in the morning to finding your workplace in its accustomed spot. Every losing lottery ticket means you cared enough to play. Laplace assesses the advance probability you assigned to each event. Where you did not assign a precise numerical probability in advance, Laplace examines your degree of anticipation or surprise, extrapolates other possible outcomes and your extrapolated reactions, and renormalizes your extrapolated emotions to a likelihood distribution over

possible outcomes. (Hence the phrase “Laplacian superintelligence.”)

Then Laplace takes every event in your life, and every probability you assigned to each event, and multiplies all the probabilities together. This is your Final Judgment—the probability you assigned to your life.

Those who follow Bayesianity strive all their lives to maximize their Final Judgment. This is the sole virtue of Bayesianity. The rest is just math.

Mark you: the path of Bayesianity is strict. What probability shall you assign each morning, to the proposition, “The Sun shall rise?” (We shall discount such quibbles as cloudy days, and that the Earth orbits the Sun.) Perhaps one who did not follow Bayesianity would be humble, and give a probability of 99.9%. But we who follow Bayesianity shall discard all considerations of modesty and arrogance, and scheme only to maximize our Final Judgment. Like an obsessive video-game player, we care only about this numerical score. We’re going to face this Sun-shall-rise issue 365 times per year, so we might be able to improve our Final Judgment considerably by tweaking our probability assignment.

As it stands, even if the Sun rises every morning, every year our Final Judgment will decrease by a factor of $0.999^{365} = 0.7$, roughly -0.52 bits. Every two years, our Final Judgment will decrease more than if we found ourselves ignorant of a coin-flip’s outcome! Intolerable. If we increase our daily probability of sunrise to 99.99%, then each year our Final Judgment will decrease only by a factor of 0.964. Better. Still, in the unlikely event that we live exactly 70 years and then die, our Final Judgment will only be 7.75% of what it might have been. What if we assign a 99.999% probability to the sunrise? Then after 70 years, our Final Judgment will be multiplied by 77.4%.

Why not assign a probability of 1.0?

One who follows Bayesianity will *never* assign a probability of 1.0 to *anything*. Assigning a probability of 1.0 to some outcome uses up *all* your probability mass. If you assign a probability of 1.0 to some outcome, and reality delivers a different answer, you must have assigned the *actual* outcome a

probability of *zero*. This is Bayesianity's sole mortal sin. Zero times anything is zero. When Laplace multiplies together all the probabilities of your life, the combined probability will be zero. Your Final Judgment will be doodly-squat, zilch, nada, nil. No matter how rational your guesses during the rest of your life, you'll spend eternity next to some guy who believed in flying saucers and got all his information from the Weekly World News. Again we find it helpful to take the logarithm, revealing the innocent-sounding "zero" in its true form. Risking an outcome probability of zero is like accepting a bet with a payoff of negative infinity.

What if humanity decides to take apart the Sun for mass (stellar engineering), or to switch off the Sun because it's wasting entropy? Well, you say, you'll see that coming, you'll have a chance to alter your probability assignment before the actual event. What if an Artificial Intelligence in someone's basement recursively self-improves to superintelligence, stealthily develops nanotechnology, and one morning *it* takes apart the Sun? If on the last night of the world you assign a probability of 99.999% to tomorrow's sunrise, your Final Judgment will go down by a factor of 100,000. Minus 50 decibels! Awful, isn't it?

So what is your best strategy? Well, suppose you 50% anticipate that a basement-spawned AI superintelligence will disassemble the Sun sometime in the next ten years, and you figure there's about an equal chance of this happening on any given day between now and then. On any given night, you would 99.98% anticipate the Sun rising tomorrow. If this is really what you anticipate, then you have no motive to say anything except 99.98% as your probability. If you feel nervous that this anticipation is too low, or too high, it must not be what you anticipate after your nervousness is taken into account.

But the deeper truth of Bayesianity is this: You cannot game the system. You cannot give a humble answer, nor a confident one. You must figure out exactly how much you anticipate the Sun rising tomorrow, and say that number. You must shave away every hair of modesty or arrogance, and ask whether you expect to end up being scored on the Sun rising, or failing to rise. Look not to your excuses, but ask which excuses you expect

to need. After you arrive at your exact degree of anticipation, the only way to further improve your Final Judgment is to improve the accuracy, calibration, and discrimination of your anticipation. You cannot do better except by guessing better and anticipating more precisely.

Er, well, except that you could commit suicide when you turned five, thereby preventing your Final Judgment from decreasing any further. Or if we patch a new sin onto the utility function, enjoining against suicide, you could flee from mystery, avoiding all situations in which you thought you might not know everything. So much for that religion.

Ideally, we predict the outcome of the experiment in advance, using our model, and then we perform the experiment to see if the outcome accords with our model. Unfortunately, we can't always control the information stream. Sometimes Nature throws experiences at us, and by the time we think of an explanation, we've already seen the data we're supposed to explain. This was one of the scientific sins committed by nineteenth century evolutionism; Darwin observed the similarity of many species, and their adaptation to particular local environments, before the hypothesis of natural selection occurred to him. Nineteenth century evolutionism began life as a post facto explanation, not an advance prediction.

Nor is this a trouble only of semitechnical theories. In 1846, the successful deduction of Neptune's existence from gravitational perturbations in the orbit of Uranus was considered a grand triumph for Newton's theory of gravitation. Why? Because Neptune's existence was the first observation that confirmed an *advance* prediction of Newtonian gravitation. All the other phenomena that Newton explained, such as orbits and orbital perturbations and tides, had been observed in great detail before Newton explained them. No one seriously doubted that Newton's theory was correct. Newton's theory explained too much too precisely, and it replaced a collection of ad hoc models with a single unified mathematical law. Even so, the advance prediction of Neptune's existence, followed

by the observation of Neptune at almost exactly the predicted location, was considered the first grand triumph of Newton's theory at predicting what no previous model could predict. Considerable time elapsed between widespread acceptance of Newton's theory and the first impressive *advance* prediction of Newtonian gravitation. By the time Newton came up with his theory, scientists had already observed, in great detail, most of the phenomena that Newtonian gravitation predicted.

But the rule of advance prediction is a morality of science, not a law of probability theory. If you have already seen the data you must explain, then Science may darn you to heck, but your predicament doesn't collapse the laws of probability theory. What does happen is that it becomes much more difficult for a hapless human to *obey* the laws of probability theory. When you're deciding how to rate a hypothesis according to the Bayesian scoring rule, you need to figure out how much probability mass that hypothesis assigns to the observed outcome. If we must make our predictions in advance, then it's easier to notice when someone is trying to claim every possible outcome as an advance prediction, using too much probability mass, being deliberately vague to avoid falsification, and so on.

No numerologist can predict next week's winning lottery numbers, but they will be happy to explain the mystical significance of last week's winning lottery numbers. Say the winning Mega Ball was seven in last week's lottery, out of 52 possible outcomes. Obviously this happened because seven is the lucky number. So will the Mega Ball in next week's lottery also come up seven? We understand that it's not certain, of course, but if it's the lucky number, you ought to assign a probability of higher than $1/52$...and then we'll score your guesses over the course of a few years, and if your score is too low we'll have you flogged ...what's that you say? You want to assign a probability of exactly $1/52$? But that's the same probability as every other number; what happened to seven being lucky? No, sorry, you can't assign a 90% probability to seven and also a 90% probability to eleven. We understand they're both lucky numbers. Yes, we understand that they're *very* lucky numbers. But that's not how it works.

Even if the listener does not know the way of Bayes and does not ask for formal probabilities, they will probably become suspicious if you try to cover too many bases. Suppose they ask you to predict next week's winning Mega Ball, and you use numerology to explain why the number one ball would fit your theory very well, and why the number two ball would fit your theory very well, and why the number three ball would fit your theory very well ...even the most credulous listener might begin to ask questions by the time you got to twelve. Maybe you could tell us which numbers are unlucky and definitely won't win the lottery? Well, thirteen is unlucky, but it's not absolutely *impossible* (you hedge, *anticipating* in advance which excuse you might need).

But if we ask you to explain *last week's* lottery numbers, why, the seven was practically inevitable. That seven should definitely count as a major success for the "lucky numbers" model of the lottery. And it couldn't possibly have been thirteen; luck theory rules that straight out.

Imagine that you wake up one morning and your left arm has been replaced by a blue tentacle. The blue tentacle obeys your motor commands—you can use it to pick up glasses, drive a car, etc. How would you explain this hypothetical scenario? Take a moment to ponder this puzzle before continuing.

(Spoiler space ...)

How would I explain the event of my left arm being replaced by a blue tentacle? The answer is that I wouldn't. It isn't going to happen.

It would be easy enough to produce a verbal explanation that "fit" the hypothetical. There are many explanations that can "fit" anything, including (as a special case of "anything") my arm's being replaced by a blue tentacle. Divine intervention is a good all-purpose explanation. Or aliens with arbitrary motives and capabilities. Or I could be mad, hallucinating, dreaming my life away in a hospital. Such explanations "fit" all outcomes equally well, and equally poorly, equating to hypotheses of complete ignorance.

The test of whether a model of reality “explains” my arm’s turning into a blue tentacle is whether the model concentrates significant probability mass into that *particular* outcome. Why that dream, in the hospital? Why would aliens do that particular thing to me, as opposed to the other billion things they might do? Why would my arm turn into a tentacle on that morning, after remaining an arm through every other morning of my life? And in all cases I must look for an argument compelling enough to make that particular prediction in *advance*, not mere compatibility. Once I already knew the outcome, it would become far more difficult to sift through hypotheses to find good explanations. Whatever hypothesis I tried, I would be hard-pressed not to allocate more probability mass to yesterday’s blue-tentacle outcome than if I extrapolated blindly, seeking the model’s *most* likely prediction for tomorrow.

A model does not always predict all the features of the data. Nature has no privileged tendency to present me with solvable challenges. Perhaps a deity toys with me, and the deity’s mind is computationally intractable. If I flip a fair coin there is no way to further explain the outcome, no model that makes a better prediction than the maximum-entropy hypothesis. But if I guess a model with no internal detail or a model that makes no further predictions, I not only have no reason to believe that guess, I have no reason to care. Last night my arm was replaced with a blue tentacle. Why? Aliens! So what will they do tomorrow? Similarly, if I attribute the blue tentacle to a hallucination as I dream my life away in a coma, I still don’t know any more about what I’ll hallucinate tomorrow. So why do I care whether it was aliens or hallucination?

What might be a *good* explanation, then, if I woke up one morning and found my arm transformed into a blue tentacle? To claim a “good explanation” for this hypothetical experience would require an argument such that, contemplating the hypothetical argument *now*, *before* my arm has transformed into a blue tentacle, I would go to sleep worrying that my arm *really would* transform into a tentacle.

People play games with plausibility, explaining events they expect to never actually encounter, yet this necessarily violates

the laws of probability theory. How many people who thought they could “explain” the hypothetical experience of waking up with their arm replaced by a tentacle, would go to sleep wondering if it might really happen to them? Had they the courage of their convictions, they would say: I do not expect to ever encounter this hypothetical experience, and therefore I cannot explain, nor have I a motive to try. Such things only happen in webcomics, and I need not prepare explanations, for in real life I shall never have a chance to use them. If I ever find myself in this impossible situation, let me miss no jot or tittle of my valuable bewilderment.

To a Bayesian, probabilities are anticipations, not mere beliefs to proclaim from the rooftops. If I have a model that assigns probability mass to waking up with a blue tentacle, then I am nervous about waking up with a blue tentacle. What if the model is a fanciful one, like a witch casting a spell that transports me into a randomly selected webcomic? Then the *prior probability* of webcomic witchery is so low that my *real-world* understanding doesn’t assign any significant weight to that hypothesis. The witchcraft hypothesis, if taken as a given, might assign non-insignificant likelihood to waking up with a blue tentacle. But my anticipation of that hypothesis is so low that I don’t anticipate any of the predictions of that hypothesis. That I can conceive of a witchcraft hypothesis should in no wise diminish my stark bewilderment if I actually wake up with a tentacle, because the real-world probability I assign to the witchcraft hypothesis is effectively zero. My zero-probability hypothesis wouldn’t help me *explain* waking up with a tentacle, because the argument isn’t good enough to make me *anticipate* waking up with a tentacle.

In the laws of probability theory, likelihood distributions are fixed properties of a hypothesis. In the art of rationality, to *explain* is to *anticipate*. To *anticipate* is to *explain*. Suppose I am a medical researcher, and in the ordinary course of pursuing my research, I notice that my clever new theory of anatomy seems to permit a small and vague possibility that my arm will transform into a blue tentacle. “Ha ha!” I say, “how remarkable and silly!” and feel ever so slightly nervous. *That* would be

a good explanation for waking up with a tentacle, if it ever happened.

If a chain of reasoning doesn't make me nervous, in advance, about waking up with a tentacle, then that reasoning would be a poor explanation if the event *did* happen, because the combination of prior probability and likelihood was too low to make me allocate any significant real-world probability mass to that outcome.

If you start from well-calibrated priors, and you apply Bayesian reasoning, you'll end up with well-calibrated conclusions. Imagine that two million entities, scattered across different planets in the universe, have the opportunity to encounter something so strange as waking up with a tentacle (or—gasp!—ten fingers). One million of these entities say “one in a thousand” for the prior probability of some hypothesis *X*, and each hypothesis *X* says “one in a hundred” for the likelihood of waking up with a tentacle. And one million of these entities say “one in a hundred” for the prior probability of some hypothesis *Y*, and each hypothesis *Y* says “one in ten” for the likelihood of waking up with a tentacle. If we suppose that all entities are well-calibrated, then we shall look across the universe and find ten entities who wound up with a tentacle because of hypotheses of plausibility class *X*, and a thousand entities who wound up with tentacles because of hypotheses of plausibility class *Y*. So if you find yourself with a tentacle, and *if* your probabilities are well-calibrated, then the tentacle is more likely to stem from a hypothesis you would class as probable than a hypothesis you would class as improbable. (What if your probabilities are poorly calibrated, so that when you say “million-to-one” it happens one time out of twenty? Then you're grossly overconfident, and we adjust your probabilities in the direction of less discrimination and greater entropy.)

The hypothesis of being transported into a webcomic, even if it “explains” the scenario of waking up with a blue tentacle, is a poor explanation because of its low prior probability. The webcomic hypothesis doesn't contribute to explaining the tentacle, because it doesn't make you anticipate waking up with a tentacle.

If we start with a quadrillion sentient minds scattered across the universe, quite a lot of entities will encounter events that are very likely, only about a mere million entities will experience events with lifetime likelihoods of a billion-to-one (as we would anticipate, surveying with infinite eyes and perfect calibration), and not a single entity will experience the impossible.

If, somehow, you really did wake up with a tentacle, it would likely be because of something much more probable than “being transported into a webcomic,” some perfectly normal reason to wake up with a tentacle which you just didn’t see coming. A reason like what? I don’t know. Nothing. I don’t anticipate waking up with a tentacle, so I can’t give any good explanation for it. Why should I bother crafting excuses that I don’t expect to use? If I was worried I might someday need a clever excuse for waking up with a tentacle, the *reason I was nervous about the possibility* would be my explanation.

Reality dishes out experiences using probability, not plausibility. If you find out that your laptop doesn’t obey Conservation of Momentum, then reality must think that a perfectly normal thing to do to you. How could violating Conservation of Momentum possibly be perfectly normal? I anticipate that question has no answer and will never need answering. Similarly, people do *not* wake up with tentacles, so apparently it is *not* perfectly normal.

There is a shattering truth, so surprising and terrifying that people resist the implications with all their strength. Yet there are a lonely few with the courage to accept this satori. Here is wisdom, if you would be wise:

*Since the beginning
Not one unusual thing
Has ever happened.*

Alas for those who turn their eyes from zebras and dream of dragons! If we cannot learn to take joy in the merely real, our lives shall be empty indeed.

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